OECD ECONOMIC SURVEYS

AUSTRALIA

1997

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ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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BASIC STATISTICS OF AUSTRALIA

THE LAND

Area (1 000 sq.km)	7 682.3	Population of major cities, 1995 (1 000):	
Agricultural area, 1986-87, per cent of total	61	Sydney	3 773
Urban population, 1991, per cent of total	85	Melbourne	3 218
		Brisbane	1 489
		Perth	1 263
		Adelaide	1 081

THE PEOPLE

Population, June 1995 (1 000)	18 054	Civilian employment, 1995 (1 000)	8 217
Number of inhabitants per sq.km	2.4	*of which:* Agriculture	411
Natural increase, 1995 (1 000)	132	Industry [1]	1 885
Net Migration, 1995 (1 000)	84	Other activities	5 921

PARLIAMENT AND GOVERNMENT

Composition of Parliament following latest elections:

Party	Senate	House of Representatives
Australian Democrats	7	–
Australian Labor Party	29	49
Independent	1	5
Greens	2	–
Liberal Party of Australia	31	75
National Party of Australia	5	18
Country Liberal Party	1	1
Total	76	148

Present government: Liberal/National Party coalition
Next general elections for House of Representatives: March 1999

PRODUCTION

Gross domestic product, 1995 (A$ million)	470 390	Gross fixed capital formation, 1995 Percentage of GDP	20.4

GENERAL GOVERNMENT SECTOR, PER CENT OF GDP, 1995

Current disbursement	36.2	Current revenue	34.7
Current transfers	14.7	*of which:* Direct taxes	17.4

FOREIGN TRADE

Main exports, 1994, per cent of total:		Main imports, 1994, per cent of total:	
Food, beverages and tobacco	21.5	Food, beverages and tobacco	4.5
Raw materials	20.5	Raw materials	2.7
Fuels	16.6	Fuels	4.8
Machinery and transport equipment	11.7	Machinery and transport equipment	46.2
Other manufactured products	29.7	Other manufactured products	41.8

THE CURRENCY

Monetary unit: Australian dollar	Currency unit per US dollar, average of daily figures:	
	Year 1995	1.3496
	October 1996	1.2620

1. Including mining, electricity, gas and water, and construction.

Note: An international comparison of certain basic statistics is given in an annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Australia by the Economic and Development Review Committee on 28 October 1996.

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After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 18 November 1996.

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The previous Survey of Australia was issued in May 1995.

Assessment and recommendations

A robust upswing with a solid record of job creation...

The Australian economy has entered its sixth year of recovery, with real output now some 17 per cent above its previous cyclical peak, a growth performance among the best in the OECD area. Although the current upswing is not as strong as its predecessor of the 1980s, due to a slower take-off, it is steadier than earlier expansion periods. Concerns about a ''jobless'' recovery in the early years of the upswing dissipated as employment began to grow strongly from 1993 onward, and the rate of unemployment fell from an annual peak of nearly 11 per cent in 1993 to 8.6 per cent in 1995. An easing in economic growth in 1995 led to slower growth in employment, which kept the unemployment rate from falling below the 8½ per cent mark in a sustained way in spite of declining labour force participation. Employment growth was weak during late 1995 and early 1996 despite continuing solid output growth. Nevertheless in the four years to mid-1996 the total number of employed increased by 703 thousand or 9 per cent, a solid record of job creation.

...in a low inflation environment

What distinguishes the current upswing from the earlier ones is the absence of strong inflationary pressures. In the first half of the 1990s underlying inflation was running at an annual average rate of 2½ per cent. This compares with 10¼ per cent and 7¾ per cent in the 1970s and the 1980s, respectively. As well, clear downward shifts have been observed in inflation expectations. The low inflation envi-

ronment has been nurtured through mounting competitive pressures worldwide in tradeable goods and services, scheduled tariff reductions, deregulation of sheltered domestic service sectors and increasingly flexible labour markets. And the consistently anti-inflationary slant given to monetary policy since the late 1980s, which was formalised by the adoption of an explicit inflation target in 1993, undeniably has played a key role in bringing it about. This is most clearly demonstrated by the Reserve Bank of Australia (RBA) raising the cash rate cumulatively by 275 basis points in three pre-emptive moves during the second half of 1994 to obviate any lasting worsening of inflation and inflation expectations in the wake of a sharp acceleration in the growth of non-farm output.

Sustained output growth is projected with unemployment falling and inflation remaining low

By the middle of 1996, a number of indicators suggested that there was room for some easing of monetary policy. In particular, the strong Australian dollar had substantially tightened overall monetary conditions while estimates of expected inflation were subsiding. Moreover, the Commonwealth Government announced a tight budget, which implied a small near-term contractionary fiscal impulse on the economy. Against the background of the prevailing high rate of unemployment, this was seen as risking an undue slowing of output and employment. Reflecting the improved inflation outlook and the scope for stronger growth the Reserve Bank eased monetary policy in July and November 1996, lowering the cash rate by one percentage point overall to 6½ per cent. With this policy mix and on other customary technical assumptions, notably that of unchanged exchange rates, output is projected to grow at 3¼ per cent over the next two years, around estimates of long-term potential growth. Employment growth may pick up gradually, from 1¼ per cent in 1996 to 2 per cent in 1998, reducing the rate of unemployment gradually, to around 8 per cent by 1998. The small reduction in the

unemployment rate over the outlook period is not expected to put upward pressure on wages. Nonetheless, with the effects of currency appreciation waning inflation is likely to edge up towards 3 per cent by 1998.

Wages a risk to the inflation outlook but the RBA remains vigilant...

Wage increases of 5 to 6 per cent concluded in private-sector enterprise agreements pose some upside risk to inflation projections. They risk raising unit labour costs by more than 2 to 3 per cent unless accompanied by efficiency gains well above trend productivity growth. The RBA has explicitly recognised the policy significance of wage developments and indicated on various occasions that it would not hesitate to raise interest rates if price pressures looked to be moving out of line with the inflation target.

...and the recent public statement added to credibility of monetary policy

The commitment of the Reserve Bank to meet its inflation target has gained additional credibility by the recent *Statement on the Conduct of Monetary Policy* which sets out the mutual understanding of the Treasurer and the Governor of the RBA on the operation of monetary policy in Australia. The *Statement* comprises the new Government's official endorsement of the RBA's 2 to 3 per cent inflation target and its formulation which allows for short run variation in underlying inflation over the cycle. Another essential point is the new Government's official recognition of the Reserve Bank's independence and its sole responsibility for monetary policy matters. In addition, the *Statement* emphasises the need for transparency and accountability and, therefore, provides for specific reports by the Bank on monetary policy – to be released at roughly six monthly intervals – which will make interest rate policy more transparent and which will include information on the outlook for inflation. Consequent upon the release of the *Statement* and announcement of the Government's fiscal consolidation plans, the Australia-US long-term interest rate differential, which is widely taken to be an indicator of policy credibil-

ity, had narrowed to 1 percentage point in September 1996, after a differential of 3 percentage points a year earlier. Altogether, the Statement is helpful in enhancing the credibility of monetary policy which, however, will ultimately depend upon low inflation actually being delivered for a number of years to come.

Refinement of the inflation targeting approach may nonetheless be warranted

The target of keeping underlying inflation on average over the business cycle in a range of 2 to 3 per cent makes sense for a country where inflation expectations still appear to be affected by the high inflation of the past, and which tends to be subject to larger terms-of-trade shocks than other countries. Although the jury is still out about the magnitude of extra net benefits of bringing inflation closer to zero, the potential for lower inflation to yield non-negligible benefits might suggest a reassessment of the numerical inflation objective by the Reserve Bank once economic agents become better accustomed to the new low-inflation environment, which should reduce the economic cost of further disinflation. In addition, further efforts would be warranted to make the measure of underlying inflation better understood by the public.

A non-cyclical rise in the external deficit is a matter for concern and a fall in government saving has been the main culprit

Another concern for policy makers, which has been the subject of intense debate in Australia, is the size of the underlying current external deficit, which has risen from a little over 2 per cent of GDP up until the late 1970s to about 4½ per cent of GDP since then. The resulting substantial increase in net external liabilities has exposed Australia to shifts in investor sentiment, potentially amplifying the impact on the domestic economy of adverse external shocks, even though a large share of foreign equity holdings has tended to limit the degree of exposure to such events. That the trend rise in the current account deficit has been due to a significant decline in national saving rather than a rise in investment is a worrying feature of the devel-

opment. National saving has fallen markedly since the 1960s, averaging only 17 per cent of GDP so far in the 1990s. This decline is greater than that which has occurred in most other OECD countries, leaving Australia with a national saving rate far below the average for OECD countries. Most of the decline in national saving is accounted for by a fall in government saving since the early 1970s. The general government budget balance has moved from a small surplus on average to deficits which have risen in each subsequent business cycle, reaching almost 5 per cent of GDP in the early 1990s. The long-term deterioration in government saving has occurred almost entirely in the Commonwealth sector.

Fiscal consolidation has been reinforced...

Consolidation of the Commonwealth budget got under way in FY 1993/94, but progress was slow relative to initial plans. By FY 1995/96, the underlying deficit (which primarily excludes from the headline deficit the proceeds of asset sales and repayments of State debt to the Commonwealth) had been halved, falling to 2.1 per cent of GDP. But by Australian standards, this was still a large deficit for the stage reached in the business cycle. In view of the contribution of the Commonwealth budget position to the long-term deterioration in national saving and the current account balance, the new Government has placed a high priority on reducing the structural budget deficit. Accordingly, the budget consolidation already in the pipeline was substantially accelerated in the 1996 Budget. Measures were announced, cutting the underlying deficit by A$ 3.9 billion (0.8 per cent of GDP) in FY 1996/97 and A$ 7.2 billion (1.3 per cent of GDP) in the following year. In contrast to recent experience, these reductions in the deficit are to be achieved mainly through lower outlays. Experience in some other countries suggests that this increases the likelihood that consolidation will be sustained. With these measures, the budget is projected to move into surplus by

FY 1998/99. General government finances are also projected to be in slight surplus in FY 1998/99.

...and the institutional framework is to be strengthened

The new Government also announced that legislation would be introduced establishing a new fiscal framework, known as the *Charter of Budget Honesty*, aimed at encouraging more prudent fiscal management and requiring greater transparency in reporting fiscal outcomes. This legislation introduces principles for the sound management of fiscal policy and requires each government to outline its fiscal strategy and report performance on a regular basis. Governments will be required to identify medium-term objectives for fiscal policy and any temporary fiscal measures aimed at dampening economic downturns. In the latter case, the process for reversing these measures must also be identified. These features of the legislation provide a valuable counterweight to the pressures governments face to support economic activity in the short term at the expense of rising deficits and government debt over time.

But consolidation efforts need also to focus on longer term issues

The new Government has also set a medium-term fiscal objective of achieving underlying budget balance on average over the economic cycle. This stands in sharp contrast to the experience of the last 20 years, where the underlying budget position has been in deficit, by on average 1½ per cent of GDP. While the 1996 Budget goes a long way towards eliminating the deficit over the next two years, further consolidation measures are likely to be required to achieve sufficiently large surpluses during the remainder of the present expansion to compensate for the inevitable deterioration in the budget position during the next economic downturn. As performance against the target will only be clear with the benefit of hindsight, it will be important for the authorities to take a cautious approach to implementing fiscal policy within the medium term framework. The *Charter of Budget Honesty* requires the Government to report on

a regular basis on the budgetary implications of expected demographic and other developments over a 40 year time span, including the effects of ageing of the population. The implications of these longer term developments will need to be taken into account when assessing the continued appropriateness of the medium term fiscal target.

Industrial relations reform...

The recent coexistence of solid economic growth and stubbornly high unemployment suggests that relying exclusively on economic growth to bring unemployment down to socially acceptable levels would – if not assisted by structural policies – require rates of output growth which are unlikely to be sustainable over the long run. Reform of the industrial relations system remains a priority for reducing structural unemployment. It has been broadly accepted since the late 1980s that a shift in the focus of wage bargaining from national awards to the level of the enterprise was necessary to enhance the organisational flexibility of workplaces and, hence, their capacity to adapt to economic change. While progress has been made, further changes in the industrial relations framework are clearly needed. The new Government considers that further industrial relations reform is central to its structural reform agenda. The 1996 reforms aim to facilitate the shift to enterprise bargaining by expanding the options for formalised agreement-making – providing for individual Australian Workplace Agreements and more flexible arrangements for collective certified agreements between employers and unions or employers and employees directly. Simplifying awards to a safety net of wages and conditions of employment, inserting provisions in awards which allow for their more flexible application at individual work places, and focusing increases in award wage rates on the low paid will also help give primacy to bargaining at the enterprise level. It will also be important that future ‘‘safety net’’ (centrally-determined) increases in award wages be modest and focused on the low

paid so as to maintain strong incentives for the shift towards enterprise bargaining.

...to enhance labour market flexibility

The industrial relations reforms aim to ensure the bargaining parties are treated equally. In particular, they will limit the excessive use of bargaining power by not allowing the uninvited involvement of third parties in bargaining, banning union preference clauses and compulsory unionism, restoring provisions prohibiting secondary boycotts to the *Trade Practices Act* and strengthening sanctions against unlawful industrial action. In addition, the new legislation will facilitate the creation of new unions, including enterprise unions. These measures will help ensure that wage outcomes from bargaining reflect the circumstances of individual enterprises, including their productivity performance, and so help create the conditions for stronger growth in employment without igniting inflationary pressures. Some of these measures may also reduce the potential for unions to obtain a portion of the rents generated in imperfectly competitive markets for their members. Less rent-seeking behaviour by unions would result in more appropriate relative wages between sectors or firms, increasing employment opportunities.

Intensifying product market competition through further tariff cuts...

Rent-seeking behaviour can also be limited by removing impediments to competition in product markets. Australia has undertaken a programme of structural reform aimed at improving the international orientation and competitiveness of the economy. The reductions in import protection already effected in Australia and those scheduled through to 2001 have been vital in this regard. The new Government is committed to those scheduled reductions and more generally to advancing the structural reform agenda. The Productivity Commission is currently reviewing post-2000 assistance arrangements for the automotive industry and will shortly undertake a similar review of the textile, cloth-

ing and footwear industries. The Government has also undertaken to review the current general tariff of 5 per cent by 2000 as part of its plan to implement the APEC goal of free and open trade and investment in the Asian Pacific region. It would be desirable if further reductions in industry protection, especially in the more heavily protected areas, were scheduled following completion of the reviews.

...and implementing the Hilmer Committee proposals

Similarly, implementation of the national competition policy recommended by the Hilmer Committee and agreed at the Council of Australian Governments meeting in April 1995 will enhance product market competition, especially in infrastructure services dominated by Government Business Enterprises. The legislative foundations of this reform are already in place: amendments have been made to the *Trade Practices Act 1974* extending its purview to the unincorporated sector and State and Territory Government Business Enterprises; the new competition institutions – the Australian Competition and Consumer Commission and the National Competition Council – have been created; the national access regime, governing third party access to services provided by significant infrastructure facilities with natural monopoly characteristics has come into effect; and price oversight of monopoly or near monopoly government businesses which supply goods or services has been established. Getting the most out of these reforms will require the co-operation of State and Territory governments, both in terms of implementing competitive neutrality requirements and identifying and reforming any legislation or regulation which is inconsistent with agreed competition principles. Such co-operation from the States is likely, especially as adherence to the review process is necessary to qualify for payments from the Commonwealth as a way of sharing the extra tax revenues arising from greater overall productivity stemming from these reforms. Finally, the new Government has commissioned a major

inquiry into the financial system to examine the appropriate regulatory framework for financial products in light of global and technological developments in the industry.

Improved targeting of income support...

Australia introduced during the 1980s in-work benefits for low income earners with children so as to ease the work incentive problems for this group posed by the social security system. There has been a similar response to this problem in many other OECD countries. But Australia went further in the income support reforms, providing each member of an unemployed couple with an individual entitlement to half of their income support, with spousal income below the level at which their benefit entitlement is exhausted not affecting the entitlement. These reforms appear to have been effective in improving incentives for unemployed couples to take part-time or low-paid full-time jobs.

...and streamlined labour market assistance arrangements

The new Government has proposed a radical change in employment services and labour market programmes. The goal of the proposed changes is to make labour market assistance more responsive to both jobseeker and employer needs, through the Employment Placement Enterprises (EPEs), compared with assistance that has been offered in the past. Under the current system, Contracted Case Managers have an incentive to place individuals as rapidly as possible to generate a fee, but they may not always take the most cost-effective path since they do not have to pay directly for programme slots. The proposed system, by making EPEs take on all of the costs for placement, should lead to a better identification of actual client needs at minimum cost, over time. In effect, it will become more client focused, rather than being programme driven. Another advantage of the proposed system is that it will ensure clear separation of purchasing and providing functions in deliv-

ering labour market assistance and full competitive neutrality between the EPEs.

The education and training system has evolved in response to pressures for change

Deteriorating labour market prospects for the low skilled, especially early school leavers, and government policy to improve the skills of the labour force, have created pressures on the education and training system which have resulted in considerable change. It has had to adapt to a substantial increase in demand for places in post-compulsory education and training – the retention rate until the final year of secondary school has doubled in the past decade and the number of university students has increased by over 40 per cent. One way to enable young people to stay in education and training longer was to develop more pathways from school to work. Traineeships, a form of short apprenticeship, were introduced in the mid-1980s to extend apprenticeships beyond the traditional male-dominated trades and, more recently, students in the final two years of secondary school have been given the option of combining their studies with work-based training. There also has been pressure to enhance the labour-market relevance of education and training. This has been done through increased input from employers on course content. The most important development in this regard has been in Vocational Education and Training (VET), where certification now is based on industry-determined competency standards, rather than on time served. In accommodating the increased participation in post-compulsory education and training, there has been a concern to promote social equity objectives and to contain any increase in Government expenditures. An innovative measure which advanced both objectives was the introduction of the Higher Education Contribution Scheme (HECS), which requires university students to contribute towards tuition costs, with repayment of students' loans (on which the real interest rate is zero) to finance these contributions being income contingent. Steps

have also been taken to raise efficiency in the education and training sector, notably through the introduction of competitive pressures into the allocation of some State Governments' VET budgets and through the merging of universities and the former colleges of advanced education (which were similar to Polytechnics in the United Kingdom).

But scope remains to improve the school-to-work transition...

A concern remains that about half of all school leavers in any one year enter the labour force directly, rather than continuing in some form of education and training. Despite efforts to enhance VET as a pathway from school to work, it has not shared in the expansion in tertiary education. The new Government intends to remove a number of obstacles to VET, principally through industrial relations reform and reforms to the apprenticeship and traineeship system. These reforms will enable employers and apprentices to enter into workplace agreements which will provide for more flexible employment and training arrangements. These reforms will also facilitate the development of apprenticeships and traineeships better adapted to employers' requirements and simplify the process for gaining official recognition for programmes. School-based vocational education is to be better articulated to apprenticeships and traineeships and VET qualifications are to be made more portable through the Australian Qualifications Framework, being phased in since 1995. These reforms address most of the obstacles which have impeded the development of VET. It remains to be seen, however, whether apprenticeships will prove popular in the absence of the social institutions which are found in the countries with a strong apprenticeship tradition.

...and to increase competitive pressures in the VET and higher education sectors

The lack of competitive pressures in the VET and higher education sectors is likely to have hampered efficiency and dynamism. Private providers are able to compete for at best a small proportion of public VET funds in some States and are not allowed to compete at all for public funds at the university level: the education sector is exempt from the purview of competition law. Yet there is no special feature of tertiary education which suggests that preventing private providers from competing for public funds might be efficient – indeed, private providers play a much greater role in many other countries. States should go much further in allowing private providers to compete with Technical and Further Education (TAFE) institutions for public VET funds, eventually allocating them all on a competitive basis. And the Commonwealth Government could consider allowing universities to set their fees, and transforming its grants into vouchers which students could spend at the university of their choice, including private universities. These reforms would sharpen incentives for providers to offer services to users which represent better value for money. The introduction of vouchers would not, however, be free of administrative complexities and difficulties.

Reforms in the funding of tertiary education are welcome...

A live issue in Australia has been the extent to which tertiary education should be paid for by students. The Government has traditionally funded a high proportion of expenditure on tertiary education. This has been justified mainly on the grounds that it confers external benefits on society. However, there is little agreement about the magnitude of these benefits beyond the observation that they are broadly inversely related to the level of education. Judging by the small contributions from university students towards tuition costs until this year, the Australian authorities appear to have been of the view that the overwhelming bulk of the benefits from university education were external. The new Government has proposed to increase HECS sub-

stantially from 1997, although the Government will continue to pay more than students towards university tuition costs. This move reflects the Government's view that its contributions towards university tuition costs were excessive in relation to likely external benefits.

...but anomalies remain

Currently TAFE students pay only a small up front fee, even though they may gain credit towards a degree. Increasing fees for higher level TAFE courses would remove the present anomaly between the two sectors and improve the targeting of Government subsidies for VET. Consideration would then need to be given to the introduction of HECS-type collection mechanisms if these fees approach levels similar to those payable by students in universities. Moreover, although the small tuition fees charged by TAFE institutions for courses at a level up to the equivalent of the final year of secondary school do not appear to provide a disincentive to participation in TAFE, consideration should be given to their abolition. This would bring the treatment of TAFE students into line with that accorded to secondary school students, and avoid the anomalous situation where TAFE students have to pay fees to do a course which leads to credentials similar to those obtainable free of charge in secondary schools. The proposed changes to HECS and the extension of HECS-type mechanisms to VET would improve incentives for students to make more considered choices about courses within and between sectors. They would also represent a move towards fairer sharing of the costs of higher education as these students on average have higher lifetime earnings than the average taxpayer.

A wrap-up

To sum up, Australia's economic performance in the current upswing has been impressive in many respects. High growth has been recorded for both output and employment, and the rate of unemployment has fallen significantly. And

this has taken place in a low inflation environment. This very good performance is the fruit of sustained policy efforts to make the economy function better, increase the value for money in public services and contain inflation. The new Government is committed to continuing the reform process to enhance long-term growth prospects, giving priority to placing macroeconomic policy on a sound institutional basis and reforming the labour market. Continuing high levels of unemployment and low domestic saving remain areas of concern. However, with the benefit of the delayed effects of past reforms and further initiatives of the new Government, improvements in these areas can be expected in the years ahead.

I. Recent trends and short-term prospects

Solid economic growth

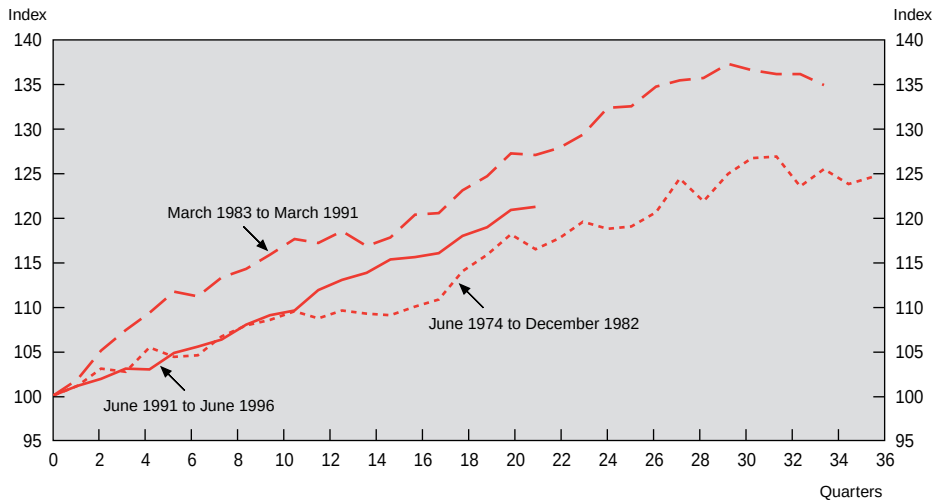
The Australian economy completed its fifth year of recovery in the middle of 1996. After rather disappointing growth of real GDP[1] of 0.6 per cent in fiscal year[2] (FY) 1991/92, the economy gathered steam: in the past three fiscal years growth averaged 4.2 per cent. And in the second quarter of 1996 seasonally-adjusted (s.a. hereafter) real GDP was 17 per cent above its previous cyclical peak of six years earlier.

Largely because of its slow take-off, the current recovery has not been as strong as the preceding upswing in the 1980s. Subdued business investment was a major impediment to faster growth during the first two years of the present upswing, caused by balance-sheet restructuring of firms after a decade of rising gearing ratios. In contrast, much lower corporate debt-to-equity ratios allowed business investment to be a major engine of growth in the early 1980s. On the other hand, the present upswing takes place in an environment of significantly lower inflation than in the preceding two decades. It also appears steadier than the expansion periods of the 1970s and 1980s (Figure 1). Although the current cycle has not finished yet, its twenty quarters of growth already represent the longest sequence of uninterrupted expansion since the early 1960s.

Economic growth accelerated vigorously in late 1993 and peaked at an annualised trend[3] rate of 6.2 per cent in the first quarter of 1994. With only little slowing during the remainder of the year, the annual average growth of GDP amounted to 4.8 per cent in calendar year 1994. But the true strength of cyclical forces is not fully reflected in total-economy GDP measures as (income-based) non-farm GDP increased by 5.7 per cent in 1994 while, hit by the drought in eastern states, farm output fell by a substantial 16 per cent.

Figure 1. **THE CURRENT EXPANSION COMPARED**

Cumulative growth of real GDP



Source: OECD, *National Accounts.*

The rapid growth in 1994 gave rise to worries of overheating and the repetition of earlier boom-and-bust cycles if the economy continued to expand at such a pace. In consequence, the Reserve Bank of Australia (RBA) rigorously tightened monetary policy during the second half of 1994. Growth in activity slowed in late 1994 and in the first half of 1995 reflecting primarily the (long awaited) downturn in the housing cycle, a moderation of the sharp increase in machinery and equipment investment and a weakening of public investment growth (Table 1). With the slowing of activity occurring at about the same time or shortly after the RBA's interest rate hikes, it is difficult to separate the impact over this period of the monetary policy tightenings on the economy from the effects of an independent slowing in some parts of the economy. There has, however, been some anecdotal evidence of more rapid initial effects of the 1994 tightenings of monetary policy, possibly reflecting announcement effects and the memory of the significant impact of the policy tightenings of the late 1980s. Altogether, economic growth turned out to be 3.5 per cent in calendar year 1995.

Table 1. **Demand and output**

Percentage changes, FY 1989/90 prices

	From previous period				From previous period, seasonally adjusted annual rate		
	Calendar years		Fiscal years[1]		1995		1996
	1994	1995	1994/95	1995/96	I	II	I
Consumption							
Private	4.3	4.2	4.7	4.3	3.2	4.8	4.3
Public	5.2	3.1	3.7	2.5	5.4	0.3	3.7
Gross fixed investment	12.2	1.8	11.2	0.5	–1.8	–1.9	8.5
of which:							
Government	1.4	–0.3	4.5	3.0	–7.4	7.5	5.7
Private	13.5	2.0	10.7	0.4	–1.5	–2.0	7.7
Dwellings[2]	11.0	–9.9	2.3	–13.6	–14.2	–14.3	–11.4
Other construction	6.6	17.1	6.1	20.8	15.1	41.6	–8.0
Equipment	21.9	6.8	21.5	5.8	5.5	–5.3	32.5
Public enterprises	6.0	6.4	20.7	–0.2	0.9	–8.4	16.7
Final domestic demand	6.1	3.5	5.9	3.1	2.5	2.5	5.1
Change in stockbuilding[3]	–0.4	0.8	0.4	0.2	–0.2	0.4	–0.3
Total domestic demand	5.7	4.2	6.3	3.3	2.0	3.4	4.5
Exports of goods and services	8.6	4.0	3.2	10.3	3.9	10.6	15.2
Imports of goods and services	15.3	9.5	17.7	5.6	9.0	–0.9	16.2
Change in foreign balance[3]	–1.1	–1.1	–2.7	1.0	–0.5	1.2	–0.1
GDP (expenditure-based)	4.5	3.1	3.5	4.3	0.9	5.9	4.4
Statistical discrepancy[3]	0.4	0.1	0.3	–0.5	0.4	–0.7	0.1
GDP (income-based)	4.9	3.2	3.8	3.8	1.7	4.5	4.6
of which:							
Farm	–16.0	3.6	–21.0	25.0	–8.9	58.4	14.3
Non-farm	5.7	3.2	4.8	3.2	2.0	3.1	4.3
GDP (average measures)[4]	4.8	3.5	4.1	4.1	1.8	5.1	4.4

1. Fiscal years begin 1st July.
2. Including real estate transfer expenses.
3. As per cent of GDP in the previous period.
4. Average of the expenditure, production and income measures of GDP.

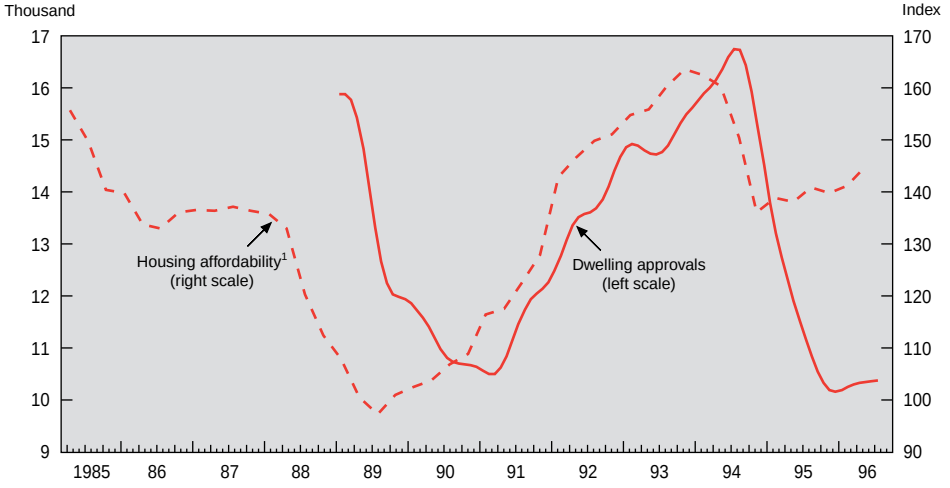
Source: Australian Bureau of Statistics and OECD estimates.

One source of the slowdown in 1995 was the completion of the residential investment cycle.[4] Housing investment had grown at annual average rates of about 11 per cent from 1992 to 1994 which raised the stock of dwellings by about 9 per cent above the previous cyclical low-point in 1991, thereby well in

excess of the population growth. The inevitable correction to excess supply began in the fourth quarter of 1994: residential investment volumes contracted from then on into 1996. The cyclical decline in demand was accentuated by higher mortgage interest rates ensuing from tighter monetary policy. The rise in mortgage interest rates amplified the fall in the ‘‘housing affordability’’ measure[5] from its peak of December 1993 (Figure 2). Altogether, housing investment fell on average by close to 10 per cent in 1995. It appears to have bottomed in the first quarter of 1996 and has shown signs of recovery since then, in line with a rising housing affordability index and a decrease in the excess supply of dwellings.

Business investment slowed during 1995, after very strong growth in 1994. The weakening was most pronounced for machinery and equipment investment which actually fell in the second half of 1995, in accordance with more subdued business confidence, lower capacity utilisation in manufacturing[6] and reports of disappointing corporate profits compared with 1994. Moreover, it appears that a number of firms decided to postpone investment decisions into 1996, when the

Figure 2. **HOUSING AFFORDABILITY**



1. Commonwealth Bank-Housing Industry Association index.
Source: Australian Bureau of Statistics and Commonwealth Bank.

economic and political outlook would become clearer. Business investment was supported in 1995 by the ongoing recovery since 1994 of private business construction, after a protracted period of sluggishness. The slowing of machinery and equipment investment occurred after a relatively brief period of recovery from recession. Government investment remained subdued over the past four calendar years: in 1995 it was about 4 per cent below its level of 1991, reflecting lower public enterprise investment (in part because of the sale of assets to the private sector), with general government investment around the same level as in 1991.

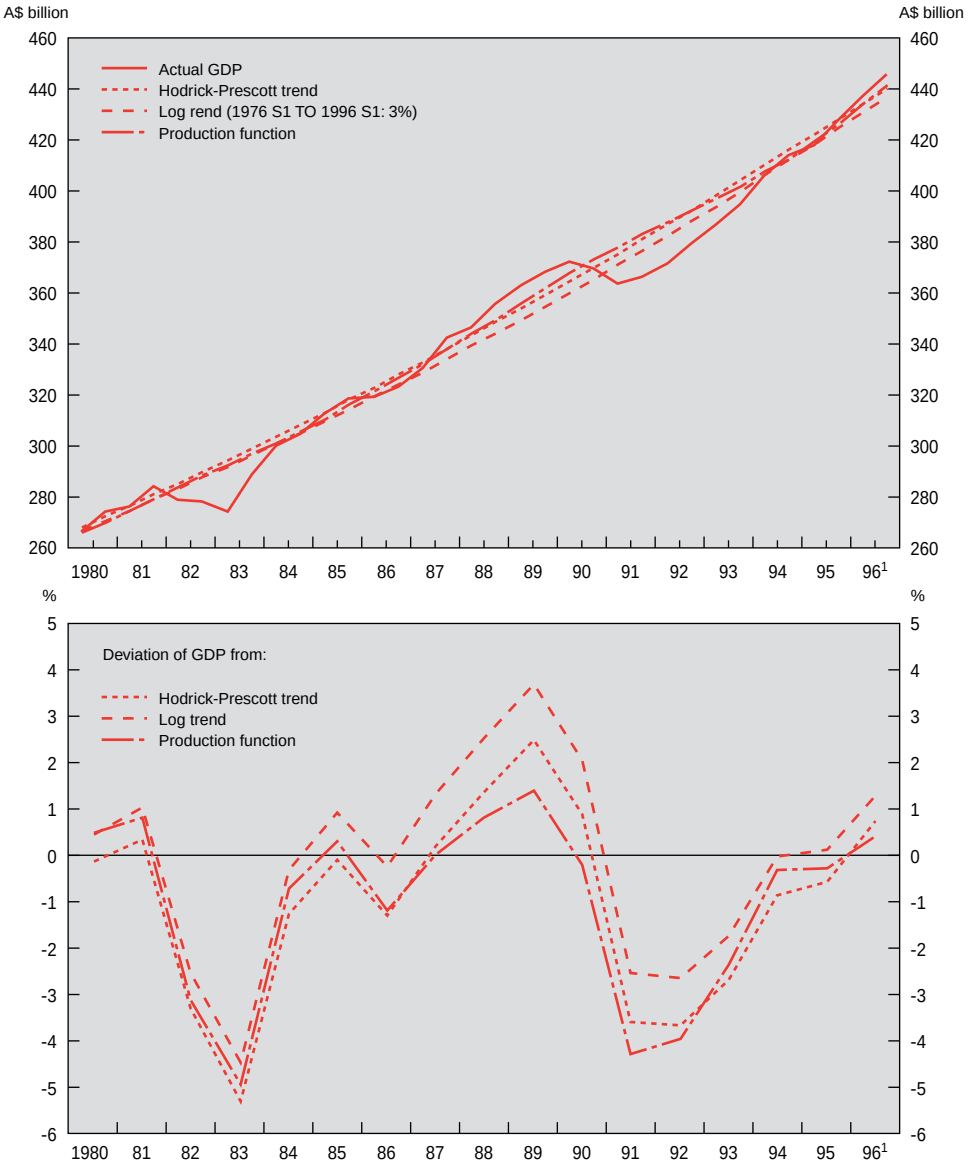
However, the recent expansion of business investment was strong enough to give a boost to the business sector capital stock. Its growth is estimated by the OECD Secretariat to have accelerated from somewhat under 2 per cent in 1993 to just below 3 per cent in 1995, thereby contributing to the return of OECD estimates of potential output growth to a rate of about 3½ per cent. On the basis of these OECD estimates the gap between the level of potential and actual output (the ''output gap'') closed in 1996[7] (Figure 3).

Household consumption was a mainstay of domestic demand right from the beginning of the economic upswing and remained so into 1996 (Figure 4). In particular in 1994 and 1995, it was bolstered by strong growth of real disposable household incomes, which were buoyed by solid growth of employment and accelerating wage growth. Household consumption grew markedly faster than personal disposable incomes in 1995, as survey indicators suggested a still favourable overall consumer sentiment, in spite of a deterioration in the second half of 1994 and only partial correction thereafter. The household saving ratio accordingly fell further in 1995 after broad stagnation in 1994, so that its trend decline since the mid-1970s continued. However, the saving rate is also likely to have been depressed by the effects of sustained drought on parts of the farm sector. Private consumption remained strong in the first half of 1996.

The pattern of economic activity in the first half of 1996 was very volatile and this makes it hard to identify a trend. GDP growth in the first quarter was extraordinarily strong: around 8 per cent (seasonally-adjusted annual rate, s.a.a.r. hereafter) and a still robust annualised rate of around 4 per cent in trend terms. Of the domestic demand components, business investment picked up vigorously, in particular machinery and equipment, in line with recovering business sentiment and the realisation of previously delayed investment plans. Demand was boosted

Figure 3. **POTENTIAL OUTPUT AND THE OUTPUT GAP**

1989-90 prices

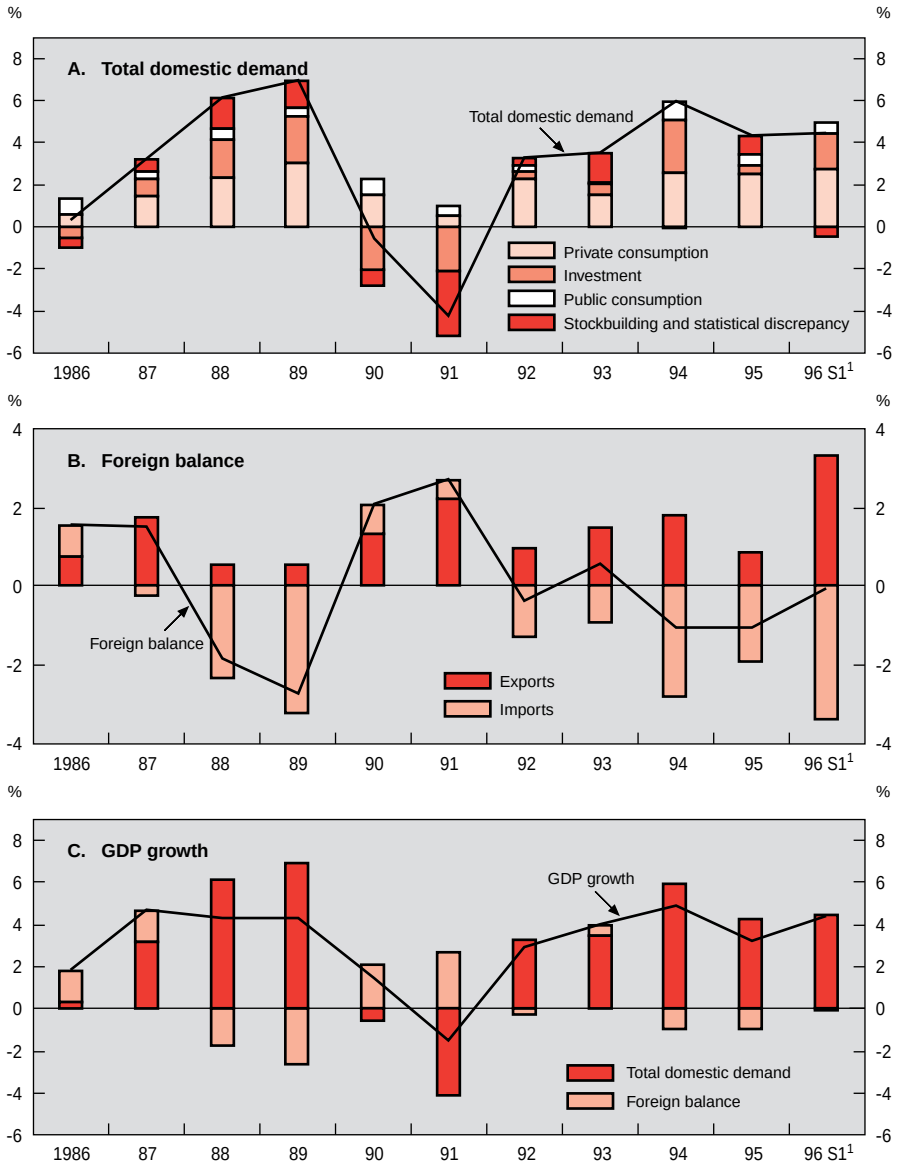


1. S2 1996 are estimates.

Source: OECD.

Figure 4. **CONTRIBUTION TO GDP GROWTH**

As a percentage change of GDP in previous year



1. Seasonally adjusted, from previous period at annual rates.

Source: Australian Bureau of Statistics, *National Accounts.*

in addition by very strong private consumption and robust public investment, especially of public enterprises. Exports of goods and services also were resilient, notwithstanding the strong Australian dollar. The strength in total demand was reflected in a rebound in imports and the buoyancy of production across a number of sectors. Profits grew strongly, with their share in GDP at an historically high level. The surge in demand and output came as a surprise to most observers with a combination of a number of short-lived influences[8] apparently exaggerating the strength of the economy. This view was confirmed by the National Accounts data for the second quarter of 1996 which showed a broadly flat GDP. It nevertheless implies a GDP growth rate of 4.8 per cent for the first half of 1996 compared with the first half of 1995 and 4.4 per cent relative to the second half of 1995 at a seasonally-adjusted annual rate.

Employment growth slows

Following broad stability in 1992 and 1993, the first two calendar years of recovery, employment responded vigorously to improved economic conditions in 1994 and 1995 and increased by 3.1 and 4.1 per cent, respectively. Part-time employment growth was stronger than full-time employment growth, although many more full-time jobs were created. With the working-age population growing at an annual rate of around 1 per cent and the participation rate picking up only hesitantly, labour force growth lagged markedly behind the growth of employment both in 1994 and 1995. Labour force participation of men, in particular, barely moved over the 1993-95 period, while female participation responded more positively to the improving labour market conditions. In consequence, the rate of unemployment fell rapidly, from an average annual peak of close to 11 per cent in 1993 to 9.7 per cent in 1994, and to 8.5 per cent of the labour force in 1995 (Table 2). In December 1995, the (s.a.) unemployment rate temporarily reached 8.1 per cent.

In the second half of 1995, the (s.a.) growth of total employment slowed markedly, probably a lagged response to the slowing expansion of output in the first half and higher real wage growth, as well as a correction to the rapid employment growth in the first half of the year. However, in spite of the solid output growth since mid-1995, employment growth continued to weaken in the first half of 1996 to under 1 per cent (s.a.a.r.). Hence, in FY 1995/96, employ-

Table 2. **The labour market**

Seasonally adjusted

	1992	1993	1994	1995	1995		1996
					S1	S2	S1
Civilian labour force[1]	0.8	0.7	1.8	2.6	2.6	2.5	1.5
of which:							
Males	0.6	0.4	1.1	1.9	1.8	1.9	1.3
Females	1.2	1.0	2.8	3.5	3.6	3.4	1.8
Employed persons[1]	–0.5	0.6	3.1	4.0	4.4	3.5	1.6
of which:							
Full-time	–2.2	0.8	2.3	3.3	3.5	3.1	1.5
Part-time	5.5	–0.2	5.9	6.0	7.3	4.7	1.9
Unemployment rate[2]	10.8	10.9	9.7	8.9	8.6	8.4	8.5
of which:							
Males	11.4	11.5	10.0	8.8	8.9	8.8	8.8
Females	10.0	10.1	9.4	8.1	8.3	7.8	8.2
Juniors looking for full-time work	33.0	31.7	30.2	27.8	27.4	28.2	27.6
Participation rate[2]	62.9	62.6	63.0	63.7	63.6	63.8	63.6
of which:							
Males	74.2	73.7	73.6	74.0	74.0	74.0	73.8
Females	51.9	51.8	52.6	53.7	53.6	53.9	53.7
Overtime (per employee), all industries (hours)[2, 4]	1.1	1.2	1.3	1.2	1.2	1.1	1.1
Average weekly hours worked[2, 3]	34.7	34.5	34.7	34.6	33.8	35.5	33.0
Job vacancies (thousand)[4]	27.2	34.2	57.2	57.3	58.1	56.4	61.4

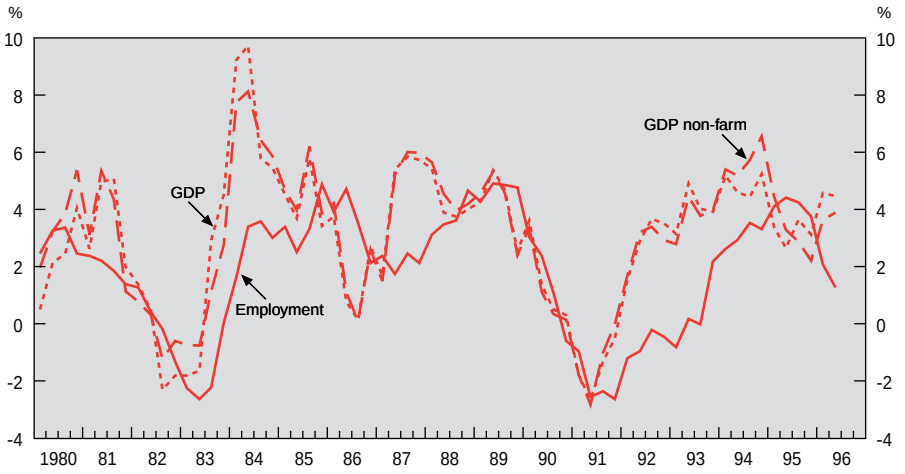
1. Percentage change from corresponding period of previous year.
2. Levels.
3. Not seasonally adjusted.
4. Quarterly data based on mid-month of quarter.

Source: Australian Bureau of Statistics and Reserve Bank of Australia.

ment rose by 2½ per cent, in contrast to output growth of 4.1 per cent in the same period and to the 4 per cent employment gain recorded in FY 1994/95 (Figure 5). The slowdown of employment growth was stronger among full-time workers. Nevertheless, employment in the middle of 1996 was some 10 per cent higher than in the trough of the recession in mid-1991. The (seasonally adjusted) share of the long-term unemployed (longer than one year) in total unemployment also fell rather steadily from its cyclical peak of 38 per cent in mid-1993 to below 30 per cent in the first half of 1996.

Figure 5. **OUTPUT AND EMPLOYMENT**

Changes over four quarters



Source: Australian Bureau of Statistics, *National Accounts,* and OECD.

The impact of the weakening employment growth since the second half of 1995 on unemployment has been cushioned by a decline in both male and female participation rates, which held the growth of the labour force broadly in line with that of total employment. However, (s.a.) monthly participation rates were very volatile so that the (s.a.) rate of unemployment jumped as high as 8.9 per cent in April 1996 but fell back to 8.3 per cent two months later. It stood at 8.8 per cent in October 1996.

The recent divergence of output and employment growth implies highly volatile aggregate labour productivity. While labour productivity growth recorded annual rates of a good 3½ per cent in 1992 and 1993, which is typical for an early phase of an upswing, it decelerated rapidly in 1994 (+1.7 per cent) and even fell in 1995 (–0.9 per cent). However, due to strong activity in the first half of 1996, labour productivity rebounded to a sound average rate of 3 per cent in FY 1995/96.

A temporary surge in inflation

With unemployment on average close to 11 per cent, wage increases were very moderate in 1993, thereby supporting price stability from the side of labour costs. But in 1994, when unemployment was still high and exceeded all published estimates of structural unemployment, clear signs of wage acceleration emerged: average weekly ordinary-time earnings of adults working full-time (AWOTE) accelerated from 2.1 per cent in 1993 to 3.4 per cent in 1994. This is probably attributable to the very fast decrease in the unemployment rate, by two full percentage points from October 1993 to December 1994. As unemployment continued to fall throughout 1995 while the inflation climate deteriorated, AWOTE accelerated further, to an average of 4.8 per cent in 1995.[9] It appears to have peaked at 5.1 per cent (s.a.a.r.) in the year to the third quarter of 1995 and to have decelerated since, to 3.8 per cent in the year to the third quarter of 1996. Average wage increases were particularly high in the private sector in 1995 (5.9 per cent), reflecting the growing impact of enterprise bargaining on wage settlements.[10] Public sector (AWOTE) pay rates were much more in line with underlying price inflation in 1995, but accelerated markedly in the first half of 1996. Award rates of pay for full-time adults, which apply to those workers not covered by enterprise agreement, continued to grow more slowly than other wage measures (Table 3).

With productivity rising strongly, aggregate unit labour costs were rather flat in the period 1992 to 1994 (Figure 6). In 1995, however, the combination of strong wage increases and weakening productivity resulted in a substantial increase in unit labour costs, which appears to have peaked in late 1995. Wage increases recorded so far for 1996 are consistent with underlying price inflation below 3 per cent in the near term, given recorded labour productivity growth and the moderating influence on prices of the recent appreciation of the exchange rate. However, future wage rises of around 6 per cent in some industries, such as banking, which were negotiated in the first half of 1996 are of concern. Settlements of executive salaries of the same order were also reported by a recent survey.[11]

During the 1992-94 period, with stable unit labour costs, inflation as measured by the consumer price index (CPI) averaged about 1½ per cent. During the same period, the Treasury underlying rate of inflation, which attempts to remove

Table 3. **Costs and prices**

Percentage change from corresponding period of previous year

	1992	1993	1994	1995	1995		1996	
					Q3	Q4	Q1	Q2
National accounts deflators[1]								
Private consumption	1.8	2.1	1.4	2.5	2.8	2.8	2.5	2.0
Total domestic demand	1.7	2.2	0.9	1.9	2.0	2.5	2.2	1.3
GDP	1.4	1.4	1.1	2.8	3.0	3.1	2.7	2.3
Exports of goods and services	1.8	1.4	–4.0	6.8	8.2	4.6	1.5	–4.1
Imports of goods and services	3.3	4.9	–5.2	1.8	2.6	1.4	–1.8	–8.8
Non-farm GDP	1.2	1.5	0.7	2.5	2.8	3.1	2.8	2.3
Consumer price index	1.0	1.8	1.9	4.6	5.1	5.1	3.7	3.1
Underlying inflation	2.5	2.0	2.0	2.7	3.1	3.2	3.3	3.1
Average weekly earnings								
All employees	2.5	2.8	3.0	2.8	2.0	2.4	2.6	3.0
Ordinary time of full-time work	3.4	2.1	3.4	4.8	5.1	4.9	4.2	3.9
Nominal non-farm unit labour costs[1]	0.6	0.7	–0.7	3.8	3.6	6.2	3.5	2.2
Award rates of pay, full-time adults (hours)	2.5	1.0	1.4	1.5	1.9	1.7	1.7	1.5

1. Derived from seasonally adjusted series.

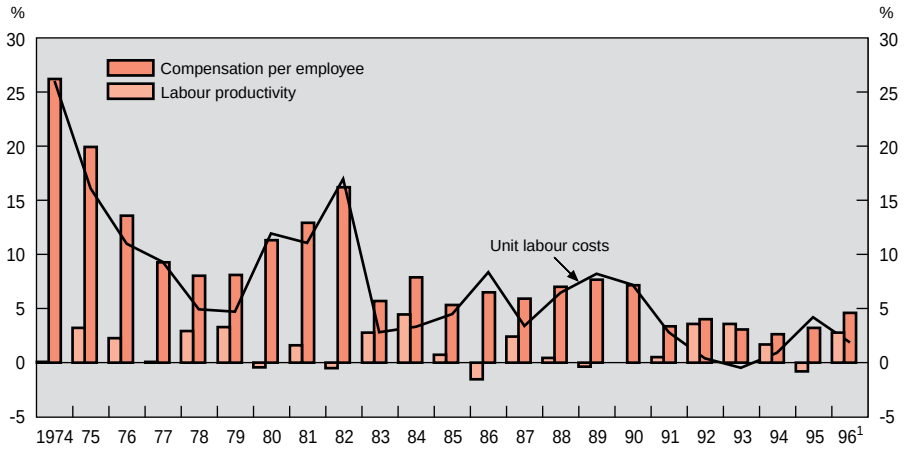
Source: Australian Bureau of Statistics, Reserve Bank of Australia and OECD.

most of the effect of changes in tax rates, charges and short-term interest rates from the CPI,[12] fluctuated around 2 per cent. These rates of inflation were below the average for the OECD as well as for the five major trading partner countries (Figure 7). A marked difference from many other low-inflation countries is the fact that during the 1992-94 period price stability was not assisted by subdued economic growth.

The underlying rate of inflation remained unchanged at about 2 per cent in late 1994 and early 1995. From the second quarter of 1995, underlying inflation accelerated, peaking at a twelve-monthly rate of 3.3 per cent in the first quarter of 1996. The acceleration in underlying inflation was in line with a pick-up in wages growth, but also reflected some effect of indirect tax increases. Also, prices of imported intermediate goods were inflated by the fall in the exchange rate during the first half of 1995. Over the course of 1995, producer prices of final manufactured goods rose by 3.5 per cent, which is likely to have fed into consumer prices. However, underlying inflation fell back to 3.1 per cent in the second quarter of 1996 and then to 2.4 per cent in the third quarter. The profile of underlying

Figure 6. **LABOUR COST AND PRODUCTIVITY**

Percentage change

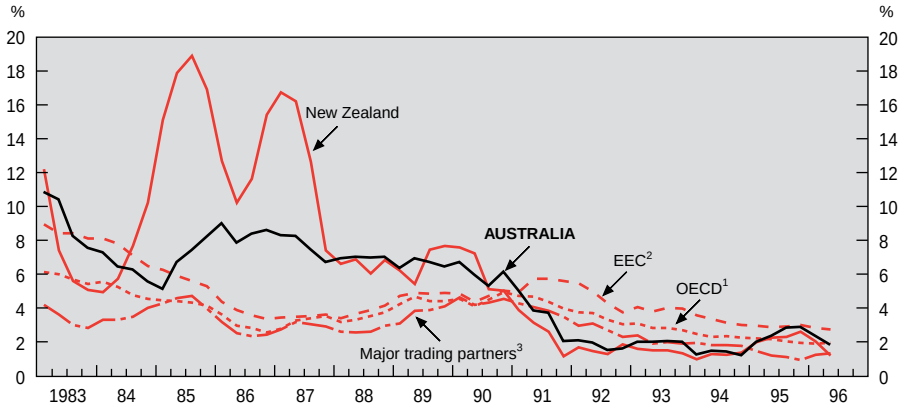


1. OECD estimates.

Source: OECD.

Figure 7. **INTERNATIONAL COMPARISON OF CONSUMER PRICES**

Change of the private consumption deflator over four quarters



1. Total OECD excludes the Czech Republic, Hungary, Turkey and Mexico and is based on GDP weights.
2. Using GDP weights.
3. Japan, United States, United Kingdom, New Zealand and Korea; calculated using trade weights.

Source: OECD.

inflation suggests that pre-emptive policy was successful in curbing the genuine inflationary pressures presaged in 1994 and in keeping inflation in check during the economy's long expansionary phase.

Headline inflation accelerated in late 1994, prior to the pick-up in underlying inflation, largely due to rising mortgage interest charges flowing from increases in the cash rate by the Reserve Bank of Australia in the second half of 1994. Further acceleration in the headline inflation rate reflected the increase in underlying inflation, while some (other than mortgage interest charges) also had an influence: domestic agricultural product prices were inflated by the drought; and tax increases by both the Commonwealth and State governments raised prices of cigarettes, tobacco and petrol from mid-1995. The increase in interest charges[13] alone made a mechanically calculated contribution to CPI inflation of 1¼ percentage point by mid-1995. Headline inflation peaked at a twelve-monthly rate of 5.1 per cent in the second half of calendar year 1995 and has decreased since, partly as the effect of the cash rate rises on the CPI dissipated and as the recent reductions in mortgage interest rates began to have an effect. In the third quarter of 1996, the twelve-monthly rate of headline inflation stood at 2.1 per cent.

Apart from the diminishing effects of special factors, the recent turn-around in underlying inflation and headline inflation is likely to be in part a result of the economy's returning to more sustainable rates of growth, which reduced the pricing power of sellers. But price discipline may also have been imposed on domestic suppliers through the increased price competitiveness of imported goods and services due to the strong appreciation of the Australian dollar, some 20 per cent in effective terms during the year ending June 1996.[14] This appreciation led to a decrease in the prices of imported goods of 6¼ per cent in the first half of 1996 over a year earlier and shows in particular in a pronounced fall of the deflator for imported goods and services in the June quarter 1996 in Table 3. As a consequence, producer prices of manufactures rose by only 1.4 per cent in the first half of 1996 over a year earlier.

Some narrowing of the current external deficit

Australia's current external deficit has widened substantially in recent years, from a recession-induced low of 3.2 per cent of GDP in 1991 to 5.3 per cent in 1995. It peaked during the first half of 1995 to an equivalent of about 6 per cent

of GDP and has declined since. At 4.2 per cent in FY 1995/96, the deficit was a little below the average of about 4½ per cent of GDP since the early 1980s. It is, however, still markedly higher than the 2½ per cent average of the 1960 and 1970s.

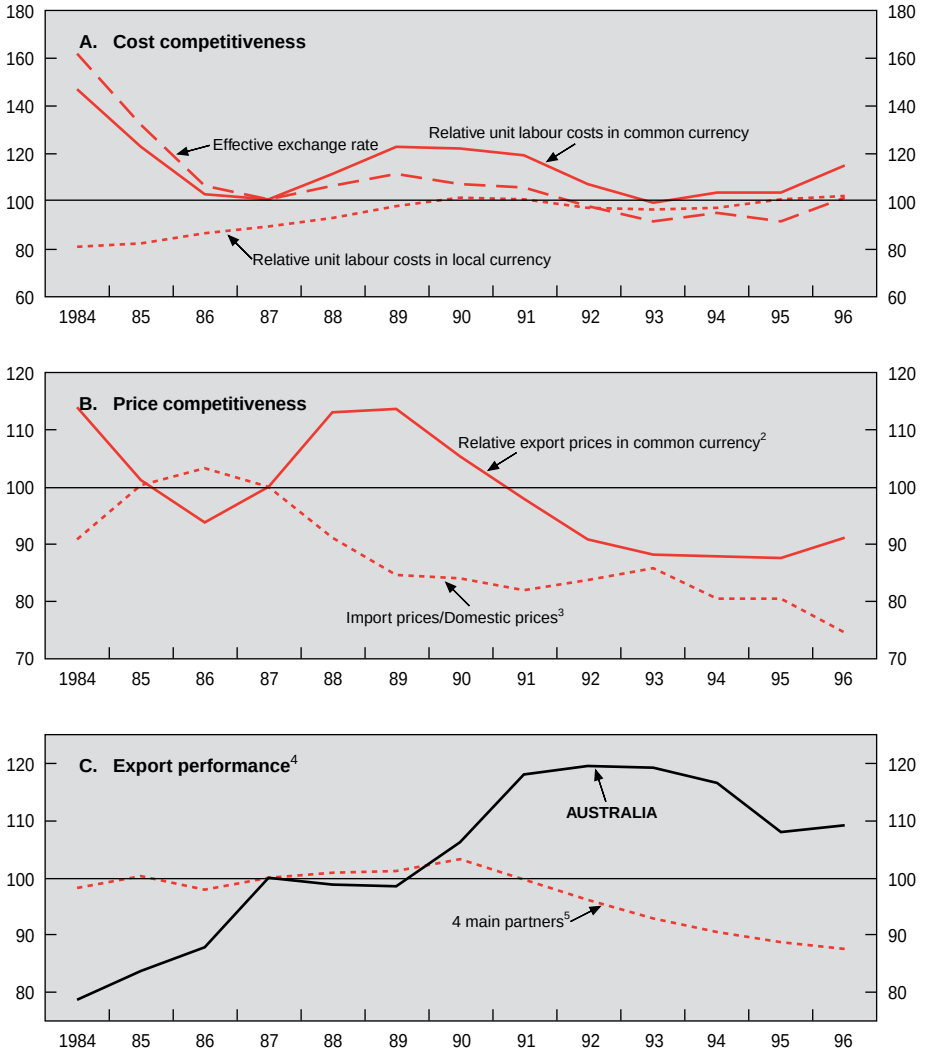
A major factor of the recent deterioration was the strong growth in domestic demand which induced soaring import volumes across broad categories. In particular, the recovery in the highly import-intensive machinery and equipment investment induced growth of imported capital goods volumes of nearly 30 per cent in 1994 and close to 20 per cent in 1995. Export volumes of manufactures also grew very strongly, in line with developments in recent years. Supported by increased price competitiveness (Figure 8), their performance in 1994 and 1995 even exceeded the 14 per cent average growth during the past ten years. However, merchandise export volumes lagged behind the strong growth of export markets in 1994 and in particular in 1995. To a large measure this was due to the adverse effect of the drought in eastern states on rural export volumes. The widening real trade gap was offset by slightly improving terms of trade in 1994. In 1995 and in the first half of 1996, strengthening world demand induced a rise in commodity prices. This entailed a further improvement in Australia's terms of trade which helped stabilise the external account.

The effects of the appreciation of the Australian dollar and the consequent changes in price competitiveness have so far been difficult to discern. After rural exports bottomed in the first half of 1995, merchandise exports strengthened in the second half when exports of resources and manufactures accelerated, a trend which continued into 1996 (Table 4). All major categories of imports slowed substantially in the second half of 1995, with the deceleration most pronounced for capital goods imports, though they recovered vigorously in the first half of 1996. The terms of trade posted further gains in the first half of 1996, which upported the positive swing in the current external account.

The deterioration in the current account was reinforced by a rising deficit in net factor incomes in 1994 and 1995, which reflected an increase in net payments due to the growth of net foreign liabilities and higher world interest rates together with increased equity returns on foreign investment in Australia. On the other hand, the small deficit in non-factor services declined somewhat in 1994 and 1995, bringing non-factor services trade to balance in the first half of 1996

Figure 8. **COMPETITIVENESS AND TRADE**[1]

Index 1987 = 100



1. 1996 figures are Secretariat estimates.
2. Manufactures.
3. Import prices of total goods and services divided by deflator of total domestic demand.
4. Ratio between export volumes of manufactures and export markets for manufactures.
5. Japan, United States, United Kingdom and New Zealand.

Source: OECD.

Table 4. **Trade in goods and services**

Percentage changes from previous year, volumes

	1993	1994	1995	1994		1995		1996
				I	II	I	II	I
Export volumes	7.4	8.6	4.2	11.7	5.7	1.3	7.2	12.8
Total goods	6.0	8.5	3.0	12.0	5.1	–0.5	6.6	13.5
Rural	4.9	9.0	–11.1	12.2	5.8	–16.2	–5.8	18.6
Manufactures	15.1	18.7	16.7	21.0	16.4	14.0	19.3	20.2
Resources	3.4	4.0	5.5	8.1	0.1	2.9	8.1	7.5
Total services	13.1	9.2	8.9	10.6	7.9	8.4	9.3	10.5
Import volumes	5.3	15.1	9.8	11.0	19.1	16.3	3.9	7.3
Total goods	6.7	16.4	11.3	11.7	20.9	18.8	4.7	7.7
Consumer goods	3.0	11.4	7.4	6.9	15.9	12.6	2.7	2.7
Capital goods	6.6	29.3	19.8	19.6	38.3	31.9	9.9	20.0
Intermediate goods	8.5	13.2	8.9	10.6	15.6	15.5	3.0	3.4
Total services	0.6	10.7	4.1	8.7	12.7	7.5	0.9	5.8

Source: Australian Bureau of Statistics.

(Table 5). The strong non-factor service export performance is largely due to buoyant net revenues from tourism but was also supported by significant contributions from education and business-related services.

As a consequence of the continued large current account deficits, net external liabilities as a percentage of GDP rose by 1.3 percentage points in the year

Table 5. **Current account trends**[1]

A$ billion

	1993	1994	1995	1994		1995		1996
				I	II	I	II	I
Exports	62.0	64.0	71.3	32.1	31.9	34.7	36.6	38.5
Imports	62.3	68.5	76.9	32.4	36.0	38.8	38.1	39.2
Trade balance	–0.3	–4.4	–5.6	–0.3	–4.1	–4.1	–1.5	–0.7
Investment income, net	–13.7	–17.5	–19.0	–8.4	–9.1	–9.3	–9.7	–9.9
Non-factor services, net	–1.3	–1.0	–1.0	–0.4	–0.7	–0.8	–0.2	0.1
Services, net	–15.0	–18.5	–20.0	–8.8	–9.7	–10.1	–10.0	–9.8
Transfers	0.4	0.3	0.8	0.0	0.2	0.3	0.5	0.7
Current balance	–14.8	–22.7	–24.8	–9.1	–13.7	–13.9	–11.0	–9.7
As per cent of GDP	–3.6	–5.1	–5.3	–4.1	–6.1	–6.0	–4.6	–3.9

1. OECD definitions.

Source: Australian Bureau of Statistics.

to June 1996, to the equivalent of 58.6 per cent of GDP, which is high by OECD standards. With the trend towards increased foreign equity investment in Australia continuing in 1995 and into 1996, net foreign debt fell from 39.9 per cent of GDP in June 1995 to 38.6 per cent twelve months later. It, thus, was substantially below the peak of 42.7 per cent reached in the third quarter of 1993. The bulk of increased equity reflected new foreign acquisitions of Australian companies and the valuation effect of higher share prices in Australia. The aggregate servicing ratio of Australia's net external liabilities fell in 1995/96; total servicing costs fell from 20.8 per cent of exports of goods and services to 18.5 per cent in the year ending in June 1996.

The outlook to 1998

Current economic indicators and policy assumptions

Current economic indicators point to continued buoyant growth with a shift in the structure of total domestic demand. Growth in consumption expenditures will likely slow somewhat in the near future, but remain robust: trend growth in (nominal) retail trade has slowed since late 1995, consumer confidence remains high; but new motor vehicle registrations have turned down, with trend growth declining since February 1996 to –3¼ per cent (a.r.) in September 1996. Private business investment, on the other hand, appears to be set to grow vigorously. The most recent Capex Survey (taken in July/August 1996) estimate is around 24 per cent higher than the first estimate (taken six months earlier) for FY 1996/97 and is 15 per cent higher than the comparable estimate for FY 1995/96. Expectations for both construction and equipment investment are strong. A turning point in the housing cycle also appears to be imminent: private building approvals, finance approvals and commencements appear to have reached, or be close to, a trough. Job vacancies data are consistent with a modest pick up in employment growth.

It is assumed for the projections that the short-term official interest rate remains at 7 per cent, the rate to which it was lowered in July 1996 (see Chapter II). Fiscal policy settings are those of September 1996 and thus reflect the 1996 Commonwealth Budget. The structural budget balance of general gov-

ernment is to improve by 2 per cent of GDP between 1995 and 1998 (see Chapter II). The other main assumptions underlying the projections are that:

- real non-oil commodity prices rise in 1996 and 1997 but are stable in the following year, contributing to a small increase in Australia's terms of trade;
- growth in Australian export markets is projected to be around 8 per cent per year;
- nominal exchange rates remain unchanged from their levels of 14 August 1996, implying an appreciation of the effective exchange rate of 9½ per cent in 1996;
- the average price of internationally traded oil is US$19.30 per barrel in the second half of 1996 but falls to US$18.30 over the remainder of the outlook period.

The projections embody information available at 11 September 1996.

Prospects

Output growth is projected to ease to 3½ per cent in the second half of this year, and broadly to maintain this rate over the outlook period (Table 6). This slowing mainly reflects the passing of the one-off effect on growth of the return to normal production conditions in the agricultural sector following the easing of the drought in eastern States. Abstracting from this factor, the outlook is for steady growth, as reflected in the stable growth rates projected for total domestic demand. Growth in consumption expenditure is projected to decline in line with that of household real disposable income, but to remain buoyant. This should be offset by higher growth in private investment expenditure and a turn in the housing cycle. The increase in private investment expenditure occurs against a background of positive business confidence and high business profits. In the case of dwelling investment, the projected turn in the housing cycle occurs in a context of some reduction in the large excess supply of dwellings built up during the latest housing upturn having already been effected and a substantial improvement in housing affordability.

Employment growth is projected to pick up gradually, from 1¼ per cent in 1996 to 2 per cent by 1998. With rising labour force participation, this will reduce unemployment only slowly to around 8 per cent by 1998. The small reduction in unemployment over the outlook period is not expected to put upward

Table 6. **Short-term prospects**

Percentage changes

	Percentage share of GDP in 1993 at current prices	Calendar year			
		1995	1996	1997	1998
A. Demand and output at constant 1989/90 prices					
Consumption					
Private	62.3	4.2	4.1	2.8	2.8
Public	18.0	3.1	2.3	0.8	–0.2
Gross fixed investment	20.0	1.8	5.0	8.1	6.2
of which:					
Government	2.1	–0.3	5.2	1.5	1.5
Private					
Total	17.8	2.0	5.0	8.8	6.6
Dwellings[1]	6.4	–9.9	–9.9	2.0	4.0
Other construction	2.5	17.1	9.9	18.0	10.0
Equipment	6.6	6.8	14.9	9.0	7.0
Public enterprises	2.4	6.4	6.1	11.0	6.0
Final domestic demand	100.3	3.5	4.0	3.6	3.1
Change in stockbuilding[2]	0.4	0.8	0.0	0.1	0.1
Total domestic demand	100.7	4.2	3.9	3.7	3.2
Exports of goods and services	19.1	4.0	11.0	7.7	8.0
Imports of goods and services	19.5	9.5	9.7	9.5	8.7
Change in foreign balance[2]	–0.4	–1.1	0.3	–0.4	–0.2
Statistical discrepancy[2]	–0.4	0.1	–0.1	0.0	0.0
GDP(I) at constant prices[3]	100.0	3.2	4.1	3.3	3.0
GDP(A) at constant prices[4]		3.5	4.1	3.3	3.0
B. Other items					
Private consumption deflator		2.5	2.1	2.5	2.8
Employment		4.1	1.3	1.9	1.8
Unemployment rate (per cent)		8.5	8.4	8.1	8.0
General government financial balance[5]		–2.2	–1.7	–1.1	–0.1
Current balance (A$ billion)		–24.8	–20.4	–21.2	–21.7
Current balance[5]		–5.3	–4.1	–4.0	–3.9

1. Including real estate transfer expenses.
2. Contributions to growth.
3. Income measures. Includes statistical discrepancy.
4. Average of the expenditure, production and income measures of GDP.
5. Per cent of GDP.

Source: OECD.

pressure on wages as it occurs slowly (there should not be any speed limit effects) and does not bring the unemployment rate to a level below estimates of the non-accelerating inflation rate of unemployment (NAIRU). Nevertheless, growth in unit labour costs in the business sector is projected to pick up as productivity growth eases back from the high rate over the past year. Initially, the

pass-through of the recent exchange rate appreciation into retail prices will help to hold down inflation. But as this effect wanes, inflation will likely rise to rates broadly in line with the increase in unit labour costs, or some 3 per cent by 1998, the upper end of the official medium-term band.

Growth in export volumes is projected to fall back from the very high rate recorded in 1996 as the effect of the return to normal production conditions in the agricultural sector passes. Also contributing to this slowing is the loss of competitiveness resulting from the exchange rate appreciation of the past year. Import growth, on the other hand, is projected to remain strong, reflecting the outlook for continuing buoyant growth in total domestic demand. The current account deficit is projected to fall to 4 per cent of GDP in 1996 and subsequently to be around, or under, 4 per cent of GDP.

Risks surrounding these central projections include the uncertainties associated with the international environment and the effects of fiscal consolidation. Export demand could be somewhat weaker or stronger than expected depending on future economic trends in major OECD export markets as well as in non-OECD economies. With the fiscal consolidation, there are uncertainties about the extent to which the direct contractionary effects of the consolidation might be attenuated by positive confidence effects, notably through a fall in long-term interest rates. Moreover, there are uncertainties surrounding the outlook for wages and inflation, although recent outcomes suggest the inflation outlook has improved slightly. The slowing of average ordinary-time earnings growth since late 1995 is encouraging. At the same time, wage outcomes in enterprise agreements – which are forward looking and cover about 30 per cent of the labour force – have increased from around 4 per cent in late 1994 to around 5 per cent in the first half of 1996. Future inflation will be strongly influenced by wage and productivity trends, so the uncertainties surrounding wages also apply to the inflation outlook. Should recent trends in enterprise agreement outcomes continue and spread more widely or large increases in award rates of pay occur,[15] inflation could become incompatible with the authorities' medium-term objective. This would necessitate a tightening in monetary policy, resulting in a less buoyant outlook for growth and unemployment towards the end of the projection period.

II. Economic policy

After five years of quite solid economic growth combined with low underlying inflation, the challenge to monetary policy is to preserve the low inflation environment which is conducive to high sustainable growth in output and employment. Given the maturity of the business cycle, fiscal policy has to step up efforts to reduce structural budget deficits in order to promote national savings. This should lower the structural savings-investment imbalance and relax the external constraint on stronger economic growth. It would also make room for fiscal saving needed to finance the expenditure bulge related to the ageing of the baby-boom generation.

Monetary policy

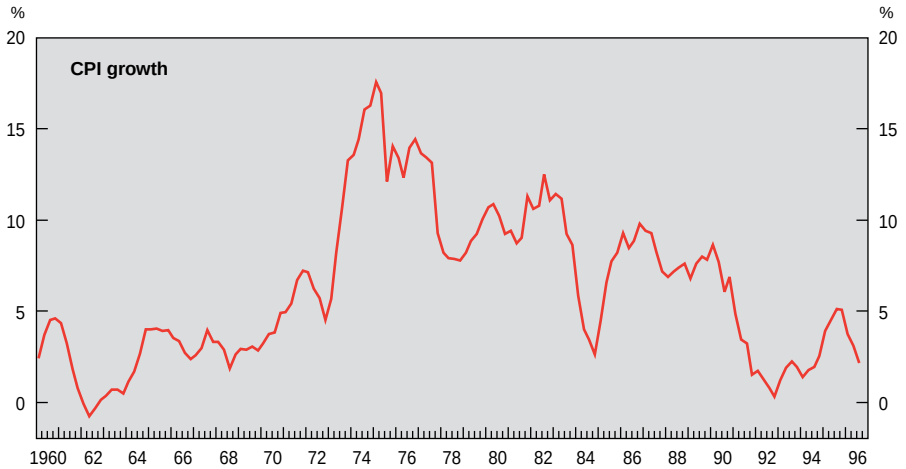
The objective of monetary policy

The objective of the Reserve Bank of Australia (RBA) is to foster sustainable growth in output and employment and to keep the underlying rate of inflation on average between 2 to 3 per cent over the course of the business cycle.[16] This inflation objective was introduced by the RBA in 1993, as one element of the growing public emphasis on the need to sustain a low-inflation environment. It has been formally endorsed by the Treasurer of the new Government in a joint statement with the Reserve Bank Governor in August 1996, the ''Statement on the Conduct of Monetary Policy''. This joint statement sets out the Treasurer's and the Governor's mutual understanding of the operation of policy in Australia. A key element of the Statement is the Government's formal endorsement of the RBA's self-imposed 2-3 per cent inflation objective.

This target was introduced only *after* a low level of inflation had been achieved, and was adopted to lock-in this low inflation (Figure 9); this differs

Figure 9. **THE CHANGED INFLATION ENVIRONMENT**

Changes over four quarters



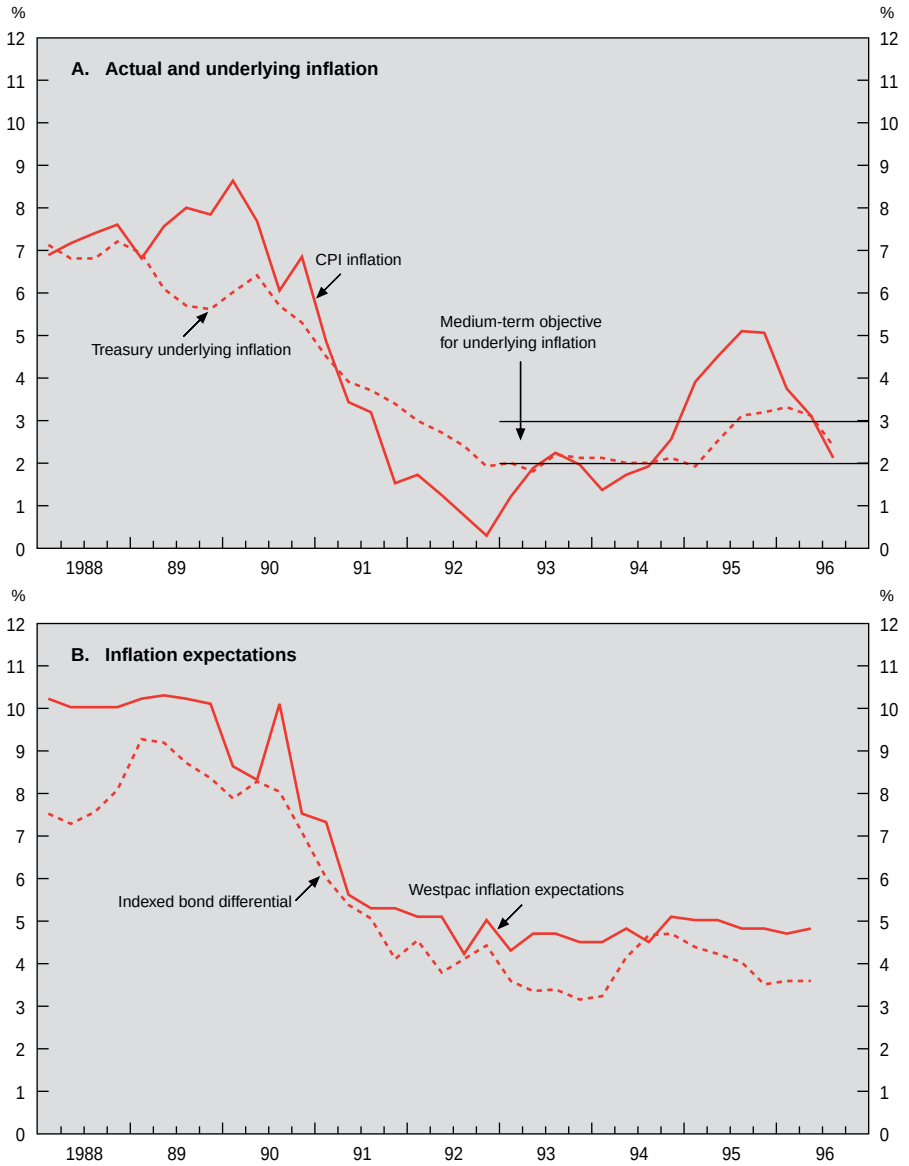
Source: OECD, *Main Economic Indicators.*

from the approach of some countries which have used inflation targets to condition inflation expectations, in order to support the transition from a high to a low inflation environment.[17] Australia is now one of the group of eight countries[18] which adhere to an inflation targeting approach to monetary policy in order to create a nominal anchor for expectations of the private sector.

The focus of the RBA's approach is on the underlying rate of inflation in the definition of the Commonwealth Treasury. This measure, published by the Australian Bureau of Statistics, excludes selected items permanently from the measure of the overall change in the CPI (''headline'' inflation) to reduce or remove interest rate effects, seasonal price effects and the effects of volatile items such as petroleum. The 2-3 per cent range defines the *mean or central tendency* of the inflation objective – similar to the United Kingdom's 2½ per cent (or below) or Finland's 2 per cent – and is specified as an average over the course of the cycle, so that deviations outside this range may be tolerated as long as they are temporary (see below). The inflation target has been met so far, with

Figure 10. **THE PERFORMANCE OF MONETARY POLICY**

Percentage change over twelve months earlier



Source: Reserve Bank of Australia; University of Melbourne, Institute of applied economic and social Research; OECD, *Main Economic Indicators*.

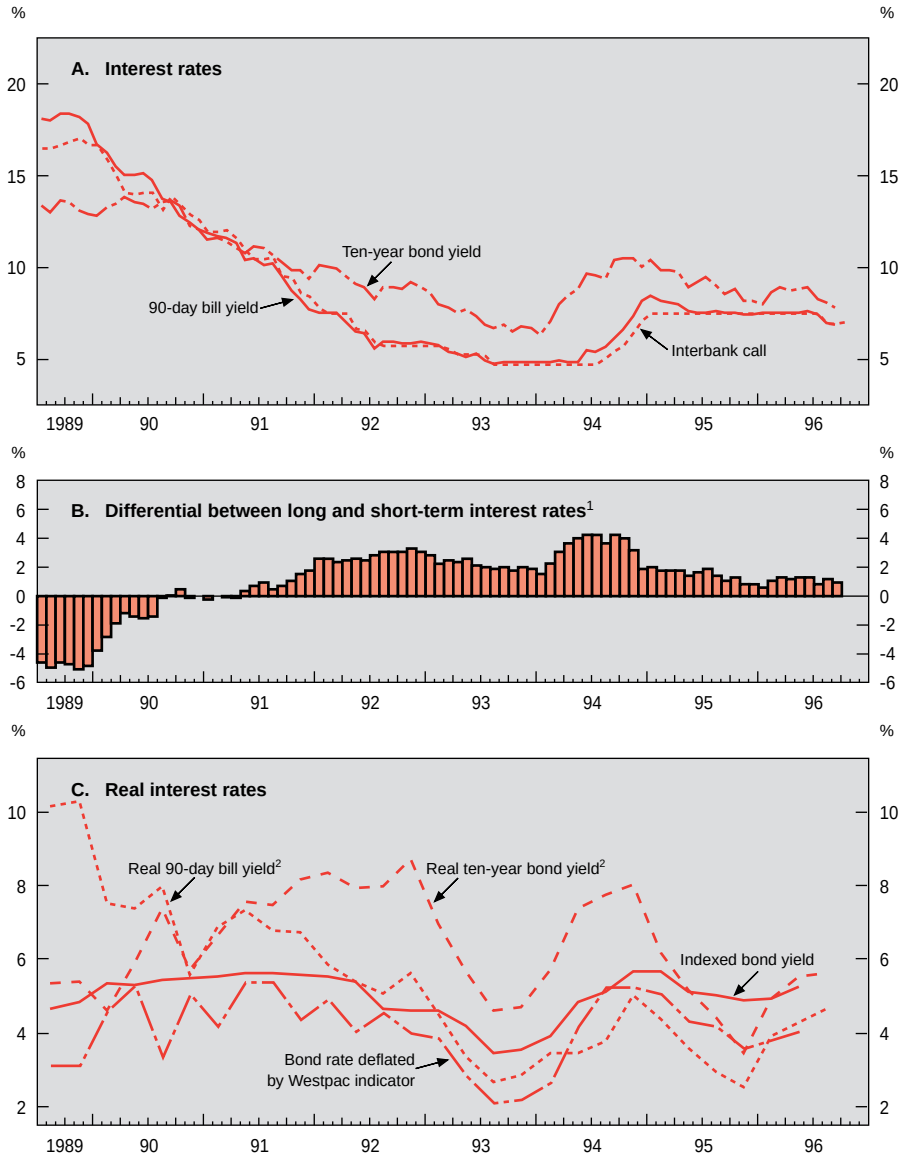
underlying inflation averaging 2½ per cent over the last five years (Figure 10, Panel A), and deviations above 3 per cent being only temporary.

The recent setting of monetary policy

Successful inflation control requires a strong commitment to a forward-looking approach to the use of monetary policy instruments. In the Australian context, monetary policy therefore needs to be tightened before an anticipated increase in underlying inflation beyond 3 per cent takes place and may only be eased when there are solid indications for underlying inflation falling back into the 2 to 3 per cent range. An initial test of early tightening was clearly passed when in response to the sharp acceleration of economic growth in the first half of 1994 the cash rate – the key policy interest rate – was raised in three steps from August to December 1994 by a cumulative 2¾ percentage points to 7½ per cent. Hence, monetary tightening began at a time of low inflation (around 2 per cent) and high unemployment (about 9½ per cent) and nearly one year ahead of the (rather moderate) acceleration in underlying inflation in June 1995.

Initially, the interest rate on three-month bank bills responded more than proportionately to the increase in the cash rate, going up by about 3 percentage points,[19] from 5½ per cent in mid-1994 to 8½ per cent in early 1995 (Figure 11). This is likely to have reflected the perception of many market participants that further cash rate rises were in prospect. A correction of this ''overshoot'' began at the beginning of 1995, when money market interest rates began to decline in the United States and Japan. Also, signs of a slowing in the Australian economy emerged, which eased financial markets' concerns about domestic developments and dissipated expectations of further monetary tightening. In consequence, Australian three-month money market interest rates declined by nearly 1 percentage point from their peak in January 1995 to around 7½ per cent in May 1995, the level of the cash rate. This fall in money market interest rates was accompanied by a marked weakening of the effective exchange rate during the first half of 1995, probably in large part attributable to the widening current external deficit. The depreciation by an actual rate of nearly 12 per cent during the first six months of 1995 combined with the decline in money market interest rates implies a substantial (temporary) relaxation of overall monetary conditions. But with growing confidence that the slowdown in output growth would keep the pick-up in underlying inflation modest and rather short-lived, the RBA abstained

Figure 11. **INTEREST RATES**



1. Differential between ten-year bond yield and 90-day bill yield.
2. Nominal rates less the change in the consumer price index from the previous year.

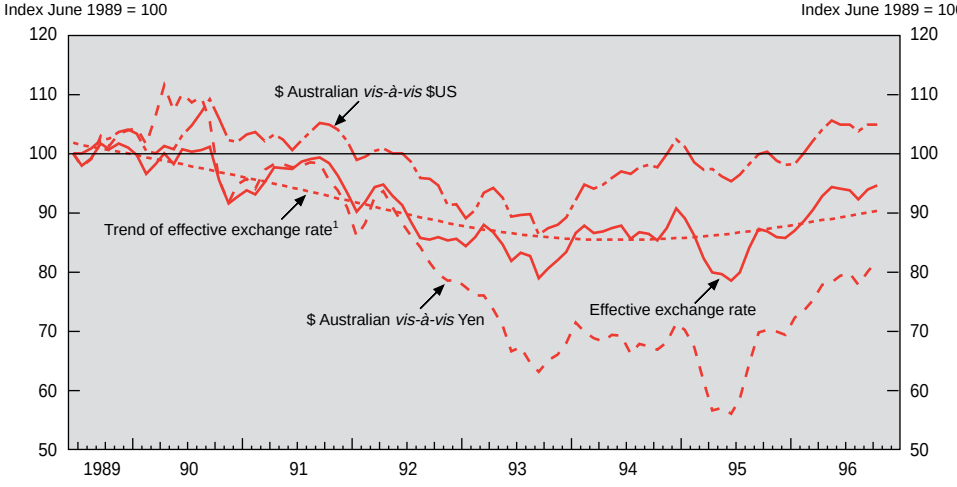
Source: Reserve Bank of Australia; University of Melbourne, Institute of Applied Economic and Social Research; OECD, *Main Economic Indicators.*

from further tightening. Hence, the cash rate remained unchanged during 1995 and the first half of 1996 and the three-month bank bill stayed at around 7½ per cent for more than one year, until July 1996.

A strengthening exchange rate

While monetary policy settings were held steady from late 1994 until mid-1996, monetary conditions changed substantially. A narrowing current account deficit, firming commodity prices and confidence that this trend would prevail in the future boosted demand for Australian dollar investments. In particular Japanese investors were attracted by the high yields of Australian financial assets relative to Yen investments. In consequence, the effective exchange rate of the Australian dollar rebounded from its trough in mid-1995 and appreciated by about 20 per cent during the following twelve months (Figure 12). By mid-1996, it attained its highest level in about four years and was some 4.3 per cent above its average level over the past ten years, but well below its peaks of the late 1980s.

Figure 12. **EXCHANGE RATES**



1. Low-frequency component of a Hodrick-Prescott filter (param. = 25 000).
Source: OECD, *Main Economic Indicators.*

Short-term nominal interest rates remained unchanged between mid-December 1994 and July 1996. The exchange rate, however, changed markedly over that time, easing first and then strengthening from mid-1995. This suggested some firming of monetary conditions from mid-1995 to mid-1996 but the strengthening of the exchange rate over this period coincided with an improvement in commodity prices and the terms of trade.

The firming of monetary conditions in the first half of 1996 is reflected in the long-short interest rate differential (as a proxy of the slope of the ''yield curve''). However, the changes in this measure of monetary conditions result not only from changes in short-term interest rates (which directly reflect policy), but also the forces operating on long-term yields. To the extent that these reflect changing price expectations, they are an important element in assessing monetary conditions: but to the extent that they reflect changing risk assessments and uncertainty (as seems to have been the case, for example, in February 1994) or perceptions of policy more generally (which may be the case since March 1996), then the interpretation of changes in the slope of the yield curve are less clear. That said, the long-short spread was over 4 percentage points immediately before the tightening of policy in July 1994, declined to around 1 percentage point with the tightening, and has remained much the same since then. Measures of real short-term interest rates depend on the deflator used, but generally show some small fall during 1995 (as underlying inflation and inflation expectations rose). Real short-term interest rats increased in the first half of 1996, but fell slightly later on in response to cuts in the cash rate. Nominal long bond yields have moved down since end-1994, probably more rapidly than most measures of underlying or expected inflation.

Credit markets

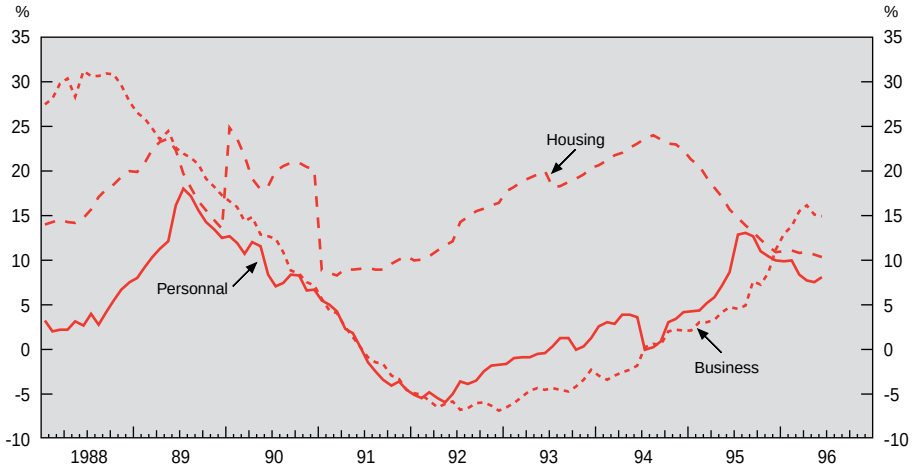
The increase in the cash rate by 2¾ percentage points in the second half of 1994 induced a rise in bank lending rates for business and housing loans, but by about 1 percentage point less. This is attributable to the fact that banks raised deposit rates by less than the increase in the cash rate and that the interest rate margin between loans and deposits narrowed somewhat. However, (variable) business and housing loan rates did not follow the softening of money market interest rates during the first half of 1995 and remained unchanged until May 1996 at the levels attained at the end of 1994. Competition in the housing

loan market intensified in the first half of 1996, due largely to greater competition from lenders such as mortgage managers. This induced a reduction in most banks' standard variable interest rates on housing loans from 10½ to 9¾ per cent by the middle of 1996. With the recent cuts to the cash rate most banks' standard variable interest rates on housing loans have fallen to 8¾ per cent.

Notwithstanding increased credit costs, the growth of total credit to the private sector accelerated from a twelve-monthly rate of 7½ per cent in mid-1994 to 9¾ per cent twelve months later and to 12½ per cent in mid-1996. This occurred in spite of a large and rather steady deceleration of housing credit from around 23 per cent in mid-1994 to 11¾ per cent in the middle of 1996. The slowing in lending for housing is likely to be primarily a reflection of a turning housing construction cycle after several years of strong expansion and was only reinforced by higher lending rates. Private sector borrowing was buoyed by vigorous credit demand of the business sector – which accounts for just over half of total outstanding credit – after a protracted period of subdued business borrowing (Figure 13). The pick-up in business borrowing reflected the cyclical upswing

Figure 13. **CREDIT TO THE PRIVATE SECTOR**

Twelve month growth



Source: Reserve Bank of Australia.

and the completion of balance sheet restructuring of the business sector which followed the overborrowing of the late 1980s. Personal credit demand other than for housing also expanded strongly, along with robust household consumption and a longer-term trend to higher levels of household indebtedness, in line with developments in other countries. On the other side of the consolidated balance sheet of financial intermediaries, the quickened growth in credit extended to the private sector showed in an acceleration of the monetary aggregate M3, from around 7 per cent during the financial year 1994/95 to 9 per cent in the course of 1995/1996. Growth of ''broad money''[20] was slightly stronger.

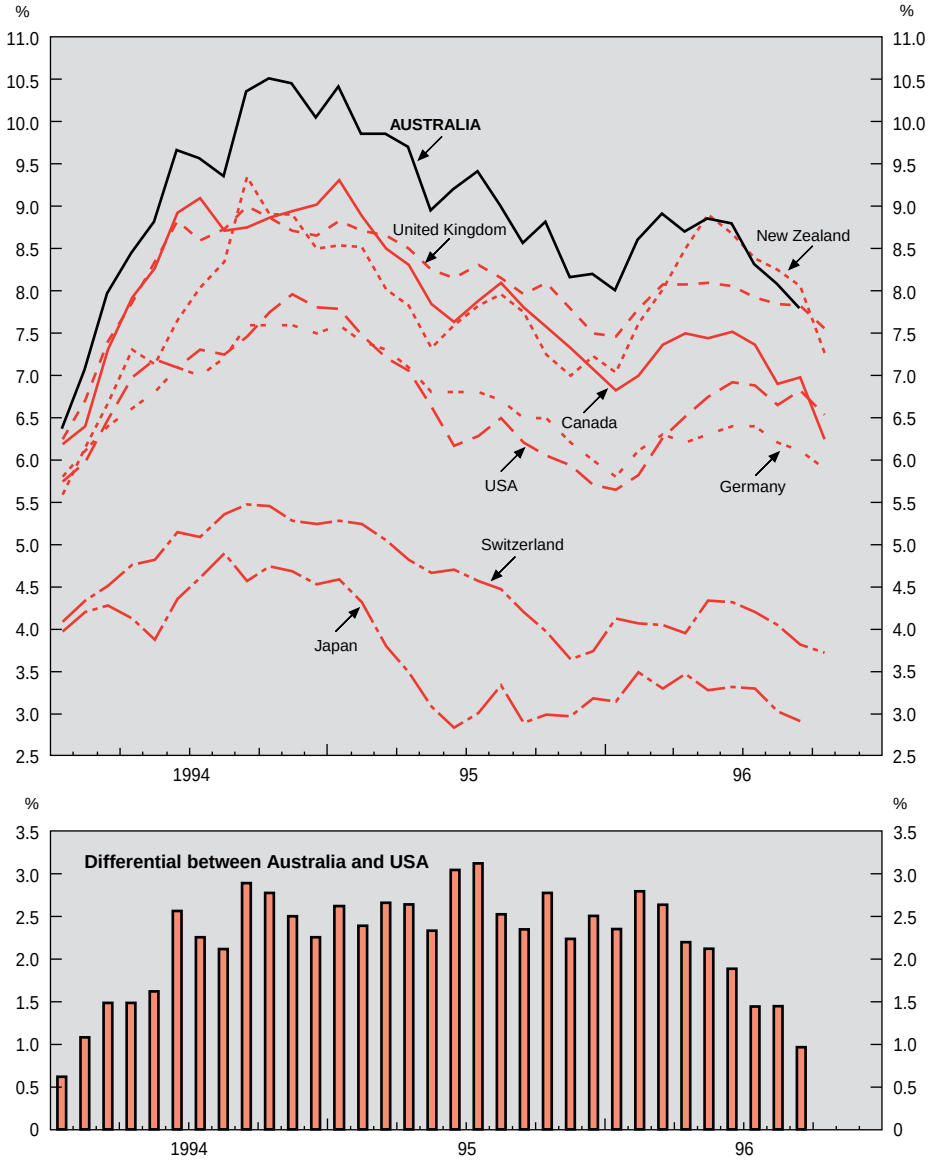
Long-term bond rates

Interest rates on ten-year bonds had gone up dramatically during 1994, from some 6¼ per cent at the beginning of 1994 to above 10 per cent at year-end.[21] This increase was much bigger than that in the United States, raising the Australia-US long-term bond rate differential from a low ½ percentage point at the beginning of 1994 to around 2¾ percentage points in September and October 1994. When bond rates in the United States and in many other countries turned around in late 1994 and trended downward throughout 1995, Australian bond yields followed, although with a delay, apparently associated with concerns about the current account deficit and the perceived risk of a depreciation of the Australian dollar. Hence, the long-term bond yield differential with the United States widened temporarily to more than 3 percentage points by mid-1995, but returned to 2½ percentage points at the end of 1995 (Figure 14). The Australian ten-year bond rate fell to 8 per cent in January 1996, 1½ percentage point above its trough of October 1993.

When the United States bond rates picked up again during the first quarter of 1996, Australian rates initially followed, rising by close to 1 percentage point in February and March. But while US bond rates continued to climb in the second quarter of 1996, the Australian ten-year bond yield only stayed in the 8¾ to 9 per cent range until mid-1996, revealing some degree of uncoupling from developments in United States capital markets. Hence, the A$-US$ bond rate differential narrowed to 2 percentage points by mid-1996 and – after the release of good inflation data in April and the tight Commonwealth Budget for 1996/97 – further to around 1 percentage point in September 1996. This is in strong contrast to an average differential of about 4½ percentage points during the second half of the

Figure 14. **INTERNATIONAL COMPARISON OF LONG-TERM INTEREST RATES**

Ten-year bond yields



Source: OECD, *Financial Statistics.*

1980s. Also, the previously substantial differential between Australian and New Zealand bond rates had virtually disappeared by mid-1996. All this suggests that financial markets have become much more confident about Australia's inflation prospects than about a year ago.

Stabilised inflation expectations

That inflation expectations stabilised recently is supported by several indicators. In response to the vigorous economic growth in 1994, private sector forecasters raised their projections of headline and underlying inflation for 1995 and 1996. For example, the Consensus Economics forecasts of 1995 for headline inflation in 1996 was consistently above 4 per cent while the predictions for underlying inflation ranged between 3 to 4 per cent. As from the beginning of 1996, inflation forecasts both for headline and underlying inflation were lowered; underlying inflation is now expected to be on average within the RBA's 2 to 3 per cent target range in 1997. Also, expected inflation as proxied by the difference between the nominal yield on non-indexed ten-year Treasury bonds and the real yield on index-linked Treasury bonds of about the same maturity[22] rose by nearly 1½ percentage points to 4¾ per cent during 1994. It declined thereafter and in mid-1996, at around 3½ per cent, it was less than ½ percentage point above its trough of late 1993 (Figure 10, Panel B). Another indicator, the Westpac/Melbourne Institute Survey of consumers' inflation expectations also revealed some small increase in late 1994 and early 1995, but remained remarkably flat thereafter.

Summarising the analysis so far, the tightening of monetary policy during the second half of 1994 has exerted a stabilising effect on wage and price inflation. There has been suggestions that the lags in monetary policy may have been rather shorter, in this episode, than they were in the late 1980s, possibly reflecting, in part, announcement effects and the memory of the sharp rise in short-term interest rates in the late 1980s. The sharp rise in bank bill futures in the second half of 1994 suggests that financial markets not only recognised that policy had been tightened, but incorporated further tightening into their interest rate forecasts. The exchange rate channel of transmission of monetary policy is likely to be important for Australia in line with empirical work embodied in the OECD INTERLINK model, but in the most recent episode the exchange rate

actually fell for the six months following the tightening phase of monetary policy in the second half of 1994.[23]

The stabilisation of inflation expectations is probably the most important achievement of monetary policy since 1994. This is the more so as there is evidence that the concept of underlying inflation is not yet generally understood by the public. Hence there was a risk that the sharp increase in headline inflation in 1995 could have ratcheted up expected inflation. It thus could have spilt into wage and price setting decisions, thereby undermining the Reserve Bank's efforts to bring underlying inflation rapidly back into the 2 to 3 per cent target range. This risk has not materialised thus far. The high level of the Westpac/Melbourne Institute measure of expected inflation suggests that monetary policy has still some way to go to entrench low inflation in people's minds, although the measure does not appear able to distinguish small shifts in expectations (survey results tend to ''cluster'' around 5 per cent or 10 per cent). The National Australia Bank survey (of businesses) suggests a more optimistic view, with almost 90 per cent of respondents expecting inflation during the remainder of the 1990s to average less than 4 per cent, including 23 per cent expecting less than 3 per cent.

The Reserve Bank's response to tightened monetary conditions

By the middle of 1996, a number of indicators suggested that there was room for some easing of monetary policy: the strong Australian dollar had tightened overall monetary conditions substantially while short-term indicators showed signs of weakening economic activity; estimates of expected inflation were subsiding; the twelve-monthly rate of underlying inflation had declined to slightly above 3 per cent, which underpinned expectations that it would safely return into the 2 to 3 per cent target range in the second half of 1996. Moreover, the Commonwealth Government announced a tight budget, which implied a small near-term contractionary fiscal impulse on the economy. Although economic growth averaged 4.1 per cent in 1995/96, the Reserve Bank concluded[24] that the pace of growth by mid-1996 left the Australian economy with the capacity to grow a little faster without threatening the inflation objective. Against the background of projections that inflation would soon fall comfortably into the 2-3 per cent band, the RBA lowered the cash rate by ½ percentage point to around 7 per cent in late July 1996. It was the first cut in official interest rates

since July 1993. The RBA reduced the official cash rate further in early November 1996, to 6½ per cent.

There remains, however, a substantial inflation risk from the side of labour costs. The twelve-monthly growth of average weekly ordinary-time earnings of adults working full-time (AWOTE) fell from a rate a little above 5 per cent in the September 1995 quarter to 3.8 per cent in the September quarter of 1996, which made it broadly consistent with the RBA's inflation target. But wage settlements of 5 per cent and above in private-sector enterprise agreements remain a source of concern. However, the RBA has made explicit on various occasions that it would not hesitate to raise interest rates if wage increases grew out of line with the inflation target.[25]

Clarifying the Reserve Bank's independence

In August 1996, on the occasion of the announcement of the new Governor of the Reserve Bank as from September, the Treasurer and the Governor designate released a *Statement on the Conduct of Monetary Policy* which set out their mutual understanding of the operation of monetary policy in Australia. The Statement notes that the Reserve Bank Act provides a transparent mechanism for the resolution of policy differences between the Reserve Bank and the Government, arguing that, although they have never been used, the politically demanding procedures required in such a case reinforce the Bank's independence.[26] To underline the independence of the RBA, the government pledged to no longer make parallel announcements of monetary policy adjustments, when the Reserve Bank changes the overnight cash rate. The government, however, reserves the right to comment on monetary policy from time to time. In addition, the *Statement* emphasises the need of transparency and accountability and, therefore, provides for specific reports of the Bank on monetary policy – to be released at roughly six monthly intervals – which will make interest rate policy more transparent and which will include information on the outlook for inflation. Moreover, the Governor will report twice a year to Parliament on the conduct of monetary policy. In sum, the *Statement on the Conduct of Monetary Policy* moves the operating framework of monetary policy in a direction which provides greater freedom to the Bank in pursuing its objectives as well as making it fully accountable for the outcome.[27]

Is the Reserve Bank's inflation objective optimal?

A comparison of countries which have adopted explicit inflation targets shows that the concrete formulation of the targets can vary considerably. This is a result of differing institutional frameworks and inflation histories, but also of differing views about what measures of inflation is used, what rate of inflation should be targeted and what variation around this is allowed. The majority of inflation-targeting countries uses as target variable a consumer price index adjusted for the variation of items which do not signal inflation in the sense that it would call for monetary policy action; examples are direct effects of interest rates on household debt, changes in indirect tax rates or in government charges and very volatile price items. In this vein, the Reserve Bank uses the long-standing Treasury measure of underlying inflation which excludes from the CPI mortgage and consumer debt charges as well as numerous volatile items, items with a marked seasonal pattern and goods and services mostly provided by the public sector. As with all underlying measures, this measure has its advantages and disadvantages. For example, the Treasury underlying measure only retains about 50 per cent of the CPI basket. In addition, while the underlying rate abstracts from most volatile price movements, it can be affected by some government charges, *e.g.* the increased sales tax on cars in July 1995 boosted the measure of underlying inflation.[28] However, by not re-calculating the basket every quarter, the measure has a stable, known base and is not subject to credibility problems. Moreover, this measure of underlying inflation converges to the headline rate in the long run and analysis shows that it consistently performs in a similar fashion to other underlying measures. It is important, nevertheless, for the public to be clear on the distinction between headline and underlying CPI.

Another issue is the quantification of the target. The RBA has specified a target for underlying inflation, namely 2 to 3 per cent, to be achieved on average over the business cycle. The interpretation of this target differs from inflation ranges chosen in other countries. In the case of Australia, the target range identifies the medium-term *mean* or *central tendency* – and not a band in the sense that the rate of underlying inflation should stay inside it at all times. This formulation allows for the natural short-run variation in underlying inflation over the cycle while also preserving a clearly identifiable benchmark performance over time. This approach is also appropriate given that the error margin of inflation forecasts, which are needed to set monetary policy in a forward-looking

manner, can be quite large. This is particularly so for an economy like Australia, which tends to be subject to larger term-of-trade shocks than many other OECD countries. For example, Debelle and Stevens (1995) report a 95 per cent confidence interval of about 1.2 percentage points on either side of a central forecast over a period of one to two years, based on standard price equations for Australia.[29] Hence, in light of these forecasting errors, defining a target range within which inflation could be held with high likelihood would imply a rather wide band of at least 2½ to 3 percentage points. But such a wide band could run counter to the credibility of the approach.

Apart from the width of a band, another issue is its level. This is related to the problem of establishing just what is ''price stability'': how to measure inflation correctly, taking account of changes in quality, new products, substitution within the ''basket'', and outlet substitution effects.[30] The choice of the target level is also related to the complex issue of the welfare cost of inflation; even a low inflation rate will introduce some distortions, but it can be argued that a low rate of inflation may smooth relative price changes, facilitate falls in real wages if necessary, and allow negative real interest rates if required. Moreover, if the target is lower than the existing rate of inflation, there is an output cost in the transition to lower inflation. The RBA assesses the welfare gains of bringing inflation below the 2 to 3 per cent band as low[31] relative to the associated potentially high short-term loss of output and employment.[32] However, the economic cost of further disinflation may diminish the more economic agents get acquainted with the low-inflation environment. A case could then be made for a reappraisal of the inflation target.

Fiscal policy

Almost five years after the recession of the early 1990s ended, the Commonwealth Government's budget is still in deficit[33] to the tune of 2 per cent of GDP. Although this deficit is not as large as those in many other OECD countries, it is nevertheless high by Australian standards for this stage of the economic cycle. Action to address the budget deficit was required to ensure that over time the Commonwealth budget does not detract from national saving, especially in view of the relatively low level of private saving. A structural deterioration in the current account deficit since the early 1980s, underpinned by a structural deterioration in the

Commonwealth budget balance, has led to a substantial rise in Australia's external liabilities. In general, increasing dependence on foreign savings may lead to a higher risk premia being built into domestic interest rates and increased vulnerability to shifts in investor sentiment, potentially amplifying the impact on the domestic economy of adverse external shocks. To reduce this structural dependence on foreign saving, and to ensure that the government's finances are in better shape to meet the future budgetary pressures associated with population ageing, the new Commonwealth Government announced a substantial fiscal consolidation programme in its 1996 Budget. This reinforces the budgetary consolidation already under way, especially this year and next year. The measures announced cut A$ 3.9 billion (0.8 per cent of GDP) from the Commonwealth budget deficit in FY 1996/97 and A$ 7.2 billion (1.3 per cent of GDP) by the following year. With these measures, the Commonwealth underlying budget balance is projected to move into surplus by FY 1998/99. The new government also announced a medium-term fiscal objective of a balanced budget in underlying terms over the course of the economic cycle and that legislation would be introduced to improve the discipline, transparency and accountability applying to the conduct of fiscal policy (*Charter of Budget Honesty*).

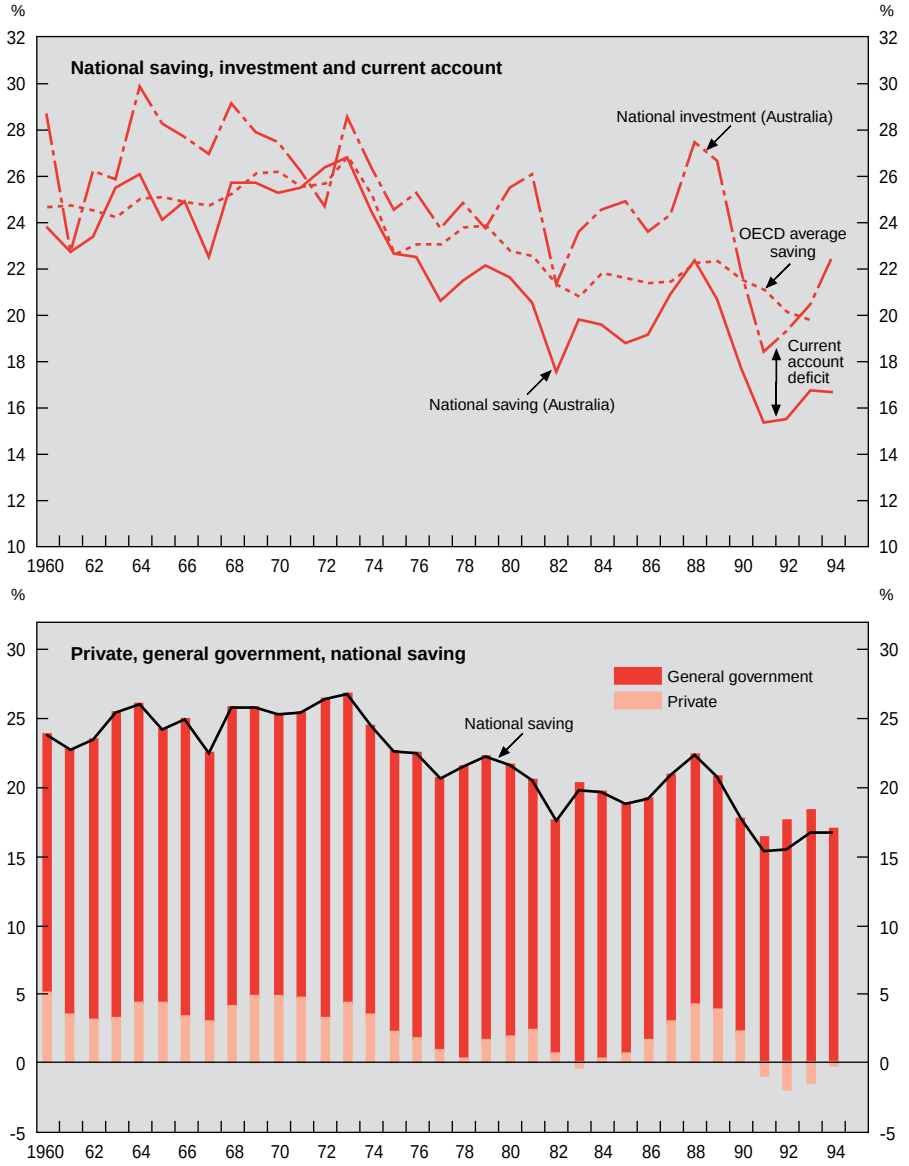
Background: Long-term trends in national saving and investment

Concerns about budget deficits in Australia are more easily appreciated in the context of long-term trends in national saving and investment. National saving has fallen markedly since the 1960s, averaging only 17 per cent of GDP so far in the 1990s (Figure 15). This decline is greater than that which has occurred in most other OECD countries, leaving Australia with a national saving rate significantly below the OECD average. Most of the decline in national saving is accounted for by a fall in general government saving since the early 1970s (see Figure 15); it is also since this time that national saving in Australia has generally been significantly lower than the OECD average. Although national investment also has declined, it has not done so to the same extent as national saving. The difference between national saving and national investment, the current account deficit, has widened in structural terms from a little over 2 per cent of GDP up until the late 1970s to about 4½ per cent of GDP since then.

The decline in saving in the general government sector has reduced its net lending from a small surplus on average up until the early 1970s to deficits most of the time since then (Figure 16). These deficits have risen over each of the

Figure 15. **NATIONAL SAVING AND INVESTMENT**[1]

As per cent of GDP

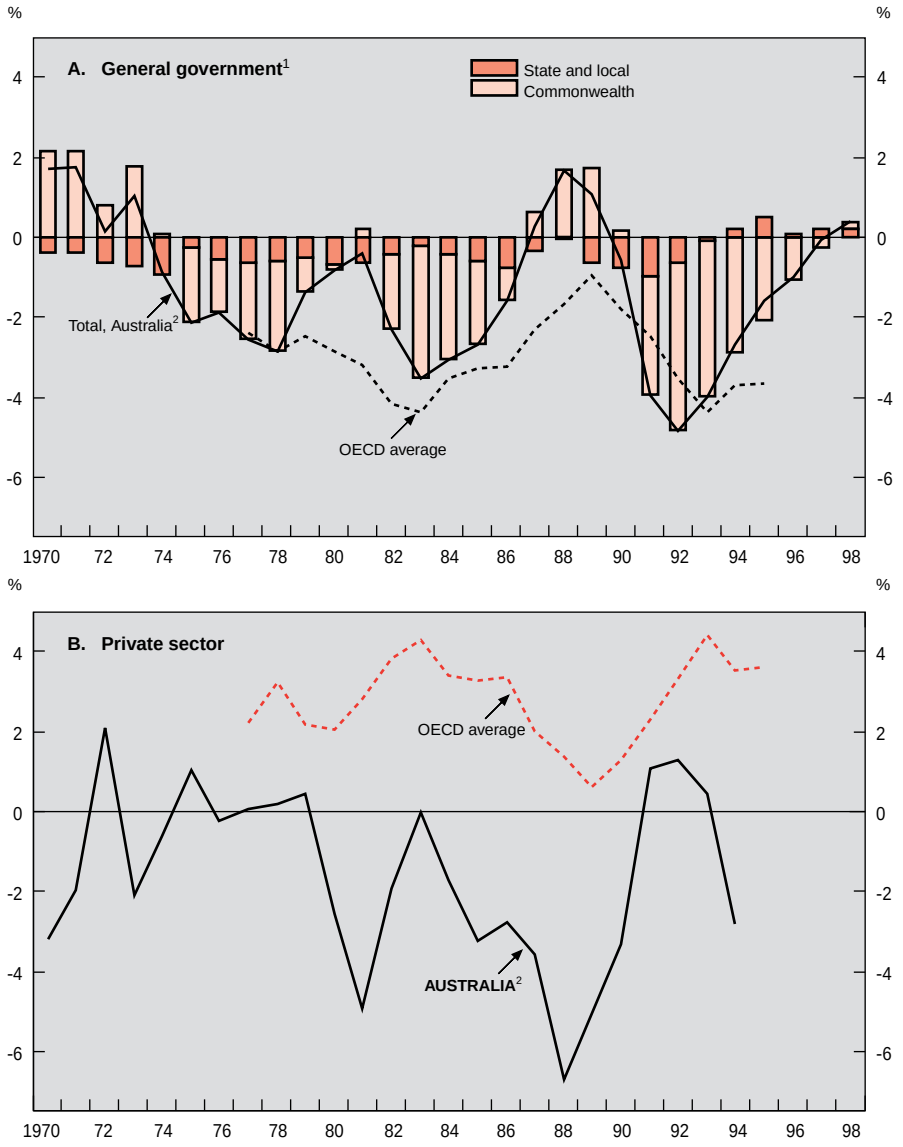


1. Data for Australia refer to fiscal year begining on the 1st July of the indicated year.

Source: OECD, *National Accounts.*

Figure 16. **NET LENDING**

As per cent of GDP



1. Data for FY 1996/97 to FY 1998/99 are on a Government Financial Statistics basis.
2. Fiscal years.

Source: OECD, *National Accounts, and estimates.*

subsequent business cycles, reaching almost 5 per cent of GDP in the early 1990s. Although general government net lending remains higher (*i.e.* budget deficits remain lower) in Australia than in most OECD countries, the past structural deterioration in the budget has contributed to the deterioration in the current account deficit and the accumulation of external liabilities. In addition, over the past decade private sector net lending has been negative in Australia but positive for the average of OECD countries. The public sector is therefore unable to draw on private saving without adding to external liabilities. This accumulation can result in a downgrading of credit ratings, as occurred in the mid-1980s, and increases the vulnerability of the economy to adverse shifts in investor sentiment.

The long-term deterioration in general government net lending reflects developments in the Commonwealth sector; the States/local government sector has moved from small deficits in recent decades to small surpluses since FY 1994/95 (see Figure 16). The Commonwealth Government was in budget surplus until the early 1970s, but since then has been in underlying deficit in all but five years. Moreover, deficits on average have been significantly higher in the 1990s than in the previous two decades: deficits in the five years since the last trough in economic activity averaged 3 per cent of GDP in the 1990s compared with 1½ per cent of GDP in the 1970s and 1980s (Table 7).

Table 7. **Commonwealth budget over economic cycles**

Per cent of GDP

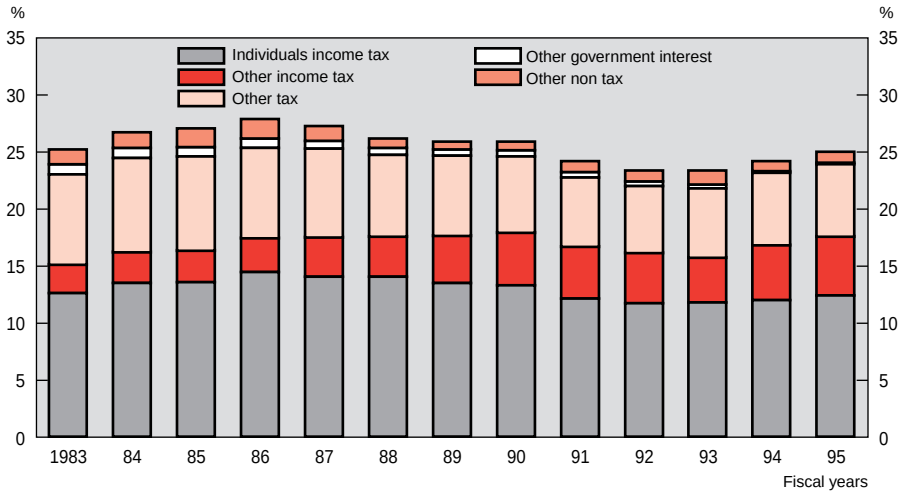
Fiscal years	Revenue	Underlying[1]	
		Outlays	Balance[2]
Five year averages			
1961/62-1965/66	20.8	18.7	2.1
1974/75-1978/79	24.2	25.6	–1.4
1983/84-1987/88	26.8	28.6	–1.7
1991/92-1995/96	24.0	27.3	–3.2
In fifth year since trough			
1965/66	22.1	19.5	2.6
1978/79	23.8	25.9	–2.1
1987/88	27.3	26.9	0.5
1995/96	25.0	27.2	–2.1

1. Underlying outlays exclude net advances (which include equity asset sales).
2. May not add due to rounding.

Source: Budget Statements 1996-97, Budget Paper No. 1.

Figure 17. **COMMONWEALTH REVENUE**

As a percentage of GDP



Source: Budget Statements, Budget Paper No. 1, various issues.

The move into deficit in the mid-1970s reflected a substantial increase in outlays which was not matched by sufficiently higher revenues (see Table 7). By contrast, the increase in the budget deficit in the 1990s largely reflects declining revenues as a percentage of GDP although once again, there was some increase in outlays. However, for the first time since the 1960s, expenditure as a share of GDP was actually lower than that for the same period in the previous recovery period.[34] This decline in revenues is mainly attributable to falling indirect tax revenues[35] (Figure 17). The decline in indirect tax revenues, which has occurred in spite of the increases in indirect tax rates in recent years, reflects a decline in customs duty revenue as levels of protection have fallen and a structural weakness in the indirect tax base: the principal indirect tax, the wholesale sales tax, does not apply to the rapidly growing services sector of the economy.

The FY 1995/96 Commonwealth budget out-turn

The process of budget consolidation which got under way in FY 1993/94 continued in FY 1995/96, with the underlying budget deficit falling 0.8 per cent

Table 8. **Commonwealth budget developments**

	FY 1994/95			FY 1995/96						FY 1996/97		
	Actual	Real growth	GDP	Actual	Real growth	GDP	Budget forecast	Actual less budget forecast		Estimate	Real[1] growth	GDP
	$ million	Per cent	Per cent	$ million	Per cent	Per cent	$ million	$ million	Per cent	$ million	Per cent	Per cent
Revenue												
Individuals income	54 647	6.6	12.0	60 414	7.6	12.4	62 520	–2 106	–3.4	65 940	6.1	12.8
Companies	15 588	21.1	3.4	18 252	14.0	3.8	17 140	1 112	6.5	19 700	5.0	3.8
Other	6 420	39.0	1.4	6 804	3.2	1.4	6 890	–86	–1.2	6 990	–0.1	1.4
Total income tax	76 656	11.5	16.8	85 470	8.5	17.6	86 550	–1 080	–1.2	92 630	5.4	18.0
Indirect	27 099	9.3	5.9	28 928	3.9	6.0	30 690	–1 762	–5.7	30 260	1.7	5.9
Other	1 916	8.9	0.4	1 960	–0.4	0.4	2 021	–61	–3.0	2 138	6.1	0.4
Total tax revenue	105 671	10.9	23.2	116 358	7.2	23.9	119 261	–2 903	–2.4	125 028	4.5	24.3
Total non-tax revenue	4 743	–30.4	1.0	5 302	8.8	1.1	5 187	115	2.2	5 132	–5.9	1.0
Total revenue	110 413	8.1	24.2	121 660	7.3	25.0	124 448	–2 788	–2.2	130 160	4.0	25.3
Outlays												
Final consumption expenditure	20 452	0.5	4.5	21 497	2.3	4.4	21 391	106	0.5	22 359	1.1	4.3
Personal benefit payments	42 400	1.6	9.3	45 541	4.6	9.4	43 769	1 772	4.0	47 317	1.0	9.2
Interest	8 103	21.4	1.8	9 219	10.8	1.9	10 148	–929	–9.2	9 884	4.2	1.9
Subsidies	2 585	–0.4	0.6	2 764	4.1	0.6	2 944	–180	–6.1	3 074	8.0	0.6
Grants to other governments	29 710	3.8	6.5	31 136	2.0	6.4	31 334	–198	–0.6	32 259	0.8	6.3
Other current payments	16 524	9.7	3.6	17 919	5.6	3.7	17 692	227	1.3	17 696	–3.9	3.4
Total current outlays	119 774	4.1	26.3	128 076	4.1	26.3	127 278	798	0.6	132 589	0.6	25.8
Capital outlays on goods and land	452	2.0	0.1	625	34.6	0.1	471	154	32.7	338	–47.4	0.1
Capital transfer payments	3 408	–16.3	0.7	3 277	–6.4	0.7	3 470	–193	–5.6	2 939	–12.9	0.6
Underlying capital outlays	3 860	–14.4	0.8	3 902	–1.6	0.8	3 941	–39	–1.0	3 277	–18.3	0.6
Contingency reserve	–	–	–	–	–	–	–	–	–	–57	–	–
Total underlying outlays	123 634	3.5	27.1	131 978	3.9	27.1	131 219	759	0.6	135 810	0.1	26.4
Underlying balance	–13 221		–2.9	–10 317		–2.1	–6 771	–3 547		–5 649		–1.1
Memorandum items:												
Net advances	–1 592		–0.3	–5 272		–1.1	–7 465	2 193		–6 123		–1.2
Headline balance	–11 629		–2.5	–5 046		–1.0	694	–5 740		474		0.1

1. From FY 1995/96 outcome.
Source: Budget Statements 1995/96 and 1996/97, Budget Paper No. 1.

of GDP to A$ 10.3 billion (Table 8). This reduction was achieved through increased income tax revenues: both other sources of revenue and total underlying outlays were constant as a proportion of GDP. The stability in outlays reflected the offsetting effects of low real growth in final consumption expenditure (2.3 per cent) and grants to and through the State governments (2 per cent) on the one hand, and high real growth in personal benefit payments (4.6 per cent) and in debt interest payments (10.8 per cent) on the other.

This out-turn for the budget deficit was, however, considerably higher (+A$ 3.5 billion) than forecast in the 1995 Budget owing to revenue shortfalls (see Table 8). The shortfall in revenue was mainly in income tax receipts from individuals, which were depressed by lower-than-forecast earnings and employment growth, and in indirect tax receipts, which were undermined by weak growth in sales of taxable goods – especially motor vehicles – and by the non-passage of a sales tax measure on building materials. Although the outturn for outlays was close to the budget forecast, this reflected the offsetting effects of substantially higher than forecast personal benefits payments (mainly to the unemployed) and much lower than forecast debt interest payments.

The FY 1996/97 Commonwealth Budget

Consolidation of the Commonwealth Government's budgetary position was considerably reinforced in the 1996 Budget. Measures were announced cutting from the budget deficit A$ 3.9 billion (0.8 per cent of GDP) in FY 1996/97 and A$ 7.2 billion (1.3 per cent of GDP) by the following year (Table 9). For the most part, these reductions are to be achieved through lower outlays. The largest savings are to come from reduced transfers to the States, increased contributions from university students towards the cost of their education (HECS) and a major refocusing of labour market programmes toward those of maximum effectiveness at least cost. Revenue is also to be increased, with the principal measures being a reduction in tax concessions for R&D expenditures and a surcharge on high income individuals' contributions to superannuation schemes. With these and other measures,[36] outlays are forecast to fall steadily as a percentage of GDP while revenue is to remain at current levels (Figure 18). The budget deficit is forecast to fall to A$ 5.6 billion (1.1 per cent of GDP) in FY 1996/97 and to be eliminated by FY 1998/99.

Table 9. **Budget measures and forecasts**

A$ million

	1996/97	1997/98	1998/99
FY 1996/97 budget underlying balance			
No policy change	–9 558	–8 702	–5 414
Per cent of GDP	–1.9	–1.6	–0.9
Spending measures			
Additional expenditures			
Private health insurance incentive	6	489	494
Family tax initiative	248	483	484
Other	1 266	1 054	1 211
Total	1 520	2 026	2 189
Savings measures			
State fiscal contributions	–619	–640	–300
AUSTUDY and HECS	–189	–496	–705
Labour market programmes	–575	–956	–130
Other	–3 066	–5 131	–5 901
Total	–4 449	–7 223	–7 036
Total impact on outlays	–2 929	–5 197	–4 847
Revenue measures			
Revenue reducing			
Family tax initiative	–147	–441	–591
Other	–215	–145	–482
Total	–362	–586	–1 073
Revenue raising			
R&D measures	59	718	630
Superannuation measures	0	523	521
Tariff concession system and policy by-law system	313	344	369
Other	969	956	1 077
Total	1 341	2 541	2 597
Total impact on revenues	979	1 955	1 524
Total effect of measures	–3 908	–7 152	–6 371
FY 1996/97 budget underlying balance			
Estimate	–5 649	–1 548	957
Per cent of GDP	–1.1	–0.3	0.2

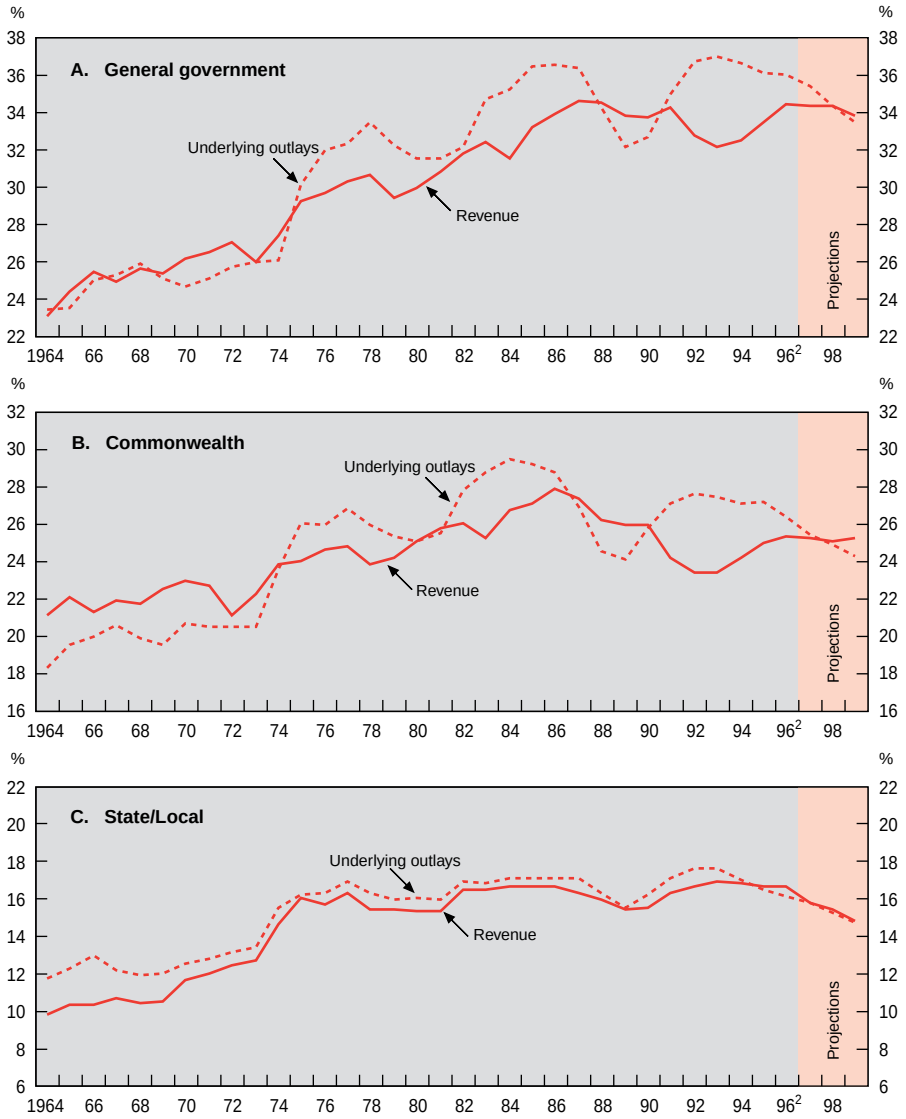
Source: Budget Statements 1996-97, Budget Paper No. 1.

The outlook for States/local government

The fiscal position of States/local government has been improving in recent years, with net lending rising from a deficit of 1 per cent of GDP in FY 1991/92 to an estimated surplus of 0.5 per cent of GDP in FY 1995/96 (see Figure 16).[37]

Figure 18. **GENERAL GOVERNMENT REVENUES AND UNDERLYING OUTLAYS**[1]

As per cent of GDP



1. Outlays excluding net advances.
2. Estimates.

Source: The Commonwealth Treasury.

This budget consolidation has been achieved through greater reductions in underlying outlays rather than changes in revenue as a percentage of GDP (see Figure 18). However, the budget surplus is forecast to fall almost to zero in FY 1996/97, mainly reflecting reduced transfers from the Commonwealth, and to remain broadly unchanged thereafter. For the period FY 1995/96 to FY 1998/99, revenue is projected to fall by 1.8 per cent of GDP while outlays are projected to fall by 1.4 per cent of GDP. Both own source revenue and Commonwealth grants are to decline. The projected decline in own-source revenue (as a share of GDP) is greater than that in Commonwealth grants and is based mainly on expectations that tax bases will grow more slowly than economic activity. This reflects the relative importance of taxes which relate to specific aspects of economic activity, such as motor vehicle registration fees and stamp duties. With respect to outlays, the decline reflects reforms implemented over a number of years to improve the efficiency of the public sector (including ongoing savings from agency restructuring) and lower interest payments associated with declining net debt as a percentage of GDP (assisted by the proceeds of asset sales).

The outlook for general government

Developments in general government net lending mainly reflect those in Commonwealth net lending. The large deterioration in the Commonwealth's finances in the early 1990s drove general government net lending to a peak deficit of almost 5 per cent of GDP in FY 1992/93, while the subsequent improvement in Commonwealth finances contributed the bulk of the decline in the deficit to 1½ per cent of GDP by FY 1995/96. The general government deficit is projected almost to be eliminated by FY 1997/98, with small surpluses projected for subsequent periods (see Figure 16). With the projected reductions in outlays as a percentage of GDP at both the Commonwealth and State/local levels of government, general government outlays are projected to fall by some 3 per cent of GDP to 33 per cent of GDP by FY 1998/99 (see Figure 18). This will give Australia the lowest ratio of government expenditure to GDP among OECD countries.[38] General government revenues are projected to fall slightly, reflecting developments in State finances.

The stance of fiscal policy

Fiscal policy by the OECD is projected to have a first-round contractionary effect on aggregate demand over the next few years: the general government's budget balance is projected to improve by 2 per cent of GDP between FY 1995/96 and FY 1998/99, mainly reflecting developments in the Commonwealth budget. As growth is assumed in the 1996 Budget to be around OECD estimates of the trend rate over the next few years, the projected reductions in the deficit are structural. The measures announced in the 1996 Budget reinforce the budget consolidation which was already in the pipeline.

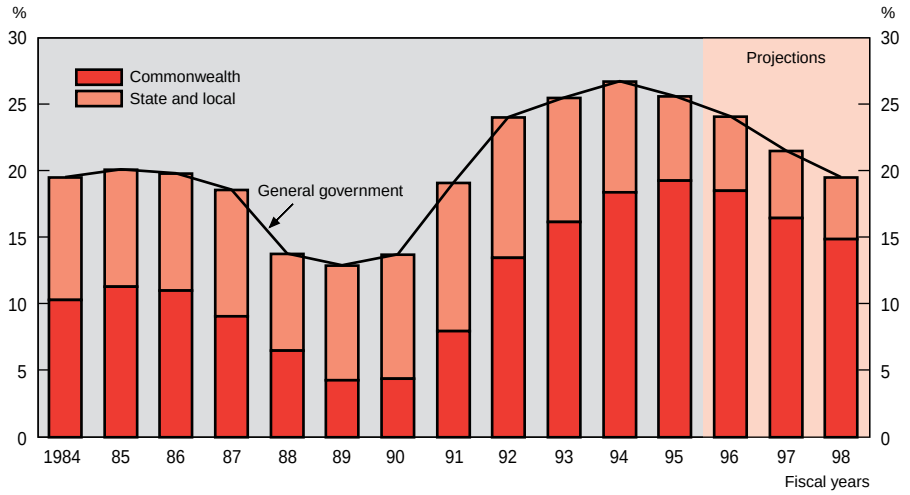
The direct effects of these cuts in the deficit would be to reduce output from the level it otherwise would have reached. Counteracting these effects, however, the budget consolidation could contribute to lower interest rates or a lower exchange rate than otherwise, stimulating economic activity and thereby providing an offset to the direct contractionary impact of the consolidation. Moreover, expenditure could be boosted by a rise in confidence and/or reductions in private saving.[39] The 1996 Budget forecasts are based on the assumption that such effects, while significant, do not fully offset the direct short-term effects of budget consolidation, resulting in a small net contractionary impact on activity in FY 1996/97. The OECD Secretariat shares this judgement.

Debt

General government net debt has risen sharply in recent years, from 12 per cent of GDP in FY 1989/90 to 25 per cent of GDP in 1995/96, but nevertheless remains much lower than in most other OECD countries (Figure 19); general government gross debt has increased from 24 per cent of GDP to 34½ per cent of GDP over the same period. This increase has entirely reflected the rise in Commonwealth Government net debt, which reached 19 per cent of GDP in 1995/96 and would have been even higher had it not been for asset sales.[40] With the fiscal consolidation programme announced in the 1996 Budget and planned assets sales, Commonwealth general government net debt is projected to fall to 15 per cent of GDP and total general government net debt to 19½ per cent of GDP by FY 1998/99. The Commonwealth Government's medium-term objective is for underlying budget balances to sum to zero over the economic cycle. This will ensure that net debt does not rise from one economic cycle to the next and that, as a percentage of GDP, net debt declines over time.

Figure 19. **GENERAL GOVERNMENT NET DEBT**

As per cent of GDP



Source: The Commonwealth Treasury.

Charter of Budget Honesty[41]

The government also announced its intention to introduce legislation to establish a new fiscal framework, known as the *Charter of Budget Honesty*. This legislation incorporates many of the recommendations of the National Commission of Audit, is intended to improve the discipline, transparency and accountability applying to the conduct of fiscal policy. The adoption of the new framework through legislation has the objective of applying the new framework to all future governments.

In the new framework, fiscal policy is to be formulated against the set of principles of sound fiscal management contained in the legislation. These principles will require a government to take into account the impact of fiscal policy on: government debt and managing fiscal risks; national saving; the stability and integrity of the tax base; and inter-generational equity. To assist public evaluation of fiscal policy, governments will be required to present a *Fiscal Strategy Statement* each year. This should: outline fiscal objectives and strategic priorities;

specify expected fiscal outcomes and indicators against which fiscal policy will be set and assessed; and identify temporary fiscal measures aimed at dampening economic downturns and indicate the process for their reversal. Assessment of a government's fiscal performance will be facilitated by the following reporting requirements: economic and fiscal outlook reports twice a year (at budget time and mid-year); a pre-election report on the economic and fiscal outlook; and an inter-generational report every five years on the long-run implications of government policies. Moreover, in preparing these reports, governments will be required to have regard to external reporting standards such as those of the Australian Bureau of Statistics (ABS) and the accounting profession. Divergences from these standards will have to be explained.

Assessment

The 1996 Budget represents a fiscal-policy watershed in Australia. For the first time, the principles of sound fiscal management contained in the *Charter of Budget Honesty* legislation (presently before the Senate) have been rigorously followed. One implication of this change is that a clear medium-term objective for fiscal policy has been set: namely, that the underlying budget should be in balance on average over the course of the economic cycle. This is important as fiscal policy in the past all too often has focused on supporting the economy in the short-term at the expense of rising deficits and government debt over time. Giving effect to this medium-term objective calls for a swift reduction in underlying budget deficits and indeed, a move into significant underlying surplus during the remaining years of the present economic expansion. While the 1996 Budget goes a long way towards eliminating the underlying budget deficit over the next two years, further consolidation measures are likely to be required to achieve sufficiently large surpluses during the last years of the present expansion for government to be able to withstand the inevitable conjunctural deterioration in its financial position during the next economic downturn and achieve the medium-term fiscal objective.

The medium-term fiscal objective chosen reflects the government's concerns about the impact of government saving on national saving and the structural current account deficit. The deterioration in government saving since the early 1970s accounts for much of the decline in national saving and rise in the structural current account deficit since then. Underlying budget balance on aver-

age over the course of the economic cycle will raise national saving and ensure that government net lending has no direct impact on the structural current account deficit – this will entirely reflect private net lending. As noted in the 1996 Budget Papers, the increase in government saving does not mean that there will be a corresponding increase in national saving and decline in the structural current account deficit. The private sector is likely to respond in part by lowering its saving, either because lower future tax burdens are anticipated or because of liquidity constraints. And domestic investment could increase in response to possible declines in real interest rates and/or improvements in business confidence, further reducing the decline in the structural current account deficit. Nevertheless, international experience suggests that these effects are unlikely to offset fully the rise in government saving, so that national saving can be expected to rise and the structural current account deficit to fall.

Underlying budget balance on average over the economic cycle represents a better fiscal performance than observed of late in most other OECD countries (or indeed Australia). Like other OECD countries, Australia faces the medium-term prospect of large increases in government expenditures as the ''baby-boom'' generation enters retirement. Some measures have already been taken or are planned to contain the growth in these expenditures, such as the introduction of compulsory superannuation contributions and the move in the 1996 Budget to increase private contributions towards the cost of long-term care for the elderly. However, it is likely that these measures alone will not be sufficient to avoid some increase in government expenditures on the elderly as a proportion of GDP. Fiscal policy formulation will be assisted by the *Charter of Budget Honesty* which will require governments to examine the budgetary implications of expected demographic and other developments over a 40 year time span and produce inter-generational reports on a regular basis. The implications of these longer term developments will need to be taken into account when assessing the continued appropriateness of the medium term fiscal target which, as indicated above, can be expected to result in a decline in net debt as a proportion of GDP over time.[42]

Ideally, greater fiscal consolidation should have occurred at an earlier stage of the current economic expansion. The 1996 Budget endeavours to make up for a late start by front-loading budget consolidation. Fortunately, the economic outlook is propitious for accelerated budget consolidation – any short-term con-

tractionary effects on the economy are likely to occur during a period of robust growth. Moreover, this should reduce the burden on monetary policy of achieving the medium-term inflation objective.

The focus on expenditure reductions in the consolidation programme also is well placed. A recent study[43] of experience in a number of OECD countries suggests that this focus increases the likelihood that the consolidation will be sustained. Nevertheless, an important factor in the structural deterioration in the Commonwealth budget during the past decade has been the structural weakness in the indirect tax base. This reflects the fact that the base for the principal indirect tax, the wholesale sales tax, excludes the large and fast-growing service sector. Tax reform to repair this structural weakness would have the added advantage of reducing the efficiency cost (*i.e.* the excess burden) of taxation. Experience with the introduction of a goods and services tax in other countries suggests that public acceptance is greater when there are simultaneous and at least matching cuts in other taxes (for example, as in New Zealand).

III. Implementing the OECD Jobs Strategy

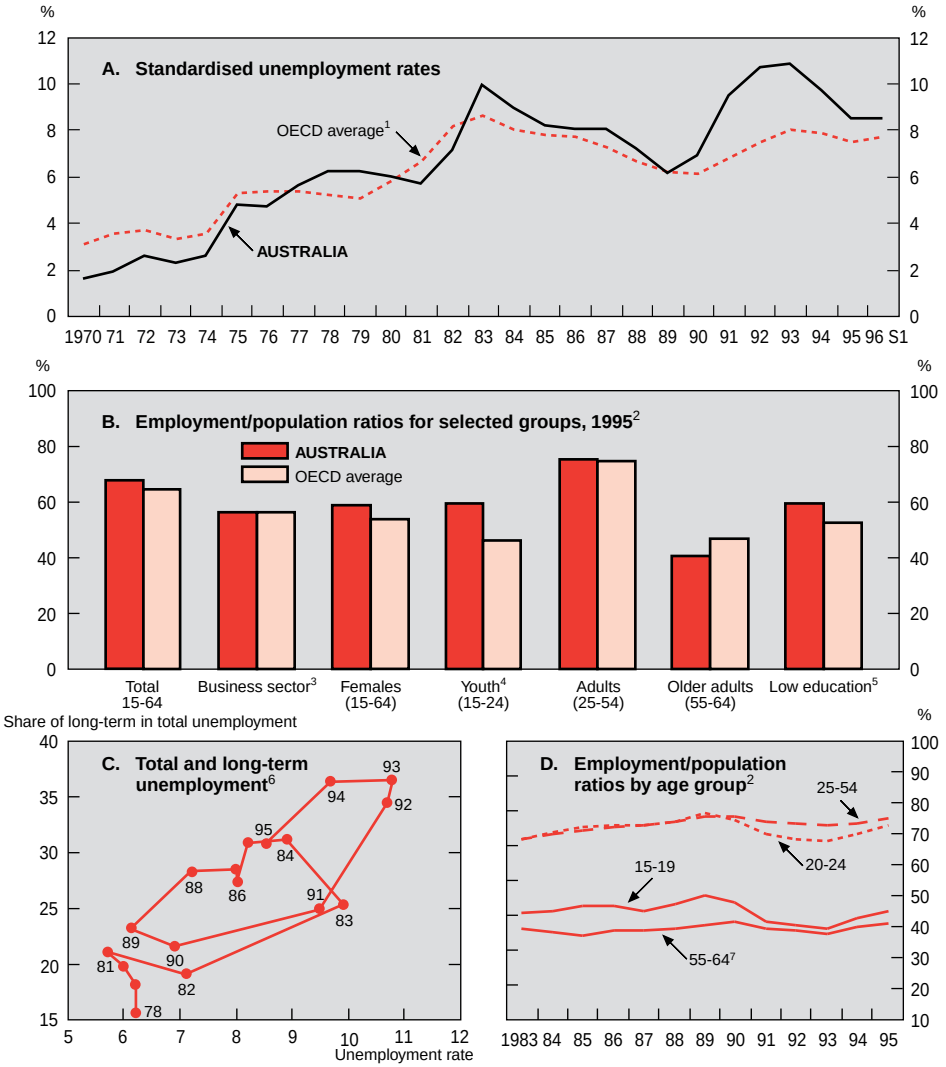
Introduction

Starting from below 2 per cent at the beginning of the 1970s, the Australian aggregate rate of unemployment broadly followed the OECD average over the last two decades (Figure 20, Panel A). This changed abruptly in the early 1990s when it jumped to a post-war record of almost 11 per cent (year average) in 1993 despite a slackening in labour force growth. This was in marked contrast to the rapid employment growth experienced in the previous recovery and led to fears of a so-called ''jobless recovery''. But these fears diminished when employment growth picked up strongly in 1994 and 1995 as the recovery became more firmly entrenched. Despite a cyclical rebound in labour force growth, the unemployment rate fell to around 8¼ per cent in mid-1995.[44] Since then, however, no further major inroad into unemployment has been recorded, in spite of economic growth in the 1995-96 period somewhat above potential output growth.

Upward spikes of unemployment in the mid-1970s, the early 1980s and at the beginning of the 1990s were reversed only slowly and incompletely so that the unemployment rate has ratcheted up over successive business cycles. The Australian experience accordingly resembled that of the European OECD countries. In consequence, various proxy estimates of the structural (non-cyclical) rate of unemployment, in spite of the substantial differences among the measures, all show an upward movement over the last 20 years (Table 10).

Concerns over low productivity growth and high structural unemployment led the previous government to embark on a series of microeconomic reforms, including the liberalisation of financial markets, reduction of border protection, deregulation of sheltered sectors and workplace reform. The latter comprised new industrial relations legislation and the introduction of enterprise agreements and award simplification. The further industrial relations reforms of the new

Figure 20. **KEY FEATURES OF THE LABOUR MARKET**



1. For 18 countries excluding Austria, the Czech Republic, Denmark, Greece, Hungary, Iceland, Luxembourg, Mexico and Turkey.
2. Defined as the percentage of each population group that is employed.
3. Business sector employment as a per cent of working age population.
4. The minimum age for youth differs across countries (either 14, 15 or 16) therefore, any comparison with the OECD average should be made with care.
5. Completed less than upper secondary school, 1994.
6. Long-term unemployment is defined as individuals looking for work for one year or more.
7. Calculated with unemployment for 55 and over.

Source: OECD, *Main Economic Indicators*, Labour Force Statistics and estimates.

Table 10. **Indicators of the structural rate of unemployment**[1]

Per cent of total labour force

	1970-79	1980-86	1987	1988	1989	1990	1991	1992	1993	1994	1995
NAIRU[2]			6.3	6.3	6.3	6.7	6.9	7.1	7.5	7.4	7.4
NAWRU	3.9	7.9	8.1	8.1	8.1	8.2	8.3	8.5	8.6	8.7	8.7
Beveridge curve indicator[3]	5.4	7.2	7.7	7.5	7.2	7.1	7.1	7.3	7.6	7.9	8.3
Okun curve indicator[4]	3.9	7.8	8.7	8.8	8.8	8.8	8.8	8.7	8.5	8.2	8.0
Hodrick-Prescott filter[5]	4.4	6.3	7.2	7.4	7.6	7.8	8.0	8.1	8.3	8.5	8.7
Actual unemployment rate	3.9	7.7	8.0	7.0	6.1	7.0	9.5	10.7	10.9	9.7	8.5

1. For the methodology of the NAWRU, the Beveridge curve and the Okun curve indicators, see Elmeskov (1993).
2. Rolling estimates for the 13 years to the year shown, based on the wage equation of the Treasury Macroeconomic (TRYM) Model of the Australian Economy. See Downes and Stacey (1996). Note that the NAIRU estimates have a very wide confidence interval. For example, at the end of the sample period and at a 95 per cent confidence level the NAIRU estimate ranges from 5.2 to 9.6 per cent.
3. The vacancy data used to construct the Beveridge curve indicator have been adjusted for a break in the series in 1983. Vacancies have also been lagged by two quarters to remove some cyclical dynamics.
4. The capacity utilisation rates used to construct this indicator are lagged two quarters.
5. Low-frequency component of a Hodrick-Prescott filter, $\lambda = 25\ 000$. See King and Rebello (1989).

Source: OECD.

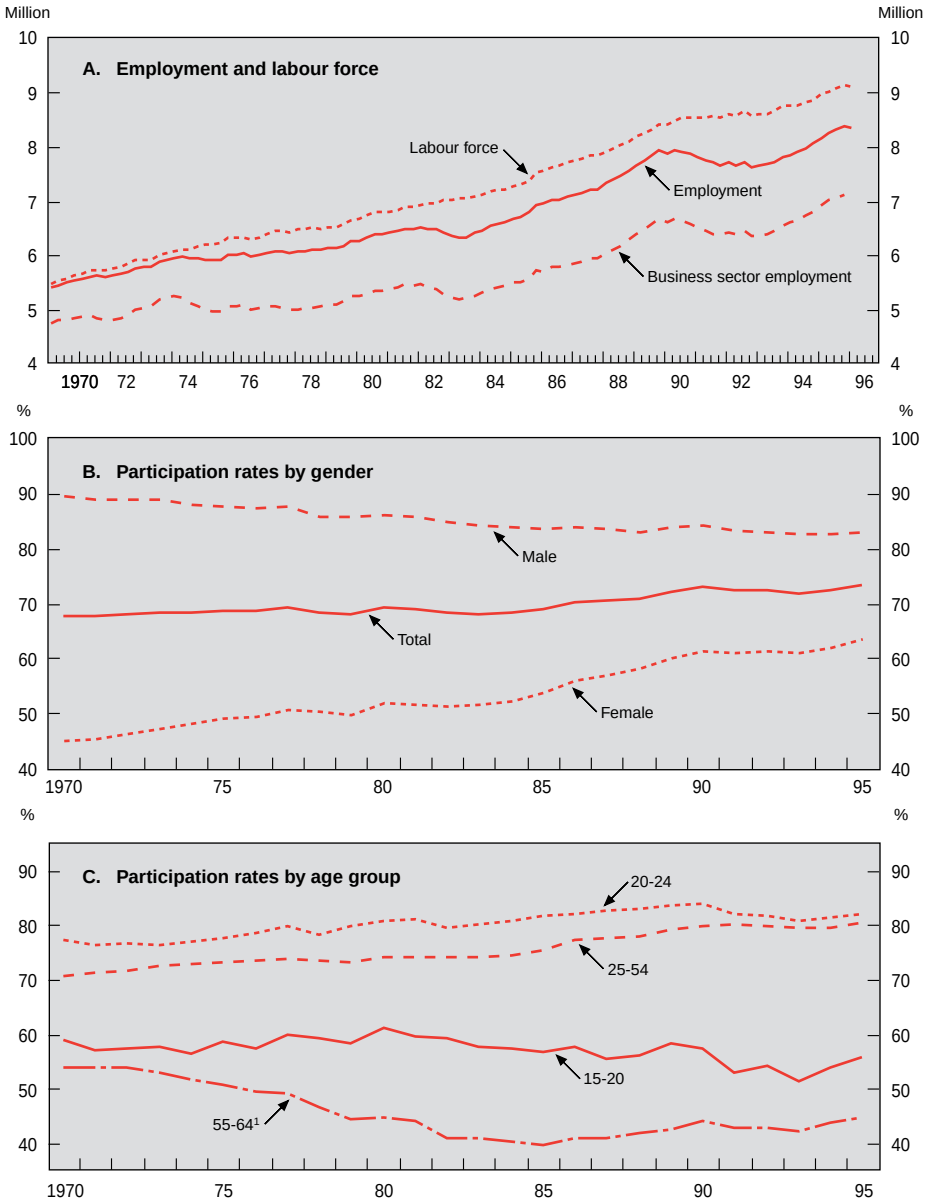
Coalition Government, which aim to make a more decisive shift to enterprise bargaining and achieve greater simplification of awards are outlined below.

This Chapter follows up the general structural policy recommendations of the *OECD Jobs Study* in the specific context of Australia. The structure of the chapter is as follows. The first section reviews the Australian labour market and employment performance. The policy requirements for Australia to emerge from the Study are discussed in the second part, while the following section sketches out some very recent policy initiatives. The final section concludes with an assessment of the scope for further action.

Labour market and employment performance

The ratcheting up of the unemployment rate over successive cycles took place against a background of exceptionally strong employment growth (Figure 21, Panel A). Indeed, employment growth in Australia over the latter part of the 1980s outpaced all other OECD countries, averaging over 3 per cent per annum, roughly double the rate recorded in the 1970s. Nevertheless, the unemployment rate has remained above the OECD average in all years except 1989.

Figure 21. **LABOUR FORCE TRENDS**



1. Labour force 55+ over population 55-64.

Source: OECD, *Main Economic Indicators* and *Labour Force Statistics.*

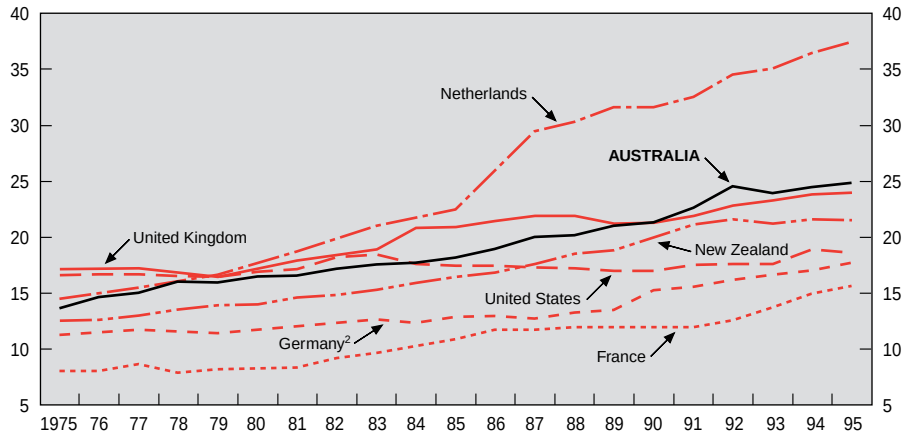
Labour force growth, however, has also generally been among the strongest in the OECD area, with average annual growth in the 2 to 3 per cent range in the 1960 to 1990 time period (Figure 21, Panel A). This primarily reflects growth in the working-age population, particularly among prime-age workers, and is mainly the result of natural increases in the population although net migration flows have also been relatively large in Australia, peaking at about 173 thousand people in 1988. Unlike other countries, changes in participation rates only had a negligible impact over most of this period. This changed over the latter part of the 1980s when increased participation rates of women contributed about a third to labour force growth. Migrants make up about a quarter of the labour force and this share has been relatively constant over time.[45]

Thus, up until 1985, the participation rate remained in the 69 to 70 per cent range, with increases in the participation rates of women offset by declines in those of men (Figure 22, Panel B). Over the latter part of the 1980s and early 1990s, however, the participation rate rose to the 73 to 74 per cent range – slightly above the OECD average – as declines in male participation rates levelled off and those of women continued to rise, particularly among married women. The decline in male participation rates was largest for men aged 60-64 years, for whom they fell about 22 percentage points over the last two decades to the 60 per cent range.[46]

Youth participation rates have generally hovered in the 70 per cent range over the recovery period. At the same time, however, there have been increases in full-time participation in education. Youth activity rates remain high in comparison to most OECD countries, but similar to North America, where there is more of a tradition for youths to combine both school and work. As in many OECD countries, youth unemployment rates are high, about double those of other age groups.

Employment/population ratios in Australia are close to the OECD average for adults, but higher for youths and lower for older workers (see Figure 20, Panel B). Youth ratios have been falling over time (Figure 20, Panel D), but rose again in 1994 and 1995 when employment posted large gains. A notable feature of the Australian labour market has been the increase in the share of part-time employment (Figure 22). Over the 1980s, Australia recorded one of the largest increases in the OECD area, and currently about one-quarter of employment is in

Figure 22. **SHARE OF PART-TIME WORKERS IN TOTAL EMPLOYMENT**[1]



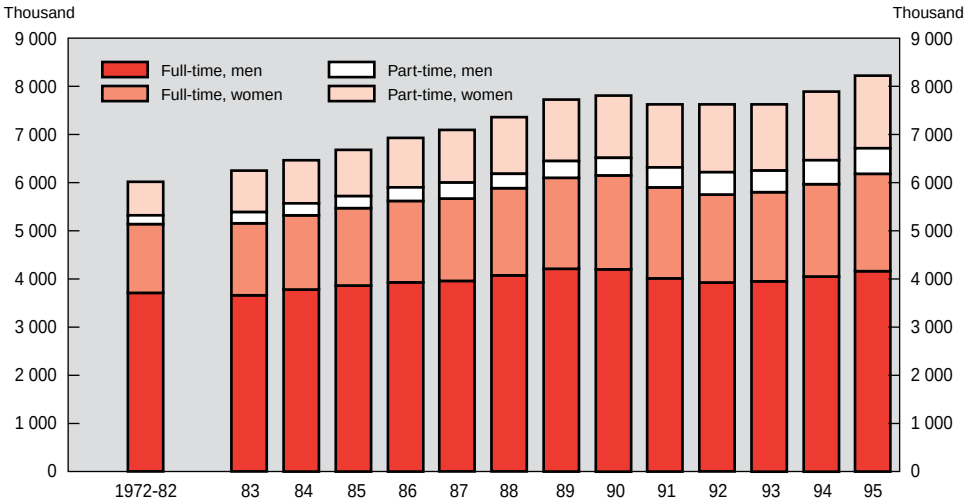
1. Break in series in 1987 (change in definition).
2. Western Germany up to end-1990 and the whole of Germany afterwards.

Source: OECD, *Employment Outlook.*

part-time work. The increase reflects rises in the share of both women and men in part-time work, although women make up about 75 per cent of part-time workers (Figure 23). Over the recent recovery period, part-time employment growth has been much stronger than full-time employment growth, and part-time work accounted for about half the employment gains. This is related to employment growth in retail trade where over 80 per cent of employment gains were in part-time jobs.[47] Not all part-time work, however, is voluntary. The number of involuntary part-time workers has doubled over the last 10 years, and amounted to almost 7 per cent of the labour force in 1993, one of the higher shares in the OECD area.[48] Currently, about one-quarter of part-time workers are involuntary in the sense that they would prefer to work more hours.

Casual employees[49] have also been increasing in number, comprising 24 per cent of employment in 1995, almost 8 percentage points higher than 10 years before. Since about 70 per cent of casual work is part-time, similar influences to the growth in part-time work are probably operating behind its increase. Moreover, casual work has increased in all industries and tends to be found in small

Figure 23. **PATTERNS OF EMPLOYMENT**



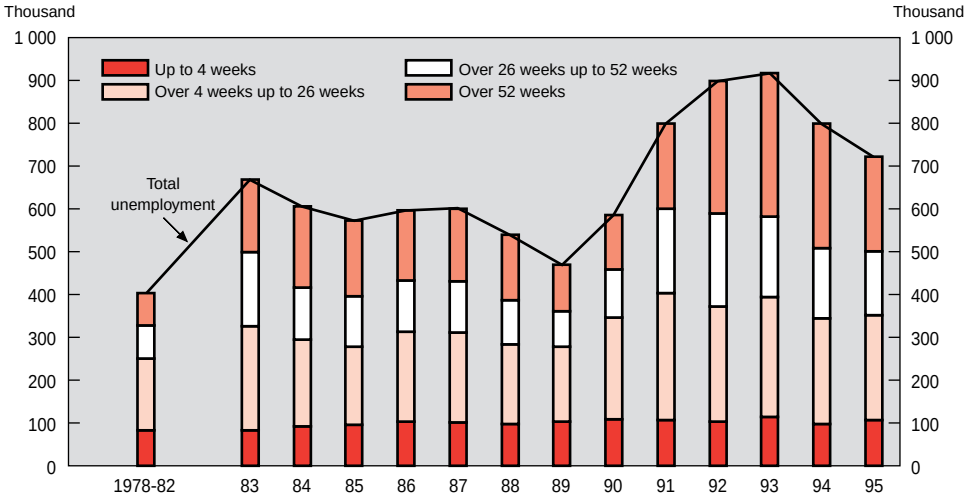
Source: Australian Bureau of Statistics, and OECD.

firms in the private sector which are non-unionised. Casual employment also tends to be concentrated among women and youths.[50]

An important feature of the latest recession and subsequent recovery has been the changing pattern of male and female unemployment rates. The latter have typically been higher than those of men, but since 1991, the unemployment rate for men has surpassed that of women. This reflects the job shakeout in industry and loss of full-time jobs, which predominantly affected men, and the expansion of job opportunities in the service sector that have favoured employment of women. In fact, new jobs have mainly come in the service sector and this was the only broad sector where the level of employment in 1995 was higher than in 1990, prior to the recession. While services have gained about half a million jobs, employment in industry is still about 100 000 below its 1990 level. Similar to most OECD countries, the structure of employment in Australia has moved overwhelmingly to services over the last 20 years, and it currently accounts for about 72 per cent of employment.

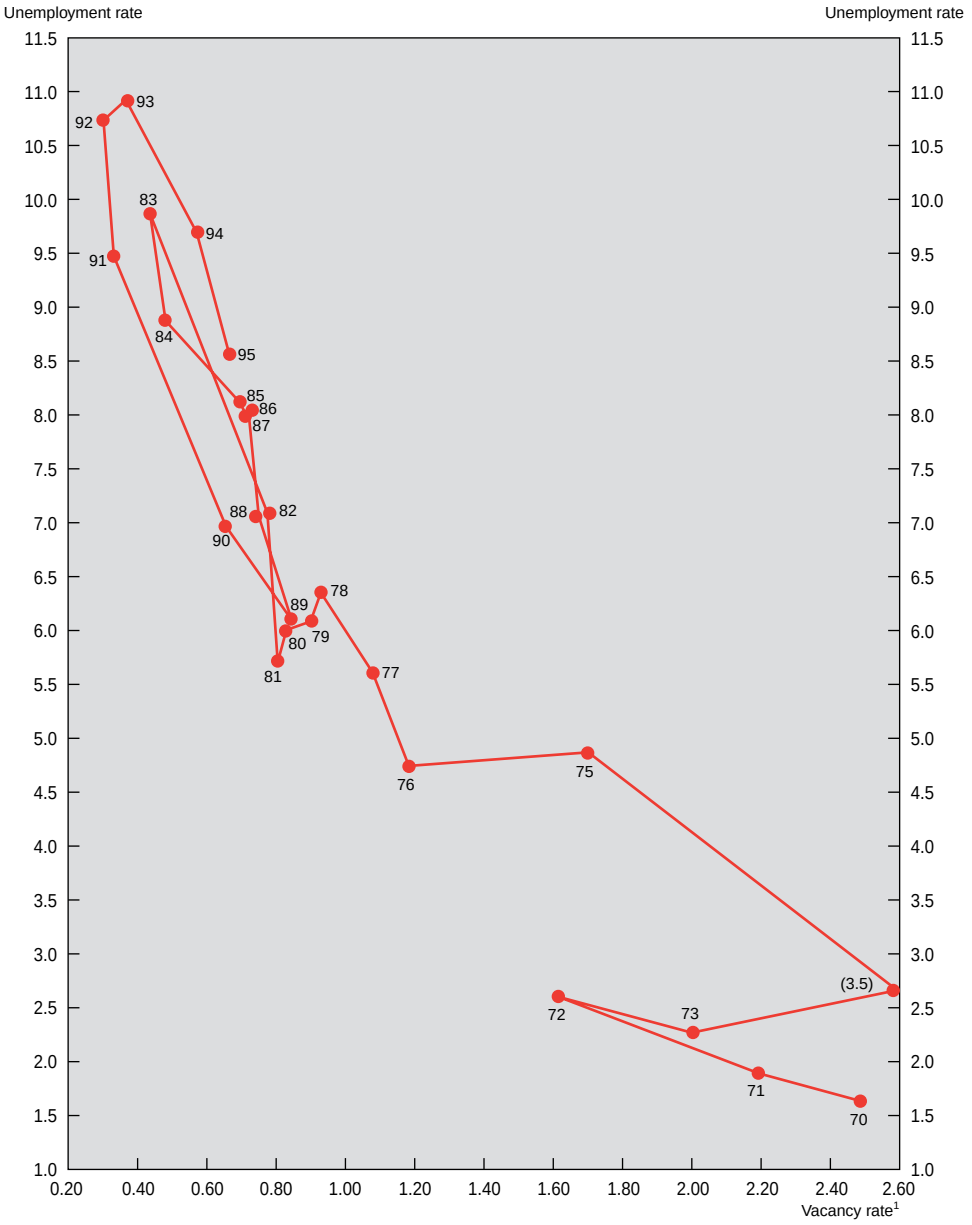
The main concern over the recovery period was the rise in long-term unemployment (Figure 24 and Figure 20, Panel C). Australia occupies a middle-ground among OECD countries, with larger inflows and outflows from unemployment than in most European countries, but smaller than those in North America. The incidence of long-term unemployment therefore lies between that found in the two regions. Over the latter part of the 1980s, long-term unemployment fell only 40 per cent below its 1984 level – despite rapid employment growth – and then began to increase sharply in the early 1990s, reaching a record peak of 37 per cent of total unemployment in 1993 before falling back to the 30 per cent range.[51] Furthermore, over half of the long-term unemployed were unemployed longer than 2 years. Long-term unemployment tends to be associated with lower education attainment: about two-thirds of the long-term unemployed do not have post-school qualifications, but the share of the long-term unemployed with post-school qualifications has increased, reflecting the fact that the unemployed are increasingly made up of individuals with relatively more advanced qualifications.

Figure 24. **STRUCTURE OF UNEMPLOYMENT**



Source: Australian Bureau of Statistics, and OECD.

Figure 25. **THE BEVERIDGE CURVE**



1. Annual vacancy rate based on quarterly data lagged two quarters. Break in series in 1984.
Source: OECD, *Main Economic Indicators* and Australian authorities.

The recent decline in the incidence of long-term unemployment – which has been much quicker than in the previous recovery – is probably related to an important extent to the rapid employment growth. This, combined with a stable participation rate, may have helped improve the chances of employment for the long-term unemployed.[52] Although the *Job Compact* may be a recent contributing factor in reducing long-term unemployment, the fall in long-term unemployment began prior to its introduction.[53] It is also possible that the expansion of the *JobStart* wage subsidy programme (described below) in the early 1990s may have been a contributing factor to the decline.

There is also conflicting evidence on whether the upward trend in the structural rate of unemployment has begun to level off (see Table 10 above). This is so because the relative importance of the forces which caused the rise in non-cyclical unemployment is not well determined. The estimated increases in the structural unemployment rate broadly coincide with indications of repeated upward shifts in the (non-cyclical) relationship between unemployment and vacancy rates (Beveridge curve) which occurred primarily around 1974 and during the 1980s[54] (Figure 25). The marked shift in 1974 is likely to have been induced by the surge in real wages by a cumulative 13½ per cent in 1974-75, which was much in excess of labour productivity growth (see below). Since the subsequent increase in unemployment did not induce a timely correction of the jump in unit labour costs the increase in unemployment is likely to have become persistent. However, earlier analyses of the outward shift of the Beveridge curve in the 1970s emphasise the influence of the fall in migration[55] and of the increased generosity of the unemployment benefit system, but empirical support for these factors appears rather weak.[56] More recently, it has been suggested that globalisation and technological change may have reduced demand for unskilled labour and thereby have raised non-cyclical unemployment. However, studies suggest that these effects, including any short run impact of structural reform measures, are fairly weak.[57] Moreover, the long run effects of structural reform on unemployment can be expected to be clearly positive.

Policy requirements

The experience of the cyclical upswing of the Australian economy since 1992 confirms that strong economic growth alone is unlikely to be sufficient to

Box 1. **The OECD Jobs Study: synopsis of recommendations for Australia**

The OECD *Jobs Study* sets out a strategy to reduce unemployment and to improve the ability of OECD economies to cope with structural change, through enhancing the ability to adjust and to adapt, and increasing the capacity to innovate and be creative. In Australia, the shift in the focus of wage bargaining from awards to the level of the enterprise during recent years has already enhanced the organisational flexibility of workplaces and hence, the adaptation to economic change. But important barriers remain. Therefore, the new Commonwealth Government's industrial relations reforms aim at providing further impetus to the transition to enterprise bargaining, through simplifying the agreement-making process, increasing the choice in agreements, ensuring more equal treatment of bargaining parties and restricting awards to a safety net of minimum wages and conditions of employment. The reforms will provide a greater range of opportunities for direct bargaining between employers and employees, free of the uninvited involvement of third parties.

High on the Government's agenda is the improvement of the school-to-work transition. Notwithstanding recent reforms, there is continuing concern about skill mismatches and shortages, particularly with regard to vocational and technical qualifications. Also, the implementation of the national competition policy recommended by the Hilmer Committee and agreed at the Council of Australian Governments meeting in April 1995 will enhance product market competition, especially in infrastructure services dominated by Government Business Enterprises. And the goal of the proposed changes to labour market assistance and its delivery is to make it more responsive to both jobseeker and employer needs, compared to assistance that has been offered in the past, partly through increased private sector involvement.

On the basis of the review of labour market policies and performance presented in this chapter, reforms in the following policy areas would appear to offer the greatest scope to contribute to improved labour market outcomes:

Improve labour force skills and competences

- *Ensure that reforms to apprenticeship and traineeship arrangements announced in the 1996 Budget are put in place.*
- *Allocate a much larger proportion of governments' budgets for vocational education and training on a competitive and non-discriminatory basis, allowing private providers to compete on the same terms as Technical and Further Education institutes.*

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(continued)

Increase the flexibility of wages and employment conditions

- *Make a more decisive shift towards enterprise-level bargaining than has occurred to date*. The 1996 industrial relations reforms promise an environment which will be considerably more favourable to enterprise-level bargaining. It will also reform employment protection legislation, limiting the potential costs of unfair dismissal. It is vital for improved labour market performance in Australia to move more quickly towards enterprise bargaining, to enhance the flexibility in overall employment conditions.
- *Increases in award wage rates should be modest and focussed on the low paid so as to maintain strong incentives for enterprise bargaining*. This would be consistent with the intention that awards operate as a true safety net.
- *States which maintain award systems should harmonise their industrial relations legislation with that of the Commonwealth Government*. This would encourage the spread of enterprise bargaining in the State systems, where progress to date has been limited.

Active labour market policies

- *Ensure reforms to labour market assistance are implemented*. The radical programme of change to labour market assistance arrangements put forward by the Government should lead to a much greater degree of contestability and competition in the employment placement market. By providing case managers with appropriate incentives and flexibility in deciding the most appropriate forms of assistance for individuals, the new arrangements should allow for the achievement of better employment outcomes for jobseekers. The targeting of intensive employment assistance to the long-term unemployed and the ''at risk'' groups, while providing the less expensive forms of support to other jobseekers, will ensure that individuals receive the level of assistance they need and at the same time minimise deadweight costs.
- *It will be necessary for the Government to ensure that monitoring mechanisms are properly designed and implemented for the new arrangements: firstly, to ensure that job seekers are able to make informed decisions regarding their choice of Employment Placement Enterprise; and secondly, to ensure that the new arrangements deliver the expected benefits*.
- *Ensure, as proposed, that there is clear separation between the purchasing and providing functions in delivering labour market assistance and full competitive neutrality between the public and private EPEs*.

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Enhance product market competition

- *Establish a tariff reduction programme for the period beyond 2001, when the present programme expires*. Reductions in tariffs should be scheduled in the motor vehicle, footwear, clothing and textiles industries, where current levels of import protection are high.
- *States and Commonwealth regulators should cooperate to ensure that the national competition policy works effectively*. Among other things, this will involve allowing their Government Business Enterprises to participate in inter-State markets for infrastructure services, such as electricity, and exercising good-will in their reforms of legislation and regulation inconsistent with the objectives of the national competition policy.

Enhance innovative performance

- *Continue the cost/benefit analysis of technology-related programmes* and monitor the effects of changes to the R&D tax concession on business research spending. In view of the elimination of the R&D syndication programme, steps should be taken to ensure an adequate supply of venture capital for innovative start-ups.

bring unemployment back to the levels recorded in the 1970s. Moreover, it is not excluded that the unemployment rate could edge up again in future cyclical downturns, unless policy action improves the economy's ability to create new jobs and make existing jobs more viable. To this end, the *OECD Jobs Study* identified a variety of potential measures in numerous areas which will be briefly reviewed below. These measures stretch far beyond the labour market, to include policies to promote competition in goods and services markets, enhance human capital resources – discussed in greater detail in Chapter IV – and to foster the creation and diffusion of technological know-how. A synopsis of the recommendations made below can be found in Box 1.

Enhancing the ability to adjust and to adapt

Wage formation and wage flexibility

- Industrial relations

Labour market performance crucially depends on how much the wage formation process allows aggregate and relative wages to respond flexibly to changing economic conditions. Wage fixing is strongly influenced by the institutional set-up of industrial relations. In Australia, industrial awards generally specify minimum wages and conditions of work for most categories of labour. It is thus illegal to employ a worker at a wage or on other terms which are less favourable than the relevant award, irrespective of whether a worker is a union member or not. Awards are the outcome of the process of conciliation and arbitration. The terms of the award may have been reached by a settlement imposed by the Australian Industrial Relations Commission (AIRC), by a combination of conciliation and arbitration, or by conciliation alone. In five of the six States, industrial tribunals with similar powers to the AIRC make State awards.[58] In Victoria, the Employee Relations Act 1992 abolished all existing awards, with employees deemed to be covered by individual agreements where a new award or agreement is not made. The Employee Relations Commission may make new awards only if all parties consent to the tribunal exercising its powers. The Victorian system is, therefore, one of voluntary arbitration.[59] Traditionally, actual wages paid were by some margin ''over-award'': normally, an increase in the award wage led to a corresponding percentage increase in the actual wage paid. In 1990, 80 per cent of wage and salary earners were covered by an award (32 per cent by a federal award and 48 per cent by a state award).

The system of Australian industrial relations has changed considerably during the last ten to fifteen years. Up to the early 1980s, collective bargaining took place primarily at the sectoral level. But from 1983 until early 1996, wage bargaining took place in the context of incomes policy agreements, known as Prices and Incomes Accords, made between the Australian Council of Trade Unions (ACTU) and the then Federal Government. This centralised approach was facilitated by the relatively high trade union density and a high share of employees covered by collective arrangements.[60] Under these Accords,[61] the ACTU committed to deliver agreed outcomes from wage bargaining in exchange for a greater say in social policy. Initially, the Accord involved highly centralised forms of wage setting, focused on limiting overall wage increases. But towards

the end of the 1980s, the trend turned towards more decentralisation, as the Accord progressively encouraged wage bargaining at the level of the enterprise while ensuring that awards continued to provide a secure and relevant network of minimum wages and conditions of employment. The move to enterprise bargaining was considered to be necessary to increase the organisational flexibility of workplaces and hence, productivity.

The first step towards formal enterprise bargaining was taken in 1987, when the Australian Conciliation and Arbitration Commission [predecessor of the Australian Industrial Relations Commission (AIRC)]awarded a two-tiered wage increase, with the second tier being linked to productivity improvements negotiated at an enterprise level. Enterprise bargaining was given a further boost with the passage of the *Industrial Relations Act 1988*, which created a formal stream of enterprise bargaining through Certified Agreements (CAs). Once the AIRC agreed in 1991 to certify enterprise agreements subject to certain conditions, notably that wage increases were based on the actual implementation of measures designed to boost productivity, awards were relegated to the role of providing safety net increases in wages and conditions.

Further changes have since been made to accelerate the spread of enterprise bargaining. The *Industrial Relations Act 1988* was amended in 1992 to encourage the use of Certified Agreements and to make the process of certification easier. Under the new amendments, an agreement was required to meet certain simple criteria for certification, including that it did not result in an overall deterioration in employees' terms and conditions of employment.[62] And the *Industrial Relations Reform Act 1993* created a new stream of enterprise agreements – known as Enterprise Flexibility Agreements (EFAs) – which allowed non-unionised and partly unionised corporations with federal award coverage to reach agreements directly with employees, provided that a majority of employees approved the agreement; however, even in non-unionised workplaces, unions retained the right to appear before the AIRC to be heard on whether the agreement should be ratified. In recognition of the difficulties which small enterprises may have with formal bargaining, the *Reform Act 1993* also encouraged the wider use of enterprise flexibility clauses and facilitative provisions in awards, which allow agreements at individual workplaces to change the general conditions of the award to suit their needs.

By June 1996, about 62 per cent of employees covered by a federal award (amounting to some 25 per cent of all employees) were also covered by a formally registered enterprise agreement.[63] While approximately 87 per cent of public sector employees who were covered by a federal award were also covered by a registered agreement, the proportion of private sector employees covered in the federal system was only around 52 per cent. In the state systems, where there are more employees, the proportion of workers covered by formally registered collective agreements varies from State to State. Queensland had the highest proportion, with 34 per cent of state award employees covered – New South Wales, South Australia and Western Australia each had around 30 per cent coverage. Only limited information was available for Tasmania; none is available for Victoria.

A particular problem in spreading formal enterprise bargaining has been the small and medium-sized enterprise (SME) sector. Certified Agreements are generally not adapted to this sector, which is lightly or non-unionised, and EFAs have not proved to be especially well adapted either: only 177 EFAs covering about 13 000 employees were formally registered in the federal system[64] between March 1994 and April 1996. A number of factors have constrained the spread of these agreements, including that employers had both to be legally incorporated and have federal award coverage, and that many of them regarded EFAs as unnecessarily complicated and likely to encourage unwanted union intervention (for example, in proceedings before the AIRC to approve agreements).[65] Removing such impediments to the spread of decentralised bargaining at the enterprise level is one of the main aims of the new Government's 1996 industrial relations reforms (see below).

The changing industrial relations in Australia were accompanied by a steep decline in trade union membership from some 51 per cent of total employees in 1976 to 33 per cent in 1995. Union density fell both in the public and the private sectors; it is particularly low in the latter, 25 per cent in 1995. At the same time, an unprecedented decline was seen in industrial disputes: while there was no obvious change in trend in the 1970s, the number of workdays lost through industrial disputes fell sharply in the early 1980s and remained at a very low level since then. For example, the number of workdays lost through industrial disputes in 1994 was one eighth of the level of 1980.

- Aggregate wage flexibility

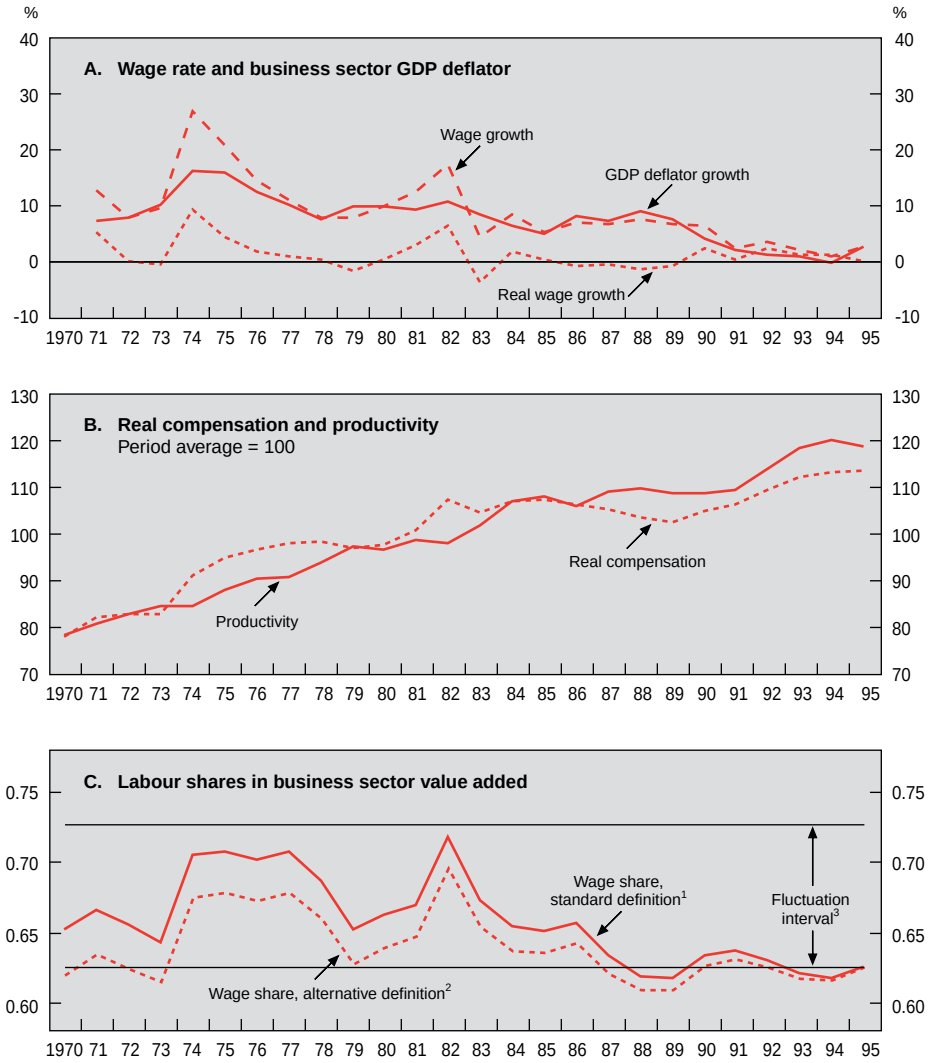
In line with findings for virtually all OECD countries, Australian average real wages respond to the state of the labour market in the long run. However, in international comparison, the magnitude of the real wage response to labour market pressures – ''real wage flexibility''[66] – appears to be either somewhat below average or average,[67] depending on the chosen specification and estimation period. The notion of comparatively low aggregate real wage flexibility is in line with the observation that strong increases in unemployment in 1973-1975 and 1982-1983 were accompanied by comparatively strong real wage growth (a cumulative 12.8 per cent in 1973-1975 and 4.3 per cent in 1982-1984). A general finding of empirical research is that the impact of long-term unemployment on real wages appears rather negligible.[68]

There is some support for a restraining effect on real wage growth exercised through the various Accords,[69] although there is a great variety of (sometimes conflicting) empirical estimates. It is noteworthy in particular, that from 1983 to 1995, real compensation per employee in the business sector rose by a cumulative rate of 8 per cent while during the same period labour productivity advanced by a cumulative 16½ per cent (Figure 26, Panel B). This contributed heavily to strong employment growth in the business sector during this period (a cumulative 34 per cent), although most of it went into part-time jobs. The specific role of the Accords in achieving these results is, however, hard to isolate clearly. Nevertheless, the cumulative 3.7 per cent real wages growth recorded in 1991-1993 suggests that, at least by this stage, wages had again become quite insensitive to unemployment.

- Relative wage flexibility

When the structure of relative demand for and relative supply of different groups of workers is changing, aggregate wage flexibility – average changes in wages of all employees in response to changing overall labour market conditions – is not sufficient. What is required in these circumstances is a fall in relative wages for groups of workers whose supply has risen or demand fallen, hence relative wage flexibility. Analysis presented in the *Jobs Study* suggests a positive correlation between the dispersion of earnings and private sector employment. This appears to hold also for Australia, where comparatively high wage dispersion is accompanied by substantial private sector employment growth.[70]

Figure 26. **REAL WAGE, PRODUCTIVITY AND LABOUR SHARES: BUSINESS SECTOR**



1. Calculated by imputing a wage compensation to self-employed equal to average compensation for wage earners.
2. Calculated by imputing a wage compensation to self-employed rising from 70 per cent of average compensation in 1970 to 100 per cent in 1995.
3. Horizontal lines show fluctuation interval over the period up to and including 1980. They have been calculated as the mean wage share, based on the first method, +/- two times its standard deviation.

Source: OECD.

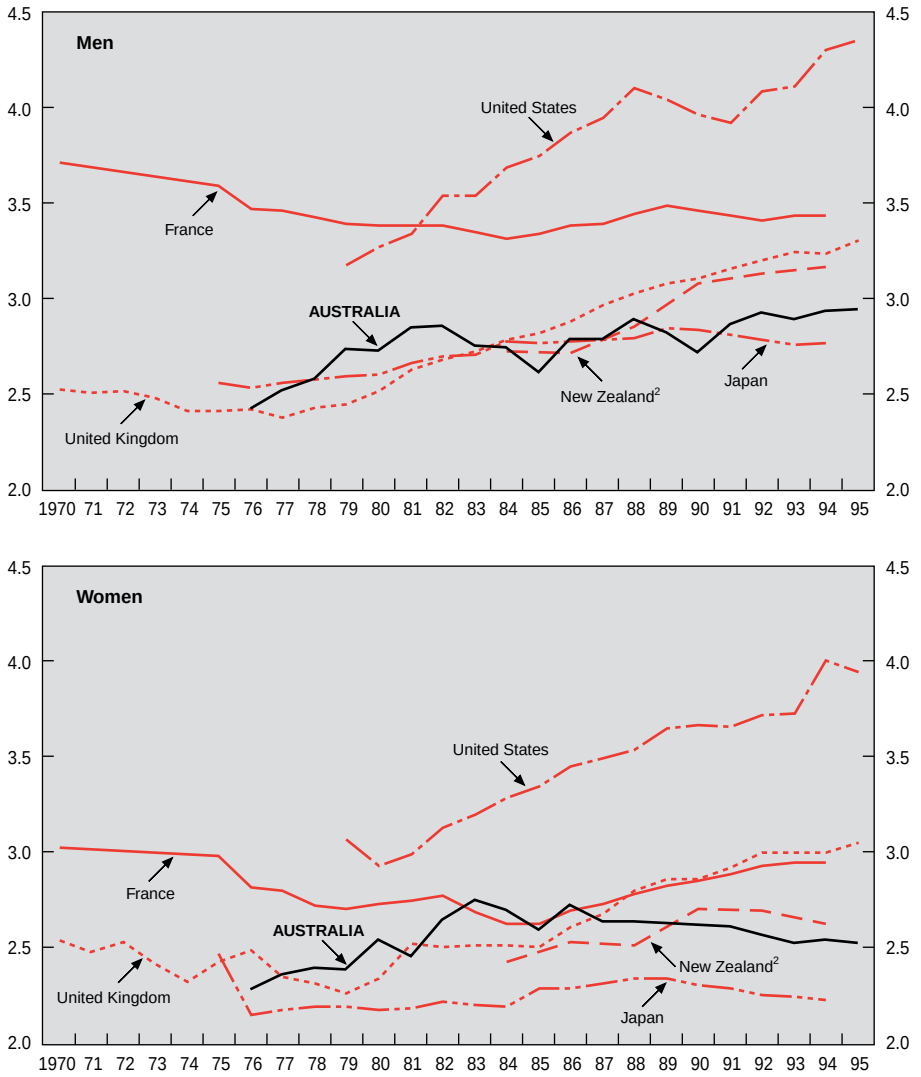
Over the past 20 years, the dispersion of earnings in Australia has, on balance, widened (Figure 27), which is in marked contrast to the unchanged or narrowing earnings dispersion experienced in a number of continental European OECD countries.[71] Among the countries for which data are available, the trend towards earnings dispersion was more pronounced only in the United States, the United Kingdom, Italy and Portugal. The coincidence of rising earnings dispersion and falling union density and collective bargaining coverage in Australia is in line with observations made in other OECD countries.[72] It is difficult, however, to draw definitive conclusions from the Australian data about the impact on earnings dispersion of the centralised approach to wage fixing under the Prices and Income Accord in the mid to late 1980s, compared to the increased focus on enterprise bargaining in the first half of the 1990s.

The notion of flexible relative wages is supported by another recent study (Coelli[73] *et al.*), which found a high measure of sectoral wage dispersion in Australia, in spite of the centralised wage setting system during much of the 1980s. Over the 1975-1990 period, Australian sectoral wage dispersion was estimated to be on the scale of that in the United States, a country which is widely considered to have a flexible labour market.[74] The study infers that Australian wages seem relatively strongly related to factors influencing labour demand and that ''the relationship between Australian sectoral wages, productivity and relative prices was among the strongest in the OECD countries for which data were available''.[75] There is, however, some evidence of an only weak tendency for local wages to adjust to regional differentials in unemployment.[76]

For the country as a whole, the degree of competitive forces in Australia would seem to be quite strong, if based on an international comparison of price levels allowing for differences in per capital income,[77] for lack of a better overall indicator of anti-competitive behaviour. However, research reported in the *OECD Jobs Study* on data for 1986 suggests that Australian wages continue to differ considerably across sectors and firms even after controlling for age, education, occupation and gender[78] which could indicate rent sharing of employers with their workers rather than reflecting productivity differentials.[79] This in turn could point to imperfect product market competition which could affect employment through wages above market-clearing levels or prolonged job search. For example, relatively high wages in the Australian mining sector coincide with scale economies in this industry, which act as a powerful entry barrier.[80]

Figure 27. **TRENDS IN EARNINGS DISPERSION**

D9/D1[1]



1. D1 and D9 refer to the upper earnings of respectively, the first and ninth deciles of employees ranked in order of their earnings from the lowest to highest, *i.e.* 10 per cent of employees earn less than the D1 earnings limit and 90 per cent earn less than the D9 earnings limit.
2. The data have been interpolated for missing years.

Source: OECD, *Employment Outlook,* 1996.

Minimum wages and social benefit entitlements may hamper the adjustment of relative wages. While there is no universal statutory minimum wage in Australia, the award system effectively defines a multitude of minimum wages for different crafts and occupations.[81] The impact of these arrangements on employment and unemployment could be substantial if these wage floors were relatively high. Indeed, 198⅝6 data for the total hourly earnings distribution for full-time workers in Australia shares with other Member countries the basic feature of being skewed towards higher wages. However, the usual discrete jump at the bottom part of the earnings distribution appears less marked in Australia than in other countries,[82] which suggests that in general there is no outstanding wage floor problem. There are, however, considerable differences for different groups of wage earners. For example, the wages of women are more compressed around a lower mean.

Overall, there is some evidence of aggregate and relative wage flexibility, but the picture is complex, and flexibility has been insufficient to prevent structural unemployment from rising over time. As noted earlier since the early 1970s, the unemployment rate has increased sharply during periods of cyclical weakness but failed to fully unwind in the subsequent upturn. Even though the Accords of the 1980s may have provided some aggregate wage flexibility it had become clear by the end of the decade that the lack of flexibility entailed by the centralised system was not sustainable in the longer term. This reflected the constraints the centralised system placed on relative wage flexibility between individual enterprises and sectors. The system did not permit wages to vary in recognition of the differing productivity performance of enterprises and the differing circumstances facing firms in product and labour markets. As a consequence, the need to move to more decentralised wage setting arrangements with the focus on enterprise bargaining was widely accepted. Furthermore, movements in aggregate real wage growth recorded in the recession and recovery of the early 1990s perhaps indicates the Accord's effectiveness in achieving aggregate wage levels consistent with lower unemployment had become limited. This suggested a decisive move towards a decentralised wage system may be more successful in achieving the aggregate real wage flexibility required to achieve lower unemployment. While there has been a shift towards enterprise bargaining in the first half of the 1990s, policies which further encourage enterprise bargaining and enhance labour market flexibility in a broad sense are needed.

Taxes and transfers

- Taxes on labour

The *Jobs Study* concluded that taxes which create a wedge between the cost of employing someone and the consumption which can eventually be financed by working can affect employment and unemployment. The effect on labour supply can be positive or negative, depending on the extent to which the reduced returns from working encourage the substitution of leisure for labour (the ''substitution effect'') as opposed to an increased supply of labour to earn any given income (the ''income effect''). Insofar as such taxes are not borne by labour through a lower real consumption wage (*i.e.* insofar as there is wage resistance), they increase the cost of employing someone and, thus, reduce demand for labour.

The overall tax wedge on labour, including income taxes, social security contributions and consumption taxes, is low in Australia and has remained relatively stable in recent decades (Table 11).[83] Similarly, the marginal tax wedge on below average incomes is among the lowest in OECD countries (Figure 28). As tax wedges on labour in Australia have changed little, they are unlikely to have been an important factor in the rise in structural unemployment since the 1970s.[84]

Although tax wedges in Australia are undoubtedly much lower than in most other countries, some caution is called for in assessing the extent of this difference. In particular, the estimates presented here exclude payroll taxes not earmarked for social security. These taxes are considerably higher in Australia than in most other countries.[85] More importantly, compulsory superannuation contributions in Australia[86] are excluded from the wedge as they are paid into private funds whereas compulsory earnings-related retirement pension contributions (which provide for pensions linked to contribution records) in European countries are included as they operate through general government. Yet, insofar as governments are expected to honour the associated pension obligations and they are actuarially fair, it should make no difference to an employee – and hence their wage demands – whether or not contributions are paid to the government or to a private fund.

- Unemployment and other social benefits

Unemployment and other social benefits alleviate the financial hardship associated with unemployment and help the unemployed to find a job which

Table 11. **Overall tax wedges,[1] single APW, 1978-94**

Per cent of gross labour costs

	Total wedge		
	1978	1985	1994
Australia	28.1	30.3	29.2
Belgium	56.9	61.5	61.3
Canada	30.9	37.0	40.0
Denmark	59.2	65.8	62.7
Finland	52.2	54.4	55.5
France	n.a.	n.a.	59.7
Germany	49.9	52.9	58.5
Iceland	n.a.	31.9	36.4
Ireland	46.0	54.6	55.0
Italy	51.3	55.5	57.5
Japan	20.8	25.6	26.3
Luxembourg	47.7	51.2	52.5
Netherlands	55.3	58.4	55.3
New Zealand	33.3	33.1	39.0
Norway	62.5	65.9	58.0
Portugal	36.4	40.3	46.7
Spain	39.5	43.4	46.6
Sweden	57.8	63.0	59.6
United Kingdom	43.8	48.1	44.4
United States	35.6	36.8	35.0
Average[2]	44.8	48.9	50.0

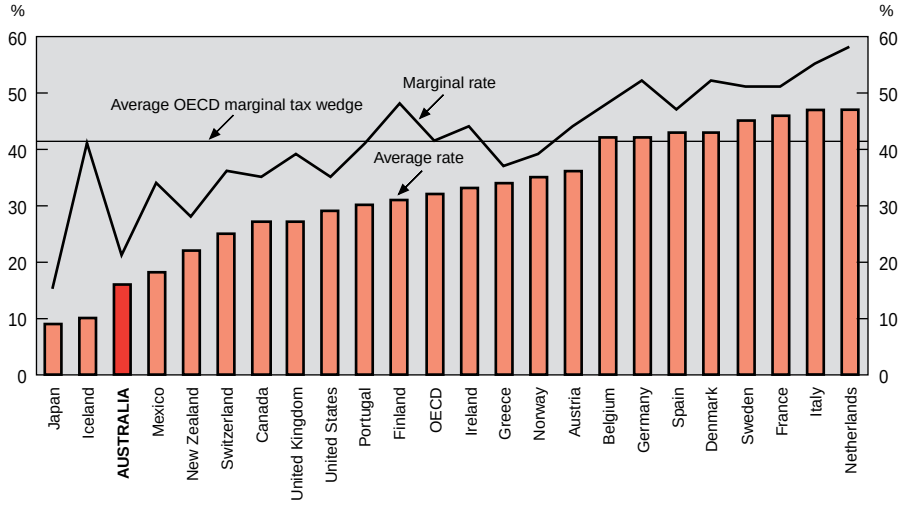
1. The overall tax wedge includes employees' and employers' social security contributions, personal income taxes and consumption taxes. Social security contributions and income taxes are calculated by applying tax rules to the level of earnings of an Average Production Worker (APW), as calculated in OECD, *The Tax and Benefit Position of Production Workers*. Consumption tax rates are calculated from aggregate tax and national income data. Non-wage labour costs other than social security contributions are not included in the calculations. Social security contributions include only those paid to the public sector; contributions to the private sector are ignored despite their importance in some countries. Social security contributions in some countries are closely linked to expected benefits; therefore, treating them in the aggregate as if they were simply taxes is a simplification. Payroll taxes which are not earmarked for social security are not taken into account in these calculations. No account is taken of ''non-standard'' reliefs, such as those for mortgage payments (see the *Tax and Benefit Position of Production Workers* for a detailed discussion of limitations). Furthermore, the tax wedges in this table only concern one point on the earnings distribution (the APW case), and someone on this income level may have consumption patterns which lead to different consumption tax payments from those derived from aggregate data.
2. Unweighted average excluding Australia.

Source: OECD.

better matches his/her competence. But they may also contribute to unemployment by prolonging periods of search between the loss of a job and finding a new one, and by encouraging workers and unions to take greater risks with jobs and by accommodating those who do not consider any extra income from working to be worth the effort. These effects may be compounded by an atrophy of skills during prolonged spells of unemployment. Important factors determining the

Figure 28. **AVERAGE AND MARGINAL TAX WEDGES IN THE EARLY 1990s**[1]

Single person receiving 66% of an average production worker's income



1. The average wedge is the sum of employees' and employers' social security contributions and personal income tax as a percentage of gross labour costs – gross wages plus employers' social security contributions. The marginal tax wedge is the percentage of any marginal increase in gross labour costs arising as a result of an increase in gross wages which is paid in employees' and employers' social security contributions and personal income tax. For Switzerland, estimates are based on the income tax legislation of the canton of Zurich.

Tax and wage data refer to the following years:
1991: Austria, France, Germany, Iceland, Ireland, Japan and the Netherlands.
1992: Belgium, Finland, Mexico, New Zealand, Norway, Portugal, Spain, Sweden, Switzerland and the United States.
1993: Australia, Canada, Greece, Italy and the United Kingdom.
1994: Denmark.
Source: OECD, *The OECD Jobs Study, Implementing the Strategy.*

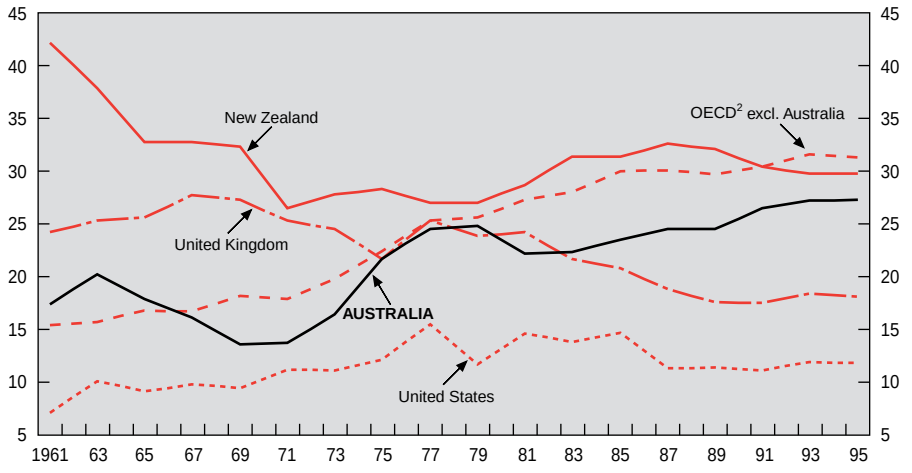
extent to which unemployment and other social benefits contribute to unemployment are likely to include the generosity of benefits, and the effectiveness with which the eligibility conditions[87] are enforced. Focusing on benefit generosity, which is more amenable to statistical analysis than the enforcement of benefit conditions, the *Jobs Study* found that cyclical rises in unemployment and the tendency for high unemployment to persist in recovery periods were often greater in countries with more generous benefit entitlements.[88] This suggests that benefits may influence unemployment with a lag, for example because increases in unem-

ployment which are due initially to a macroeconomic shock persist when benefits are high.[89]

The generosity of benefit entitlements has two dimensions: how high they are compared with potential earnings from working (the income replacement rate); and how long they last (their duration). According to a summary index of benefit generosity constructed for the *Jobs Study*,[90] the generosity of unemployment benefits in Australia has increased markedly since the 1960s, with much of the increase occurring in the 1970s (Figure 29). Similar or greater increases have occurred in most other OECD countries.

The nature of benefit generosity, however, differs considerably between Australia and most other OECD countries. For short spells of unemployment, gross replacement rates are relatively low in Australia, especially for single people (Table 12). And an unemployed person with a spouse in full-time

Figure 29. **SUMMARY MEASURE OF BENEFIT ENTITLEMENTS[1] 1961-95**



1. Unemployment benefits before tax as a proportion of the average across two earnings levels – average earnings and two thirds of average earnings – and three sets of family circumstances – single; with a dependent spouse; and with a spouse in work.
2. Excludes the Czech Republic, Hungary, Iceland, Luxembourg, Mexico, Poland, Portugal and Turkey.

Source: OECD, *The OECD Jobs Study.*

Table 12. **Unemployment benefit replacement rates by duration categories and family circumstances,[1] 1994**

Duration categories	First year			Second and third year			Fourth and fifth year			Overall average
Family circumstances	Single	With dependent spouse	With spouse in work	Single	With dependent spouse	With spouse in work	Single	With dependent spouse	With spouse in work	
Australia	29	52	0	30	53	0	30	53	0	27
Austria	35	38	21	33	37	0	33	37	0	26
Belgium	51	51	47	34	51	28	34	51	28	42
Canada	58	58	58	13	24	0	13	24	0	27
Denmark	70	72	69	70	72	69	70	72	69	70
Finland	65	65	65	47	47	40	30	30	0	43
France	58	58	58	40	40	34	25	25	0	38
Germany	35	38	35	31	34	0	31	34	0	26
Greece	44	53	44	19	19	19	0	0	0	22
Ireland	26	42	26	26	42	3	27	43	0	26
Italy	45	45	45	14	14	14	0	0	0	20
Japan	30	30	30	0	0	0	0	0	0	10
Netherlands	70	70	70	46	58	18	34	48	0	46
New Zealand	33	54	0	34	57	0	34	57	0	30
Norway	62	62	62	47	47	47	8	8	8	39
Portugal	65	65	65	39	43	39	0	0	0	35
Spain	65	65	65	30	30	30	0	0	0	32
Sweden	76	76	76	6	6	6	0	0	0	27
Switzerland	70	70	70	19	19	19	0	0	0	30
United Kingdom	18	29	18	19	30	0	19	30	0	18
United States	27	29	25	5	8	0	5	8	0	12
Unweighted average (excluding Australia)	50	53	47	29	34	18	18	23	5	31

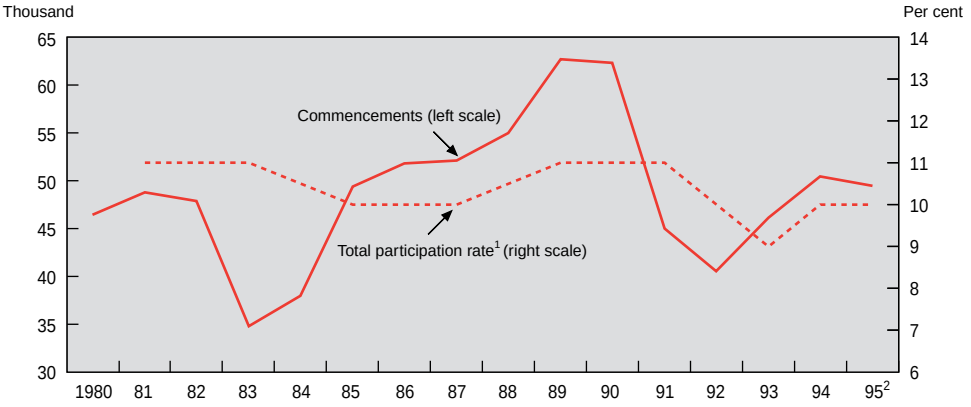
1. Benefit entitlement before tax as a percentage of previous earnings before tax. Data shown are averages over replacement rates at two earnings levels (average earnings and two-thirds of average earnings). See Annex 8 of the *Jobs Study* for further definitional information.

Source: OECD Database on Unemployment Benefit Entitlements and Replacement Rates.

employment is usually[91] not entitled to any benefit at all, in contrast to the situation in most other countries. But Australian replacement rates are the same at all durations of unemployment, whereas they fall sharply in most other countries once entitlement to insurance benefits has been exhausted (usually within the first 2 years). As a result, gross replacement rates after a long spell of unemployment are relatively high in Australia for unemployed people who are single or have a dependent spouse.

The OECD summary measure of benefit entitlements shown in Figure 30 and Table 12 does not allow for taxation, benefits for (dependent) children, social assistance and housing benefits. Allowing for these factors, unemployment entitlements are more generous in all countries than suggested by the summary measure (Table 13). The most important of these factors is additional entitlements for the unemployed with dependent children. The increase in benefit generosity for this factor is greater in Australia than in most other countries for short periods of unemployment (compare columns 3 and 4), but much greater in other countries after long periods of unemployment (compare columns 6 and 7).

Figure 30. **APPRENTICESHIPS: COMMENCEMENTS AND PARTICIPATON RATES**



1. Apprentices as per cent of population aged 15-19.
2. Break in series in 1995 due to introduction of new Statistical Standard.

Source: NCVER, Apprenticeship Statistics; Australian Bureau of Statistics, Estimated Resident Population by Sex and Age.

Table 13. **Replacement rates**[1] **for single-earner households, 1994**

	Panel A. Replacement rates at the average production worker level of earnings						
	Replacement rates in first month of unemployment: no social assistance					60th month of unemployment: including social assistance	
	Gross replacement rates (before tax)		Net replacement rates (after tax and other benefits)			Gross replacement rates (before tax)	Net replacement rates (after tax and other benefits)
	1	2	3	4	5	6	7
	Single	Couple, no children	Couple, no children	Couple, 2 children	Couple, 2 children, housing benefits	Couple, no children	Couple, 2 children, housing benefits
Australia[2, 3, 4]	22	40	49	64	71	40	71
Belgium	46	46	64	66	66	42	70
Canada	55	55	63	67	67	0	47
Denmark[5]	60	60	69	73	83	60	83
Finland	53	53	**63**	**75**	**88**	25	98
France	57	57	69	71	80	36	65
Germany	37	42	60	71	78	37	71
Ireland	23	37	49	64	64	37	64
Italy	30	30	37	47	47	0	11
Japan	37	37	**43**	**42**	**42**	0	68
Netherlands	70	70	77	77	84	0	80
New Zealand[3, 4]	26	43	48	64	70	43	70
Norway	62	62	**67**	**73**	**73**	0	83
Spain	70	70	75	75	74	0	46
Sweden[2, 5]	80	80	**81**	**84**	**89**	0	99[2]
Switzerland	70	70	77	89	89	0	89
United Kingdom[3]	16	26	35	51	77	25	77
United States[6]	50	50	60	68	68	0	17
Unweighted average (excl. Australia)	54	53	61	68	73	18	67
	Panel B. Replacement rates at 2/3 of the average production worker level of earnings						
Australia[2, 3, 4]	34	61	66	76	78	61	78
Belgium	60	60	**75**	**76**	**76**	55	91
Canada	55	55	64	67	67	0	61
Denmark[5]	86	86	92	93	95	86	95
Finland	60	60	**67**	**83**	**89**	37	100

Table 13. **Replacement rates**[1] **for single-earner households, 1994** *(cont.)*

	Panel B. Replacement rates at ⅔ of the average production worker level of earnings						
	Replacement rates in first month of unemployment: no social assistance					60th month of unemployment: including social assistance	
	Gross replacement rates (before tax)		Net replacement rates (after tax and other benefits)			Gross replacement rates (before tax)	Net replacement rates (after tax and other benefits)
	1	2	3	4	5	6	7
	Single	Couple, no children	Couple, no children	Couple, 2 children	Couple, 2 children, housing benefits	Couple, no children	Couple, 2 children, housing benefits
France	65	65	79	81	88	54	83
Germany	40	44	**60**	**70**	**77**	39	80
Ireland	35	66	67	70	70	66	70
Italy	30	30	36	45	45	0	14
Japan	43	43	**49**	**48**	**48**	0	86
Netherlands	70	70	**79**	**78**	**84**	0	95
New Zealand[3, 4]	38	64	70	80	86	64	86
Norway	62	62	**66**	**75**	**75**	0	100
Spain	70	70	74	78	77	0	66
Sweden[2, 5]	80	80	**82**	**85**	**89**	0	1 212
Switzerland	70	70	**75**	**87**	**87**	0	97
United Kingdom[3]	24	39	52	67	90	38	90
United States[6]	50	50	66	60	60	0	19
Unweighted average (excl. Australia)	55	60	68	73	77	26	80

Note: In the first month of unemployment it is assumed that families possess enough assets to be ineligible for social assistance. In the 60th month it is assumed that they no longer have such assets and so social assistance (SA) is assumed to be paid where it is higher than other benefits to which they may still be entitled. Figures in **bold** indicate those cases where families would be entitled to SA on the basis of their income, were they not to have been assumed to have been disqualified by an assets test. The replacement rates reflect a strict application of legal provisions rather than common practice, where these differ.

1. It is assumed that the worker is 40 years old, and started work at 18. The replacement rates are for the first month of unemployment, after waiting periods have been satisfied. This entitlement is then multiplied by 12 to give an annualised equivalent, on which tax is calculated. The person is fully unemployed. Social assistance is calculated according to a ''typical rate'' for the country concerned. This may involve making assumptions about housing costs.
2. Benefit amounts for couples are calculated on the basis of both spouses actively seeking work.
3. Figures for Australia, Ireland, New Zealand and the United Kingdom are for 1995. Unemployment benefit parameters for Japan are for 1996.
4. There is no social insurance in Australia or New Zealand. All figures in the Table, including columns 1-5, refer to the assistance benefit.
5. SA is only available when there is a ''social event'' such as unemployment. Low earnings are not themselves a social event.
6. The taxes and benefits are calculated using the rules applying in Detroit, Michigan. All figures include aid to families with dependent children (AFDC) and food stamps. If these are treated as being equivalent to social assistance, columns 3, 4 and 5 would read 60, 59 and 59 at the level of APW, and 59, 52 and 52 at ⅔ APW.

Source: OECD Database on Unemployment Benefit Entitlements and Replacement Rates.

The difference in the short term reflects the fact that benefits in Australia are in the nature of social assistance, and are, therefore, closely tied to family circumstances, whereas benefits in most other countries are insurance payouts. After long periods of unemployment, benefits in almost all countries are unemployment/social assistance and, as such, operate along similar lines to those in Australia: *i.e.* means-tested, open-ended and dependent on family circumstances.[92] Such benefits for the unemployed with dependent children provide similar replacement rates in Australia as in most other OECD countries. However, for the long-term unemployed without children, benefits are much higher in Australia. On balance, benefit generosity in Australia is probably even further below the OECD average than suggested by the summary index as most unemployment durations are short.

If the increase in benefit generosity in Australia and most other OECD countries has contributed to the long-term rise in unemployment, an obvious part of the solution to the unemployment problem would be to reduce the generosity of benefits. But this could increase poverty, especially in Australia where benefits only provide a social safety net (rather than assuring some – frequently higher – percentage of former earnings as in the case of unemployment insurance benefits). Another potential solution is to improve work incentives by giving in-work benefits. But these may give rise to poverty trap effects if benefits are phased out quickly as income rises or to large budgetary costs (and hence a high excess burden of taxation) if the income taper is gradual. Until recently, the response in Australia to the conflict between work incentives and equity was the same as in many other countries – namely, to provide in-work benefits with high withdrawal rates to low income earners with children. But, as a result of growth in real benefit levels relative to low wage levels since the early 1980s, by the early 1990's these payments no longer provided sufficient encouragement for some unemployed families to accept low-paid work.[93] In addition, these payments did not address work incentives for couples without children. Because the means test for unemployment benefit was on household income, there was a strong disincentive for a member of a couple to take or retain a low paid full-time job or a part-time job where their partner was unemployed. This may have been a factor explaining the relatively low employment rates of women married to unemployed men: in 1992, only about a quarter of women with unemployed husbands were employed, compared with 64 per cent of women whose husbands were

employed.[94] In response to these concerns, significant changes in income support arrangements were announced in the *Working Nation* statement and came into effect in July 1995 (see below). Although these reforms lowered marginal effective tax rates (METRs) for families with children, they remain high (Table 14),

Table 14. **Incidence and causes of high marginal effective tax rates**

One-earner couples

	METR %	Region where METR applied % of APW earnings	Tax and benefit combinations causing high METRs
Australia	90	38-62	Income tax (20%), parenting allowance (70%).
	38	62-78	Income tax (34%), low-income rebate withdrawal (4%).
	104	78-84	Income tax (34%), Medicare payments (20%), additional family payment (50%).
	85	84-100	Income tax (34%), additional family payment (50%), Medicare levy (1.45%).
France	78	57-91	*Revenu minimum d'insertion (RMI)* disregard (50%), social security (18.7%), *Contribution sociale généralisée (CSG)* (2.3%), housing benefit (16.5% average).
Germany	89	72-82	*Milderungszone* [phase out of income-tax free zone (this has now been abolished)]: income tax (51%), social security (18.3%), housing benefit (20%).
Ireland	105.5	62-76	Social security (5.5%), income tax (40%), family income supplement (60%).
Sweden	72	147-160	Income tax (20%), social security contributions (2%), local tax (31%), housing benefit (20%).
United Kingdom	97	46-65	Income tax (20%), social security (10%), family credit (70%), housing benefit (65%), Council tax benefit (20%).
	80.5	65-77	Income tax (25%), social security (10%), family credit (70%).
United States	72	62-71	Social security (7.65%), income tax (15%), local tax (5%), food stamps (24%), earned income tax credit (17.68% for family with two children).

Note: 1994 systems, except for Australia and the United Kingdom (1995). Family credit is only revised every 6 months, so the long-term METR given in the table for the United Kingdom may be substantially higher than that faced in the short term. Fewer than half of Ireland's family income supplement recipients are on earnings' levels that are exposed to the METR indicated. The benefit level, once set, is not revised downwards for 12 months, even if income increases in the meantime. The long-term rate presented in the table is substantially higher than that faced in the short term. Figures for individual taxes and benefits do not sum to the overall METR in France and the United Kingdom because benefits are withdrawn against net, rather than gross income. The 38 per cent rate for Australia is included to give a more complete impression of Australian METRs.

Source: OECD Database on Unemployment Benefit Entitlements and Replacement Rates.

exceeding 80 per cent over most of the income range from 38 per cent to 100 per cent of Average Production Worker's (APW) earnings.

As previously noted, apart from benefit generosity, the other main factor likely to have a bearing on the extent to which unemployment and related benefits contribute to unemployment is the effectiveness with which entitlement conditions are enforced. In Australia, as elsewhere, unemployment benefit recipients must satisfy an activity test. This is administered under the Social Security Act, by both the Department of Social Security (DSS) and the Commonwealth Employment Service (CES). The payment of unemployment benefits is administered by DSS. The main elements of the activity test are that the individual is actively seeking paid work, willing to take suitable work (including part-time and casual jobs) and agrees to requests from the DSS or the CES to take suitable paid work or go on approved training courses. Mechanisms are in place to strictly enforce the activity test.

Flexibility of employment and working time

- Flexibility in conditions of employment

The *Jobs Study* concluded that legal impediments to flexibility in conditions of employment could reduce the demand for labour, because they prevent employers from negotiating with their employees changes in workplace organisation which could increase productivity. Indeed, flexibility in employment conditions has become more important than ever in recent years with the globalisation of economic activity and of the radical changes in information technology. These developments, together with falling import protection[95] and the exposure of the non-traded sector to a more competitive regulatory regime, have increased competition in product markets, placing pressure on firms to respond in ways which usually call for some workplace reorganisation. Impediments to flexibility in employment conditions could also hinder job growth by blocking a supply of jobs with irregular conditions, such as for hours of employment, which may nevertheless be more convenient for some workers.

The capacity of individual employers to negotiate changes in employment conditions with their employees has been circumscribed in Australia by awards (see above). But in recent years, flexibility has increased through shifting the focus of bargaining over wages and conditions of employment to the level of the enterprise (see above). Enterprise agreements have introduced measures such as

more flexible working hours,[96] job redesign, multiskilling, new classification structures, reduced demarcations, team work, benchmarking for international best practice, quality assurance systems, consultative arrangements and special provisions for workers with family responsibilities.[97]

There also have been moves to simplify award coverage. Whereas awards previously contained lists of job classifications without any emphasis on skills and training, awards now generally contain a small number of classifications arranged in terms of skill levels. The new classification schema has increased flexibility in the conditions of employment by facilitating the relocation of workers between different activities within an enterprise.

Nevertheless, awards have remained highly prescriptive, covering aspects of the conditions of employment which go far beyond providing a safety net. For example, matters such as occupational health and safety (in addition to any legislative provisions), introduction of organisational change, training, superannuation (also by the Superannuation Guarantee legislation) and even the proportions of different kinds of workers which may be employed in some workplaces are frequently regulated by awards. However, there has been evidence of non-compliance with award conditions by small and medium-sized enterprises.

- Employment Protection Legislation

Employment Protection Legislation (EPL) is a general term usually taken to cover a number of areas, including statutory rights to severance pay and notice periods, rules and regulations pertaining to dismissal and regulations pertaining to the use of various forms of labour contracts, such as temporary work. In theory, the impact of EPL on the level of unemployment is unclear, but there is empirical evidence to suggest that if overly strict, it can affect both the level and structure of unemployment, particularly by making it difficult for outsiders – unemployed and new entrants to the labour market – to get work[98] and by encouraging hiring under contracts where unfair dismissal laws do not pertain, or are less severe. Based on a summary of data in these areas, the OECD *Jobs Study* concluded that EPL in Australia was relatively less strict compared with many countries. Indeed, a ranking of OECD countries placed Australia fourth out of 21 countries: only the United States, Canada and New Zealand had less strict EPL. Since then, however, changes in the area of unfair dismissal in the 1993 *Industrial Relations Reform Act* have probably altered this standing in the direction of making it more strict.

The Act[99] imposed a set of distinct requirements for termination of employment, which are (ultimately) heard before a newly created Industrial Relations Court, breach of which renders termination unlawful. If unlawful, the Court had the power to order reinstatement of an employee or award compensation to be paid. This moved away from the former conciliation/arbitration system under the Australian Industrial Relations Commission (AIRC), and although the AIRC still had the possibility to conciliate disagreements, the Industrial Relations Court only had to refer the matter to the Commission if it felt it appropriate. The incentive, therefore, became to litigate rather than resorting to conciliation to resolve disputes. In addition the Act allowed dismissed employees to appeal not only the merits of the dismissal but also the process followed in dismissal, *i.e.* unfair dismissal claims could also arise if certain dismissal procedures were violated. As a result of these and other changes, the Act was criticised as encouraging ''chronic recourse to legalism, complexity and technicality''.[100]

It is difficult to assess the impact of the 1993 changes on the labour market from aggregate statistics given the multiple influences at work, and the relatively short time period over which these changes have been in place. But anecdotal evidence suggests that they have created a severe disincentive to hiring workers, particularly in small firms, compared with the period prior to 1993. For example, a survey of employers ranked unfair dismissal laws as one of the top 10 areas that needed reform.[101] The sheer number of cases lodged also gives some idea of the extent of the problem: some 3 000 applications were lodged with the Court in the first nine months of the Act being passed, creating a backlog for the Court of almost 18 months.[102] In response to employer complaints over the disincentives to hiring, the government amended the Act in 1994 to take some of the bite out of unjust dismissal laws. *First*, in most cases, compensation for unjust dismissal was limited to six months' salary. *Second*, the Court's jurisdiction was limited to covering individuals under federal or state awards and earning less than $60 000 per year. *Third*, termination provisions were to exclude workers engaged for a specific task or period of time; workers serving a probationary period of not more than six months; and workers engaged on a casual basis for six months or less.[103] These changes, however, did little to limit the overall cost involved in actually terminating an employee because they did not address issues related to what could be considered a valid claim, *i.e.* reason for dismissal *versus* procedure followed in dismissal. Another important externality of the policy was its impli-

cations for active labour market policies. An important element of the *Job Compact* is placement in either subsidised or unsubsidised work, the former through the *JobStart* programme. Anecdotal evidence in this area indicates that one reason for lower take-up may have been fears by employers – particularly in small firms – that once the subsidy period ended, unfair dismissal claims could arise if firms attempted to dismiss the subsidised employee.

- Industrial Relations and Other Legislation Amendment Act 1995

In response to widespread dissatisfaction with the unfair dismissal regime included in the Industrial Relations Reform Act 1993, and its perceived negative impact on hiring, further changes were made in the Industrial Relations and Other Legislation Amendment Act 1995 (which came into effect in January 1996). Consent arbitration before the AIRC was introduced to allow parties – if they so agree – to settle the dispute through arbitration, although either party could opt to have the matter heard before the Court instead.[104] And, more importantly, the Act allows the Court and the AIRC to take into account a broad range of factors to determine if any remedy should be granted. This moves beyond the requirement to observe procedural fairness, which alone may no longer necessarily result in an order of reinstatement.[105] Nevertheless, dissatisfaction with the system remained, leading to further changes being proposed following the election of the new Government which are discussed below.

Active labour market policies (ALMPs)

Australia has a comprehensive set of ALMPs in all of the main programme areas. It currently devotes about 30 per cent of its labour market expenditures to ALMPs, similar to many other OECD countries.[106] But unlike in many other countries where the largest share of resources are devoted to training, relatively more is spent on subsidies to employment. This was in part due to the *Job Compact*, which used both job creation and wage subsidy programmes to place the eligible long-term unemployed (*i.e.* those receiving an unemployment allowance for 18 months or more) into jobs.[107]

The introduction of the *Working Nation* and the *Job Compact* led to changes in the composition of jobseekers targeted for assistance and changes in the mix of programme placements. Greater concentration was placed on the very long-term unemployed who make up the *Job Compact* group, *i.e.* unemployed for more than 36 months, but this was offset by a decline in assistance to those groups

unemployed for between 12 and 18 months. Despite this strengthened focus, the decline in the number of unemployed under the *Job Compact* was less than expected, *i.e.* from the June quarter of 1994 to the December quarter of 1995, the drop was only about 20 per cent as compared with an expected decline of 47 per cent. This was due to a higher than anticipated inflow into the *Job Compact* group, related in part to the withdrawal of some labour market assistance to the 12 to 18 month group who moved into the higher duration category.

The previous government expected that the *JobStart* wage subsidy programme would be the main ALMP used to help place the long-term unemployed into jobs (see Annex I for a description of the main labour market programmes). Under *Working Nation*, about 70 per cent of *Job Compact* employment placements were expected to be through wage subsidies: the actual proportion in 1995-96 was only about 34 per cent.[108] Targeting the very long-term unemployed was one of the factors which led to lower take-up of *JobStart*[109] (Table 15). This was due to employers' concerns about the level of skills and motivation of many of the long term unemployed (LTU). In order to provide job placements for the *Job Compact* group, the lower than expected take-up of *JobStart* was compensated for by increasing the number of placements in the so-called ''brokered'' programmes, mainly job creation projects intended primarily for *Job Compact* recipients. The most important of these was the *New Work Opportunities* (NWO) programme, which provides a mix of supervised work experience and training over a 26 week period. The advantage of these programmes is that they can provide some work skills for the long-term unemployed; the disadvantage is their higher programme cost and poorer post programme outcomes relative to *JobStart*. Moreover, there was some evidence that NWO was competing for a limited number of placements in the private sector with some other labour market programmes, such as *JobStart*. Also, the increased focus on the very long-term unemployed coincided with a fall in the main form of job search assistance (*i.e.* Job Clubs) provided to the short-term unemployed. This could have had negative longer run effects to the extent that such programmes are useful in combating increased duration. After the election of the new Government, the *Job Compact* was discontinued.

Monitoring of post-programme outcomes gives some idea of the overall effectiveness of the ALMPs. Surveys of participants' labour market status three and 12 months after programme participation indicate that overall, about one-

Table 15. **Commencements into selected labour market programmes**

1993-94 to 1995-96 (February)

Programme	1993-94		1994-95		1995-96	
	Thousand	Per cent	Thousand	Per cent	Thousand	Per cent
JobStart	145.2	26.4	95.4	16.8	64.4	12.7
National Training Wage	. .	. .	7.3	1.3	19.4	3.8
Total wage subsidies	**145.2**	**26.4**	**102.7**	**18.1**	**83.8**	**16.5**
New Work Opportunities	. .	. .	11.5	2.0	35.5	7.0
JobSkills	10.7	1.9	20.8	3.7	16.2	3.2
Landcare and Environment Action Programme	10.1	1.8	15.1	2.7	7.7	1.5
New Enterprise Incentive Scheme	4.9	0.9	5.9	1.0	4.9	1.0
Total brokered programmes	**25.7**	**4.7**	**53.3**	**9.4**	**64.3**	**12.7**
Total employment programmes	**170.9**	**31.1**	**156.0**	**27.5**	**148.1**	**29.2**
JobTrain	73.5	13.4	91.0	16.0	71.9	14.2
JobClubs	42.3	7.7	44.9	7.9	29.2	5.8
SkillShare	147.7	26.8	141.3	24.9	109.9	21.7
Other	115.9	21.1	135.1	32.8	148.5	41.3
Total training and other programmes	**379.4**	**68.9**	**412.3**	**72.5**	**359.5**	**70.8**
Total	**550.3**	**100.0**	**568.3**	**100.0**	**507.6**	**100.0**

Source: *Working Nation: Evaluation of the employment, education and training elements*, Table 4.1., Department of Education, Employment, Training and Youth Affairs (1996).

third are in unsubsidised employment (Table 16). For *JobStart*, however, about 50 per cent are in unsubsidised employment and the *New Enterprise Incentive Scheme* (NEIS) has a success rate of over 80 per cent.[110] Furthermore, in most cases the proportion in unsubsidised employment is higher after 12 months. For brokered programmes other than NEIS, success rates have been much lower. Preliminary evidence on the success of NWO, for example is not favourable: only 22 per cent of the long-term unemployed were in unsubsidised employment after three months.[111] Nevertheless, given that the programme's primary purpose is to help the most disadvantaged jobseekers, this figure should be treated with caution.

Even then, however, these results are only suggestive of programme impacts. They do not take into account deadweight losses – where some individuals might have achieved these outcomes without the programmes – nor do they account for displacement or substitution effects.[112] For wage subsidy programmes, displacement and substitution effects may be very large in the short-run and evidence from other countries suggests that they can crowd out short-run employment gains almost completely.[113] However, in the longer term, to the extent wage subsidy programmes improve the employability of disadvantaged jobseekers relative to other jobseekers they may expand the effective supply of labour.[114]

As described further in Annex I, a key element of the *Job Compact* was case management provided to jobseekers to assist them in getting work. This form of

Table 16. **Outcomes of selected active labour market programmes: three and twelve months after participation**[1]

Programme	Unsubsidised employment	
	3-month follow-up	12-month follow-up
New enterprise incentive scheme	82.8	78.5
JobTrain	33.1	40.4
Job Clubs	37.4	45.6
JobStart	52.0	58.8

1. 3-month programme outcomes refer to programme cessations September 1995, and status 3 months after leaving assistance. 12-month outcomes refer to outcomes for clients leaving programme assistance between April and June 1994.

Source: Department of Employment, Education, Training and Youth Affairs.

assistance was also provided to those unemployed for 12 months or more and those ''at risk'' of long-term unemployment. Jobseekers, to some extent, have a choice between case management offered in the public sector (through Employment Assistance Australia or EAA) and the private sector (through Contracted Case Managers, or CCMs). For jobseekers to make an informed choice between public case management the EAA and CCMs, and among CCMs, an evaluation of the relative effectiveness in placement is also important. Preliminary findings on relative placement rates between the EAA and CCMs indicate that performance appears to be converging as CCMOs (CCM organisations) gain more experience, and as more of them enter the market. Other preliminary findings indicate that outcomes for CCMs (after 10 months in operation) and the EAA are similar once adjustment is made for jobseeker characteristics. Nevertheless, given the learning curve necessary to start up this initiative, it may still be too soon to take these outcomes as a robust measure of relative effectiveness.

Given the vast number of changes involved with implementing *Working Nation* initiatives, it is inevitable that there would be some problem areas that would still need to be resolved. These include:[115]

- difficulties in accurately identifying ''at-risk'' jobseekers, and classifying them for referral to case managers;
- an inadequate focus on non-subsidised jobs as an outcome, and too much use of relatively expensive interventions that guarantee an outcome (*e.g.* broker programmes); and, while access to programmes must be approved by the CES, it may not lead to the most cost-effective path to placement since case managers do not have to pay for them;
- the administrative aspects of case management are time consuming, due in part to the complexity of the programmes on offer and how to access them, detracting from time case managers could use to find jobs for clients;[116]
- regional differences in employment opportunities, which are not reflected in funding to the CCMs;
- given that both EAA and the CES are part of DEETYA, it is unclear whether the playing field is level between the CCMs and the EAA.

These and other factors have led the Government to propose significant changes to labour market programmes and their delivery, outlined below.

Increasing the knowledge base, efficiency and innovative capacity

Upgrading skills and competences[117]

The skills and competences of the labour force have always been an important determinant of labour productivity (and therefore, of living standards) and employment. But in recent years, their importance in influencing labour force participation and unemployment rates has increased. Globalisation and technical change have reduced the demand for low-skilled workers in most OECD countries. In Australia, as in most European countries, this has mainly impacted on labour force participation and unemployment, rather than on real wage rates. During the past decade or so, unemployment rates for persons not having completed secondary school have been about one quarter higher than the national average, while for graduates, the rate has been around one half the national average. Youth have been particularly hard hit by the decline in demand for low-skilled workers; unemployment rates for persons aged 19 without vocational qualifications are more than twice the national average for 15-64 year-olds.

The deterioration in labour-market prospects for the low skilled, especially early school leavers, and government policy to improve workers' skills have led to a large increase in participation in education and training. The retention rate until the final year of secondary school has risen to over 70 per cent, double the rate of a decade earlier, and the number of university students has increased by over 40 per cent. A particular focus of policy has been to develop work-based vocational education and training (VET) pathways from school to work. Traineeships, a form of short apprenticeship, were introduced to extend work-based VET beyond the narrow range of occupations covered by traditional apprenticeships. And vocational courses with work placements have been developed for students in the final two years of secondary school. Efforts have also been made to enhance the labour-market relevance of education and training. This has been done through increased input from employers on course content. The most important development in this regard has been in VET, where certification now is based on industry-determined competency standards, rather than on the length of courses.

However, the lack of competitive pressures in the VET sector is likely to have hampered the development of courses well adapted to users' needs. Private providers are able to compete for at best a small proportion of public VET funds

in some States. Allocating a much larger store of these funds competitively would increase efficiency in the VET sector, facilitating the development of courses which are responsive to changing skill demands, convenient for students and employers and represent good value.

Even though there has been considerable progress in raising participation in post-compulsory education and training, about one half of school leavers still do not go on to further education and training as soon as they leave school. This group is currently assisted by the Youth Training Initiative (YTI) which provides case management assistance to help them find employment or return to education and training. However, the programme was only implemented in January 1995, and it is too early to evaluate its effectiveness. Ensuring that there are attractive options for these people to resume formal education and training therefore remains a policy challenge, as well as a key to reducing unemployment.

Product market competition

Product market competition is a key link between labour market flexibility and overall economic performance. Where product market competition is weak, employees often enjoy wage premia but at the expense of lower employment levels and less flexible working arrangements than in industries exposed to greater competitive conditions. However, relationships between product market reforms and labour market outcomes are complex. Product market reform in a particular sector could reduce wage levels, insofar as wage premia exist, or have a negative impact on sectoral employment, insofar as there was technical inefficiency or use of excessively labour-intensive technology. But deregulation in a particular sector also has pervasive effects on other sectors, not only by affecting relative prices and incomes, but also by facilitating growth and innovation so that overall income and employment effects are likely to be positive.

As in most other OECD countries, Australia's competition legislation aims to ensure that competition is not undermined by anti-competitive behaviour of firms. The Commonwealth Trade Practices Act (TPA) of 1974, Australia's main competition policy statute, prohibits certain anti-competitive agreements and the abuse of market power, including exclusive dealing, resale price maintenance and anti-competitive mergers and acquisitions. However, until 21 July 1996 the Act was limited in its scope, with application of the conduct rules to the professions, agricultural marketing authorities, and, most importantly, certain State and Terri-

tory government businesses being uncertain. Accordingly, the TPA may not have prevented state-owned enterprises, particularly in utilities and rail transportation from engaging in anti-competitive behaviour. Certain state legislation also acted to limit interstate competition between public utilities in the energy sector by granting monopoly rights.

But with per capita GDP having fallen from a top level in the 1960s to the bottom third of the OECD league in the 1980s, and the recognition that a large part of the problem resulted from a poor productivity record in infrastructure industries, Australia has devoted substantial efforts to intensifying competition in various economic sectors since the early 1980s. Reform has included deregulation, creation of new regulatory bodies, removing barriers to movements of goods and services across state borders through mutual recognition, and broad-brush improvement of the legal framework in which competition takes place. But until recently, most microeconomic reforms were pursued by the Commonwealth, and State and Territory governments were largely independent of these reforms. The practical effect of this rather uncoordinated approach was that different arrangements were often put in place to cope with similar issues, making progress with reforms variable and intermittent. Recognising that better co-ordination would help to maximise the benefits and limit the costs of structural reform, the Commonwealth, States and Territories agreed to examine a national approach to competition policy. Consequently, a Commission of Inquiry, headed by Professor Fred Hilmer, was established in October 1992 to develop a national competition policy.

The Commission's report,[118] released in August 1993, recommended a broad approach to national competition policy which addresses six major issues:

- anti-competitive conduct by firms;
- unjustified regulatory restrictions on competition;
- inappropriate structure of certain public monopolies;
- proprietary access to facilities vital for effective competition;
- monopoly pricing; and
- competitive advantages of some government enterprises as a consequence of their public ownership.

The Hilmer Committee's key recommendation was to apply the Trade Practices Act universally to all business activity in Australia, hence extending its application to Government Business Enterprises (GBE), Statutory Marketing

Authorities – the main source of assistance to agriculture – and unincorporated associations. The Report also proposed new policy guidelines to build a more competitive economy, in particular where the lack of competitive pressures is a result of government regulation or government ownership. These guidelines include the review of regulatory restrictions to competition, structural reform of monopolies, the granting of access rights to essential facilities (such as telecommunications networks, electricity grids and rail networks), prices oversight of GBE and ensuring that public and private enterprises compete on equal terms. The Report proposed the repeal of many government regulations which impede the functioning of markets unless it can be clearly demonstrated that such restrictions are in the public interest.

While the Council of Australian Governments (COAG) endorsed the thrust of the Hilmer-Committee recommendations at its meetings in February and August 1994, their implementation was impeded by the fact that notwithstanding the expected substantial overall gain in economic efficiency in the long run, some state governments stood to lose monopoly rents. But at the COAG meeting in April 1995 an agreement on a package of competition-enhancing measures was found, together with an arrangement on the appropriate sharing of revenues from the reforms. The COAG agreement is discussed in the section on Recent policy actions below.

Enhancing innovative performance

Australia's technology performance has improved since the 1980s as firms increased R&D spending against a background of trade liberalisation and greater international competition. Business sector R&D spending has grown at an average rate of 13 per cent per year, reaching 0.7 per cent of GDP in 1992/93 compared to 0.3 per cent in 1981. Total expenditures on R&D now account for an estimated 1.6 per cent of GDP, hence still below the OECD average of 2.3 per cent. Over half of research is funded by the government, mainly in the higher education and public research sectors. Government funding for research in higher education institutions tripled from A$ 480 million in 1983/84 to an estimated A$ 1.5 billion in 1995/96, which implies an increase in volume terms by a substantial 89 per cent. A recent study[119] by the Industry Commission estimates that 0.4 percentage point of annual real GDP growth is attributable to the current stock of R&D, which is about half of trend growth of total factor productivity.

The R&D intensity of Australian manufacturing (R&D/production) is half the OECD average, which probably owes to the fact that manufacturing has traditionally been dominated by resource and labour-intensive industries which are not large performers of research. But Australia is catching up as reflected in trade statistics. High technology exports, though still relatively low, grew by nearly 30 per cent per year between 1986 and 1993, faster than general manufacturing exports and above the world average. The share of high technology products in exports of manufactures thereby rose from 29.6 per cent in 1988 to 35.3 per cent in 1992 (hence from 7.0 to 9.3 per cent in total exports). This catch-up process is not confined to manufacturing, but is also spreading to the service sector which is both a large user and developer of technology. Commercial and engineering service industries, for example, perform around a quarter of industrial R&D in Australia. In 1992, non-manufacturing industries accounted for 40 per cent of industrial R&D compared with only 10 per cent in 1973.[120]

Business research spending has been stimulated by fiscal incentives which rank among the most generous in the OECD. Since 1992, Australia's 150 per cent R&D tax concession has become a main feature of the government's technology policy. Evaluations show that business research spending increased by 28 per cent per year after introduction of the tax concession (1984-87) slowing to about 17 per cent in the early 1990s. It is unclear, however, to what extent increased research expenditures are due to the incentive effects of the concession, and to what extent they are due to companies identifying as research and development expenditures which would previously not have been classified as research and development to take advantage of the concession. Trade liberalisation and other factors may also be partially responsible for the increased business expenditure on research and development.

Changes have been made over time to re-focus the R&D tax concession, enhance its cost-effectiveness through better targeting, reduce uncertainty and close off opportunities for misuse, partly through changing registration procedures and providing a clearer definition of eligible R&D activities. Recently, the expenditure threshold was reduced for smaller firms. In addition, the Competitive Grants Scheme, which provides financial assistance to firms undertaking R&D, has been modified to finance trial and demonstration activities that are outside the scope of the R&D tax concession. Access to competitive grants has been facili-

tated for smaller firms and greater emphasis will be given to supporting collaborative research projects.

AusIndustry is the primary agency for delivery of programmes relating to innovation, particularly to small and medium-sized enterprises (SMEs). Under the *Technology Access Program*, recently discontinued by the incoming Government, funding was provided for technology audits, feasibility studies and technology investment evaluations to assist smaller firms in the uptake and adaptation of innovations. A network of Technology Counsellors acted as the interface between firms and relevant programme providers. Work is currently proceeding on a replacement programme, taking into account the experience gained from TAP.

The government is promoting stronger links between public research and industry. To this end, administrative responsibilities for science and technology have been consolidated within the Department of Industry, Science and Tourism. Recent changes to the Commonwealth Scientific and Industrial Research Organisation (CSIRO), the largest public research agency, have resulted in greater involvement in applied research and technology transfer and an increase in its share of funding from business clients. The government intends to further improve the functioning of CSIRO and other government research organisations[121] through establishing costing and pricing guidelines. An evaluation of the Co-operative Research Centres (CRCs), which since 1990 have promoted collaborative research between industry, universities and government research agencies, showed they have contributed substantially to increasing public-private sector linkages.

Recent policy actions

Labour market reforms

Workplace Relations and Other Legislation Amendment Bill

One of the top priorities of the new Government has been to reform the legal framework for industrial relations so as to give fresh impetus to the move towards enterprise bargaining. The legislative changes which the Government considers to be necessary to remove the main obstacles to enterprise bargaining, whilst honouring its election commitment that workers entering formalised agreements would not suffer a fall in take-home pay as a result of the reforms, are

contained in the *Workplace Relations and Other Legislation Amendment Bill 1996*. The original Bill was submitted to the Parliament in May and, following passage by the House of Representatives, was sent to the Senate in June. The Government does not have a majority in the Senate and amendments to the original Bill were agreed. These amendments, while providing some additional protections for workers, did not fundamentally change the central intent of the proposed legislation.

Awards will be focused on providing a safety net of fair minimum wages and conditions, with continuing safety net adjustments focused on the low paid. Awards will contain only a range of ''allowable matters''.[122] All other matters will be governed by statute or determined at the enterprise or workplace level, whether in formal agreements or informally.[123] Transitional arrangements will apply, whereby parties to existing awards will be encouraged and assisted to simplify them to focus on allowable matters and address other issues by agreement; after 18 months, any award provisions outside the allowable matters will no longer be enforceable.[124]

To provide more effective choice and flexibility for parties in reaching enterprise agreements, the new Act provides for Australian Workplace Agreements (AWAs) and modifies Certified Agreements (CAs). AWAs are intended to provide a more effective stream for enterprise bargaining in lightly or non-unionised workplaces than EFAs, which are to be abolished.[125] AWAs are negotiated between an employer and employees either collectively or individually, but must be signed individually. Employees may appoint a bargaining agent (including a union) to negotiate on their behalf. AWAs, like CAs, will be subject to a more flexible and global no disadvantage test. This means that agreements must not result in a reduction in employees' overall terms and conditions of employment when compared with the award and any relevant laws. There is, however, scope for an agreement that did result in an overall reduction to be approved if its approval would not be contrary to the public interest, for example, where the agreement is part of a reasonable strategy to deal with a short-term business crisis, and to help in the revival of the business. The global rather than line by line no disadvantage test means that an agreement would be able to fall below any individual terms and conditions set by the award benchmark, provided that there was no overall disadvantage. The public interest test would only apply where there was an overall disadvantage.

There will be simple process for lodging agreements with the Employment Advocate[126] (EA) for approval. The EA will check that agreements meet the no disadvantage test and other statutory requirements prior to approving them. Where the EA has doubts about whether a proposed agreement meets the no disadvantage test, it can refer the agreement back to the parties for amendment. Where there is ongoing uncertainty about whether the agreement meets the test, the EA will refer the agreement, following consent by both parties, to the Commission, which will assess it against a public interest test. Once approved, AWAs will be legally enforceable. Unions will have no involvement in these agreements, unless invited. AWAs will enable those workplaces wishing to establish individual employment contracts to do so.

CAs will be made between an employer and their employees' union(s) and registered by the AIRC provided that they meet the new no disadvantage test. Scope is being provided for CAs to be reached directly with employees (*e.g.* in non-unionised workplaces) but, in such cases, relevant unions will be able to participate in negotiations and become parties to an agreement where a member requests this. The AIRC also must be satisfied that a majority of employees to whom the agreement applies have genuinely endorsed the agreement. The current precondition to certification of an agreement that there be a federal or state award in place will be repealed.

So as to promote freedom of association, the Bill will make membership of associations voluntary and will allow for the formation of new unions in areas of existing union coverage, including enterprise unions. Preference (for union members) clauses will be made illegal, as will compulsory unionism and discrimination based on membership or non-membership of a union or employer association. These changes will bring union security legislation in Australia more in line with practice in most other OECD countries.

There will be a limited right to strike or lock-out while negotiating CAs and AWAs, but not during the period of an agreement's operation. The AIRC's powers to curtail all other (*i.e.* non-protected) industrial action will be strengthened and made enforceable by injunctions. Provisions banning secondary boycotts will be restored to the *Trade Practices Act*, which will make them effective again.[127]

Unfair dismissal procedures for employees covered by federal awards or agreements[128] are to be further simplified based on the principle of a ''fair go all

around'' (*i.e.* fair to both employees and employers).[129] In deciding whether a termination was harsh, unjust or unreasonable the Commission will be required to have regard to a number of practical matters including whether there was a valid reason for the termination based on the capacity or conduct of the employee or the operational requirements of the employer 's business. The reforms will restore jurisdiction on unfair dismissals to the AIRC:[130] it will conciliate and, if that does not work, arbitrate, and decide remedies and costs. Procedural fairness will be only one of many factors to take into account when determining whether termination was harsh, unjust or unreasonable. When the AIRC determines that a remedy[131] is appropriate, it will have to consider the effect on the viability of the employer's business.[132]

In an effort to reduce the complexity and regulatory burden of Australia's dual system of state and federal industrial laws and tribunals, harmonisation of federal and state systems is being pursued through complementary legislation and other arrangements (*e.g.* between the federal and state tribunals) and through a co-operative approach to administrative arrangements and to service delivery.

Unemployment and other benefits

Significant changes in income support arrangements came into effect in July 1995, which aim at encouraging unemployed families to accept low-paid work. Since 1994, each partner in a couple in which one partner is unemployed has been entitled (subject to a means test on their joint incomes) to half of their income support: the allowance paid to the spouse of the jobseeker was named the ''Partner Allowance''. In July 1995, these entitlements were made separate, with income of one spouse below the level at which his or her individual entitlements are exhausted (A$ 248 per week from September 1996) not affecting the entitlement of the other. Other changes included:

- The replacement of the partner's allowance with Parenting Allowance for one partner in a couple with at least one child under 16, in recognition of their caring responsibilities. The Partner Allowance was henceforth to be confined to spouses born before July 1955 without dependent children and who had no recent workforce experience. All other spouses would have to apply for Job Search or Newstart allowance and satisfy the activity test for these.

– Income tests were changed to deal predominantly with the income of the individual. Each member of a couple in receipt of an allowance now has his or her income-test free area of A$ 30 per week and a 50 per cent withdrawal rate on the next A$ 40 per week. The withdrawal rate, which applies to all further income of the individual until their entitlement is exhausted and then to their spouse's income before any personal income test is applied, was reduced from 100 per cent to 70 per cent. These changes significantly reduced METRs for couples with low earnings potential, although METRs remain high.[133]

The Australian authorities have clearly taken the view that it is better for the unemployed to work part-time, and so maintain contact with the labour market, than not to work at all; and that for many of the unemployed, there is a higher probability of getting a part-time job (which may lead to a full-time job) than a full-time job. As in other countries with such arrangements, the payment of unemployment benefits is restricted to the part-time workers who were previously unemployed and continue to search for full-time work and remain available to take it if offered. But, unlike the practice in other countries, benefits may also be paid in Australia to those who were working full-time and whose hours have been sufficiently reduced to be entitled to a means-tested benefit (although benefit entitlement remains conditional on continued search for another full-time job and availability for it). Entitlement depends on the merits of each case, for example, whether the person has voluntarily accepted reduced hours. A person job sharing is not generally regarded as unemployed. Except during seasonal lows following Christmas and Easter, about 17 per cent of Australian unemployment benefit recipients work part-time (including those in full-time jobs working reduced hours), considerably more than in most other countries.

The disadvantage of paying (means-tested) unemployment benefits to part-time workers is that it reduces the incentive to work full-time. For example, an unemployed person with a family and two children who works two days a week, earning two-fifths of the APW level of earnings, would have a net replacement rate of 86 per cent, comparable with rates for those countries having such arrangements and for which data are available (Table 17).

The recently elected Government has introduced a range of measures to tighten the job search requirements placed on unemployed people and ensure only those actively seeking work are provided with income support. In the

Table 17. **The incentive to work part-time for an unemployed person with two children**

	Benefit	Percentage of net income in full-time work	
		Fully unemployed	Part-time worker earning 40 per cent of full-time weekly wages
Australia	Job-search allowance	71	86
Denmark	Unemployment	83	88
Germany	*Arbeitslosengeld* (unemployment insurance)	78	92
Ireland	Unemployment insurance/unemployment assistance	64	64
	Unemployment assistance/part-time job incentive	64	84
Netherlands	Unemployment insurance	84	91
	Social assistance with disregard	80	91
	Social assistance without disregard	80	82
Norway	Unemployment insurance	73	84
	Social assistance	83	84
Spain	Unemployment insurance	74	85
United Kingdom	Income support (less than 16 hours work)	74	78
	Family credit (more than 16 hours work)	74	79

Note: Incomes are expressed as percentages of net incomes in full-time work at APW wages. Figures are for a couple with 2 children. An earnings disregard of 15 per cent of benefit is applied for a maximum of 2 years in the Netherlands. Thereafter, there is no earnings disregard.

Source: OECD Database on Unemployment Benefit Entitlements and Replacement Rates.

recently announced new labour market assistance arrangements the co-location of income support and employment services for many clients will facilitate greater co-ordination between the provision of assistance and the enforcement of job search obligations.

Active labour market policies

In the 1996-97 Budget the Government proposed radical reforms to all areas of labour market assistance. As noted in the Budget document *Reforming Employment Assistance – Helping Australians into Real Jobs*, these reforms are ''the most significant reorganisation of labour market assistance arrangements since the establishment of the Commonwealth Employment Service in 1946''. The changes address the problems and weaknesses identified in the evaluation of *Working Nation* and reflect the Government's commitment to deliver better and more lasting employment outcomes for jobseekers, while at the same time contributing to its fiscal consolidation programme. The Government identified three key considerations underlying its labour market assistance reforms:

- to deliver better quality assistance to unemployed people, leading to better and more sustainable employment outcomes;
- to address structural weaknesses and inefficiencies in current arrangements for labour market assistance; and
- to achieve better value for money from the expenditure of public funds.

The main elements of the new arrangements, to come into full effect from December 1997, include:

- the cashing out of most current labour market programmes to create a flexible pool of funding for labour market assistance and a shift in emphasis to the needs of individuals rather than programme parameters and guidelines;
- the integration of the public contact services of the currently separate Department of Social Security (DSS) and Commonwealth Employment Service (CES) networks. These services will be delivered by the Service Delivery Agency, a statutory authority established within the Social Security portfolio. It is intended that the Agency will add to its initial functions over time and become a one-stop-shop for a wide range of Government services;

- a fully competitive market for employment placement services where private and community employment placement enterprises will operate in competition with a corporatised public employment placement enterprises;
- payment arrangements for the employment placement enterprises where fee payments and future contracts are clearly related to the enterprise's performance in placing their clients into real and sustainable jobs; and
- the maintenance of the focus of assistance on the long-term unemployed and those most at risk of becoming so.

Reforms are to be implemented in two distinct steps. Abstracting from much of the detail, the broad thrust of the changes, which are expected to be fully implemented by December 1997, is described below. The first step will be to simplify current programmes and their delivery, reduce complexity and administrative costs, but maintain the existing infrastructure, described above. From December 1996, the main changes will be:

- Enhancing the case management process by cashing out of some labour market programmes to provide additional flexibility to contracted case managers to determine how to help their clients.
- Simplification of the current set of ALMPs. This will reduce administrative costs and complexity, enhancing the case management process by providing additional flexibility to the case manager in determining how to help the client.
- Reduction in funding for relatively costly programmes, namely the brokered programmes, with more emphasis placed on programmes found to be relatively cost-effective, such as *JobStart.*
- Increased emphasis on outcomes in unsubsidised jobs; therefore, case management fees will no longer be paid for placements into brokered programmes.
- Increase in the market share of CCMs to about 50 per cent of all new case management placements through a new tender by ESRA.

In the second stage, all areas of labour market policy will change significantly. For example, as of December 1997, the CES will cease to exist. Some of its functions will be taken over by a new Service Delivery Agency (SDA), which will commence operation in mid-1997. Most of the remaining functions will be transferred to a public Employment Placement Enterprise (EPE) to compete on an

even footing with other EPEs (EPEs are described below). The SDA will take on vacancy registration, registration of jobseekers for benefits, and assessment of jobseekers for referral to EPEs. The SDA will also take on the income support role from DSS, which implies for the unemployed, a more direct link between job search activities and income support. The SDA will provide self-help facilities for jobseekers *via* touchscreens which will display a national vacancy database managed by DEETYA. Jobseekers who access the touchscreens will either self refer to open vacancies or follow up the contact of the EPEs displayed on the ''closed'' vacancy. Formal job brokerage services will not be undertaken by the SDA. They will be contracted out to EPEs. EPEs will, in effect, take over case management and by December 1997, the current system of CCMs/EAA (and ESRA) will cease to exist.

EPEs will undertake three main functions: labour exchange services (matching jobseekers with vacancies), jobsearch assistance, and intensive employment assistance, the latter targeted on the long-term unemployed or those at risk to it (as assessed by the SDA using an assessment mechanism developed by DEETYA when the unemployed register for a benefit). With respect to intensive employment assistance, EPEs will be responsible for determining the needs of the jobseeker, coming up with an action plan, and administering activity tests. In the first instance, EPEs will be drawn from the private and community sectors, similar to the set-up of CCMs, and jobseekers will be allowed to choose among them. In addition, there will be a public EPE, fully separate from the SDA, which will compete on an even basis with other EPEs for clients. It will also serve as a ''last-resort'' option in regions where no other EPEs exist.

Significant changes will also take place in the area of labour market programmes. Except for the retention of a few labour market programmes, most will cease to exist as of December 1997. They will be ''cashed-out'', creating a pool of funds for contracting labour market services by DEETYA. EPEs will be paid a fee, and will purchase suitable interventions for their clients (*e.g.* training) directly. Furthermore, there will be significant changes in payments to EPEs compared with CCMs. A full fee will be paid only after a client has been off allowances for longer than six months (compared to 13 weeks for CCMs), which should lead to longer duration outcomes. The objective is to undertake the contracting process on the basis of full price competition as soon as possible.

These changes imply a new role for DEETYA. Many of its current operational responsibilities will be distributed to the SDA and the public and private EPEs. There will be a clear split between purchaser and provider. DEETYA, on behalf of jobseekers, will be the purchaser of services which will be provided by the EPEs. It will also regulate and manage the tendering and contracting out process, and maintain its policy development and evaluation roles.

Training and education

Further efforts are required to enhance vocational education and training (VET) as a pathway from school to work. The new Government intends to remove a number of obstacles to VET, principally through industrial relations reform and reforms to the apprenticeship and traineeship system. These reforms will make apprenticeships more attractive to employers by enabling them to negotiate more flexible employment and training arrangements, by facilitating the development of apprenticeships and traineeships better adapted to their requirements and by simplifying the process for gaining accreditation for programmes. School-based vocational education is to be better articulated to apprenticeships and traineeships and VET qualifications are to be made more portable through the Australian Qualifications Framework being phased in since 1995. These reforms address most of the obstacles which have impeded the development of VET. It remains to be seen, however, whether apprenticeships will prove popular in the absence of the social institutions found in the countries with a strong apprenticeship tradition.

In order to remove disincentives to undertake education and training in the income support system, the Government has announced plans to introduce a Youth Allowance from January 1998, consolidating assistance to unemployed young people and students into a single allowance which will be more flexible and simpler for young people and their families to understand. The proposed Youth Allowance would apply to eligible 16 to 20 year-olds, and full-time students up to the age 25 years.

Competition policy

In April 1995, the Commonwealth, State and Territory governments agreed to establish a national competition policy on the basis of the recommendations of the Hilmer Committee, and to work co-operatively on competition issues within

their jurisdictions. It was decided to implement a package of measures (see Box 2), with the Commonwealth's *Competition Policy Reform Act 1995* as the legislative centre-piece. The *Act inter alia* amends the *Trade Practices Act* (TPA) and extends price surveillance powers to cover State- and Territory-owned businesses. Two new institutions were established – the Australian Competition and Consumer Commission and the National Competition Council to oversee the implementation of the reforms. The *Act* was passed by the Commonwealth Parliament on 30 June 1995, received Royal Assent on 20 July 1995 and commenced in three stages, the final one on 21 July 1996. Another element of the reform package, the intergovernmental *Conduct Code Agreement*, sets out the basis for extending the competitive conduct rules of the TPA to the unincorporated business sector and to state government business activities.

The *Competition Principles Agreement* sets basic principles which address key recommendations of the Hilmer Committee. One of these principles involves a *legislation review* by the year 2000 to identify and reform any restrictions on competition that are inconsistent with the objectives of the legislation or which do not yield net benefits to the community. The *principle of structural reform of monopolies* mandates that before a government deregulates a market supplied by a public monopoly, it must separate the regulatory and business functions of the public monopoly. A *statutory access regime* will provide third party access to services provided by significant infrastructure facilities with natural monopoly characteristics. The *competitive neutrality principle* specifies that significant[134] government businesses should not enjoy any net competitive advantage merely as a result of public sector ownership; this will involve subjecting government businesses to the same tax and regulatory regimes applicable to private business. A complaints mechanism will be put in place as from 1 July 1997, which will be administered by the Productivity Commission and will respond to complaints that Commonwealth organisations are not complying with the competitive neutrality arrangements where required. Also, *price oversight* of monopoly or near monopoly government businesses which supply goods or services will be established.

According to estimates of the Industry Commission,[135] the full implementation of the Hilmer proposals (and of related reforms) would raise annual real GDP in the long run, once all adjustments have taken place, by 5.5 per cent, or A$ 23 billion per year[136] (in 1993-94 dollars). Higher economic activity will – of course – induce an increase in revenues of the Commonwealth while some state

Box 2. **Elements of the 1995 national competition policy package**

In April 1995, the Council of Australian Governments (COAG) agreed to implement a package of measures designed to extend pro-competitive policies to previously exempt sectors of the economy [unincorporated enterprises, Government Business Enterprises (GBEs) and the professions].

The Commonwealth's *Competition Policy Reform Act 1995* is a key element of the competition policy package. The Act:

- amends the competitive conduct rules of Part IV of the Trade Practices Act (TPA) and the provisions that exempt specific forms of conduct from these rules;
- inserts provisions into the TPA extending the coverage of the competitive conduct rules to the unincorporated sector and to State and Territory GBEs;
- creates a new section of the TPA (Part IIIA) establishing a new national regime for access to services provided by means of ''nationally significant'' infrastructure facilities;
- amends the Price Surveillance Act to extend prices oversight to State- and Territory-owned business enterprises;
- creates two new institutions responsible for overseeing and providing advice on the implementation of the policy package. The Australian Competition and Consumer Commission (ACCC) has been formed from the merger of the Trade Practices Commission (TPC) and the Prices Surveillance Authority and will primarily be responsible for administering the Prices Surveillance Act and the TPA. The National Competition Council (NCC) will be formed to provide coordination of reform efforts, to ensure that the Commonwealth, States and Territories meet their commitments under the Agreement, and to assess whether jurisdictions have met the reform obligations set out in the Agreements.

The competition policy package also consists of three intergovernmental agreements.

- The *Competition Principles Agreement* establishes agreed principles on structural reform of public monopolies, competitive neutrality between the public and private sectors, prices oversight of government enterprises, a regime to provide access to essential facilities, a programme of review of legislation restricting competition and consultative processes for appointments to the NCC.
- The *Conduct Code Agreement* sets out the basis for extending the application of the TPA and the consultative processes for making modifications to the competition law and appointments to the ACCC. It also commits each State and Territory to pass the required application legislation enabling the Commonwealth's new legislation to take effect.

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– Under the *Agreement to Implement the National Competition Policy and Related Reforms*, the Commonwealth will provide payments in return for States and Territories meeting agreed obligations set out in the Agreement, the Conduct Code Agreement plus reform commitments in electricity, gas, water and road transport.

Source: Commonwealth of Australia (1995), Implementing the National Competition Policy: Access and Price Regulation, Information Paper of the Industry Commission, Canberra, November.

governments will lose monopoly rents. Therefore, the inter-governmental *Agreement to Implement the National Competition Policy and Related Reforms* will provide phased compensation payments of the Commonwealth to the States and Territories – reaching A$ 600 million per year by 2001 – conditional on the progress they make in implementing the agreed competition policy reforms. As of November 1996, all States and Territories had passed the necessary legislation through their Parliaments to apply the Competition Code within their jurisdiction. This legislation is one condition necessary for each State and Territory to qualify for competition payments from the Commonwealth.

Technology and innovation

The new Government has implemented a wide-ranging review of the costs of programmes with a view to re-orienting selected science and technology policies. Although the R&D tax concession will remain a main component of government policies to promote business research, the premium rate for deductions has been reduced from a maximum of 150 per cent to a maximum of 125 per cent. The R&D tax deductions for partnerships of companies has been eliminated. A new R&D funding programme – START – will allocate A$ 340 million over four years in competitive grants to support research and development. There are also plans to restructure the technology development and diffusion schemes of AusIndustry to ensure they address market failures.

Assessment and scope for further action

Analysis presented above indicates that past reform efforts moved the Australian economy a long way towards a more flexible and adaptable one. The areas where most progress has been made include increasing competition in the tradeables sector and improving the tax/transfer system which is now among the best in the OECD – it is well-targeted and inexpensive. However, there is a clear need to reinvigorate the reform agenda. Recent policy actions will provide further impetus in that direction. The areas where – despite the significant progress made – there is much further scope include the decentralisation in determining pay and working arrangements, improving the efficiency and effectiveness of education and training and exposing the infrastructure service provision to greater competitive pressures. The changes being implemented by the new Government represent important steps forward and should contribute to improving living standards and reducing structural unemployment.

The continued adjustment of the industrial relations system remains a priority area. The gradual shift in the focus of wage bargaining from awards to the level of the enterprise during recent years has already enhanced the organisational flexibility of workplaces and hence, the adaptation to economic change. But important barriers remain. Therefore, the 1996 industrial relations reforms aim at providing further impetus to the transition to enterprise bargaining, through simplifying the agreement making process, increasing the choice of agreements, ensuring more equal treatment of bargaining parties and restricting awards to a safety net of minimum wages and conditions of employment, focusing increases in award wage rates on the low paid. Enterprise agreements will be able to be negotiated collectively or individually, and free from uninvited union involvement. The award system will operate as a genuine safety net – and should not present disincentives for entering into enterprise bargaining – provided that future increases in award pay rates are modest and focused on the low paid. While it might have been preferable from an economic point of view to have moved more quickly away from the award system towards enterprise bargaining, as occurred in New Zealand, this does not appear to have been a politically viable alternative in Australia: the present strategy seems to be the most that could be done in view of the political constraints faced by the Government.

The new legislation also includes further reforms to the law on unfair dismissals. Concerns about this aspect of industrial relations legislation date from 1993, when changes made by the *Industrial Relations Reform Act 1993* placed special emphasis on procedural fairness, raising the possibility of claims being decided in favour of the applicant based on technicalities rather than substance. The reformed arrangements should clarify further and limit the potential costs of unfair dismissal. The parties to a dispute will be encouraged to settle it among themselves. If that fails, the AIRC will conciliate and arbitrate, taking into account the situation of both the firm and the worker. The principle of a ''fair go all round'' once again will be the corner-stone of unfair dismissals legislation in Australia.

The new legislation also aims to provide employee choice about representation, and to remove uninvited third party involvement in the bargaining process. The Act will facilitate the creation of new unions, including enterprise unions, ban union preference clauses and compulsory unionism and restore the provisions banning secondary boycotts to the *Trade Practices Act*, thereby making them effective again. Moreover, the sanctions against unlawful industrial action are to be strengthened. It is anticipated that as a result of the changes unions' involvement will be restricted to those areas where it is genuinely sought by employees, and that unions will become more democratic, member-focused and responsive bodies.

Some of these measures may also reduce the potential for unions to extract a portion of the rents generated in imperfectly competitive markets for their members. Less rent seeking behaviour by unions would result in more appropriate relative wages between sectors or firms affected, increasing employment.

Rent-seeking behaviour can also be limited by removing impediments to competition in product markets. The reductions in import protection already effected in Australia and those scheduled through to 2001 have been vital in this regard. The Government is committed to those scheduled reductions. Further, the Productivity Commission is currently reviewing post-2000 assistance arrangements for the automotive industry and will shortly undertake a similar review of the textile, clothing and footwear industries. The Government has also undertaken to review the current general tariffs of 5 per cent by 2000 as part of its plan to implement the APEC goal of free and open trade and investment in the Asian Pacific region.

Similarly, implementation of the national competition policy recommended by the Hilmer Committee and agreed at the Council of Australian Governments meeting in April 1995 will enhance product market competition, especially in infrastructure services dominated by Government Business Enterprises. Getting the most out of these reforms will require the co-operation of State and Territory governments, both in terms of allowing their business enterprises to participate in the national market (this is particularly important in establishing a single electricity market in the eastern States) and of identifying and reforming any legislation or regulation which is inconsistent with the objectives of the competition legislation or which does not yield net benefits to the community. Such co-operation from the States is likely, especially as adherence to the review process is necessary to qualify for Competition Payments from the Commonwealth.

Reinforcing the national competition policy are measures to increase competition in a number of infrastructure services. These include: telecommunications; electricity; rail; gas; coastal and trans-Tasman shipping; the waterfront; and aviation. In some cases, the impediments to competition in these sectors will be removed as a result of the industrial relations reforms in the *Workplace Relations and Other Legislation Amendment Bill*.

Taken as a whole, the planned industrial relations reforms and the competition reforms should increase relative wage flexibility in the long run towards the levels observed in North America and the United Kingdom. In view of the Australian commitment to a comprehensive social safety net, it is unlikely, however, that relative wage flexibility will ever match that observed in the United States. This underlines the conflict in social security systems between providing an adequate social safety net but not unduly raising reservation wages and unemployment. The level of what is perceived to be an acceptable social safety net has risen considerably in Australia during the past two decades, as it has in most other OECD countries but notably has not in the United States or the United Kingdom. Nevertheless, the level of generosity of unemployment benefits in Australia remains below the OECD average, particularly with regard to shorter duration unemployment. This, combined with ongoing measures to tighten the administration of the activity test means that it is unclear that the unemployment benefit system, when taken as whole, has contributed significantly to the rise in the structural and long-term unemployment. As in many other countries, the main work incentive problems created by the benefit system concern families with low

income earning potential. This is primarily the case for couples where only one partner is actively seeking employment. The introduction of separate unemployment benefit entitlement for couples in 1995 has increased the incentives for both partners to gain employment – and in situations where both partners are seeking employment the work incentives are comparable to that of single people. Moreover, the indefinite duration of benefits does not seem to pose an incentive problem for single unemployed people as replacement rates are low.

Australia introduced during the 1980s in-work benefits for low income earners with children so as to ease the work incentive problems posed for this group by the social security system. There has been a similar response to this problem in many other OECD countries. But Australia went further in the income support reforms of *Working Nation*, providing each member of an unemployed couple with an individual entitlement to half of their income support, with spouse income below the level at which their benefit entitlement is exhausted not affecting the entitlement. These reforms appear to have been effective in improving incentives for unemployed couples (with or without children) to take part-time or low paid full-time jobs. And there is much to be said for the philosophy underlying these arrangements, according to which it is better for the unemployed to maintain contact with the labour market, than not to work at all. However, a disadvantage of paying (means-tested) unemployment benefits to part-time workers is that it reduces the incentive to work longer hours.

Comprehensive means-testing of social security benefits in Australia has constrained expenditure on them to levels far below those in most other countries. This has enabled Australia to combine a comprehensive social safety net with low tax wedges on labour income and hence, an ‘‘employment-friendly’’ taxation environment.

The goal of the proposed changes to labour market programmes and their delivery is to ensure labour market assistance has a clear focus on real job outcomes and genuinely makes a difference to those assisted. The initiatives will be more responsive to both jobseekers’ and employers’ needs. Clients should benefit from higher standards of service, more flexibility and customised service and ultimately better and more lasting employment outcomes.

Although many aspects of the EPEs are new, the Employment Services Regulatory Authority, drawing on its experience over the past two years in accrediting contracted case managers and letting tenders for case management

services to private and community providers, should be able to provide DEETYA (as purchaser in the new market) with some guidance for assessing the suitability of EPEs as potential providers of labour market assistance, and in deciding on appropriate fees for providing quality intensive employment assistance.

The case management process has significant advantages, including targeting assistance on employer and jobseeker needs. It can also allow the costs of placement and unemployment in general to become more transparent. Under the current system, Contracted Case Managers have the incentive to place individuals as quickly as possible to generate a fee and take on a new client, but they may not always take the most cost-effective path since they do not have to pay directly for slots on DEETYA programmes. The proposed system, by forcing EPEs to take on all of the costs for placement, should lead to a better identification of actual client needs at minimum cost, over time. In effect, it will become more client focused, away from being programme driven.[137] Since fees will be allowed to vary by region in the longer term, some tailoring to specific markets should also take place. Where a sufficient market does not exist to attract EPEs, the public EPE will be paid to take over as provider of last resort. Another advantage of the proposed system is that it will ensure that there is clear separation between the purchasing and providing functions in delivering labour market assistance and full competitive neutrality between the EPEs.

These proposals, once implemented, will require careful monitoring to ensure that they are achieving their stated aims. For example, a crucial element, which will require a lot of oversight is data on relative placement records of EPEs over time to funnel this information to jobseekers so that they can make informed decisions. This will be important to determine whether the market-based system has led to quality outcomes. This may be lacking if EPEs do not have the requisite skills to help jobseekers and/or if the appropriate tools (*i.e.* job clubs, etc.) do not exist. It will take some time for this ''market'' to develop and it is unclear to what extent private or community sector involvement in high unemployment rate regions will be forthcoming, with the possibility that the public sector may be left to take over. In addition, feedback from EPEs to DEETYA will be important in developing and assessing policy initiatives.

Substantial progress has been made in increasing participation in education and training, especially in the final years of secondary school and in university. Nevertheless, about half of all school leavers in any one year leave school to

enter the labour force directly, rather than continuing in some form of education and training. Improving the school-to-work transition for this group, which experiences high unemployment, remains a priority. Despite the efforts which have been made to expand work-based vocational education and training (VET) pathways from school to work, participation in VET has not increased significantly. The new Government intends to remove a number of obstacles to expanding work-based VET through its industrial relations reform and reforms to the apprenticeship and traineeship system. These reforms will make apprenticeships more attractive to employers by enabling them to negotiate more flexible employment and training arrangements better adapted to their requirements and by simplifying the process for gaining accreditation for programmes. The extension of school-industry arrangements to allow students to commence apprenticeships while at school is another valuable initiative.

Whether work-based VET expands following these reforms remains to be seen. A constraint could be the absence of the social institutions which underpin apprenticeships in countries with a strong apprenticeship tradition. So long as obstacles to the development of work-based training which can be removed efficiently are dealt with, this pathway from school to work will develop to an extent which is efficient for Australia. It may be that the best way to ease the school-to-work transition is to ensure that employment and training arrangements are flexible and that there are many opportunities to continue education and training beyond school.

IV. Education and training

Introduction

Economic developments during the past two decades have contributed to pressure for far-reaching reforms in Australia's system of education and training (the system is described in Annex II). Employment prospects for the low skilled have deteriorated markedly, especially for early school leavers. This, together with the greater availability of part-time jobs which enable students to earn while studying, improved government student income support arrangements and changing community attitudes to education and training, has increased the attractiveness of completing secondary school and of obtaining tertiary qualifications. The education and training system has been called upon to accommodate the resulting substantial increase in student numbers and to offer programmes better adapted to mass education at the upper secondary and tertiary levels. At the same time, increasing competitive pressures in the economy have led to calls for improved efficiency in all areas of activity, including the education and training sector. Governments have taken steps to introduce greater competition amongst suppliers and to make them more accountable for outcomes. The Commonwealth Government has also required university students to contribute towards the cost of their courses with a view to enhancing equity, to providing a source of funds for the considerable expansion of enrolments in higher education and to increasing efficiency (through a more considered evaluation of the prospective costs and benefits of their education).

The most intense pressures for change have come to bear on the Vocational Education and Training (VET) sector. Greater participation in VET, which has been relatively low in Australia compared with most other OECD countries, has been judged in numerous Government reports to be vital to increasing the skills and productivity of the Australian workforce as well as easing the school-to-work

transition problems presently encountered by the young; the unemployment rate for the 15-24 age group is more than twice that of prime-age workers. Key challenges in this sector have been to extend work-based VET beyond the traditional male-dominated occupations in which apprenticeships have been available and to shift the emphasis of such VET from the length of courses to obtaining competencies. At the same time, it has been recognised that providers of VET, most of which are (public) Technical and Further Education (TAFE) institutions, must become more responsive to their clients' needs: in particular, it has been necessary to increase employer input into course content.

Changing workplace demands also have figured amongst the pressures for reform of education and training. Employees are increasingly required to be flexible, to perform a wider variety of tasks and to adapt to constantly changing workplace demands; in extreme cases, employees are confronted with the need to change careers. Life-long learning has become the norm for many employees and is becoming so for a growing proportion of the workforce. Meeting these challenges requires individuals to have a solid foundation in literacy, numeracy and analytical skills and to be able to build on them through connected pathways to further education. Adapting the education and training system to meet these challenges has been a priority of reform in recent years.

This chapter reviews the changes which have been made to the education and training sector in response to these challenges and assesses the extent to which the objectives of the reforms have been achieved. Problems which have been encountered with the reforms are discussed and some outstanding issues are signalled.

Traditional features of the education and training system

Up until the early 1980s most students (65 per cent) did not complete secondary school. The system catered well for the minority of students headed for professional jobs. It provided apprenticeships for a minority of school leavers, mostly in traditional male-dominated trades.[138] Apart from hairdressing, there were few opportunities for girls that included work-based training, though there was an expanding range of preparatory and other skilled courses, such as in clerical work, provided in TAFE. In all about 8 per cent of 15 year-olds and 23 per cent of 16 and 17 year-olds were enrolled in TAFE, of which a significant

proportion were apprentices. Overall, only about one half of 17-19 year-olds were participating in formal education and training, with the participation rate in the TAFE sector being around twice that in higher education (Table 18). Most TAFE courses were part-time and generally of much shorter duration than courses in universities, where the large majority of young students took full-time courses.

School curricula were mainly general (as opposed to vocational), although some trades-related subjects oriented to TAFE were offered in junior secondary school. Senior secondary schooling was mainly academic, with the curriculum largely dictated by the needs of university entry. Nevertheless, there was diversity in curricula, both in schools and in TAFE, as this was a State responsibility. Alternative forms[139] of senior secondary schooling and assessment were permitted in some States. School and university staff and State department officials predominated over industry representatives in school curriculum and assessment bodies.

The conditions of employment of apprentices, including wages, were determined under industrial awards or agreements between employer and employee groups. Employers and unions were involved in a range of advisory networks and TAFE councils to advise on training policy. TAFE colleges contributed to the training of apprentices by providing formal courses, for about 10 per cent of the total time of a four year apprenticeship (COSTAC, 1985). There was ongoing concern for both the nature of apprenticeships and the number of apprentices. Commonwealth subsidies to employers of apprentices had commenced in the

Table 18. **Educational participation rates of 17-19 year-olds**

Per cent

	Schools	TAFE[2]	Higher education	Total
1975	13	20	11	45
1985	16	24	11	50
1990	20	25	15	60
1995 estimates[1]	24	26	17	66

1. Preliminary estimates, changes in collections affect comparability over time.
2. Technical and Further Education.

Source: Department of Employment, Education and Training, *Education Participation Rates, Australia*.

early 1970s. Encouragement was given to new forms of apprenticeship and to pre-apprentice courses in TAFE. National TAFE core curricula were developed in a number of trades.

A feature of post secondary education and training in Australia is that it was nearly wholly government provided and funded. Formal qualifications were the preserve of TAFE colleges and public higher education. Some private providers, especially private business colleges, provided training for fees, and formal training was provided by employers such as Telecom and regarded as of high quality. Non-government provision of education was a major feature only of primary and secondary schooling. Non-government schools began to receive government subsidies in the late 1960s and these grew in the 1970s.

Meeting the challenge

Substantial changes have occurred in the system for post-compulsory education and training in response to the pressures which have come to bear on it. There has been a large expansion in participation in post-compulsory education and training, calling for programmes better adapted to a wider variety of abilities. More pathways from school to work have been required and there have been pressures for the labour-market relevance of courses to be enhanced. Moreover, education institutions have been faced with the challenge of producing their outputs more efficiently and meeting social equity objectives. The ways in which the education and training sector has evolved to meet these challenges are discussed in this section.

Increased participation in post-compulsory education and training

The retention rate to the final year of secondary school (Year 12) rose from 46 per cent in 1985 to 77 per cent by 1992 but has since fallen slightly in the recent economic recovery, to 72 per cent in 1995. In terms of *age-participation rates*, there has been a doubling for students aged 17, the normal age in the final year of secondary school (Table 19). There also has been a substantial increase in the participation rate for 16 year-olds; nevertheless, there are still about 7 per cent of 15 year-olds and over 20 per cent of 16 year-olds not at school, with only a minority of them enrolled in TAFE.[140] The participation rate in VET for students aged 17-19 has increased slightly (see Table 18) while that of younger

Table 19. **Schools: Age-participation rates**

	15 years	16 years	17 years	18 years	19 years	15-19 years
1982	87	58	30	5	2	36
1985	90	65	38	7	1	41
1992	93	81	60	15	3	49
1995	93	79	59	12	2	48

Source: Australian Bureau of Statistics, *Schools Australia*, and Department of Employment, Education and Training, *Education Participation Rates*.

students has declined, reflecting higher school retention rates (see Table 19). The proportion of students completing secondary school or equivalent level vocational studies in Australia is now close to the average for most OECD countries[141] (Table 20).

In contrast to the experience in some other English-speaking countries, notably the United States and the United Kingdom, apprenticeship has survived relatively well in Australia.[142] The number of apprentices has been stable since the early 1980s, and their ratio to the older teenage population has been maintained; their ratio to the younger teenage population has fallen, reflecting increased retention rates until the end of secondary school[143] (Figure 30). The number of new apprenticeships appears to have moved roughly in line with employment in the traditional apprenticeship trades: metal, electrical, building, vehicle and food. The decline in numbers in the early 1990s largely reflects the impact of the recession. Nevertheless, the numbers of apprentices remain much lower than in Germany, where the dual system covers areas such as clerical and retailing which until recently were only covered in Australia by shorter traineeships.

Increasing the proportion of students attaining a level of education at least equivalent to completing secondary school has long been an objective of policy. The Finn Committee (1991) set a target, which was endorsed by the Commonwealth and State Governments, of increasing the proportion of 19 year-olds participating in formally recognised education and training or, if not participating, having completed Year 12 or an initial post school qualification, from about 70 per cent in 1990 to 95 per cent by 2001. By 1995, the figure was over 80 per cent (though changes in data classification affect the comparison) (Table 21). This increase reflects a rise in the proportion of 19 year-olds participating in

Table 20. **Upper secondary graduation rates**[1]

	Total	General	Vocational and apprenticeship
North America			
Canada	70.8	n.a.	n.a.
Mexico	25.0	21.5	3.6
United States	73.6	n.a.	n.a.
Pacific Asia			
Australia	n.a.	67.6	n.a.
Japan	92.1	67.4	24.7
New Zealand[2]	63.9	39.0	24.8
European Union			
Austria[2]	81.9	13.9	67.9
Belgium[2]	96.7	33.9	64.4
Denmark[2]	82.3	45.3	37.0
Finland	93.4	43.9	48.6
France	80.8	36.3	46.4
Germany	88.5	22.6	65.9
Greece[2]	75.1	53.0	22.2
Ireland	97.5	90.0	7.6
Italy	76.2	17.3	58.0
Netherlands[2]	69.4	33.3	35.3
Portugal	n.a.	38.0	n.a.
Spain[2]	67.9	44.6	23.1
Sweden	74.6	25.6	50.5
United Kingdom	n.a.	n.a.	n.a.
Other OECD countries			
Czech Republic	77.6	13.4	64.2
Hungary[2]	80.8	18.4	60.8
Norway	102.0	41.7	60.2
Switzerland	82.1	19.2	62.8
Turkey	38.4	23.6	15.1
Country mean	76.9	36.8	42.2
Other countries			
Korea	91.3	54.6	36.7
Poland	90.3	22.3	68.0
Russian Federation	90.5	46.8	43.7

1. Ratio of upper secondary graduates to population at theoretical age of graduation by type of programme (1994). First educational programmes.
2. Data refer to 1993.

Source: OECD *Education at a Glance*, 1996.

Table 21. **Monitoring the Finn targets for 19 year-olds**

Per cent

	1990	1991	1992	1993	1994	1995
Participating in education	41	46	48	51	52	51
Not attending						
With qualifications	12	11	12	10	10	11
Completed Year 12	18	19	18	17	18	20
Total	71	76	78	78	80	82

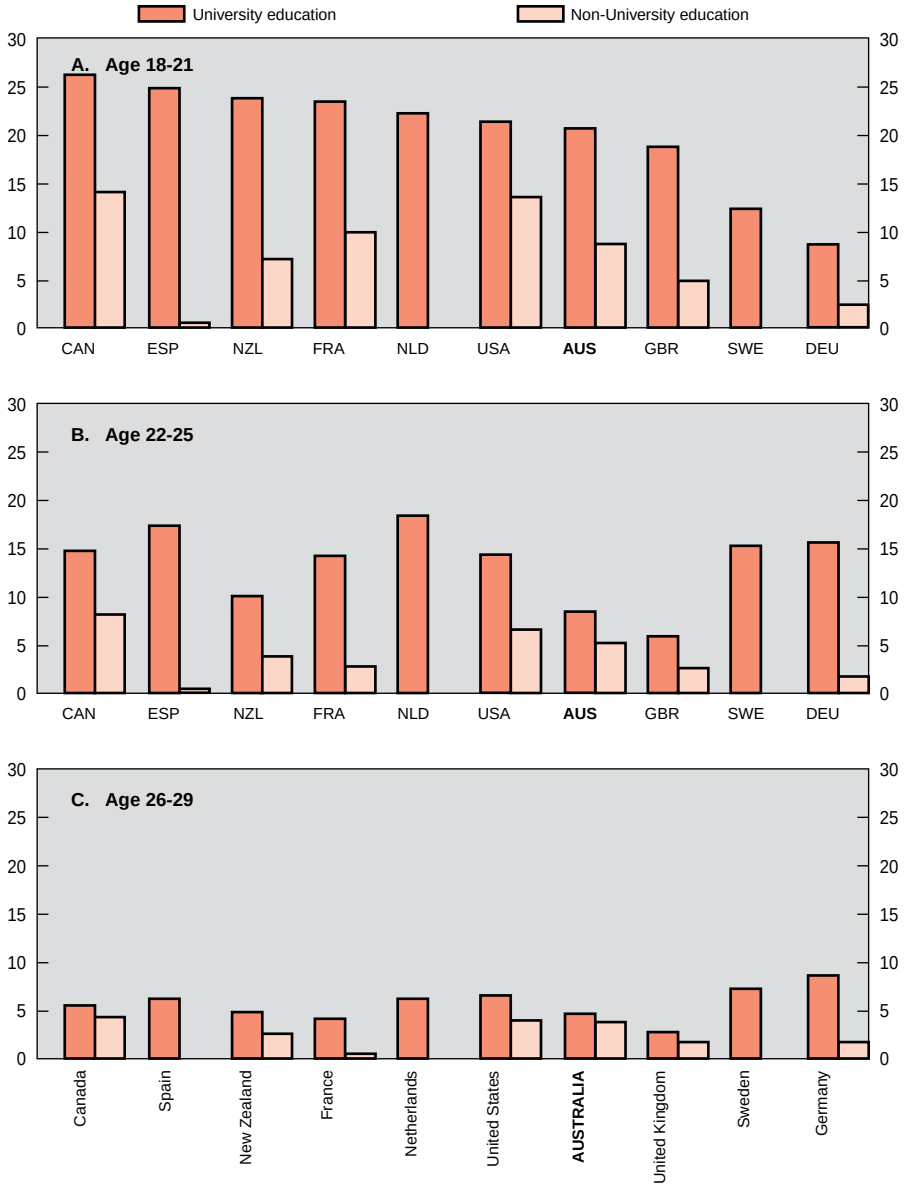
Source: Derived by Australian National Training Authority from the ABS Transition from Education to Work Survey.

education and training; the proportion of 19 year-olds not participating in education and training who have completed secondary school or a vocational qualification has been stable.

The higher education sector has expanded markedly: the number of students leapt by 44 per cent between 1988 and 1995, to over 600 000. This was far more than had been provided for in indicative planning in 1987, where a small rise to a little more than 450 000 in 2001 had been forecast.[144] The fastest growth has been among fee paying overseas students (to 10 per cent of equivalent full-time students), masters coursework students and research students. The participation rate for 17 to 19 year-olds rose from 11 per cent to 17 per cent in the last decade (see Table 18). However, due in part to demographic fluctuations and the shortage of places in the early 1990s, the growth in student numbers has been faster among persons in their twenties than among teenagers. Participation in university education is higher than in most other OECD countries and approaches North American levels, especially for persons aged 18-21 (Figure 31). Non-university tertiary enrolments are also high in Australia, although this may reflect data comparability problems across countries.

In its effort to expand training in industry the Commonwealth Government supported a range of measures including award restructuring and enterprise agreements within which provision for training or rewards for training can be made.[145] It introduced the Training Guarantee in 1990 which required firms with a wage bill of A$ 200 000 or more to spend an amount rising to 1.5 per cent of its gross wages on training. The Training Guarantee was suspended in 1994 and abolished in 1996. The proportion of wage and salary earners receiving

Figure 31. **NET ENROLMENT RATES IN TERTIARY EDUCATION, 1994**



Source: OECD, *Education at a Glance.*

Table 22. **Wage and salary earners in training**[1]

Thousands

	Study or training courses undertaken				Informal workplace learning[2]	Some training undertaken	Total wages and salary earners
	Studied	In-house	External	Total			
1989	1 124	2 338	658	3 205	4 814	5 300	6 705
1993	1 316	2 214	836	3 324	5 792	6 072	7 079

1. In the previous 12 months. Persons can undertake more than one type of study or training.
2. Informal workplace learning includes: teaching self, watching other people, asking questions of fellow workers and being shown how to do the job. Teaching self was reported by 47 per cent of all employed respondents as being the most important kind of informal learning they undertook.

Source: Australian Bureau of Statistics, *Training and Education Experience, Australia 1993*.

on-the-job informal training rose during the years of the Training Guarantee (Table 22). The positive effects on training expenditure were strongest among medium sized businesses and least in those industries which were not well advanced in developing a training culture. Larger firms generally had undertaken greater expenditures than required and the administration of the Guarantee was unpopular with business.

Modifying upper secondary school curricula for mass participation

The need to reform school structures and curricula for the wider range of abilities and aptitudes among senior school students and for destinations other than higher education was highlighted in a number of reviews of schooling in the 1980s.[146] Recommendations were made for a wider range of subject offerings at varying levels, including technological and vocational subjects. Considerable progress has been made in implementing these recommendations. Many senior school students now take courses, including in vocational subjects, which provide preparation for destinations other than higher education.

More pathways from school to work

The transition from school-to-work is difficult for many young Australians. About half of all school leavers in any one year still enter directly into the labour force. Many of them experience a series of casual jobs punctuated by spells of unemployment. A recent survey by the Australian Bureau of Statistics (ABS) found that among 19 year-olds in the labour force, the unemployment rate was

27 per cent for those who had not completed secondary school and had no vocational qualification and 18 per cent for those who had completed only secondary school.[147] However, for those who held a skilled vocational qualification, the unemployment rate was only 7 per cent.[148] Statistics like these help to explain the priority which governments have accorded to expanding the work-based VET route from school to work.[149] The challenge, addressed in the major reforms to entry level training, has been to extend work-based training to non-traditional areas and to create new combinations of on- and off-the-job and institutional training, including training in secondary schools.

The extension of structured and recognised work-based training to a wider range of occupations than covered by traditional apprenticeships was proposed in the Kirby Report (1985) and supported by the Commonwealth. This led to the introduction of traineeships, usually for one year, alongside the (four year) apprenticeship system. A target of 75 000 traineeship places was set for the end of 1988 but numbers fell far short of this. Traineeships, in a number of forms, are now an established part of the training system. They have extended the range of occupations for work-based training and participation by young women. The numbers never approached the original target and commencements usually have been below 20 000 per annum. Traineeships did not achieve the recognition with employers or unions given to traditional apprenticeships. Traineeships were less attractive to young people in that they were paid only for time on-the-job and, being for one year, led to a lower level of qualification than apprenticeships. Moreover, completion of a traineeship did not in general increase earnings, contributing to high drop-out rates.[150] Following the Carmichael report (1992), which recommended the merging of apprenticeship and traineeships into an articulated progression of awards under the Australian Vocational Training System (AVTS), a variety of new schemes was developed. With additional finance under *Working Nation* (Commonwealth of Australia, 1994), and vigorous promotion, the numbers of trainees leapt in FY 1995/96 to 37 000, including increased numbers in retailing and clerical areas not covered by traditional apprenticeships.

Some progress has been made in expanding work-based VET into more highly skilled occupations than those traditionally covered by apprenticeships and this trend is likely to continue with the implementation of the industrial relations reforms. Expansion of structural work-based training arrangements into industry sectors not traditionally covered by apprenticeship is occurring with the

development of competency standards and new training arrangements in those sectors. There is considerable scope to expand apprenticeships in those occupations, in which employment opportunities are growing quickly

Vocational education has been expanding rapidly in the final two years of secondary schooling. Many of the programmes involve dual recognition for credit for both school and TAFE qualifications and/or industry placements.[151] A stimulus to the recent expansion was the introduction under *Working Nation* of the Australian Student Traineeship Foundation (ASTF), which is funded to support school-industry partnerships. Ainley and Fleming (1995) estimated that about half Australia's secondary schools offered programmes in 1995 involving some workplace learning and that about 9 per cent of senior students (Years 11 and 12) participated in the programmes.[152] An indicator of the rapidity of growth in courses involving work placements is that about half of the programmes identified were started in 1995.

Surveys of employers show a strong commitment to the concept of workplace learning for Year 11 and 12 students. However, it will be necessary to improve the co-ordination of competing requests for work placements if that support is to be translated into continued expansion. The Government has provided funding in the 1996 Budget to expand the co-ordination of work placement activity in school-industry programmes. Results to date from these school work-based programmes are encouraging. School leavers who participated in work placements tend to have lower unemployment and higher participation in post-school education and training.

Other measures to boost participation in VET-pathways from school to work have included additional funding from the Commonwealth Government for TAFE institutions since 1992,[153] the establishment of the Australian National Training Authority (ANTA) to manage this growing financial contribution and to oversee continuing reform, and reforms in some States to make the TAFE sector more efficient (see below). Nevertheless, the growth in participation in VET has been much less than desired.

The new Government is particularly concerned about the lack of progress in raising participation in apprenticeships. It considers that this is due in part to the lack of flexibility in the industrial relations system in providing wage arrangements which are appropriate to the particular types of work-based training arrangements which are considered optimal in particular areas of employment. It

is necessary for a diversity of wage arrangements to be available which can reflect different mixes of training and work, different durations and forms of employment relationship (for example, part-time or full-time), and different types of training (for example, on- or off-the-job). In the past, the award system has impeded the implementation of new training arrangements because of the time taken to introduce the new industrial arrangements necessary to match the new training arrangements. Another factor which has impeded growth in apprenticeships is their relatively low social status.

The new Government intends to address these problems through its wider industrial relations reform and through reforms to the apprenticeship and traineeship system. These reforms will enable employers and apprentices to agree Australian Workplace Agreements (AWAs) and Certified Agreements (CAs) in which the apprentice is only paid for time spent on the job in productive work; wage top ups will be provided to ensure that full-time trainees and apprentices will have an income that does not fall below a minimum amount.[154] Agreements may also spell out clearly employers' responsibilities to provide training; and monitoring arrangements are to be put in place to ensure that employers do provide this training. Other apprenticeship and traineeship reforms include: simplifying the process of gaining accreditation for programmes; giving users greater choice (from 1998) over the nature of government-funded off-the-job training; the extension of school-industry arrangements to allow students to commence apprenticeships and traineeships while at school, with a variety of arrangements of school and work days; and the development of part-time traineeships. Employer incentives payments have been revised to provide assistance based on skill outcomes, encourage progress to higher skill level training and to target assistance to small and medium employers.

Many obstacles to the expansion of apprenticeships will be removed by these reforms. Whether they remove the binding constraints on expanding apprenticeships remains to be seen. Other English-speaking countries which already have more wage flexibility than Australia, and in some cases also have strong employer input into the content of competency-based vocational qualifications, have not been noticeably more successful in developing apprenticeships. It could be that the binding constraint is the absence of certain social institutions which are present in countries, such as Germany, with a strong tradition in apprenticeship. These institutions include the ''meister''[155] and a powerful local

chamber of commerce which, together with unions, runs the system. It may be that the best way to ease the school-to-work transition in Australia is to ensure that employment and training conditions (including wages) are flexible and that there are many opportunities to continue education and training beyond school. In this context, the general concept of alternation of education and work, of which traditional apprenticeships are but one form, should be encouraged.

Improving the labour-market relevance of education and training

A major thrust of reforms in recent years has been to make education and training programmes more relevant to the needs of the clients and in particular to industry and to less advantaged young people. This thrust has been supported by reforms to funding methods (see below) and to the framework for curriculum, assessment and qualifications.

In order to prepare students better for the world of work, attention has been given to incorporating employment-related key competencies in school curricula.[156] The key competencies are summarised as: collecting, analysing and organising information, communicating ideas and information; planning and organising activities; working with others and in teams; using mathematics; solving problems; and using technology. Pilot projects integrating key competencies into education and training systems have been under way since 1993, as have information campaigns directed at teachers, parents and business. Another area of focus for increasing the labour-market relevance of secondary education has been foreign languages. There has been a considerable increase (from a low base) in the proportion of students studying Asian languages.

Ensuring that students develop strong literacy and numeracy skills at school is an important aspect of preparing them for further education and the labour market, where required levels of proficiency are rising. A recent review of the evidence available from State assessments[157] indicates, with few exceptions, stable or slightly rising levels of achievement.[158] Nevertheless, the Government is concerned that standards may not be adequate to meet employers' needs and to enhance social cohesion (especially by increasing employment amongst the low skilled). Accordingly, it announced a National Literacy and Numeracy Strategy in the 1996 Budget. The first step in this strategy, to undertake a national survey of literacy in schools, already under way, will provide data for the development of the strategy.

For the time being, no recent data are available which include Australia in international comparisons of educational achievement.[159] Data which are available show an average performance amongst participating countries with no clear trends over time.[160] In the science studies for fourteen year-olds conducted in 1970-71 and 1983-84, Australia ranked respectively third and sixth out of 10 countries. In mathematics, Australian students ranked higher than in the United States in years 8 and 12 in the early 1960s but below England, Scotland and Japan. However the Australian students were younger and at year 12 were a larger proportion of the population. A study in 1978 showed a slight decline in performance of 13 year-olds but an increase for year 12 students despite the increase in the proportion of the population at school.

In the VET sector the major reform was to base certification on industry determined competency standards, rather than on time served in education and training. Vocational Competency Standards provide the specification of the knowledge and skill and the application of that knowledge and skill to the standard of performance required in employment (National Training Board, 1991). A system of State and National Industry Training Boards (ITABs) was established, with employers and union membership, to assist with advice on industry standards across all occupations covered by training in the VET sector. The establishment of national competency standards is particularly important for the development of a training market for it defines a ''product'' for which suppliers can compete. There has been criticism of the effects of competency based teaching and assessment accompanying the standards.[161] The operation of the framework within which training is recognised has also been criticised for being excessively bureaucratic and complex, discouraging enterprises and private providers of training from participating in the system (Taylor, 1996). The Commonwealth and State and Territory Governments have commenced reforms of the regulatory systems across Australia which will ensure a more simple and transparent national system of mutual recognition based on nationally endorsed industry standards and national qualifications within the Australian Qualifications Framework.

Some ITABs have had difficulty in carrying out some of their roles, in particular the offering of advice on the quantity of training needed for particular industries. In part this is because while representing a grouping of industries they have to advise on training for occupations which are found across a wide range of

industries. The data for such tasks are inadequate and the ITABs lack the range of personnel to grapple with the data available. In addition the government has reported that industry has not seen the ITABs as meeting their needs (Kemp, 1996*b*). New industry training organisations will be developed and the views of the existing advisory bodies, business groups and enterprises are to be sought on the structure. Within the Australian National Training Authority (ANTA) a business-led committee will be responsible for competency standards.

The new Government's reforms to the apprenticeship and traineeship system also aim to enhance the labour-market relevance of VET by allowing employers to determine from 1998 the nature of Commonwealth Government funded off-the-job training. They will be able to decide what the nature of the training should be, where it will be delivered and by whom; this will expose the traditional providers in TAFE to increased competitive pressure.

Universities continue to receive strong input from employers on the content of degrees in the professions (such as law, medicine, accountancy and engineering). Various schemes of collaboration between universities and industry have been supported and universities have more actively sought industry funds for courses, consultancies and research. Competency standards have been developed for 19 occupations by professional bodies. Universities are not, however, obliged to incorporate competency standards into curricula or assessment techniques.

Improving the articulation between educational sectors

To ensure the continued upgrading of the workforce and for reasons of equity there has been substantial pressure since the beginning of this decade to improve articulation across levels of qualifications and sectors of education and training. This has been facilitated by changes in the structure of qualifications under the Australian Qualifications Framework and the increased emphasis on competency standards as the basis for awards in VET. The move to recognise prior learning, whether on the job or in another sector of education, and the agreement by State and Commonwealth Governments to a National Framework for the Recognition of Training (NFROT) has meant that competencies, however achieved, should be recognised across Australia.

The ease of articulation within the VET sector appears to be increasing, though until recently dissatisfaction was being expressed with NFROT. There has been progress in the recognition of VET courses by universities for credit

transfer purposes, and a number of innovative programmes have been encouraged (including the development of fully articulated courses by higher education institutions which offer both VET and university programmes). However, progress has been hampered by a number of factors, including the fact that most VET teaching and assessment incorporates competency standards while university teaching does not. Nonetheless there continues to be an increase in the total share of offers of university places made by higher education institutions to TAFE students (with corresponding increases in applications for places by TAFE students) and an even larger number of university students who enrol in a VET course after completing, or attempting, a university course.

Enhancing equity

Increased participation of students from lower socio-economic backgrounds also has been an important objective of policy in recent years. Measures to further this objective have included the introduction of AUSTUDY in 1987 and increased funding for TAFE institutions from the early 1990s. AUSTUDY provides financial assistance to full-time students aged 16 and over, subject to family and personal means tests; less than half of such students qualify.[162] And students at TAFE institutions tend to be from less privileged socio-economic backgrounds than university students.

Insofar as a larger proportion of each age cohort now completes secondary school and indeed continues in formal education, considerable progress has been made in extending participation to students from more modest socio-economic backgrounds. Nevertheless, the retention rate until the final year of secondary school for students from low socio-economic backgrounds remains relatively low, although it has risen somewhat relative to that for the rest of the population.[163] Such students who do stay until the end of secondary school tend to be relatively less successful in the Year 12 assessment[164] and thus have fewer post-school options.[165]

The increase in retention rates has been an important factor in extending university studies to students from a wider range of socio-economic backgrounds than in the past. Indeed, among students who have successfully completed secondary school, socio-economic background has little impact on whether students actually enrol in higher education.[166] Socio-economic background, however, does have an effect on the type of university in which students enrol.[167] This is because

students for a low socio-economic background tend to do less well in the Year 12 assessment.

Social equity also has been supported through the maintenance of relatively small differences in resource levels per student across government schools and TAFE institutes within States. There are also very small differences among States in school expenditures, though apparently somewhat larger for TAFE. The provision of higher education funding by the Commonwealth means similar funding for institutions with similar course mix across the nation. This equality in funding levels is in marked contrast with the experience across and within states in the United States.

Nevertheless many ''other'' (*i.e.* non-Catholic) non-government schools operate with significantly higher resource levels per student than government schools, while Catholic schools have lower resource levels than government schools. The extra resources for the ''other'' schools come from high tuition fees; although these schools also receive government grants, they are much lower than those to government and Catholic schools, being inversely related to their overall resource levels.[168] Because tuition fees are high, students mostly come from affluent backgrounds. For a variety of reasons, retention rates until Year 12 are very high in these schools.[169] Their graduates tend to do well in the Year 12 assessment and consequently, take a disproportionate number of places in the most prestigious universities.

There are also programmes in schools for the less advantaged framed largely in terms of improving their educational participation and outcomes. The Commonwealth's Literacy and Special Learning Needs Programmes and various complementary State activities, incorporate a range of extra resources including support for English as a Second Language, literacy, and for rural students and students with a disability. Special efforts have been made to increase the retention rate until the final year of secondary school for Indigenous Australians. This rate had risen to 31 per cent by 1995, but remains far below that for the rest of the population.

There has been considerable success in raising female participation in upper secondary school and tertiary education. By 1995, 78 per cent of girls were remaining to the end of secondary school compared with 67 per cent of boys. Females now have a notably higher rate of entry into higher education but a

lower rate of entry into TAFE and a particularly low rate of entry into traditional apprenticeships.

Social equity objectives also have been important in the decision to shift a greater part of the costs of university education onto students and their families. As graduates earn more than the average taxpayer, the financing of tuition from tax revenue is regressive. The Higher Education Contribution Scheme (HECS) charge was introduced in 1989 partly to reduce the scale of this transfer. In 1996, the charge was set at 23 per cent of the average cost of courses with payments being made either up -front for a discount or, more typically, once the student's earnings exceeded a specified amount; in 1996, HECS amounted to A$ 2 442 per year of study for a full-time student. The real interest rate on deferred repayments is zero. The income-contingent aspect of repayments was intended to reduce the risk of deterring students from low socio-economic backgrounds. In fact, the introduction of HECS does not appear to have affected the participation of this group in higher education. HECS are to rise in 1997 to about 30 per cent of the cost of a course in medicine and 50 per cent of the cost of an Arts course.[170] These charges will cover a lower share of tertiary expenditures than fees do in the United States[171] and Japan, but higher than in most other countries (see Annex III). Some HECS exemption scholarships will be provided on merit to students from low income backgrounds. Repayment of HECS now will be required at a lower level of income than in the past, speeding the recovery rate and tapping income groups who might not otherwise have been required to repay their debt.[172]

In the TAFE sector, fees are low, usually amounting to less than a tenth of tuition costs,[173] with exemptions for students from low income backgrounds[174] and there is no HECS equivalent. However, it is by no means obvious that social equity objectives are best pursued by such untargeted subsidies. Not all TAFE students are from low socio-economic backgrounds and some do courses which lead to high earnings. Experience with HECS has demonstrated that it is possible to obtain a contribution from students towards the cost of their courses without discouraging the participation of students from low income backgrounds. Consideration could be given to increasing fees for higher level TAFE courses to improve the targeting of Government subsidies for VET. If fees were increased to similar levels as apply in higher education, consideration would need to be given to the introduction of HECS-type arrangements for the collection of these fees.

Enhancing efficiency

Two main approaches have been taken to increase the efficiency of the provision of education and training. One is a reform of the organisational structure and management of public education and training. The second is the increase in competitive pressures resulting from making an increasing proportion of public funds available on a competitive basis among both public and private providers.

Organisational reforms

At a national level separate government departments were merged to establish the Commonwealth Department of Employment, Education and Training (DEET) in 1987. Such a merger, the first of its kind among OECD countries, facilitated the designing of education and training policies that are better geared towards the needs of the labour market. A related development has been the appointment of industrial leaders rather than educators to head the major committees of review in recent years. With the agreement of the States, the Australian National Training Authority (ANTA) commenced work in 1993 to co-ordinate VET and in particular to allocate increased Commonwealth funds to the States. The operation of the ANTA agreement has been reviewed and reforms carried out to reduce the degree of bureaucratic[175] complexity in its regulatory role in VET and to increase the input of business into its operations.

Reforms to higher education announced in 1988 involved merging the two sectors of higher education, colleges of advanced education and universities. The reforms aimed at expanding the number of graduates closer to levels in economies such as United States and Japan, with an emphasis on areas of particular economic need, especially technologies and management. Prior to the unification the colleges were not funded for research and their teaching was supposed to be more applied and oriented to the labour market. The two sectors had increasingly overlapped in functions of teaching and research. Economies of scale in teaching and research were hence thought to be important and were the main reason given for specified minimum enrolments for institutions to become members of what is now called the Unified National System. The effect of the size requirement was to promote mergers that halved the number of institutions to which the Commonwealth would provide funding on the basis of pre-agreed outcomes. The resulting managerial economies to the Commonwealth were perhaps rather important.

At various levels of education there has been a move to increased specification of outcomes. Governments also exercise influence over the broad distribution of activities, quality assurance requirements and equity objectives. In universities, financial incentives were also provided to speed up the adoption of quality assurance procedures. Performance indicators are being developed as a means by which government authorities monitor the providers. In VET the establishment of industry competency standards has been an important part in the development of a training market for it defines a ''product'' for which suppliers can compete.

There has also been devolution of the methods by which the outcomes are reached. In higher education, the universities have been operating as autonomous institutions in raising private revenue, in allocating their expenditures and in making contracts in Australia and internationally. In the Unified National System the merged institutions were permitted increased autonomy in the methods used to achieve the outcomes agreed in profiles with the Commonwealth. The TAFE institutions in some States have been given similar autonomy with the role of the State authority increasingly being to allocate funds among public and private providers, rather than as manager of the TAFE sector. At school level there is also a movement to greater autonomy in management of government schools, accompanied by more extensive accountability requirements, again with variation across States. Across the sectors there have been labour market reforms that are leading to the development of enterprise agreements and some individual contracts, though most employees still work mainly under award conditions in 1996. There also has been a trend to short term and casual employment (*e.g.* DEET, 1995*b*).

Increasing competitive pressures

TAFE institutions have competed in tenders since the mid-1980s with private providers for the provision of training under Commonwealth labour market programmes. There has been continuing encouragement to TAFE institutions to undertake fee-for-service activities, competitive tendering, preferred supplier activities and industry-provider partnerships. The more recent development has been the active promotion by ANTA of the concept of a training market. ANTA (1994) in its national strategy placed the development of a market for training as a prime means of increasing the responsiveness of VET providers to the needs of industry and students. In particular, ANTA has required State authorities to make

available an increasing part of the Commonwealth funds for open tender by private and public providers; A$ 65 million was so allocated in 1996. Following a review of training reforms (Allen Consulting, 1994), it funded pilot projects for ''user choice'' in apprenticeship and traineeship training which is to be the basis of the provision of training for all apprenticeships and traineeships from the beginning of 1998.

Unlike VET, universities have not been required to tender for part of operating grants though the idea has been floated (NBEET, 1993), and public funds have not been available to private universities (which are few in number and cater for a small number of students). Universities have, however, engaged in strong competition among themselves for full-fee overseas students since the late 1980s and more recently full-fee postgraduate students. Part of the research funding received in the Commonwealth's operating grant is determined by the research performance of the university. Academics compete for research grants. The need for increased contestability for research funds is argued by the Productivity Commission (1996).

In primary and secondary schooling the pressure for government schools to respond to parents' needs is affected by few zoning restrictions on students enrolments and by the existence of a large and growing non-government school sector (Table 23). The non-government share of students has grown from about 21 per cent in 1977 to 29 per cent in 1995. Most of the growth was in ''other'' (*i.e.* non-Catholic) non-government schools, where enrolments have increased by 40 per cent since 1985 to almost 10 per cent of all enrolments. The growth of such schools represents a saving to public expenditure – though there are a range of other subsidies to non-government schools to be considered (National Catholic Education Commission, 1996). The Commonwealth Government announced in 1996 the abolition of a policy that restricted the access of new non-government schools to public funds, and projects a further increase in this sector's share of school enrolments.

On balance, judging by aggregate indicators such as school retention rates and participation rates and the (admittedly limited) evidence on student performance, education and training outputs have risen significantly, while spending per student has declined. Hence, there is little doubt that the overall efficiency of the system has improved. In the absence of adequate statistical data, particularly in VET, it is however difficult either to ascertain in more detail the relative per-

Table 23. **School enrolments by type of school**

	Government	Non-government			Total
		Catholic	Other	Total	
Thousands					
1985	2 231	573	202	775	3 006
1995	2 208	608	294	901	3 109
Per cent					
1985	74.2	19.1	6.7	25.8	100.0
1995	71.0	19.5	9.4	29.0	100.0

Source: Australian Bureau of Statistics, *Schools Australia*.

formance of various sectors in aggregate terms or to evaluate improvements in quality. It seems undeniable that the policy changes described above have contributed to the improved overall efficiency of the system, though it is difficult to be precise about the extent of their contribution.

Education, training and earnings

A rise in human capital through education and training should in theory lead to higher productivity and real earnings. At an aggregate level research available indicates that over the two decades to 1990 the increase in labour quality accounted for 28 per cent of the increase in real wages for full-year full-time workers[176] with that contribution higher for male workers (40 per cent).[177]

The question arises as to what extent the increased supply of skilled workers has reflected developments in labour demand. In flexible labour markets, an increase in the demand for skills would tend to be reflected in a wage premium for them, which would encourage more workers to acquire them, eventually reducing the premium. In Australia, however, earnings by level of educational attainment have been relatively stable over the past two decades (Table 24). A recent study,[178] which considered the distribution of full-year full-time earnings by qualification level from 1968-69 to 1989-90, shows that earnings of degree holders fell relative to those of individuals who left school at 17 until 1985-86 but rose marginally in subsequent years. For less well-educated workers the opposite trend occurred – a slight relative increase to the mid 1980s but decline

Table 24. **Earnings by level of education**[1]

		Males							Females						
		15-24	25-34	35-44	45-54	55+	Total	Adjusted total[2]	15-24	25-34	35-44	45-54	55+	Total	Adjusted total[2]
Degree	1968-69	1.31	1.66	1.88	2.21	1.69	2.05	2.09	1.78	1.75	1.89	1.45	1. .32	1.75	1.89
	1973-74	1.43	1.53	1.59	1.81	1.80	1.82	1.84	1.57	1.67	1.76	2.04	2.07	1.89	1.90
	1978-79	1.38	1.42	1.53	1.66	1.82	1.75	1.72	1.49	1.49	1.31	1.79	1.92	1.63	1.63
	1981-82	1.32	1.24	1.43	1.84	1.97	1.67	1.64	1.43	1.30	1.45	1.78	1.40	1.55	1.56
	1985-86	1.28	1.35	1.37	1.98	1.59	1.58	1.62	1.19	1.49	1.18	2.09	1.54	1.54	1.54
	1989-90	1.47	1.37	1.42	1.67	1.86	1.66	1.67	1.35	1.41	1.55	1.71	1.94	1.59	1.63
Certificate (non-trade), diploma[3]	1968-69	1.26	1.24	1.39	1.36	1.35	1.49	1.52	1.36	1.33	1.31	1.23	1.14	1.30	1.40
	1973-74	1.25	1.21	1.24	1.22	1.27	1.43	1.39	1.36	1.36	1.44	1.48	1.18	1.48	1.45
	1978-79	1.24	1.11	1.15	1.11	1.22	1.29	1.27	1.11	1.18	1.05	1.26	1.31	1.22	1.21
	1981-82	1.18	1.15	1.05	1.33	1.40	1.32	1.30	1.05	1.14	0.99	1.17	0.80	1.13	1.13
	1985-86	0.97	1.12	1.19	1.51	1.32	1.33	1.35	0.99	1.20	0.92	1.52	1.23	1.17	1.17
	1989-90	1.14	1.07	1.14	1.17	1.23	1.25	1.27	1.08	1.11	1.15	1.34	1.27	1.20	1.21
Technician/trade/other	1968-69	1.20	0.99	0.96	0.96	0.74	1.04	1.06	1.08	1.10	1.30	0.95	1.32	1.15	1.20
	1973-74	1.23	0.99	0.89	0.95	0.86	1.07	1.06	0.98	1.08	1.12	1.23	1.07	1.12	1.12
	1978-79	1.15	1.00	0.94	0.89	0.91	1.06	1.05	1.02	0.98	0.92	1.11	1.14	1.05	1.04
	1981-82	1.23	0.95	0.87	0.99	1.03	1.06	1.04	0.93	0.97	0.77	0.92	0.64	0.95	0.92
	1985-86	1.12	1.08	0.88	1.10	0.97	1.06	1.08	0.98	1.00	0.80	1.53	0.90	1.05	1.04
	1989-90	1.19	0.99	0.94	0.92	0.87	1.04	1.05	0.90	0.91	0.94	1.05	1.09	0.94	0.98
Left school aged 16	1968-69	0.84	0.90	0.93	0.99	0.80	0.88	0.91	0.89	1.00	1.07	0.84	0.79	0.87	0.97
	1973-74	0.89	0.92	0.86	0.98	0.90	0.90	0.91	0.90	0.88	0.92	1.03	0.99	0.93	0.93
	1978-79	0.89	0.94	0.93	0.92	0.91	0.93	0.93	0.95	1.00	0.87	1.06	1.09	0.99	0.98
	1981-82	0.90	0.90	0.87	1.12	1.42	0.97	0.98	0.84	0.93	0.80	0.93	0.59	0.88	0.87
	1985-86	0.82	0.95	0.93	1.19	1.29	0.97	0.98	0.89	1.04	0.77	1.15	1.02	0.96	0.94
	1989-90	0.84	0.90	0.94	0.93	1.15	0.95	0.93	0.93	0.90	1.03	1.02	0.95	0.98	0.97
Left school aged 15 or younger	1968-69	0.86	0.81	0.78	0.79	0.66	0.85	0.87	0.81	0.91	0.84	0.77	0.67	0.83	0.90
	1973-74	0.90	0.85	0.74	0.80	0.77	0.89	0.89	0.83	0.84	0.85	0.91	0.89	0.91	0.90
	1978-79	0.84	0.92	0.82	0.81	0.79	0.93	0.92	0.88	0.91	0.77	0.98	0.90	0.94	0.93
	1981-82	0.91	0.88	0.81	0.93	0.99	0.97	0.96	0.84	0.92	0.74	0.93	0.60	0.91	0.88
	1985-86	0.86	0.93	0.83	0.97	0.80	0.92	0.93	0.80	0.91	0.69	1.09	0.78	0.89	0.88
	1989-90	0.85	0.89	0.82	0.83	0.83	0.92	0.92	0.85	0.86	0.87	0.92	1.09	0.92	0.92

1. Full-time, full-year earnings relative to completing secondary school (matriculated or left aged 17 plus).
2. Adjusted for demographic shifts.
3. The classification has not remained constant. This category comprises non-degree tertiary in the earlier surveys.

Source: Australian Bureau of Statistics, *Income Distribution Surveys*.

since then. Thus, there has not been a marked increase in the relative earnings of skilled workers, in contrast to the experience in the United States.

The stability of earnings by education category is remarkable considering the higher unemployment rates since the mid-1970s, the large changes in industry composition and in the supply of educated workers.[179] These data can be interpreted as showing that there was no obvious shortage in supply of educated workers. Alternatively, it suggests an economy where relative earnings do not reflect changes in relative demand and supply of workers of different skill and education levels, but rather, adjustment takes place in terms of employment levels. The increased demand for post-compulsory education, especially higher education, is likely to have been induced by the deterioration in employment prospects for low skilled workers, among other things.[180]

As in most other OECD countries, there has been a large decline in the male full-time employment ratio. When full-time male jobs are grouped according to earnings in 1983, middle and low skilled occupations account for all of the decline in the employment-population ratio since 1977.[181] Alternatively, job skill levels can be assumed to match full-time earnings directly, allowing the skill levels of jobs to change as the distribution of full-time earnings change. On this basis, there has been considerable growth in the low-skilled employment-population ratio, the high-skilled ratio has held steady and the middle-skilled ratio has fallen sharply. This suggests that any increase in demand for labour has been restricted to low and high skilled labour and that there has been a ''disappearing middle''. Yet another study which analysed employment changes by skill deciles, as measured by qualification levels, found that the fastest growth was in the top deciles and that growth in the middle deciles was average.[182] A possible reconciliation of these partly conflicting findings may lie in the changes in the distribution of earnings *within* occupations – and that the earnings of poorly educated persons were falling within occupations.

Several studies have used the data on earnings by qualification level to make estimates of private rates of return (*e.g.* Maglen, 1995). For persons with degrees these show a similar trend to earnings differentials as considered by Gregory (1995): in general, there has been a downward trend with some rates slightly up in the late 1980s. Private rates for persons with degrees, with adjustments for the likelihood of unemployment and assumptions about full-time study, length of course and method of payment of HECS have also been calculated.[183] University

graduates, compared with persons who left school at 17, had private rates of return of about 13 per cent in the late 1980s. While lower than for earlier periods, they are still quite attractive rates.

Private rates of return on apprenticeships appear to vary considerably across trades and by gender, according to a recent study. In comparison with males without post-secondary schooling, the average private rate of return to the six main male apprenticeships has been estimated to be 46 per cent.[184] The private rates of return range from more than 85 per cent for electrical mechanics and compositors to 27 per cent for cooks. The relatively high rate of return to male apprenticeships may be due mainly to the low level of earnings foregone during training as earnings do not appear to be greatly enhanced by the training.[185] The low level of earnings foregone occurs in some trades, such as electrical mechanics, because students in their third and fourth years of an apprenticeship tend to earn more than the average school leaver. On the other hand, for males in some apprenticeships, including vehicle mechanics and hairdressers, the estimated lifetime earnings streams are lower than for unqualified persons, and estimates of rates of return could not be made. Three trades were considered for females (namely garments tradespersons, hairdressers and cooks) and only for cooks were earnings higher than for unqualified persons.

Training in the workplace appears to make positive contribution to earnings. Those who were undertaking training provided by employers in 1993 in general earned more than those who did not.[186] Chapman and Tan (1992) found in analysing longitudinal data that formal training increased wages and had a larger effect in firms experiencing technological change as proxied by growth in multifactor productivity. Such data are, however, only a broad guide to what sorts of training are most beneficial.

To sum up, while it is difficult to detect any evidence of excess demand for high skilled workers, demand for workers with low qualifications has been in decline. The rapid employment growth in low-paid jobs has probably been met by workers with higher qualifications. It should be noted that even the most recent empirical works reported above examined data up to 1990, in the absence of more recent detailed information, and hence are of limited value in assessing the impact of the policy changes that have taken place since the late 1980s.

Outstanding issues

An issue which all OECD countries are having to confront is the extent to which expenditure on education should be publicly financed. Public funding can be rationalised on efficiency and equity grounds. Education conveys external benefits to society, such as greater ease of communication in a literate society and a shared sense of community. In societies such as Australia, with a developed welfare state, welfare expenditure may also be reduced by education insofar as it contributes to lower unemployment. The external benefits of education are generally considered to be greatest for primary education and lowest for tertiary education.[187] Hence, on efficiency grounds, the government's share of expenditure on university education should be lower than for school education, as indeed is the case in Australia and other OECD countries (see Annex III). Equity and efficiency objectives can also be advanced through public funding of education, especially by ensuring that children from modest social backgrounds have greater access to education services than would otherwise be the case.

As there is little agreement about the magnitude of external benefits of education, except that they are likely to be higher for schooling than for university studies, a variety of views about the efficient degree of public funding can be held. The decision of the new Government to raise HECS could be interpreted as reflecting a judgement that the external benefits of university education did not justify public subsidies for undergraduate courses amounting to over 80 per cent of tuition fees.[188] Another factor underlying this decision was undoubtedly a concern about the implications for social equity of providing such large subsidies to individuals with higher lifetime earnings than the average taxpayer. Whether HECS have been increased by enough or by too much is a matter of judgement.

Currently higher education students must make a contribution to the cost of their courses through HECS while TAFE students pay a smaller up-front fee. Therefore, even though TAFE courses are funded at a lower level than higher education courses, the public subsidy for TAFE is greater in proportionate terms. It is possible that these different funding arrangements could distort student decisions about where to study (and students may seek to minimise HECS debts by enrolling in TAFE courses in the first instance and then seeking credit towards a university degree at a later date). Increasing fees for higher level TAFE courses would remove this possible distortion and allow Government to improve the

targeting of its subsidies. As noted above, if fees were increased to similar levels as apply in higher education consideration would need to be given to the introduction of HECS-type arrangements for the collection of these fees.

Whether the share of lower levels of education which is publicly financed is efficient is also a matter of judgement. But at least public expenditure on these levels of education does not have perverse implications for social equity. The population of school students and TAFE students doing courses below diploma level is likely to have lifetime earnings similar to the average for their age-cohort, especially in view of the Finn objective of universal participation up to Year 12 or its vocational equivalent. Given that the Government considers that school education should be available free of charge up until the end of secondary school, there is a case also for making equivalent level vocational education available free of charge, unless the Government considers that the external benefits of vocational education are lower or that social equity objectives would be served by retaining this discriminatory treatment. Even though there is no evidence that they have had a disincentive effect on participation in these TAFE courses (mainly because they are small and generally they are waived or paid for by the Commonwealth Government for disadvantaged young people), charging fees for TAFE courses which lead to much the same credentials as are obtainable free of charge by completing secondary studies at school seems difficult to justify on consistency grounds.

How public funds are allocated to providers also has important implications for efficiency. The greater the extent to which funds are allocated administratively, without recourse to competitive markets, the less likely it is that resources will be used efficiently. Governments have begun to develop internal markets in which public providers must compete for public funding. A step in this direction was the creation of the Unified National System of Higher Education in 1988, which made the allocation of an increasing proportion of public funds for research on the basis of research performance. Some States have introduced an internal market for funding for TAFE institutions. An increasing proportion of public funds for VET are open to tender by public and private providers and it is intended under ‘‘user choice’’ in 1998 that the allocation of funds to providers of apprenticeship and traineeship training will be determined by employers and students. States and Territories have agreed to implement User Choice for all apprenticeship and traineeship training. Competition among public schools and

private schools has increased as schools need to ensure retention of adequate numbers to sustain their curriculum range. The elimination of most zoning limits on student mobility has also increased competition among public schools.

Internal markets could be made more competitive and hence, responsive to clients' needs. A building block for increasing competition is greater labour-market flexibility. This would make it possible to link employees' financial rewards more closely to their contribution towards their institution's objectives. Better quality staff could be attracted with higher pay and under-performing staff could be made redundant more easily. To produce a higher education sector that is more responsive to the needs of students and society, these labour-market reforms would need to be coupled with more flexible funding mechanisms for university teaching. One mechanism for achieving this would be to allow universities to set their own fees and to give students a voucher equivalent in value to their present grant.[189] This would give each university an incentive to offer courses well adapted to client requirements at the lowest possible price, just as producers in other markets seek to offer clients the best value for money. Well run universities would expand and poorly run ones would contract, eventually going out of business. The same kinds of reforms could be made in the rest of the public education system. At least at the school level, especially in non-metropolitan regions, such a reform could provoke concerns about the social equity implications of greater variety in the quality of schools. However, this could be addressed by vouchers which are inversely related to the school's level of resources. It is recognised that a number of administrative details would need to be resolved before a voucher system could be introduced in either of the sectors.

All States and Territories have introduced processes for allocating some VET funding in a competitive manner. The amount of funds put out to open competitive tender varies considerably among States and Territories but generally remains low. Substantially increasing the proportion of funds allocated in this way would be likely to reduce the cost of training and increase the diversity of choice. For the same reasons, the Commonwealth should consider allowing providers currently outside the unified national system to compete for public funding for university teaching.

List of acronyms

ABS	Australian Bureau of Statistics
ACE	Adult and Community Education
ACER	Australian Council for Educational Research
ACVETS	Australian Committee on Vocational Education and Training Statistics
AGPS	Australian Government Publishing Service
ANTA	Australian National Training Authority
AQF	Australian Qualifications Framework
ASF	Australian Standards Framework
ASTF	Australian Student Traineeship Foundation
AVETMISS	Australian Vocational Education and Training Management Information Statistical Standard
AVTS	Australian Vocational Training System
DEET	Department of Employment, Education and Training
DEETYA	Department of Employment, Education, Training and Youth Affairs
HECS	Higher Education Contributions Scheme
ITB	Industry Training Board
MCEETYA	Ministerial Council on Education, Employment, Training and Youth Affairs
NBEET	National Board of Employment Education and Training
NTB	National Training Board
TAFE	Technical and Further Education
VET	Vocational Education and Training

Notes

1. If not indicated otherwise, all references to GDP are to GDP(A), which is the average of the expenditure, income and production measures of GDP.
2. Fiscal years start 1 July.
3. The Australian Bureau of Statistics (ABS) calculates trend estimates by smoothing the seasonally-adjusted series using a statistical procedure based on centred Henderson moving averages. The purpose of this practice is to extract the underlying behaviour of the series by purging irregular components from the data which may have remained even after seasonal adjustment.
4. Residential investment encompasses dwellings and real estate transfer expenses, which are separately recorded in the Australian National Accounts. The causes of the recent housing investment boom were discussed in last year's OECD *Economic Surveys, Australia,* Chapter I.
5. Housing affordability is measured by the ratio of average household disposable income to the (''qualifying'') income required to meet payments on a typical dwelling (expressed as an index). In calculating qualifying income a deposit of 20 per cent with repayments equal to 30 per cent of income is assumed using a conventional 25 year loan. An increase in the Index represents an improvement in affordability.
6. Both the NAB and the ACCI-Westpac survey measures of manufacturing capacity utilisation indicate a peak in the second half of 1994 and a trend decrease since then.
7. However, there is a range of conceptual approaches to defining potential output and the output gap, which result in varying quantitative estimates. This illustrates the need to assess all indicators in coming to a view on the level of excess capacity in the economy and, in particular, to avoid mechanically applying any one indicator of spare capacity to the operation of monetary policy.
8. An example of such temporary phenomena are flow-on effects from the recovery in the rural sector to the associated wholesale, transport and storage industries.
9. Another widely used wage index, the survey-based average weekly earnings (AWE) of all employees, which provides a broader measure of wage developments than AWOTE, tended to accelerate less. But AWE comprises full-time, part-time, casual and junior employees and includes overtime earnings, which makes it responsive to compositional changes in the labour force. For example, the recent slower growth in AWE than in AWOTE reflects faster growth in part-time than in full-time employment and a fall in overtime worked. The measure of average earnings derived from the National Accounts also increased by less than

AWOTE in 1995. This series is also affected by compositional changes, as well as factors such as redundancy payments to workers. For these reasons AWOTE is often preferred as an indicator of underlying wage pressures.

10. AWOTE may also have been inflated in 1995 as overtime and penalty rates are being incorporated in ordinary conditions of some enterprise agreements, raising AWOTE but not labour costs.

11. The Cullen Egan Dell Survey of earnings of skilled workers.

12. See also the discussion of Monetary Policy in Part II.

13. Given the proven sensitivity of mortgage interest rates to cash rates, the substantial weight of mortgage costs in the CPI and the fact that most mortgage loans are of floating-rate nature, cash rate changes have a pronounced and rapid effect on the CPI.

14. See the discussion of Monetary Policy in Part II.

15. The Australian Council of Trade Unions is pursuing a ''Living Wage'' claim in the Australian Industrial Relations Commission which in the first year would provide an increase of around 8.75 per cent in all award minimum rates of pay, with a minimum award safety net adjustment of A$ 20 per week. This claim is expected to be heard in December 1996.

16. See, for example, Reserve Bank of Australia (1994), *Report and Financial Statement*, p. 3.

17. See also OECD (1995), *Economic Surveys, Australia*, Chapter II.

18. See OECD (1996), *Economic Outlook*, No. 59, June, Table 11.

19. The three-month bank bill rate had already risen by around ½ percentage point to about 5½ per cent in the June 1994 quarter, as financial markets anticipated an increase in the cash rate.

20. Broad money equals M3 plus borrowings from the private sector by non-bank financial institutions less the latter's holdings of currency and bank deposits. In general, broad money follows movements of M3.

21. For a brief discussion, based on empirical estimates, of why in late 1994 Australian nominal long-term interest rates were so high by international comparison, see Chapter II of last year's OECD *Economic Surveys, Australia*. The section *inter alia* concludes that investors seem to demand a risk premium in proportion to a long-term average of past inflation relative to expected inflation, which proxies monetary policy credibility by the past record of inflation control. This interpretation of the data is in line with the notion that a poor past record of controlling inflation makes gaining inflation credibility a slow process.

22. Main imperfections of this measure of expected inflation are discussed in OECD (1995), *Economic Surveys, Australia*, footnote 15.

23. Relatively long lags between changes in the cash rate and the effects on output and prices are reported in Grenville, S.A. (1995), ''The monetary policy transmission process: what do we know? (And what don't we know?)'', *Reserve Bank of Australia Bulletin*, September.

24. See Statement of the Governor (1996), *Reserve Bank of Australia Bulletin*, August, p. 7.

25. See Statement of the Governor (1996), *op. cit*.

26. For OECD countries with independent central banks, some have provision for policy override by Governments (*e.g.* New Zealand) while others do not (*e.g.* Switzerland, Germany).

27. The objectives for monetary policy set out in the Reserve Bank Act include both inflation and employment. In this context, it may be recalled that the Central Bank laws of Germany and Switzerland, countries which have a particularly favourable long-run inflation record, also oblige their Central Banks to take account of objectives other than price stability when making monetary policy decisions.

28. See Reserve Bank of Australia (1995), ''The focus on underlying inflation'', *Reserve Bank of Australia Bulletin*, July and Commonwealth Treasury of Australia (1995, Summer) Economic Round-Up: ''Treasury's underlying rate of inflation''.

29. ''Monetary policy goals for inflation in Australia'', *Research Discussion Paper No. 9503*, Reserve Bank of Australia, March. The reported mean error is based on the – unrealistic – assumption that future values of the regressors are known. Allowing for the forecast errors of the regressors would likely result in bigger confidence intervals.

30. On these issues see, for example, Edey, M. (1994), ''Costs and benefits of moving from low inflation to price stability'', *OECD Economic Studies*, No. 23, Winter.

31. See Grenville, S. (1996), ''Recent developments in monetary policy: Australia and abroad'', *The Australian Economic Review*, 1st quarter.

32. Such losses would be high, for example, if the short-run Phillips curve were very flat near zero inflation. This would make approaching zero inflation increasingly costly. However, M. Feldstein (1996) argues that the cost of reducing inflation is a ''one-time'' loss of output and employment while the benefit of a lower inflation rate is of a permanent nature. Hence the present value of this permanent gain could easily outweigh the one-time cost of disinflation. See ''The costs and benefits of going from low inflation to price stability'', *National Bureau of Economic Research*, Working Paper No. 5469, February.

33. All references to the ''budget balance'' and to ''outlays'' are in underlying terms unless otherwise stated. This means that net advances, which consist primarily of asset sales and net repayments of debt by the States, are excluded.

34. The decline in both revenues and outlays as a percentage of GDP partly reflects the decline in interest receipts from the States and payments on their behalf as States have progressively refinanced their debts to the Commonwealth since the second half of the 1980s. Insofar as this is so, the recent declines in outlays and revenues as a percentage of GDP overstate the underlying trends. Abstracting from these interest receipts and payments, the decline in outlays and revenues as a percentage of GDP in the 1990s was about ½ per cent of GDP less than shown in Table 7.

35. These are included in the category ''other tax'' shown in Figure 17. The stability in income tax receipts as a percentage of GDP reflects lower receipts from individuals and higher receipts from companies.

36. There were also measures which will add to the budget deficit. The most important of these are: the Family Tax Initiative (A$ 0.9 billion by FY 1997/98); and the Private Health Insurance Incentive (A$ 0.5 billion by FY 1997/98).

37. This paragraph draws heavily from the 1996 *National Fiscal Outlook*.

38. For which general government data are available.

39. The notion that increases in public savings may be offset by reductions in private savings is known as Ricardian equivalence. Recent studies by the IMF and OECD suggest that the private sector tends to compensate partially for changes in public sector saving. See IMF, *World Economic Outlook*, May 1996 and OECD, *Economic Outlook 59*, June 1996.

40. All government asset sales in Australia affect net debt as equity participations in businesses are not included in government financial assets. This is not in accordance with SNA accounting rules, according to which governments' equity participations, such as in the Commonwealth Bank of Australia, should be classified as financial assets. If this treatment were followed, the sale of such assets would have no effect on government net debt but would reduce gross debt to the extent that the sale proceeds were used to retire debt.

41. This section draws heavily on the description of the ''Charter of Budget Honesty'' in *Budget Statements 1996-97*, Budget Paper No. 1, Appendix C.

42. These considerations played an important part in the New Zealand government's decision to aim to maintain structural budget surpluses at around 3 per cent of GDP.

43. See Alesina, A. and Perotti, R. (1996), ''Fiscal adjustments in OECD countries: Composition and Macroeconomic effects'', *IMF Working Paper*, July.

44. The number of discouraged workers also follows a cyclical pattern indicating that unemployment may be understated in a downturn. If they were included as unemployed, this would add another 1½ percentage points to the unemployment rate. See OECD (1995), *Employment Outlook*, Chapter 2.

45. See OECD (1995), *Trends in International Migration, Annual Report 1994* for further details.

46. Over the 1970s, participation rates of men aged 60-64 had the largest declines and have since levelled off, but much of the decline over the 1980s was in the 55-59 age group. The decline in manufacturing and in full-time job opportunities for older workers were probably important reasons for reduced participation, leading many older workers to leave the labour force. See Kenyon, P. and M. Wooden (1996), ''Labour supply'', in: K. Norris and M. Wooden (eds.), *The changing Australian labour market*, EPAC Commission Paper No. 11, Canberra. But on the supply-side, increased wealth and access to state retirement pensions, and more recently access to superannuation schemes from age 55 have undoubtedly played the most important role.

47. Despite the increase in the share of part-time work, average weekly hours worked have increased, reflecting increases in the share of men working full-time at very long hours. See Australian Bureau of Statistics (1996), *The Labour Force*, February, and Norris, K. and M. Wooden (1996), ''The changing Australian labour market: an overview'', in: K. Norris and M. Wooden, *op. cit.*

48. See OECD (1995), *Employment Outlook*, Chapter 2. International comparison of the extent of involuntary part-time employment, however, is indicative only, due to differences in data definitions.

49. Casual employees are legally defined as employees who do not have an ongoing contract of employment. For statistical purposes, the ABS defines casual employees as employees not entitled to vacation or sick leave.

50. There is little evidence on how much of casual work is involuntary, *i.e.* individuals who would prefer permanent work. A relatively old survey from 1986 places the number at 42 per cent, but it is unclear whether this has grown in line with the number of involuntary part-time workers. See Borland and Norris, 1986, and J. Romeyn (1992), *Flexible Working Time: Part-time and Casual Employment*, Industrial Relations Research Monograph No. 1, Department of Industrial Relations, Canberra.

51. See Chapman, B.J. (1993), ''Long-term Unemployment in Australia: Causes, Consequences and Policy Responses, A Discussion Paper'', prepared for the Department of Employment, Education and Training, December.

52. As Borland and Norris note, this would imply more jobs going to the unemployed rather than new entrants. See Borland, J. and K. Norris (1996), ''Equity'', in: K. Norris and M. Wooden (1996), *op. cit*.

53. One concern might be that Active Labour Market Policies (ALMPs) have simply increased churning by turning long -term unemployed back into short-term unemployed following participation in a programme. There is not yet enough evidence on this issue.

54. See Fahrer, J. and Pease, A. (1993), ''The unemployment/vacancy relationship in Australia'', *Research Discussion Paper, No. 9305, Reserve Bank of Australia*, June. Their detailed analysis suggests that much of the outward shift of the unemployment-vacancy relationship in the 1980s remains invisible in a standard diagrammatic presentation of the Beveridge curve due to the offsetting effects on the curve from the declining labour force participation of men, and the very large increases in female employment (p. 26).

55. Competition of migrants – often low-skilled – with indigenous workers will tend to add to wage flexibility.

56. See Fahrer, J. and Pease, A. (1993), *op. cit*., p. 4.

57. The OECD *Jobs Study* found that, in general, globalisation and technological change were not an important contributor to the rise in structural unemployment in many OECD countries, and this finding has been confirmed in Australia's case. See EPAC (1996), *Future Labour Market Issues for Australia*, Commission Paper No. 12 which concludes ''the argument that ... lower protective barriers have increased aggregate unemployment is ... incorrect. Reduced tariffs can have a negative impact on employment, especially in protected industries. However, there are also offsetting positive effects, because reduced costs for businesses and consumers create demand and jobs elsewhere''. It also notes that ''Microeconomic reform and new technologies can cause job losses in specific industries and occupations... However, the expected higher productivity should have beneficial effects on employment and the community overall... While job displacement due to technology and reform occurs in the short run, these losses will be more than offset by longer run increases in output and employment''.

58. See OECD (1990), *Economic Surveys, Australia*, Chapter III for a discussion of industrial relations and Annex II on centralised wage fixing. Progress in the reform of industrial relations is reported in the OECD *Surveys* of Australia of 1992, 1994 and 1995.

59. On 11 November 1996, the Victorian and Commonwealth Governments announced an in-principle agreement as the basis for the referral of the State of Victoria's industrial relations powers to the Commonwealth. While the mechanics and technical details of the referral are yet to be finalised, under the proposed reforms, Victorian employers and employees will be able to enter into federal awards or federal agreements. They will be able to continue to operate under their current State agreements or informal arrangements, so long as they are not subject to a federal award or do not otherwise choose to make a federal agreement. No new formal State agreements will be possible. For those without a federal agreement or a federal award, the AIRC will be able to make safety net adjustments to Victorian minimum wages.

60. See OECD (1994), *Employment Outlook*, Chart 5.1, Tables 5.7 and 5.8 for international comparisons of union density and collective bargaining coverage rates.

61. Between 1983 and 1995, there were eight versions of the Accord.

62. The other main conditions were that an agreement: *i)* must be negotiated through a single bargaining unit comprising all relevant unions (although the AIRC was able to permit exceptions); *ii)* specifies its period of operation and includes dispute settling procedures.

63. The data discussed in this paragraph refer to formally registered agreements. There is an unknown number of informal agreements; insofar as these do not contravene award conditions of employment and are properly documented, they may be enforceable in civil law.

64. These agreements mainly applied to small workplaces. Nevertheless, at least one-third of the agreements had one or more unions bound.

65. Other factors were: that employers had to be already within the federal industrial relations system or had to move into it; and that agreements had to pass a ''no disadvantage test'' under complex outdated awards which the AIRC will generally only approve an agreement resulting in a reduction of any entitlements and protections for employees if it considers that such a reduction, viewed in the context of the overall package, is not contrary to the public interest.

66. A customary measure of aggregate short-term real wage flexibility is the semi-elasticity of real wage changes with respect to the rate of unemployment, usually estimated as the coefficient on the labour-market slack variable in an expectations-augmented Phillips curve.

67. See the literature quoted in OECD (1994), *The OECD Jobs Study, Evidence and Explanations, Part II: The adjustment potential of the labour market*, Chapter 5. However, estimates of T. Tyrväinen (1995), *Real wage resistance and unemployment: multivariate analysis of cointegrating relations in 10 OECD countries*, *OECD Jobs Study* Working Paper Series, No. 10, which differ from the Phillips-curve framework suggest a comparatively strong real wage response to labour market slack in Australia: a one percentage point rise in the unemployment rate would lower the level of the real product wage by 4 per cent in the long run. On the other hand, the Treasury Macroeconomic (TRYM) model of the Australian economy suggests a fall in quarterly wage inflation by only about 0.1 percentage point for every percentage point the unemployment rate is above the NAIRU, which suggests little aggregate wage flexibility. However, the model also includes an effect from a *change* in the unemployment rate: a one percentage point increase in the unemployment rate leads to 0.06 percentage point decrease in quarterly wage inflation, also a small effect. *Cf.* Common-

wealth Treasury (1996), *Documentation of the Treasury Macroeconomic (TRYM) Model of the Australian Economy*, Canberra.

68. For estimates of the effects of short-term and long-term unemployment on real wage growth see Table 2 of Elmeskov, J., and MacFarlan, M. (1993), ''Unemployment persistence'', *OECD Economic Studies*, No. 21/Winter. The presented tests for a unit root in the unemployment rate and for unemployment persistence lends some support to the notion of hysteresis, but the discriminatory power of the tests applied is too weak to distinguish between slow and no labour market adjustment.

69. See the literature quoted in Norris, K. and M. Wooden (1996), ''The changing Australian labour market: an overview'', in: K. Norris and M. Wooden (eds.), *op. cit.*, pp. 64-65 and 75-76. According to a hypothesis proposed by Calmfors, L. and J. Driffill (1988), ''Bargaining Structure, Corporatism, and Macroeconomic Performance'', *Economic Policy*, No. 11 Australia's shift from predominantly sectoral wage bargaining in the 1970s to more central bargaining in the 1980s and towards decentralised bargaining in the 1990s should have enhanced aggregate real wage flexibility.

70. See Chart 5.1 of OECD (1994), *The OECD Jobs Study, Part II, op. cit.*

71. See OECD (1996), *Employment Outlook*, Table 3.1, which provides data for 16 OECD countries.

72. *Ibid.*, Chapter 3, in particular Table 3.3.

73. Coelli, M., Fahrer, J. and Lindsay, H. (1994), ''Wage dispersion and labour market institutions: a cross country study'', *Research Discussion Paper, No. 9404, Reserve Bank of Australia*, June.

74. Moreover, the study did not find a strong relationship between the degree of centralisation of wage setting institutions and *sectoral* wage dispersion across countries. It, thus, concludes that ''... policies to decentralise wage setting may not necessarily experience an increase in labour market flexibility'' (*Ibid.*, p. 23).

75. *Ibid.*, p. 33.

76. See OECD (1994), *The OECD Jobs Study, Part II, op. cit.*, Tables 5.6 and 5.7.

77. See OECD (1994), *The OECD Jobs Study, Part II, op. cit.*, Chart 5.5.

78. *Ibid.*, Chapter 5, Table 5.20.

79. Also, recent OECD Secretariat estimates (ECO/CPE/WP1(95)6) of mark-ups of prices over marginal costs in 36 manufacturing sectors of 14 OECD countries identified Australia as one of the three countries with the highest observed mark-ups for total manufacturing (together with France and Finland), which could indicate a relatively low degree of competition in this industry. However, the estimation period extends from 1971 to 1990 and, therefore, only insufficiently accounts for the tariff reductions of the 1980s, which exposed the industry to greater competitive pressures from abroad. Besides, manufacturing currently accounts for less than 16 per cent of total value-added in Australia. The above quoted OECD document singled out the Australian services sector as one of the least regulated among OECD countries.

80. Chang, C.-A., and Miller, P.W. (1996), ''The inter-industry wage structure: Evidence from the 1991 Australian Census'', *Australian Bulletin of Labour*, Vol. 22, No. 1, March, suggest

that ‘‘... the considerable wage premium of 43 per cent in the mining industry may be connected to rent-sharing in this industry’’.

81. The complexity of the system is highlighted in McGaurr, D. (1994), ‘‘Australian wage setting and unemployment: the case for a labour force income policy’’, *Australian Bulletin of Labour*, Vol. 20, No. 3, p. 230-231.

82. See OECD (1994), *The OECD Jobs Study, Part II, op. cit.*, Charts 5.9 and 5.10.

83. The average, not the marginal, overall tax wedge is used as, in principle, it contributes to higher real product wages and unemployment where there is wage resistance; see Tyrväinen (1995), *op. cit*., p. 15. Indeed, the marginal overall tax wedge could reduce the incentive to push for higher real consumption wages.

84. However, Tyrväinen (1995), *op. cit*., comes to the opposite conclusion. But his methodology is controversial and he used a tax wedge (for a single -earner couple on an Average Production Worker’s (APW) earnings with two children) in his study which had increased by more than most.

85. The effective rate of payroll tax (defined as payroll tax revenue as a share of wages, salaries and supplements) in Australia was 2.9 per cent in FY 1993/94 (see Ryan, M. (1995), ‘‘What future for payroll taxes in Australia?’’, *Treasury Research Paper*, No. 10, Canberra, Table A3).

86. Presently, employers are required to make a minimum contribution to a superannuation fund on behalf of their employees of 6 per cent of employees’ gross earnings. Under the Superannuation Guarantee, this contribution rate is scheduled to rise to 9 per cent by FY 2002/03. In addition, the previous Government developed in consultation with the ACTU measures which provided for: the phased introduction through industrial agreements and awards of a requirement for employees to contribute 3 per cent of their earnings to superannuation by 2000; and the Government to make means-tested superannuation contributions matched to those made by employees (and by the self employed out of their after-tax income), in lieu of proceeding with the second tranche of personal tax cuts announced in the 1992 *One Nation* Statement (see the *Saving for our Future* Statement, 1995).

87. For unemployment benefits, these conditions generally are that claimants must be available for full-time work, actively seeking it and willing to accept any suitable job offer they receive.

88. *Jobs Study*, p. 178.

89. *Ibid*.

90. This index is the average of gross benefit replacement rates (*i.e.* before-tax benefit entitlements relative to gross earnings) at APW earnings and two thirds of APW earnings for single people, one-earner and two-earner couples and at three durations of unemployment. The duration categories are the following: first year; second and third years; and fourth and fifth years.

91. An unemployed person with a spouse in full time employment is entitled to benefit in their own right where their own earnings, and/or those of their spouse do not exceed the income test threshold. This threshold is, however, set below the level of most full time wages.

92. The means-testing of all benefits in Australia has made them more efficient in reducing poverty than transfers in countries with extensive social insurance systems: the reduction in the Sen poverty index indicator (this is the simple average of: the proportion of the population earning less than half of median earnings; the gap between the average earnings of this group and median earnings; and the Gin coefficient for this group) in Australia as a ratio of working-age related transfer expenditures was 1.5, among the highest for the 11 countries participating in the Luxemburg study and far higher than the average (0.9) for these countries. See Förster, M. (1994), ''Measurement of low incomes and poverty in a perspective of international comparisons'', OECD *DEELSA Occasional Paper*; Danish Ministry of Economic Affairs (1995), *Unemployment Benefits and Social Assistance in Seven European Countries: a Comparative Study*, Copenhagen.

93. See Commonwealth of Australia (1994), *Working Nation: Policies and Programs*, Canberra, May, p. 146.

94. Data refer to June 1992. See *Dependency-based Payments: Married Women in the Social Security System, Policy Discussion Paper No. 1*, October 1993, Department of Social Security. There is some debate whether this reflects aspects of the social security system or the fact that couples tend to share similar characteristics, both personal and regional. See Bradbury (1995), ''Added, Subtracted or Just Different: Why do the Wives of Unemployed Men have Such Low Employment Rates?'' in *Australian Bulletin of Labour*.

95. This has been a more important factor in Australia than in most other OECD countries and is set to remain so over the next few years as Australian tariffs are reduced to levels more in line with those in other industrial countries.

96. This frequently has involved rolling penalty rates of pay for overtime (*i.e.* premia for working non-standard hours) into annualised pay with greater flexibility on when these hours are actually worked.

97. See Commonwealth of Australia (1994), *Working Nation: Policies and Programmes, op. cit.*, p. 29.

98. Evidence presented by S. Scarpetta (''Assessing the Role of Labour Market Policies and Institutional Settings on Unemployment: A Cross-country Study'', *OECD Economic Studies*, No. 26, forthcoming) shows a positive link between the strictness of EPL and both long-term unemployment and youth unemployment rates. Evidence of a positive link between long-term unemployment and the strictness of EPL was also found in the 1993 OECD *Employment Outlook*, and the reasons for this are discussed in further detail in the OECD *Jobs Study, op. cit.* This link is consistent with insider/outsider arguments.

99. The purpose of some provisions of the Act was to meet Australia's obligations under the International Labour Office's termination of employment convention and termination of employment recommendation. But it also extended the sphere of influence of federal over state awards, particularly where State legislation was less generous or did not follow the same procedures (outlined below), since the Act would apply to virtually all employees in Australia.

100. See ''Legislative Changes in 1993'', by Robert Reitano in the *Journal of Industrial Relations*, March 1994.

101. See Australian Chamber of Commerce and Industry, February 1996 for a survey of the areas. Unfair dismissal ranked 8th on the list out of 30 areas.

102. See ''Major Tribunal Decisions in 1994'', by Phillippa Weeks in the *Journal of Industrial Relations*, March 1995.

103. One alternative would be to look at groups where unfair dismissal does not apply and see if there is any growth in their numbers. In the Australian context, this would be changes in the share of casual workers, who until 1994 were exempt from unfair dismissal laws. But their numbers have been increasing over time, and this began well before changes to unfair dismissal laws. Furthermore, with the Reform Act making it possible for workers of job tenure greater than six months to receive due process for unfair dismissal, it would be more interesting to examine the length of the contract to see if there is any shift to contracts of less than six months duration. It is unclear what impact this will have on casual worker numbers in the future. It is unlikely that growth in part-time work is related to EPL: to the extent that it is not casual, it would be covered under unfair dismissal procedures and to the extent that it is voluntary, it probably reflects mutually beneficial arrangements between firms and workers.

104. Nevertheless, arbitration is likely to be more attractive to the parties in most cases than a court case on account of the AIRC's less formal proceedings.

105. See ''Legislative Changes in 1995'', Robert Reitano, *Journal of Industrial Relations*, 1996.

106. See the 1996 *Employment Outlook*, Table T.

107. A *Job Compact* ''job'' offer is defined as either placement into an unsubsidised job, or placement into an employment programme, which includes wage subsidies such as *JobStart*, or broker programmes such as *New Work Opportunities*. In addition, clients are provided with intensive case management, training and support to ensure that they are job ready. The *Job Compact* was described in detail in last year's Survey.

108. See Table 4.4, *Working Nation: Evaluation of the employment, education and training elements*, EMB Report 2/96.

109. For both JobStart and the National Training Wage (NTW, see Annex I), employers who take on individuals unemployed between 26-51 weeks receive $150 for 13 weeks of employment; at high risk or unemployed 52-77 weeks, $150 for 20 weeks; 78-155 weeks, $200 for 26 weeks; 156-259 weeks, $230 for 26 weeks and for 260 weeks plus, $325 for 26 weeks of employment. In addition, employers are eligible for further payments under NTW if they provide certified training.

110. When outcomes are confined to only the long-term unemployed, results are less positive: 49 per cent for *JobStart*, 31 per cent for *JobTrain* and 34 per cent for *Job Clubs*. And when confined to only the *Job Compact* group, they are even less favourable, *e.g.* 41 per cent for *JobStart*. See *Working Nation: Evaluation of the employment, education and training elements, op. cit.*

111. See *Working Nation: Evaluation of the employment, education and training elements, op. cit.*

112. Although analysis using matched datasets of programme participants and similar non-participants also indicate programme impacts are positive, they still do not indicate whether

the individual was better off participating in the programme than he/she would otherwise have been because selection bias is not accounted for. However, for programmes targeted specifically at disadvantaged jobseekers rather than the general population of jobseekers, selection bias may result in an underestimate of the true impact of the programme, especially in the case of the Job Compact where all those in the eligible group were to receive assistance.

113. See the 1993 *Employment Outlook*, Chapter 3 and Fay (1996), *Enhancing the Effectiveness of Active Labour Market Policies: Evidence from Programme Evaluations*, DEELSA Occasional Paper No. 8.

114. See OECD (1995), *Economic Surveys, Australia*, and OECD (1996), *Enhancing the Effectiveness of Active Labour Market Policies*.

115. Some of these are identified in *Working Nation: Evaluation of the employment, education and training elements, op. cit.*

116. There have been recommendations to simplify programmes. See the CES Advisory Committee Review of *Review of Labour Market Programmes*, August 1995.

117. For a more in-depth discussion, see Chapter IV.

118. Hilmer, F.G., M. Rayner and G. Taperell (1993), ''The Independent Committee of Inquiry'', *National Competition Policy*, AGPS, Canberra.

119. Commonwealth of Australia (1995), *Research and Development, Industry Commission*, Report No. 44, Canberra.

120. See OECD (1996), *Science, Technology and Industry Outlook: 1996*.

121. These are the Australian Nuclear Science and Technology Organisation (ANSTO), the Australian Institute of Marine Science (AIMS) and the Australian Geological Survey Organisation (AGSO).

122. These include: classifications; rates of pay for different categories of employees; piece rates, tallies and bonuses (where applicable); ordinary time hours of work (and the spread of such hours); additional loadings for working overtime, casual or shift work; penalty rates; annual leave and leave loading; long-service leave; personal/carer's leave, including sick leave, family leave, bereavement and compassionate leave; public holidays; parental leave; allowances; jury service; dispute settling procedures; basis of employment, including regular full-time and part-time employment (with *pro rata* conditions), casual employment and shift work; notice of termination and redundancy pay; stand down provisions; and matters incidental to the application of these matters and essential for the effective operation of the award. Allowable award matters are to be simplified and their detailed application at workplace will be a matter suitable for agreements.

123. Matters which would generally be for determination at the enterprise or workplace level include: authorised stop work meetings; clothing; occupational health and safety (in addition to any statutory provisions); introduction of change; union picnic day; training; study leave; and trade union training leave.

124. In exceptional circumstances, the Australian Industrial Relations Commission will be able to arbitrate and make orders on matters in dispute that do not fall into the scope of the allowable matters. The circumstances in which such orders may be made are limited and any

order will cease to be in force after two years. Such orders will not form part of the benchmark for the no disadvantage test for agreements.

125. Transitional arrangements will be developed to enable parties to existing EFAs to translate to AWAs or CAs.

126. This statutory officer will be established through a separate part of the Act. The EA's functions include: providing advice to employees and employers, especially small businesses, on the new Act, in particular on AWAs and minimum conditions; receiving and filing AWAs; handling alleged breaches of AWAs and of the freedom of association provisions (see below); assisting employees in prosecuting breaches where required; and providing aggregated statistics on AWAs.

127. The Industrial Relations Reform Act 1993, which came into effect on 30 March 1994, transferred the secondary boycott provisions to the Industrial Relations Act. This made the provisions ineffective as the Industrial Relations Act provided for a 72 hour period of conciliation by the Industrial Relations Commission, which allowed unions to instigate rolling stoppages.

128. These cover Federal government employees and Territory (government) employees. All employees not covered by a federal award or agreement will have to access state unfair dismissal laws.

129. The ''fair go all round'' principle operates in the State tribunals, which also have the power to determine unfair dismissal cases under State awards.

130. A filing fee of A$ 50 will be imposed to discourage frivolous claims.

131. This may be reinstatement (and lost pay) or payment in lieu.

132. This is intended to help small business.

133. For example, a one-earner couple with two children now faces METRs of 80 per cent for the income range A$ 70-A$ 225 per week and approximately 90 per cent for the income range A$ 230-A$ 380 per week compared with 100-110 per cent before.

134. The definitions of ''significant businesses'' and various other details are spelt out in the *Commonwealth Competitive Neutrality Policy Statement* (1996), Treasurer Press Release No. 39 of 27 June.

135. Consumers would gain A$ 9 billion per year, hence A$ 1 500 per household and per year. Real after-tax wages would be 3 per cent higher than otherwise and 30 000 extra jobs would be created. See Commonwealth of Australia (1995), *The growth and revenue implications of Hilmer and related reforms: A report by the Industry Commission to the Council of Australian Governments*, Canberra, March.

136. This does not mean that economic growth would be 5.5 per cent higher, but that as a result of these reforms in the long run, the level of real GDP would be A$ 23 billion higher than otherwise.

137. While there is some room for collusion between the EPE and employer to split fees, this should be minimised for two reasons: first, the main client group will be the long-term unemployed which employers are usually reluctant to take on at any cost, as evidenced by the fall in take-up of *JobStart* subsidies; and second, under proposed arrangements, to get full fees, EPEs will have to generate jobs of greater than six months. Since employers

typically want someone who can do the job; it is unlikely they would accept someone for this length of time simply to get a fraction of the fee.

138. These are the metal, automotive, building and electrical trades.

139. For example, in Victoria in the early 1980s a considerable number of schools provided an alternative final year school certificate which was recognised for entry by some colleges of advanced education and technical colleges. Assessment, which was school based, involved much less reliance on formal examinations and greater consideration of tasks completed. Later reforms to the State-wide assessment system adopted some of these procedures but also eliminated the alternative structures.

140. Selected VET Statistics show that about 3 per cent of 15 year-olds and about 9 per cent of 16 year-olds are enrolled in VET. Some of these are full-time school students also enrolled in TAFE. Trainee and apprenticeship commencements are less than 1 per cent of the population aged 15 and are about 2.5 per cent of the population aged 16.

141. The lack of Australian data in Table 20 for graduations from vocational programmes makes comparison a little difficult. Nevertheless, given that only a minority of students who leave school before Year 12 participate in VET leading to a qualification equivalent to completing secondary school, allowing for them would probably still leave the overall graduation rate in Australia slightly below the average for OECD countries.

142. See Gospel (1993).

143. Only one-third of apprentices in 1995 left school at 16 or younger, compared with two-thirds a decade earlier. By 1995, 60 per cent of persons under contract of training (mostly apprentices) had completed year 12. See Australian Committee on Vocational Education and Training Statistics (ACVETS) (1996).

144. Similarly, the increase in the number of new graduates, from 88 000 in 1987 to nearly 140 000 in 1994 (DEET 1996), has been much greater than forecast: it was supposed to rise to 125 000 by the end of the century (Dawkins 1988).

145. See Teicher and Grauze (1996).

146. Reviews of schooling were made in most States (*e.g.* Blackburn (1984) in Victoria).

147. Kemp (1996*a*).

148. *Ibid.*

149. The major focus of reform of VET came in the late 1980s and was outlined in a series of reports (Deveson, 1990; Finn, 1991; Carmichael, 1992; Mayer, 1992). Targets for the expansion of enrolment by young people were set in the Finn report and endorsed by Commonwealth and State Ministers (see above). The new Government intends to build on these initiatives with reforms to the apprenticeship and traineeship system.

150. See Sweet (1995).

151. See Sweet (1995) and Ministerial Council on Education, Employment, Training and Youth Affairs (MCEETYA) (1996) for more details. State school and VET authorities have been active in the last few years in developing arrangements for work-based and other VET courses taken by school students to count towards a VET Certificate as well as for the school certificate used for entry to university.

152. These work placements are generally unpaid, although legislation in some States specifies a minimal rate of payment (*e.g.* A$ 5 a day in Victoria, though some employers pay more) and coverage under workers' insurance.

153. This funding is to support progress towards the Finn targets and other reforms to VET.

154. The Government has announced that it will introduce a Youth Allowance from January 1998 which will bring into line assistance to students, job search allowances and guaranteed minimum wages for trainees.

155. A meister has an advanced level of skill in his occupation, is trained to train apprentices and is responsible for doing so.

156. See Karmel (1985), Finn (1991) and Mayer (1992).

157. Assessment is a State responsibility.

158. Examples of results in recent years include: relatively stable results in NSW in year 3 and 5 literacy and numeracy performance and improvement for those of non-English speaking background; improvement in years 5 and 9 mathematics in Queensland with a slight fall at year 7; slight improvement in year 3 and year 10 mathematics and improvement at year 7 in Western Australia (Steering Committee 1995).

159. However, the results of a new international study of achievement will be released later this year. They will be summarised in OECD (1996), *Education at a Glance*.

160. See Maglen *et al*. (1994) for a summary of the results for studies in the 1960s, 1970s and early 1980s.

161. The criticisms include the tendency to narrow the learning experience to a predetermined set of competences, to underrate the importance of less measurable affective learning for young people and to emphasise separate components of skill to the neglect of overall competency.

162. The scheme is likely to be rolled into a Youth Allowance scheme in 1998 along with other allowances for youth, including the unemployment benefit.

163. Based on post-code data, students from the three lowest socio-economic deciles had an average school retention rate of 57 per cent for males and 70 per cent for females, compared with 75 per cent for males and 83 per cent for females from the three highest deciles. See MCEETYA (1996).

164. There is, however, no evidence that working-class students of high academic ability fare less well.

165. See MCEETYA (1996).

166. See Long *et al*. (1995).

167. *Ibid.*

168. In 1994 the average government grant per student in non-government schools (including Catholic schools) was about A$ 2 800 per student compared with about A$ 5 400 spent by governments per student in government schools. On average, Catholic schools received A$ 3 000 per student compared with A$ 2 200 for ''other'' non-government schools.

169. In 1995, they were 95 per cent, compared with 75 per cent in Catholic schools and 67 per cent for government schools. Transfers to non-government schools for the final years of secondary school inflate this difference.

170. The base level of the annual HECS charge is to be A$ 3 300 and will apply broadly to Arts, Humanities, Social Studies, Visual/Performing Arts, Education and Nursing. Most other disciplines will attract an annual charge of A$ 4 700 (Mathematics, Computing, other Health Sciences, Agriculture, Architecture, Sciences, Engineering Administration, Business and Economics). The higher cost and higher earning professional courses (Law, Medicine, Dentistry and Veterinary Science) will be charged A$ 5 500.

171. In the United States, however, there is considerable variation in fees, with very high fees in the numerous private universities and relatively low fees in public universities in some States for residents.

172. The net present value of HECS as a proportion of the operating grant per student is estimated to increase from 15 per cent in 1996 to the following rates in 1997: 19 per cent for students in the lowest HECS category, based on average earnings for Arts graduates; 31 per cent for students in the middle category, based on average earnings for Engineering graduates; and 35 per cent for students in the highest category, based on average earnings for Medicine graduates. The operating grant used to make these calculations is the payment by the Government to universities per equivalent full-time student unit. This figure does not reflect the actual size of the government subsidy received by students undertaking different courses.

173. Total student fees and charges in 1995 raised about A$ 146 million, about 4 per cent of total revenue of public VET institutions or 6 per cent of government recurrent grants to VET. See Australian Committee of Vocational Education and Training Statistics (ACVETS) (1996).

174. Students aged less than 18 dependent on their families are also exempt.

175. It has been pointed out that the combination of education and training with employment and the increased presence of economists in DEET compared with the former Department of Education increased the attention given to economic analysis in policy development.

176. See Gregory (1995).

177. Labour quality is measured by weighting different education categories of workers by their relative full-year earnings.

178. See Gregory (1995).

179. This and the next two sentences draw heavily from Gregory (1995), pp. 307-308.

180. Other factors include the significant social and cultural changes which have occurred (the current generation of young people has parents who themselves were beneficiaries of earlier rises in educational participation; smaller family sizes; educational zeal of immigrants) and reforms to education to make it more relevant to a wider span of young people.

181. See Gregory (1995). High skilled occupations are those which had earnings in the top quintile; middle-skilled occupations had earnings in the next two quintiles; and low-skilled occupations had earnings in the bottom quintile.

182. See Karmel (1995).

183. See Maglen (1995).

184. See Dockery and Norris (1996).

185. See Sweet (1990), and Dockery and Norris (1996).

186. This could, however, reflect the choice of employees to receive training. Analysis over time is necessary. See McKenzie and Long (1995).

187. See Psacharopoulas (1995).

188. The average net present value of HECS as a proportion of the average cost of tuition for an undergraduate student was 16-18 per cent, depending on the discount rate used (see above).

189. This proposal was made by Karmel (1996).

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Annex I

Active labour market policies (ALMPs) and case management

This annex provides additional information on some of the main Australian ALMPs, and further details on the case management process which underpins the Job Compact.

Labour market programmes

In the area of wage subsidies, there are two main programmes: JobStart and the National Training Wage.

- JobStart is the main wage subsidy programme, offering subsidies to employers that vary with the duration of unemployment of the jobseeker, lasting for up to 6 months. To make the long-term unemployed more attractive to employers, subsidy rates were changed under *Working Nation* so that employers who take on those who have been unemployed the longest would receive the highest subsidy rates.
- The National Training Wage (NTW) was introduced in 1994 and gives subsidies to employers who take on individuals and provide recognised training. Participants are paid a National Training Award. Employers can also receive additional payments upon commencement and completion of the training.
- Subsidy rates were eventually harmonised for both programmes to the following: employers who take on individuals unemployed between 26 and 51 weeks receive A$ 150 for 13 weeks of employment; for those assessed as high-risk or unemployed for 52 to 77 weeks, employers receive A$ 150 for 20 weeks; for 78 to 155 weeks, $200 for 26 weeks; for 156 to 259 weeks, A$ 230 for 26 weeks and for 260 weeks plus unemployed, A$ 325 for 26 weeks of employment.

In addition to the NTW, Australia also offers training through the JobTrain programme.

- JobTrain provides formal training to participants through both accredited and non-accredited training courses, lasting about 8 to 10 weeks. Training can either be full- or part-time.

Australia also runs a number of so-called ‘‘broker’’ programmes. Broker programmes are those which are contracted out to various organisations – the brokers – to

provide work experience and training to participants. In effect, they are mainly job creation projects with some training provided. They include:

- Landcare and Environment Action Programme (LEAP), which provides 26 weeks of formal training and practical experience to youths in environmental based projects.
- JobSkills, which provides a combination of work experience and training over a 26 week period to individuals over 21 years of age.
- New Work Opportunities (NWO) programme, which provides jobs and work experience to individuals mainly in rural or remote areas with limited work opportunities.
- The New Enterprise Incentive Scheme (NEIS). It is slightly different from the other broker programmes in that it helps unemployed people to become self-employed. NEIS participants are given training in small business management, mentor support during the first year of operation and income support equivalent to the Job Search Allowance for up to one year.

Case management

An integral part of the *Working Nation* programme is intensive case management provided to the long-term unemployed, and those at risk of becoming so. There are four key features to case management: provision of case management services through both the public and the private and community sectors; identification of those at risk of long-term unemployment; assignment of a client classification level followed by referral to a case manager; and the signing of a case management activity agreement. Intensive case management is a key element of the Job Compact which is available to jobseekers in receipt of unemployment allowance for 18 months or more.

a) When unemployed jobseekers sign on for benefits, the Commonwealth Employment Service determines whether they are at-risk of long-term unemployment. ''At-risk'' is determined through the use of the Jobseeker Screening Instrument (JSI), which screens jobseekers based on individual characteristics that are correlated with long-term unemployment. These characteristics include age, educational attainment, English speaking ability, disability and geographical location. Commonwealth Employment Service (CES) staff are allowed to supplement the JSI findings with their own judgement concerning other employment barriers which a jobseeker might display, including low self-esteem or poor motivation. If considered as being ''at-risk'', clients are placed in case management and become eligible for programmes offered by the Department of Employment, Education, Training and Youth Affairs (DEETYA), whereas other individuals must usually wait until 12 months of unemployment have passed before being placed in case management.

b) The unique aspect of case management is that it allows jobseekers – to a certain extent – to choose between publicly provided case management (through Employment Assistance Australia, or EAA, the Government's case management provider) and that provided by the private and community sectors (called contracted case managers, or CCMs), regulated by a new agency, the Employment Services Regulatory Authority (ESRA) which accredits, monitors and pays the CCMs. Through this process, contestability in the placement of the long-term unemployed has been introduced into the CES, the goal of which is to lead to new ways of helping to place them in jobs in a cost-effective manner.

c) Prior to being referred to a case manager, all jobseekers are assigned a client classification level, which reflects the relative level of disadvantage of the individual. CCMs are paid a set fee that varies with the degree of disadvantage (the EAA is not paid on the basis of outcomes). The fee, however, does not reflect differences in local labour market conditions. Currently four different fees are paid reflecting the different classification levels, and they comprise an upfront payment when the jobseeker and case manager agree on a plan of action (the Case Management Activity Agreement) and a final payment once a successful outcome is reached. Successful outcomes include enrolment in full-time training, and placement in a subsidised (such as a broker programme) or unsubsidised job for a period of 13 weeks.

d) Once in case management, jobseekers and case managers sign a Case Management Activity Agreement (CMAA), which outlines the agreed-upon steps that each will undertake to help re-integrate the jobseeker back into work. An essential element of the Job Compact is the reciprocal obligation on both the Government and eligible jobseekers. The Government must offer an employment place to the jobseeker; the jobseeker must, in turn, accept a reasonable offer of a job or lose entitlement to income support for a period. This conditional guarantee is reflected in the Case Management Activity Agreement for Job Compact participants and distinguishes the Job Compact from other employment guarantees that have existed in OECD countries, such as Finland.* Breaches in the CMAA can lead to recommendations by case managers to the CES that benefit payments be halted for a period of time. Evidence to date suggests that case managers have predominantly used the threat of a breach to ensure compliance with the CMAA.

* In theory, no country has allowed unconditional guarantees to jobs because jobseekers traditionally have had to fulfil job search requirements to get benefits, and thus a job guarantee. But in practice, when unemployment is high, enforcement of job search usually falls. The CMAA is a monitored process intended to avoid this effect.

Annex II

Institutional arrangements in the education and training sector*

General context

Australia has a federal political structure comprising six state and two territory governments as well as a federal (Commonwealth) government. The Australian Constitution does not grant the Commonwealth responsibility for education. The Commonwealth, however, has played an increasingly important role in funding education, particularly at university level, and also in co-ordination. The states and territories administer their own school and Technical and Further Education (TAFE) systems. While there is no national education system as such, there are few significant differences among the state systems and, over the last decade in particular, state Education Ministers in association with the Commonwealth have worked to minimise problems associated with differences in year levels and nomenclature.

Educational structure and enrolments

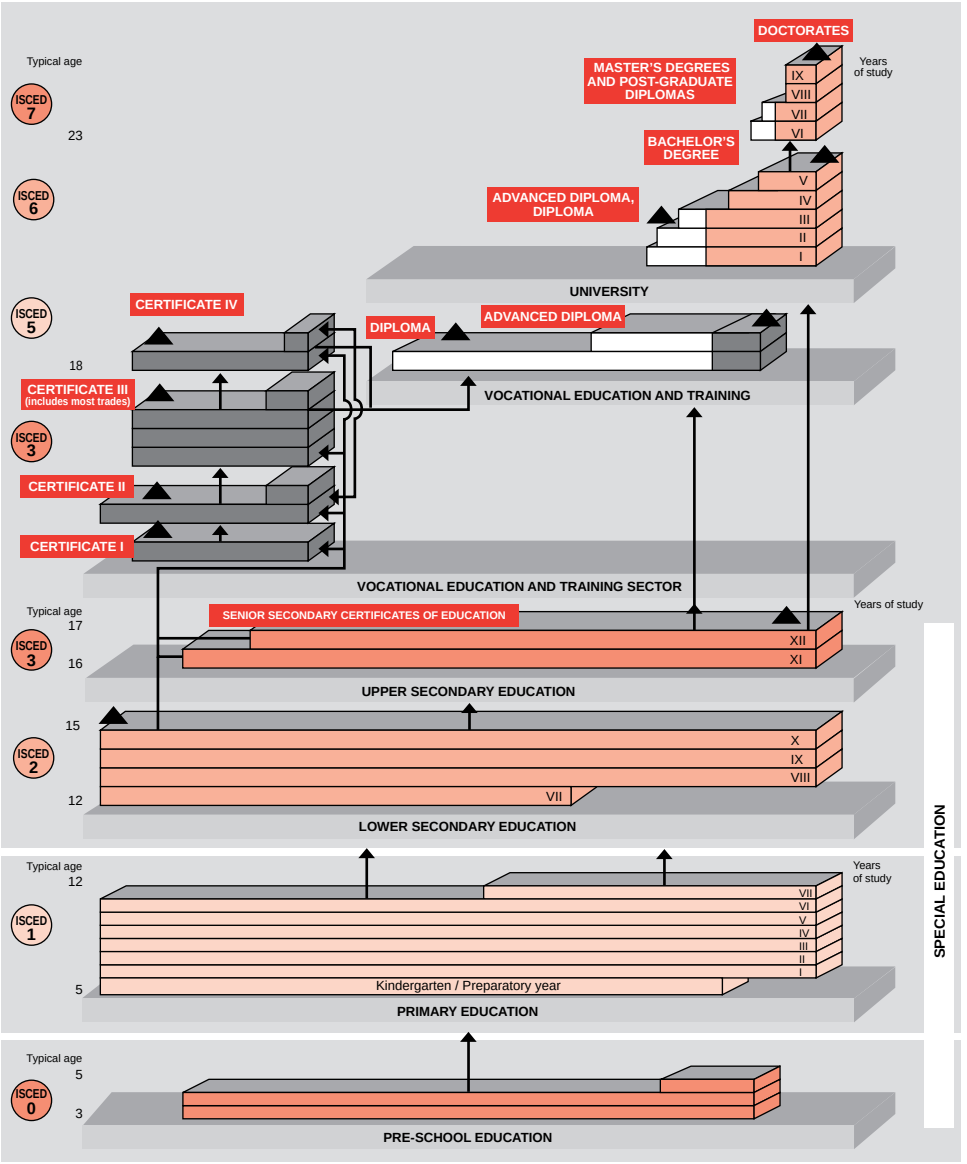
Early childhood education is available, normally on a part-time basis, before commencing primary school. Most children start primary school at the age of 5, and primary education covers 7 or 8 years depending on the state concerned. Secondary education commences at around age 12 and continues for 6 or 5 years (depending on the length of primary education). There are around 1.8 million primary and 1.3 million secondary students.

Education is compulsory between the ages of 6 and 15 (16 in Tasmania) and about three-quarters of students now complete Year 12, the final year of secondary education, twice the proportion of 1980. The growing diversity of the senior secondary school population has stimulated major reviews of secondary curriculum, pedagogy and assessment.

The tertiary sector comprises colleges of Technical and Further Education (TAFE), private vocational education providers, and universities. Vocational education and training is mainly delivered through the TAFE college system, although private providers,

* This annex consists of extracts from the description of Australia's education and training system in OECD (1996), *Education at a Glance*.

Figure A1. **AUSTRALIAN EDUCATION SYSTEM**



Note: Details of the ISCED (International Standard Classification of Education) are given in *Education at a Glance.* Dark grey indicates programmes specifically designed for part-time attendance, white indicates vocational education and training programmes, the triangles indicate recognised exit points of the education system and arrows indicate common students flows. The size of the graphical elements provides no indication of the size of the enrolment in the corresponding educational institutions.

Source: OECD, *Education at a Glance.*

senior secondary schools and universities also play a role. About 300 TAFE colleges provide a wide variety of courses, including pre-employment programmes, apprenticeships, off-the-job training, retraining programmes, and adult education. TAFE enrols more than a million students, although most of these are part-time and in relatively short courses. Vocational education and training in Australia is being reorganised to extend entry-level training to a wider range of enterprises. Australian governments are committed to quality in vocational education and training and this has found expression in client surveys, pilot projects on quality assurance and a best practice demonstration programme.

The higher education sector went through substantial change during the late 1980s. The former ''binary system'' consisting of universities and colleges of advanced education was replaced by a Unified National System (UNS) consisting of only universities. Under the binary system, universities were seen as centres of scholarship and research, while colleges were seen as possessing more vocational objectives. Associated with the establishment of the UNS was the amalgamation of many former colleges of advanced education with universities to achieve sizes where economies of scale could be realised both in terms of cost-efficiency and educational effectiveness. This resulted in a decrease in the number of higher education institutions which are now part of the UNS, from 78 in 1982 to 36 in 1995, and an increase in the average size of each institution from 4 000 in 1982 to 17 000 in 1995. The university sector as a whole provides a full range of undergraduate and post-graduate programmes, although not all institutions offer programmes in every discipline area.

Changes to the structure of the higher education sector occurred in a climate of rapid growth in student enrolments. Over the decade to 1995, the number of students enrolled in higher education increased by 63 per cent to 604 000. Within this environment of strong growth, there was a significant shift to post-graduate studies with the number of post-graduate degree students growing by 178 per cent to 74 000 in 1995. The number of female students grew more quickly than males, with women currently accounting for 54 per cent of all students compared to 49 per cent in 1985. A feature of student attendance at higher education institutions in Australia is the high proportion of students who attend on other than a full-time basis. In 1995, some 41 per cent of students attended higher education either part-time or externally.

About 50 per cent of Year 12 graduates enter university within a year or two of completing secondary school. Entry to higher education is normally based on academic results in either school or external examinations and is restricted by the number of places funded by the government. About one in seven commencing undergraduate students is now admitted on the basis of special entry provisions, rather than on the basis of school performance.

Education assistance and income support is provided through Commonwealth programmes aiming to promote equality in educational opportunities. Particular areas of focus are persons on low incomes, Aboriginal and Torres Strait Islander people and those living in geographically isolated areas. Expenditure on AUSTUDY, the major student

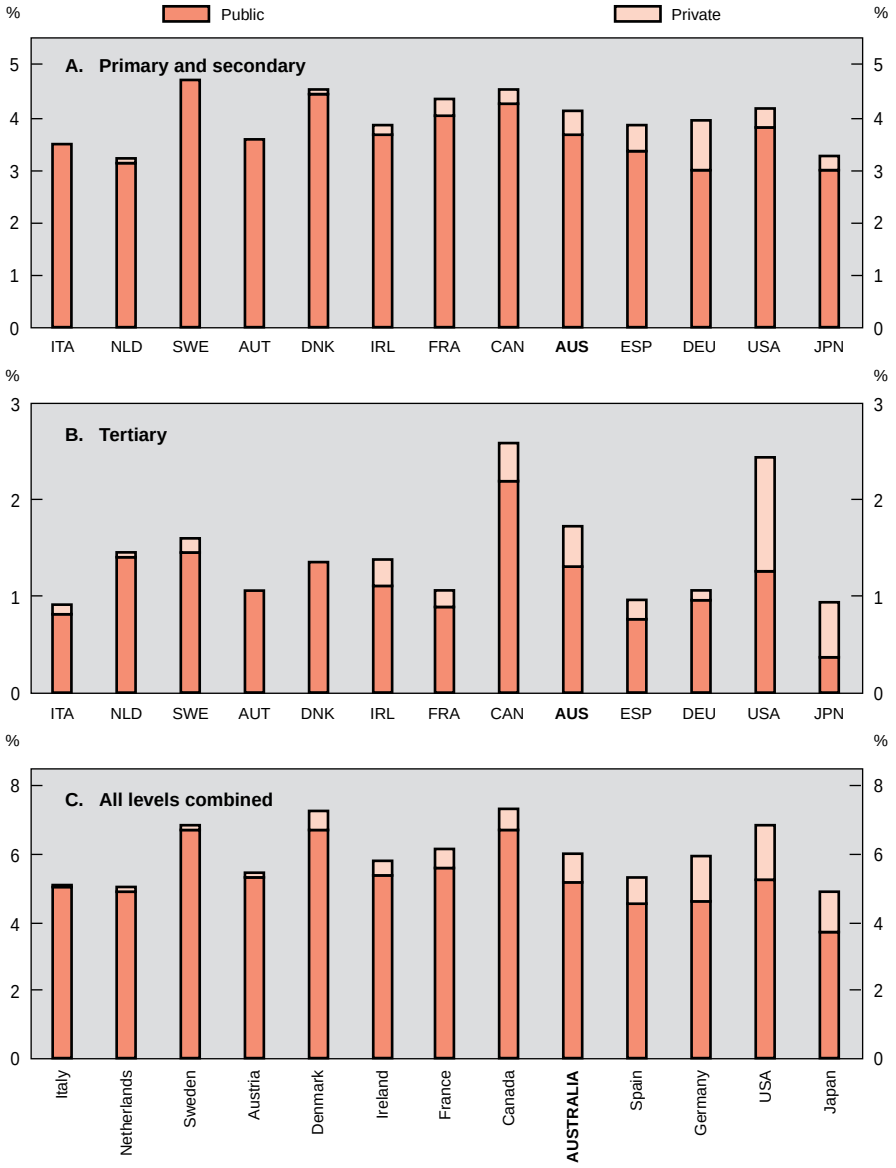
income support scheme, amounted to A$ 1 573 million in FY 1995/96 or an average A$ 3 266 per recipient. Expenditure on ABSTUDY, providing income support for Aboriginal and Torres Strait Islander students, amounted to a further A$ 128 million in FY 1995/96. In addition, expenditure under the Assistance for Isolated Children scheme was A$ 28 million in FY 1995/96. Most allowances under the schemes of assistance are subject to a means-test, which includes an income and an assets test.

Annex III

Government funding and provision of formal education

Government has the dominant role in education and training in Australia, as in most other OECD countries. It finances about 90 per cent of expenditure on formal education, with this share being somewhat higher in school and non-university tertiary education and somewhat lower in university education (Figure A2). Public institutions, including the universities established under State legislation, provide most formal education (Figure A3). Compared with other OECD countries, government's share of total education expenditures is slightly below average, with the share in tertiary expenditure being a little higher and the share in school education lower than the average; however, differences in national accounting and statistical collection systems affect the data so that small differences in international comparisons are difficult to interpret. Private provision of formal education services in Australia is relatively high for secondary schooling, catering for about 30 per cent of students, but low in tertiary education.

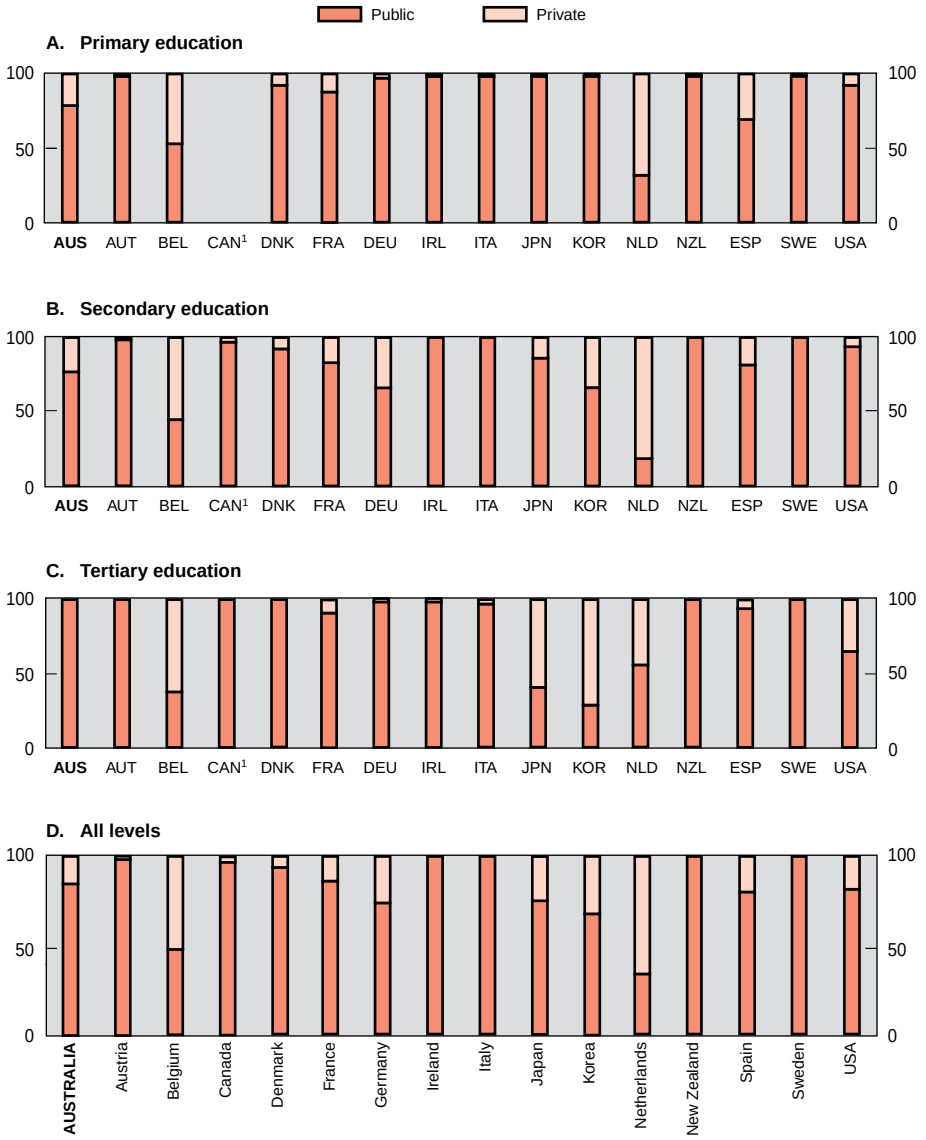
Figure A2. **PUBLIC AND PRIVATE EXPENDITURE LEVELS BY LEVEL OF EDUCATION, 1993**
As per cent of GDP



Source: OECD, *Education at a Glance.*

Figure A3. **EXPENDITURE SHARES OF PUBLIC AND PRIVATE EDUCATIONAL INSTITUTIONS**

1993



1. Included in secondary education.

Source: OECD, *Education at a Glance.*

Annex IV

Calendar of main economic events

1995

April

At the 1995 Premiers' Conference and Council of Australian Governments meeting it was agreed that financial assistance grants to the States and Territories would continue to be increased for inflation and population growth over the period FY 1995/96 to FY 1997/98.

The Council of Australian Governments (COAG) also agreed to a national competition policy package based on the Hilmer Report (the independent inquiry into national competition policy headed by Professor Fred Hilmer, released on 25 August 1993) which provides for uniform protection of consumer and business rights and increased competition in all jurisdictions.

The package consists of Commonwealth, State and Territory legislation and three inter-governmental agreements, viz. the Competition Principles Agreement; the Conduct Code Agreement; and the Agreement to Implement the National Competition Policy and Related Reforms.

The reforms:

- update the competitive conduct rules in trade practices legislation;
- extend those rules to all State and local government business enterprises and unincorporated businesses;
- provide a legislated process for third party access to the services of significant infrastructure facilities; and
- encourage competition across a broad range of business activities through a programme of regulation review, prices oversight, application of competitive neutrality principles and procedures for structural reform of monopolies.

The Council signed inter-governmental Agreements to implement the package. The Commonwealth would make Competition Payments to the States and Territories, commencing with A$ 200 million (in 1994-95 prices) in FY 1997-98. The real per capita guarantee for financial assistance grants and the making of Competition Payments would be conditional on the States and Territories complying with their obligations under the Agreement to Implement the National Competition Policy and Related Reforms.

The Loan Council agreed on Loan Council Allocations (no policy change estimates of headline public sector deficits/surpluses plus certain other items) nominated by the Commonwealth and each State and Territory for FY 1995/96. It considered that the aggregate of nominated Loan Council Allocations was consistent with macroeconomic policy objectives.

May

The Treasurer delivered the FY 1995/96 Budget. It foreshadowed a headline surplus of A$ 0.7 billion or 0.1 per cent of GDP for FY 1995/96, a turnaround in the budget balance from a deficit of 2.7 per cent of GDP in FY 1994/95. The FY 1995/96 Budget was consistent with the Government's earlier announcement, in January 1995, of its intention to tighten policy settings significantly in order to accelerate the pace of fiscal consolidation.

June

Accord VIII was agreed by the Australian Council of Trade Unions (ACTU) and the Government to take over from Accord Mark VII, which was due to expire in March 1996. Accord VIII included a commitment to achieving wage outcomes which are consistent with inflation of 2-3 per cent over the course of the economic cycle and foreshadowed further annual safety net wage adjustments over the life of the agreement (until March 1999).

The Second Corporate Law Simplification Bill was released for public exposure. The Bill will make setting up and running a company simpler. A Plan of Action was also released for the Third Corporate Law Simplification Bill, which simplifies provisions dealing with company officers, related party transactions, fund-raising and take-overs.

The Melbourne/Adelaide Standard Gauge Railway Line – the final link in the A$ 429 million project to standardise the national rail network was opened.

Plans were announced to bring the national rail track under single management control. An access price system for the Commonwealth's rail track was announced, allowing competition between the National Rail Corporation and private operators.

The Committee on Darwin Report was released.

July

The *Competition Policy Reform Act 1995*, which is the centrepiece of the national competition policy reforms, was passed by the Commonwealth Parliament on 30 June 1995, and received Royal Assent on 20 July 1995. This Act amends the *Trade Practices Act 1974* and the *Prices Surveillance Act 1983* to implement legislative aspects of the reforms.

August

The Treasurer and Minister for Finance announced the final budget outcome for FY 1994/95. A budget deficit of A$ 11.6 billion was recorded for FY 1994/95,

A$ 520 million lower than the revised deficit estimate of A$ 12.1 billion included in the FY 1995/96 Budget.

On 17 August, the amended competitive conduct rules of the Trade Practices Act came into force.

October

The Australian Industrial Relations Commission (AIRC) handed down a decision confirming the availability of a third A$ 8 safety net adjustment available at enterprise level from 22 September 1995 and at award level from 22 March 1996.

November

On 6 November, the new competition institutions – the Australian Competition and Consumer Commission (ACCC) and the National Competition Council (NCC) – commenced operations, the national access regime commenced, and amendments to the Prices Surveillance Act took effect.

The ACCC performs the functions of the former Trade Practices Commission and Prices Surveillance Authority, plus new functions relating to access to the services of infrastructure facilities.

The NCC has an advisory role in the legislated access and price oversight regimes and may assist governments with legislation review, competitive neutrality and structural reform issues in accordance with an agreed work programme. The NCC is also responsible for advising the Federal Government in respect of State and Territory compliance with the conditions necessary to receive ''competition policy payments''. These conditions are set out in the Agreement to Implement the National Competition Policy and Related Reforms.

December

The Treasurer released the FY 1995/96 mid-year review of the budget and revised economic outlook. The mid year review of the budget indicated a surplus in FY 1995/96 of A$ 115 million compared with a projected surplus of A$ 718 million at budget time. The revised budget surplus estimates reflected lower revenue estimates which more than offset an estimated decline in outlays.

The First Corporate Law Simplification Act 1995 came into force on 9 December 1995. The Act increases economic efficiency by simplifying the law relating to the incorporation, financial reporting and accounting procedures of companies. The Act also simplifies procedures relating to dealings in shares of a company.

The Collective Investment Bill was exposed for public comment. The Bill seeks to simplify the operation and structure of collective investment schemes to reduce operational costs and improve management responsibility.

The Prime Minister delivered a statement entitled *Innovate Australia* announcing a range of measures designed to underpin productivity growth and the creation of new industries and jobs.

1996

March

In a general election on 2 March 1996, the former Labour Commonwealth Government was replaced by a new Coalition Government.

The Treasurer of the newly elected Coalition Government released revised budget estimates for FY 1995/96 and projections for the forward three fiscal years to FY 1998/99 on a no policy change basis. The revised estimates showed that the budget for FY 1995/96 was expected to record a headline deficit of A$ 280 million compared with the surplus of A$ 720 million announced at the time of the FY 1995/96 Budget in May 1995, and the surplus of A$ 115 million expected at the time of the December Mid Year Review. The Treasurer also announced for the first time estimates of the underlying budget balance. These showed that the estimated underlying budget deficit for FY 1995/96 was estimated at A$ 9 billion, while the underlying deficit for FY 1996/97 was projected to be A$ 7.6 billion.

In releasing these revised estimates, the Treasurer announced the intention of the newly elected Government to return the budget to underlying balance by taking measures – primarily on the outlays side – to reduce the budget deficit by A$ 4 billion in FY 1996/97 and a further A$ 4 billion in FY 1997/98.

April

The Government announced a review into the performance and operations of the Australian National Railways Commission and the National Rail Corporation.

May

On 31 May 1996, the Government announced the establishment of an Inquiry into the financial system to report to the Treasurer by 31 March 1997. The Inquiry will undertake a stock-take of financial regulation, seek to establish a common regulatory framework for overlapping financial products, and propose ways and means of dealing constructively with further financial innovation. The Inquiry will make recommendations on the nature of the regulatory arrangements that will best ensure an efficient, responsive, competitive and flexible financial system, consistent with financial stability, prudence, integrity and fairness.

State and Commonwealth Ministers agreed that ‘‘user choice’’ for off-the-job training for apprentices and trainees will be fully implemented by 1 January 1998 (with progressive implementation during 1997).

The Government introduced the Workplace Relations and Other Legislation Amendment Bill 1996 into Parliament.

June

At the Premiers’ Conference it was agreed that financial assistance grants to the States and Territories would continue to be increased for inflation and population growth

over the period FY 1996/97 to FY 1998/99. This real per capita guarantee is conditional on the States and Territories complying with their obligations under the Agreement to Implement the National Competition Policy and Related Reforms.

In recognition of the fiscal challenge facing the Government, the States and Territories agreed to contribute A$ 619 million in FY 1996/97, A$ 640 million in FY 1997/98 and A$ 300 million in FY 1998/99. The need for such payments to be made by the States and Territories will be reviewed annually at future Premiers' Conferences in the light of the Commonwealth's fiscal position.

The Premiers' Conference also agreed that from 11 June 1996, wholesale sales tax will apply to general purpose road vehicles provided wholly or partly for private use as part of remuneration by Commonwealth, State, Territory and local governments.

The Council of Australian Governments agreed to an agenda of major reform to fundamentally reshape Commonwealth, State and Territory activity in the health and community services and housing fields. The Prime Minister, Premiers and Chief Ministers also signed the Trans-Tasman Mutual Recognition Arrangement. This provides for goods acceptable for sale in Australia to be sold in New Zealand (and vice versa), and for persons registered to practise an occupation in Australia to practise an equivalent occupation in New Zealand (and vice versa). It is intended to commence in 1997.

Loan Council agreed on Loan Council Allocations nominated by the Commonwealth and each State and Territory for FY 1996/97. Allowing for the expected fiscal tightening at the Commonwealth level, the aggregate of the nominations was considered to be consistent with macroeconomic policy objectives.

Legislation giving effect to the restructuring of the Housing Loans Insurance Corporation (HLIC), the *Housing Loans Insurance Corporation (Transfer of Assets and Abolition) Act 1996,* was passed in the Winter sitting period of Parliament and given Royal Assent on 24 June 1996. The restructuring is aimed at addressing the factors that have impeded the successful completion of past sale processes by placing the Corporation (a wholly Government-owned mortgage insurer) on a more comparable footing to its private sector competitors.

Ministerial Council on Road Transport agreed to intensify road reform to ensure the National Road Transport Law is enacted federally by January 1998. All states are to introduce national charges for heavy vehicles in 1996.

July

On 19 July 1996, the Commonwealth Government sold its 50.4 per cent shareholding, representing 499.1 million shares, in the Commonwealth Bank of Australia (CBA). The method of sale was a Public Share Offer (PSO), combined with the CBA buying back 100 million shares from the Commonwealth Government. The sale ended over 80 years of Commonwealth Government ownership of the CBA.

On 21 July, provisions of the Trade Practices Act, and complementary State and Territory legislation, applied the competitive conduct rules to all business activities in the Australian economy (except in the state of Western Australia).

Recognising that persons subject to the competitive conduct rules for the first time may need a period of adjustment in order to change any anti-competitive behaviour, those persons have a further 12 months from 21 July 1996 before being liable to pecuniary penalties.

The complementary State legislation for Western Australia has not yet been passed. Passage is expected in November 1996. Until then unincorporated businesses in Western Australia are largely outside the coverage of the competitive conduct rules.

The ACTU announced its intention to make a ''Living Wage'' claim with the AIRC for a safety net increase of a minimum of A$ 20 a week. Hearings of the claim by the Commission are set down for December 1996.

On 31 July 1996, the Reserve Bank announced a reduction of ½ percentage point in the cash rate to around 7 per cent.

August

The Treasurer delivered the FY 1996/97 Budget. It foreshadowed an underlying budget deficit for FY 1996/97 of A$ 5.6 billion (1.1 per cent of GDP). The improvement of A$ 4.7 billion on the underlying deficit of A$ 10.3 billion in FY 1995/96, reflected the net effect of savings measures amounting to A$ 3.9 billion in 1996-97 and A$ 7.2 billion in FY 1998/99.

The Government also announced its intention to introduce legislation to give effect to the Charter of Budget Honesty. The Charter aims to facilitate better fiscal outcomes by improving the discipline, transparency and accountability of fiscal policy. The Charter will require fiscal policy to be formulated against the principles of sound fiscal management.

On 14 August 1996 Mr Ian Macfarlane was appointed Governor of the Reserve Bank of Australia for a term of seven years commencing 18 September 1996. In announcing the appointment, the Treasurer and Mr Macfarlane issued the Statement on the Conduct of Monetary Policy recording a common understanding on key aspects of Australia's monetary policy framework. This was designed to clarify respective roles and responsibilities.

It is expected that the Statement will contribute to a better understanding of the nature of the relationship between the Reserve Bank and the Government, the objectives of monetary policy, the mechanisms for ensuring transparency and accountability in the way policy is conducted, and the independence of the Bank.

September

The Government announced the Single Aviation Market with New Zealand to commence from 1 November 1996.

November

On 6 November 1996, the Reserve Bank announced a reduction of ½ percentage point in the cash rate to around 6½ per cent.

STATISTICAL ANNEX AND STRUCTURAL INDICATORS

Table A. **Selected background statistics**

	Average 1986-95	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
A. Percentage changes from previous year at constant 1989/90 prices											
Private consumption	3.1	1.0	2.5	4.1	5.3	2.6	0.9	3.8	2.4	4.3	4.2
Gross fixed capital formation	1.9	–2.2	3.5	7.7	9.2	–8.1	–9.2	1.5	2.5	12.2	1.8
Public	0.8	8.9	–3.5	–12.4	8.2	1.7	4.9	–1.3	0.3	1.4	–0.3
Private	2.1	–3.5	4.4	10.2	9.2	–9.1	–10.8	1.9	2.8	13.5	2.0
Residential	2.5	–8.2	1.3	22.1	2.4	–11.5	–4.1	10.6	11.0	11.0	–9.9
Non-residential construction	3.2	9.5	15.3	17.0	11.0	–8.5	–20.3	–12.3	–3.5	6.6	17.1
Machinery and equipment	3.7	–4.9	4.4	11.8	12.2	–14.1	–11.0	6.1	3.4	21.9	6.8
Public enterprises	–1.6	–3.2	–0.2	–19.0	15.7	7.9	–10.7	–7.8	–10.7	6.0	6.4
GDP [1]	3.0	1.8	4.7	4.3	4.2	1.4	–1.6	2.9	4.0	4.9	3.2
GDP price deflator	4.4	6.9	7.5	8.3	7.5	4.4	2.3	1.3	1.3	1.3	2.7
Employment	2.0	3.6	2.2	3.7	4.7	1.5	–2.1	–0.7	0.4	3.1	4.1
Compensation of employees (current prices)	7.2	10.3	8.3	11.0	12.8	8.7	1.2	3.3	3.5	5.8	7.4
Productivity (GDP/employment)	0.9	–1.7	2.4	0.5	–0.5	0.0	0.6	3.6	3.6	1.7	–0.9
Unit labour costs (compensation/GDP)	4.1	8.3	3.4	6.5	8.2	7.2	2.8	0.4	–0.5	0.9	4.1
B. Percentage ratios											
Gross fixed capital formation as % of GDP at constant prices	22.6	23.7	23.4	24.2	25.3	22.9	21.1	20.9	20.6	22.0	21.7
Stockbuilding as % of GDP at constant prices	0.2	–0.4	0.0	0.8	1.4	0.4	–0.9	–0.4	0.6	0.2	0.9
Foreign balance as % of GDP at constant prices	0.6	0.5	1.9	0.0	–2.6	–0.5	2.2	1.7	2.2	1.0	0.0
Compensation of employees as % of GDP at current prices	49.5	51.2	49.2	48.4	48.7	50.0	50.3	49.8	49.0	48.8	49.4
Direct taxes as per cent of household income	21.4	22.6	24.1	23.8	23.1	21.6	20.6	19.1	19.5	19.5	20.4
Unemployment as per cent of total labour force	8.6	8.0	8.0	7.1	6.1	7.0	9.5	10.7	10.9	9.7	8.6
C. Other indicator											
Current balance (US$ billion)	–12.4	–9.2	–7.4	–10.0	–17.4	–14.8	–9.5	–10.0	–10.0	–16.7	–18.7

1. Income-based measure.

Source: Australian Bureau of Statistics and OECD Secretariat.

Table B. **Gross domestic product**

A$ million

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
	Current prices									
Private consumption [1]	150 000	165 589	184 348	206 911	225 110	235 537	248 887	260 127	275 077	293 856
Government current expenditure [1]	47 400	50 385	54 748	59 134	65 099	69 644	72 791	75 293	78 758	82 258
Gross fixed capital formation [2]	61 310	68 054	77 664	90 379	85 685	78 106	79 404	83 431	93 902	96 002
Private	42 835	49 094	61 095	70 568	64 086	57 238	59 570	64 676	74 680	75 973
Public enterprises	10 726	11 139	9 423	11 679	13 056	11 827	10 978	9 828	10 302	10 981
Government	7 749	7 821	7 146	8 132	8 543	9 041	8 856	8 927	8 920	9 048
Change in stocks	–1 332	639	2 950	5 099	173	–2 838	–1 389	1 657	1 558	4 525
Exports of goods and services	40 922	47 533	53 457	58 303	63 849	68 409	73 288	79 768	83 135	92 168
Imports of goods and services	47 060	49 862	56 145	66 780	66 815	66 008	73 696	81 291	88 696	98 974
Statistical discrepancy	–889	–507	1 250	3 408	4 342	–2 759	–3 119	–1 472	–12	555
Gross domestic product at purchasers' values [3]	250 351	281 831	318 272	356 454	377 443	380 091	396 166	417 513	443 722	470 390
Indirect taxes less subsidies	29 092	34 780	38 793	42 581	44 822	44 611	44 850	47 903	54 351	58 076
Gross domestic product at factor cost	221 259	247 051	279 479	313 873	332 621	335 480	351 316	369 610	389 371	412 314
	Average 1989/90 prices									
Private consumption [1]	189 885	194 651	202 558	213 235	218 792	220 830	229 211	234 800	244 927	255 152
Government current expenditure [1]	56 041	57 155	58 946	60 444	63 177	64 844	65 850	66 063	69 501	71 631
Gross fixed capital formation [2]	76 043	78 673	84 712	92 468	84 973	77 134	78 308	80 292	90 053	91 676
Private	54 192	57 169	66 717	72 216	63 645	56 776	58 963	62 056	71 122	72 127
Public enterprises	12 768	12 742	10 317	11 941	12 879	11 498	10 596	9 465	10 035	10 681
Government	9 083	8 762	7 678	8 311	8 449	8 860	8 749	8 771	8 896	8 868
Change in stocks	–1 378	–91	2 879	5 166	1 380	–3 273	–1 483	2 233	705	3 776
Exports of goods and services	49 713	55 334	57 130	58 985	63 907	72 229	75 808	81 400	88 362	91 910
Imports of goods and services	47 959	48 876	57 001	68 484	65 890	64 323	69 315	72 946	84 094	92 107
Statistical discrepancy	–1 105	–591	1 355	3 541	4 287	–2 632	–2 955	–1 387	–2	543
Gross domestic product at purchasers' values [3]	321 240	336 255	350 579	365 355	370 626	364 809	375 424	390 455	409 452	422 581

1. Not adjusted for the impact of the introduction of Medicare from 1 February 1986, which had the effect of transferring certain expenditures on healthcare, formely included as private in consumption expenditure, to public consumption expenditure.
2. Not adjusted to remove the impact of the sale to the private sector of public sector assets under leaseback and similar arrangements.
3. Income-based measure.

Source: Australian Bureau of Statistics.

Table C. **Income and expenditure of households (including unincorporated enterprises)**

A$ million, current prices

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
Compensation of employees	128 095	138 666	153 908	173 604	188 758	191 116	197 621	204 586	216 571	232 549
Income from property and entrepreneurship	43 796	50 581	58 212	67 091	69 121	65 335	64 253	64 331	66 449	68 806
Farm	1 148	2 903	5 067	4 470	2 020	923	2 423	3 701	3 079	3 972
Non-farm	42 648	47 678	53 145	62 621	67 101	64 412	61 830	60 630	63 370	64 834
Current transfers from government	23 980	26 303	28 480	30 651	35 215	39 955	44 493	47 872	50 215	53 433
Grants to non-profit institutions	3 217	3 263	3 801	4 120	4 780	5 323	5 812	6 326	7 010	8 263
Third party insurance transfers	1 364	1 429	1 303	1 207	1 324	1 368	1 158	1 080	1 769	2 324
Unrequited transfers from overseas	1 982	2 478	2 973	3 545	3 430	3 632	3 039	2 048	2 180	2 702
Income	202 434	222 720	248 677	280 218	302 628	306 729	316 376	326 243	344 194	368 077
less: Income taxes	34 930	40 551	44 767	48 773	49 754	48 421	47 071	49 228	51 681	57 594
of which: Direct taxes paid on income	1 472	1 698	1 968	2 250	2 500	2 667	2 846	3 205	3 681	3 797
Consumer debt interest	4 133	4 532	4 920	7 250	7 690	6 206	4 431	3 628	3 368	4 070
Transferts paid by household	748	765	796	873	960	1 003	1 021	1 007	1 079	1 169
Disposable income	161 151	175 174	196 226	221 072	241 724	248 432	261 007	269 175	284 385	301 447
Consumption expenditure	150 000	165 589	184 348	206 911	225 110	235 537	248 887	260 127	275 077	293 856
Saving	11 151	9 585	11 878	14 161	16 614	12 895	12 120	9 048	9 308	7 591
as per cent of disposable income	6.9	5.5	6.1	6.4	6.9	5.2	4.6	3.4	3.3	2.5

Source: Australian Bureau of Statistics.

Table D. **Prices and wages**

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
	Index FY 1989/90 = 100									
Price deflators										
Gross domestic product	78.0	83.8	90.8	97.6	101.9	104.2	105.6	107.0	108.4	111.3
Private consumption	78.9	85.0	90.9	97.0	102.8	106.6	108.5	110.7	112.2	115.1
Gross fixed capital formation	80.6	86.4	91.6	97.7	100.8	101.2	101.4	103.9	104.3	104.7
Exports [1]	82.1	85.7	93.4	98.7	99.8	94.6	96.5	97.8	93.9	100.1
Imports [1]	98.1	101.9	98.7	97.4	101.4	102.6	106.2	111.3	105.4	107.4
Terms of trade [1]	83.8	84.1	94.8	101.3	98.4	92.2	90.9	87.9	89.1	93.3
Consumer price index [2]										
Total	76.9	83.3	89.4	96.2	103.2	106.5	107.6	109.5	111.5	116.4
Food	76.5	83.5	89.5	96.0	103.6	106.9	107.9	109.7	111.9	117.3
Award rates of pay, adult persons	83.1	86.5	91.4	97.0	103.0	91.4	97.0	103.0	106.9	109.5
Average weekly earnings, all employees	80.7	84.9	90.6	96.9	103.3	90.6	96.9	103.3	107.2	109.9

1. Goods and services.
2. Not adjusted for the effects of Medicare.

Source: Australian Bureau of Statistics, Reserve Bank, and OECD Secretariat.

Table E. **Balance of payments**

A$ million

	1990	1991	1992	1993	1994	1995
Exports, fob	50 379	53 887	57 665	62 018	64 115	71 053
Imports, fob	49 733	49 381	55 346	62 278	68 522	77 063
Trade balance	646	4 506	2 319	–260	–4 407	–6 010
Services, net	–21 220	–18 959	–17 520	–14 853	–18 521	–19 980
Non-factor services, credit	13 564	14 536	15 516	17 718	19 109	21 117
Non-factor services, debit	–16 954	–16 586	–17 932	–18 936	–20 094	–22 147
Non-factor services, net	–3 390	–2 050	–2 416	–1 218	–985	–1 030
Investment income, credit	4 560	4 223	5 147	5 942	6 381	8 176
Investment income, debit	–22 390	–21 132	–20 251	–19 577	–23 917	–27 126
Investment income, net	–17 830	–16 909	–15 104	–13 635	–17 536	–18 950
Balance on goods and services	–20 574	–14 453	–15 201	–15 113	–22 928	–25 990
Transfers, credit	4 601	4 641	3 892	2 863	2 944	3 594
Transfers, debit	–2 333	–2 345	–2 375	–2 462	–2 690	–2 851
Unrequited transfers, net	2 268	2 296	1 517	401	254	743
Balance on current account	–18 306	–12 157	–13 684	–14 712	–22 674	–25 247
Net capital movements [1]	14 826	14 709	16 701	15 299	22 672	16 786
Net errors and omissions	3 740	–2 511	–2 710	–513	230	8 175
Changes in official reserves	–2 588	7 122	11 259	11 188	19 828	–939

1. Including changes in official reserves.

Source: Australian Bureau of Statistics, *Balance of Payments*, and OECD, *Main Economic Indicators.*

Table F. **Foreign trade by commodity**[1]

	1980	1985	1990	1993	1994	1980	1985	1990	1993	1994
	US$ million					Per cent of total				
SITC sections:										
Total exports, fob	22 072	22 675	39 589	42 702	47 474	100.0	100.0	100.0	100.0	100.0
Food and live animals	7 280	5 409	7 665	8 458	9 784	33.0	23.9	19.4	19.8	20.6
Beverages and tobacco	49	51	215	339	400	0.2	0.2	0.5	0.8	0.8
Crude materials, inedible, except fuels	6 571	6 275	10 196	8 669	9 576	29.8	27.7	25.8	20.3	20.2
Mineral fuels, lubricants and related materials	2 400	5 678	7 602	8 225	7 903	10.9	25.0	19.2	19.3	16.6
Animals and vegetable oils, fats and waxes	115	94	100	144	165	0.5	0.4	0.3	0.3	0.3
Chemicals and related products, n.e.s.	482	397	1 002	1 484	1 798	2.2	1.8	2.5	3.5	3.8
Manufactured goods classified chiefly by material	2 585	2 243	4 763	5 143	6 047	11.7	9.9	12.0	12.0	12.7
Machinery and transport equipment	1 170	1 086	3 124	4 766	5 551	5.3	4.8	7.9	11.2	11.7
Miscellaneous manufactured articles	384	443	1 008	1 355	1 626	1.7	2.0	2.5	3.2	3.4
Commodities and transactions not classified according to kind	1 035	998	3 914	4 119	4 624	4.7	4.4	9.9	9.6	9.7
Total imports, cif	20 194	23 316	38 937	42 439	49 923	100.0	100.0	100.0	100.0	100.0
Food and live animals	766	943	1 497	1 628	1 893	3.8	4.0	3.8	3.8	3.8
Beverages and tobacco	184	189	316	309	367	0.9	0.8	0.8	0.7	0.7
Crude materials, inedible, except fuels	876	690	1 096	1 045	1 208	4.3	3.0	2.8	2.5	2.4
Mineral fuels, lubricants and related materials	2 762	1 599	2 192	2 607	2 391	13.7	6.9	5.6	6.1	4.8
Animals and vegetable oils, fats and waxes	79	76	98	134	154	0.4	0.3	0.3	0.3	0.3
Chemicals and related products, n.e.s.	1 826	1993	4 006	4 656	5 385	9.0	8.5	10.3	11.0	10.8
Manufactured goods classified chiefly by material	3 552	3 774	5 960	6 215	7 343	17.6	16.2	15.3	14.6	14.7
Machinery and transport equipment	7 259	9 928	17 509	18 693	23 050	35.9	42.6	45.0	44.0	46.2
Miscellaneous manufactured articles	2 447	3 099	5 410	6 399	7 501	12.1	13.3	13.9	15.1	15.0
Commodities and transactions not classified according to kind	442	1 027	854	753	631	2.2	4.4	2.2	1.8	1.3

1. General trade.

Source: OECD, *Foreign Trade Statistics*, Series A.

Table G. **Foreign trade by area**[1]

US$ million

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
Exports, fob										
OECD Europe	3 648	4 783	5 277	5 815	6 242	5 953	6 208	5 529	5 593	6 458
of which: United Kingdom	818	1 139	1 166	1 323	1 399	1 330	1 664	1 910	1 710	1 828
OECD North America	2 784	3 465	4 136	4 469	5 011	4 901	4 474	4 335	4 265	4 295
Japan	6 065	6 789	8 882	9 761	10 205	11 537	10 713	10 440	11 613	12 181
New Zealand	1 032	1 511	1 629	1 891	1 956	2 043	2 241	2 465	3 151	3 840
Far East	5 125	6 273	8 468	9 830	11 355	13 068	14 582	15 258	17 570	20 888
Other non-OECD countries	3 502	3 280	3 811	4 459	3 860	3 588	3 383	3 615	3 963	4 183
Non-specified	381	404	463	762	956	841	835	1 057	1 299	1 120
Total	22 541	26 508	32 670	36 991	39 589	41 934	42 439	42 707	47 474	52 967
Imports, cif										
OECD, Europe	6 887	7 725	9 428	10 856	10 651	9 618	9 966	10 049	12 585	15 576
of which: United Kingdom	1 792	1 950	2 467	2 785	2 701	2 400	2 412	2 410	2 979	3 451
OECD North America	5 752	6 381	8 043	10 320	10 308	10 179	9 993	9 896	11 933	13 808
Japan	5 348	5 321	6 713	8 319	7 307	6 807	7 373	8 072	8 866	8 877
New Zealand	918	1 109	1 462	1 661	1 715	1 768	1 876	2 065	2 482	2 673
Far East	3 572	4 806	5 935	7 299	6 505	7 733	8 738	9 230	10 893	12 774
Other non-OECD countries	1 413	1 634	1 700	2 398	2 384	2 545	2 741	3 054	3 051	3 676
Non-specified	25	32	55	91	62	70	66	71	70	78
Total	23 919	27 010	33 339	40 948	38 937	38 723	40 755	42 439	49 923	57 465

1. General trade.

Source: OECD, *Foreign Trade Statistics*, Series A.

Table H. **Production structure and performance indicators**

Fiscal years[1]

	1975	1980	1985	1990	1994	1975	1980	1985	1990	1994
	GDP share					Employment share				
A. Production structure (constant prices)										
Tradeables										
Agriculture	4.9	3.9	4.3	4.3	3.3	6.8	6.6	6.2	5.6	5.0
Mining and quarrying	3.8	3.5	4.2	4.5	4.3	1.4	1.4	1.5	1.2	1.2
Manufacturing	18.3	17.9	15.5	15.1	15.5	21.6	19.7	16.5	14.7	13.8
Non-tradeables										
Electricity, gas and water	2.8	2.9	3.1	3.4	3.2	1.8	2.1	2.1	1.3	1.1
Construction	7.9	7.9	7.2	6.9	6.5	8.7	7.7	7.0	7.4	7.3
Trade, restaurants and hotels	21.7	20.8	19.5	18.5	19.0	19.8	20.1	23.4	24.9	25.5
Transport, storage and communication	5.7	6.7	7.1	7.8	8.9	7.8	7.4	7.5	6.9	6.6
Finance, insurance, real estate and business services	19.3	20.7	21.5	23.1	21.9	7.4	8.5	10.9	12.4	13.2
Community, social and personal services	11.9	12.8	12.8	13.5	13.7	19.9	22.1	20	20.9	21.9
	Productivity growth[2]					Investment share				
B. Economic performance (constant prices)										
Agriculture	–	–2.3	4.2	2.6	–2.1	9.3	10.7	8.7	5.5	5.0
Mining and quarrying	–	–0.5	2.8	6.6	4.7	6.7	6.3	6.7	8.2	11.9
Manufacturing	–	2.8	2.1	2.0	4.6	15.3	13.8	12.0	14.0	14.4
Electricity, gas and water	–	0.8	2.4	11.0	6.3	11.9	13.9	11.3	6.8	6.3
Construction	–	3.8	1.7	–1.8	0.9	2.8	3.4	4.2	4.4	3.1
Trade, restaurants and hotels	–	0.3	–2.9	–2.0	2.1	8.8	8.7	9.9	7.7	10.2
Transport, storage and communication	–	5.9	2.2	3.9	7.1	15.1	13.0	13.4	14.2	15.1
Finance, insurance, real estate and business services	–	0.0	–2.7	–0.9	–0.7	10.6	9.5	12.8	20.4	13.6
Community, social and personal services	–	0.8	3.3	0.4	1.5	4.7	5.9	6.7	7.8	9.0
	1980	1987	1988	1989	1990	1991	1992	1993	1994	1995
C. Other indicators (current prices)										
Effective rate of protection, manufacturing	22	19	19	17	16	14	13	12	10	. .
R&D in manufacturing/manufacturing GDP	. .	1.6	1.7	1.8	1.8	1.9	2.2	. .	. .	. .
Levels of net foreign debt[3]/GDP	5.6	32.6	32.2	35.0	35.9	37.9	39.3	41.0	38.1	39.9
Levels of foreign direct investment[3]/GDP	17.1	20.8	22.1	25.5	26.2	27.0	27.3	27.5	28.4	27.5

1. Beginning 1st July of the year indicated.
2. Average rate of growth between periods.
3. At 30th June of the year indicated.

Source: Australian Bureau of Statistics; Reserve Bank of Australia, *Bulletin;* OECD, *National Accounts* and *Main Science and Technology Indicators.*

Table I. **Labour market indicators**

	A. Evolution					
	Peak	Trough	1985	1990	1994	1995
Standardised unemployment rate	1993: 10.8	1981: 5.7	8.2	6.9	9.7	8.5
Unemployment rate						
Total	1993: 10.9	1981: 5.7	8.2	6.9	9.7	8.5
Male	1992: 11.3	1981: 4.7	7.8	6.7	10.0	8.8
Female	1983: 10.4	1990: 7.2	8.8	7.2	9.4	8.0
Youth [1]	1992: 19.5	1989: 10.4	14.3	13.2	16.2	14.4
Share of long-term unemployment [2]	1993: 36.7	1982: 19.0	30.9	21.6	36.0	32.0
Non-farm vacancies (thousand)	1989: 68.2	1983: 17.6	54.0	48.3	46.6	46.8
Average hours worked in manufacturing, weekly	1970: 39.5	1993: 35.6	37.0	38.2	34.7	36.0
Overtime hours per week, non-farm	1989: 1.5	1984: 1.1	1.2	1.3	1.3	1.2

	B. Structural of institutional characteristics					
	1970	1980	1985	1990	1994	1995
Labour force (percentage change) [3]	. .	2.0	1.7	3.0	1.0	2.9
Participation rate [4]						
Total	61.2	61.0	60.2	63.8	63.0	63.7
Male	83.0	77.9	75.2	75.7	73.7	74.0
Female	39.6	44.7	45.7	52.2	53.6	53.8
Employment by sector						
Per cent of total						
Agriculture	8.0	6.5	6.1	5.6	5.1	5.0
Industry	36.4	30.8	27.6	25.4	23.5	22.9
Services	55.6	62.7	66.2	69.0	71.4	72.1
Percentage change [3]						
Agriculture	. .	–0.6	0.4	1.2	–1.9	1.8
Industry	. .	–0.1	–1.2	1.5	–1.7	1.8
Services	. .	2.8	2.4	4.2	1.0	5.2
Total	. .	1.5	1.2	3.2	0.2	4.2
Part-time employment, per cent of total employed	10.6	16.4	18.2	21.3	24.1	24.5
Non-wage labour cost [5]	3.4	5.9	7.9	8.3	8.9	9.1

1. People between 15 and 24 years as a percentage of the labour force of the same age group.
2. People looking for a job since one year or more as a percentage of total unemployment.
3. Average rate of growth between periods.
4. Labour force as a percentage of the corresponding population aged between 15 and 64 years.
5. Total social contributions as a percentage of total compensation.

Source: Australian Bureau of Statistics; OECD, *Labour Force Statistics.*

Table J. **The public sector**

	1973	1980	1985	1990	1994	1995
Budget indicators: General government accounts (per cent of GDP)						
Current receipts (excluding interests)	23.6	27.7	30.5	31.9	29.7	31.3
Non-interest expenditures	23.9	29.5	32.3	29.9	32.5	32.4
Primary budget balance	–0.3	–1.4	–1.8	2.0	–2.8	–1.1
Net interest (including net capital transfers)	–0.1	–0.4	–1	–1.4	–1.2	–1.1
General government budget balance	–0.4	–1.8	–2.8	0.6	–4.0	–2.2
Structure of expenditure and taxation (per cent of GDP)						
Government expenditure	24.1	30.7	35.2	33.9	36.3	36.2
Transfers	6.9	9.5	11.2	10.9	13.2	13.4
Subsidies	1.1	1.5	1.8	1.3	1.4	1.3
General expenditure [1]						
Education	3.6	4.5	4.3	3.8	3.7	. .
Transportation	1.5	1.6	1.6	1.6	1.4	. .
Health	2.0	3.1	3.2	3.2	3.0	. .
Tax receipts [1]	25.3	28.5	30	30.8	29.9	. .
Personal income tax	10.3	12.5	13.6	13.2	12.0	. .
Corporate tax	3.8	3.5	2.8	4.4	4.2	. .
Payroll tax	1.3	1.4	1.4	1.9	2.0	. .
Taxes on goods and services	7.5	8.8	9.8	8.6	8.9	. .
of which; Specific taxes on consumption	4.7	6.4	6.1	4.7	4.6	. .
Other indicators						
Income tax elasticity [1]	1.62	1.2	1.1	0.1	1.4	. .
Income tax as a percentage of total tax [1]	40.5	44.0	45.2	43.0	40.1	. .
Net public debt as a percentage of GDP	. .	. .	. .	10.6	27.3	28.4

	Prior to		After
Tax rates (per cent)			
Top rate of income tax	49	1st Janyary 1990	47
Lower rate of income tax	21	1st January 1991	20
Corporation tax rate	39	1st July 1993	33

1. Fiscal years beginning 1st July of indicated years.
2. 1974 figure.

Source : Australian Bureau of Statistics; OECD, *National Accounts*, *Revenue Statistics of OECD Member Countries* and Secretariat estimates.

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ISSN 0376-6438